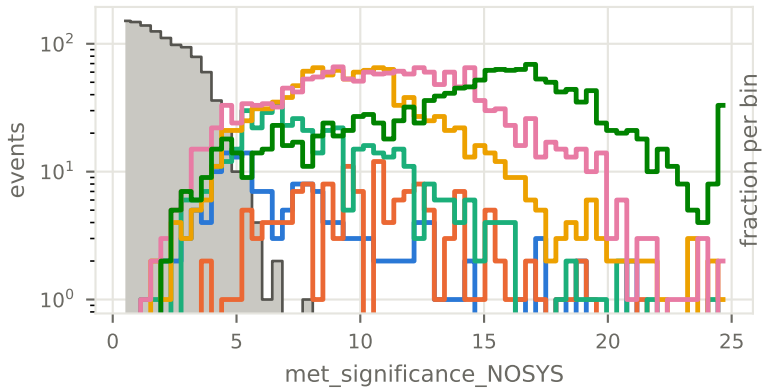
# met_significance_NOSYS

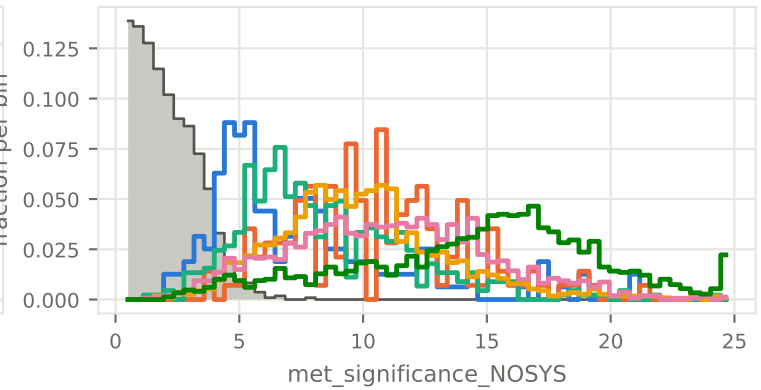
region: BARREL · rank 1 / 98

D_equal 0.898 (noise 0.051) D_pooled 0.919 min 0.831 (mH125_mS15) max 0.962 (mH125_mS55) bkg n = 1,089 sig n = 4,987

1a · each mass point vs background · raw

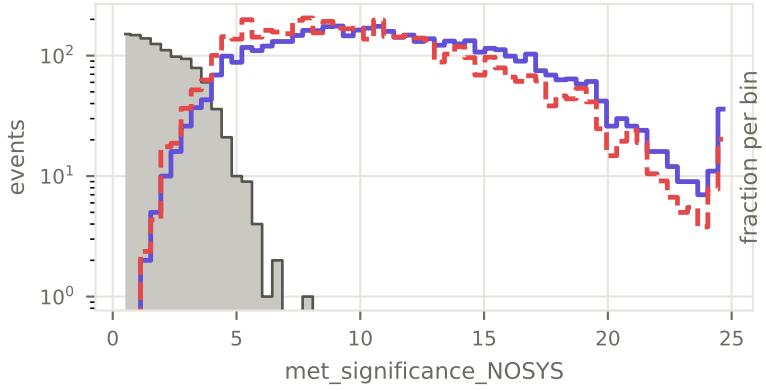



1b · each mass point vs background · normalized

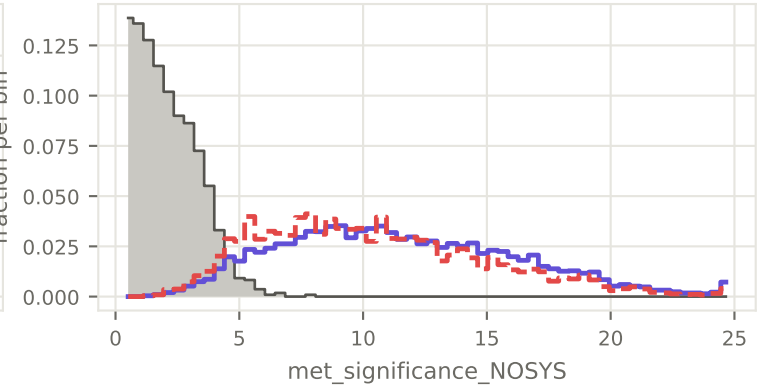



1 · each mass point
background n 1,089
mH125_mS15 D 0.831 n 159
mH125_mS55 D 0.962 n 142
mH200_mS50 D 0.887 n 449
mH400_mS100 D 0.936 n 1,144
mH600_mS150 D 0.918 n 1,608
mH1000_mS275 D 0.938 n 1,485

2a · combined signal vs background · raw

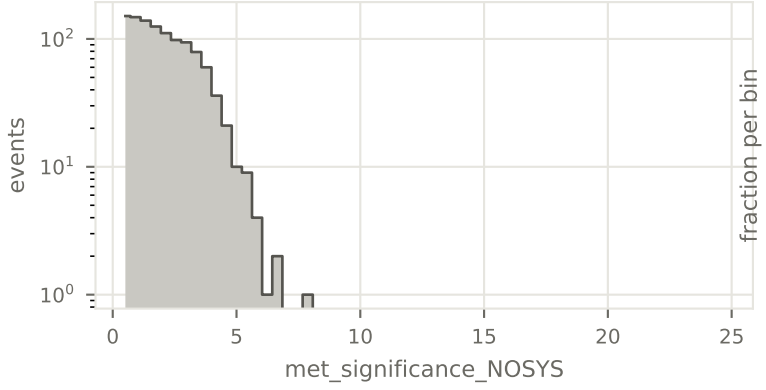



2b · combined signal vs background · normalized

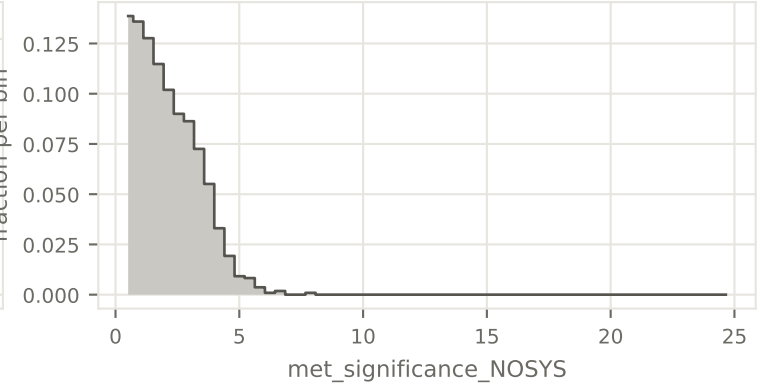



2 · combined signal
background n 1,089
combined, pooled D 0.919 n 4,987
equal weight per mass D 0.898

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0% −999 0.0%
background n 1,089

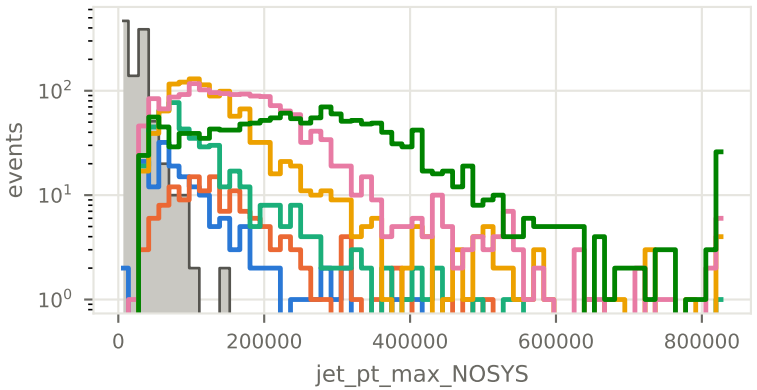
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

# jet_pt_max_NOSYS

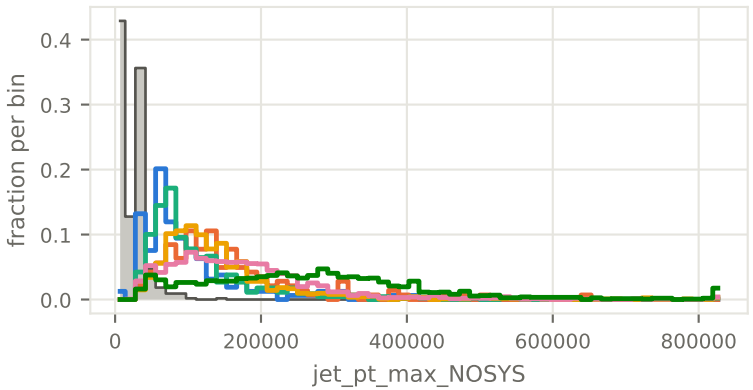
region: BARREL · rank 2 / 98

D_equal 0.873 (noise 0.051)  D_pooled 0.892  min 0.789 (mH125_mS15)  max 0.916 (mH400_mS100)  bkg n = 1,089  sig n = 4,987

1a · each mass point vs background · raw

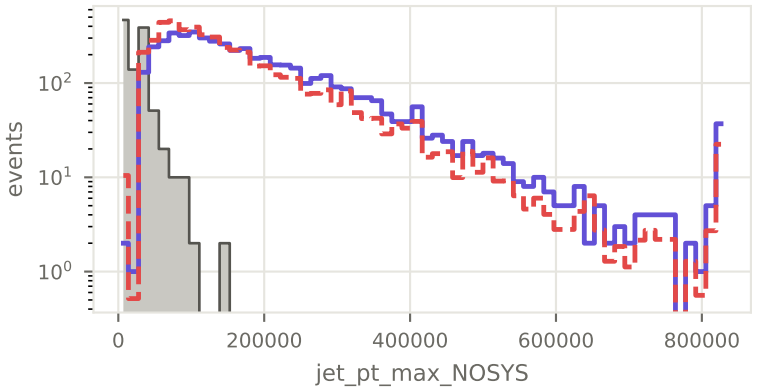



1b · each mass point vs background · normalized

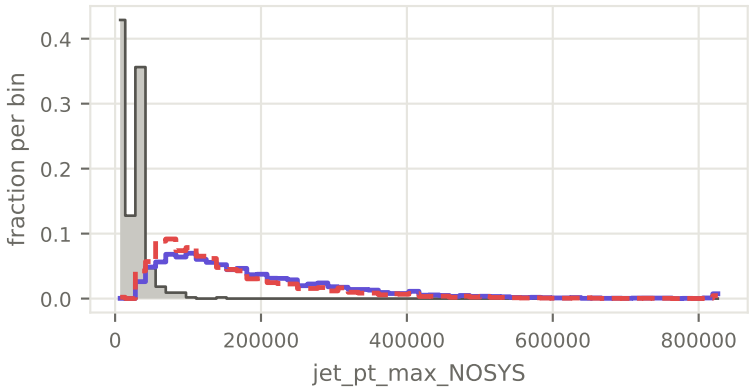



1 · each mass point
background n 1,089
mH125_mS15 D 0.789 n 159
mH125_mS55 D 0.909 n 142
mH200_mS50 D 0.877 n 449
mH400_mS100 D 0.916 n 1,144
mH600_mS150 D 0.890 n 1,608
mH1000_mS275 D 0.908 n 1,485

2a · combined signal vs background · raw

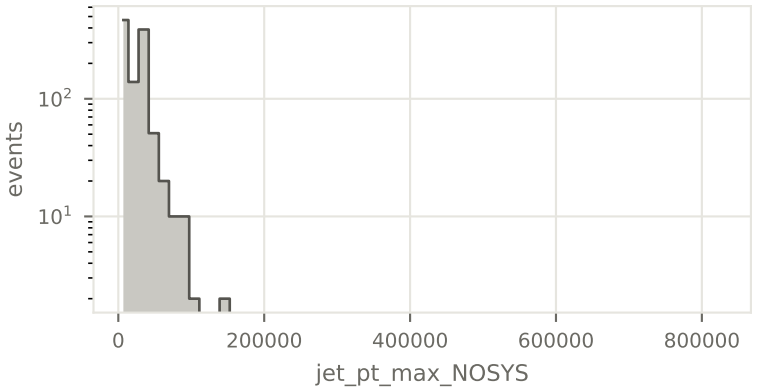



2b · combined signal vs background · normalized

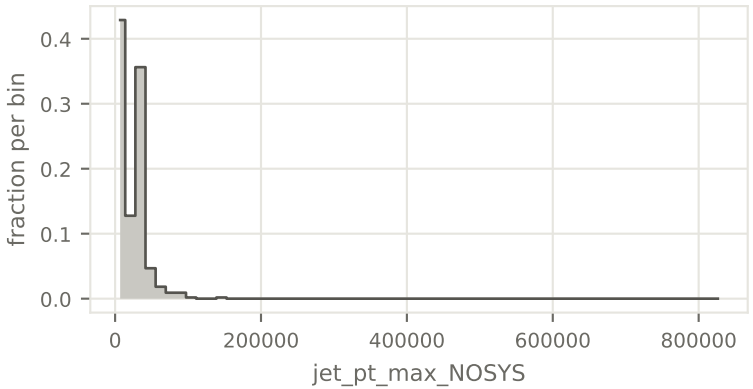



2 · combined signal
background n 1,089
combined, pooled D 0.892 n 4,987
equal weight per mass D 0.873

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0% −999 0.0%
background n 1,089

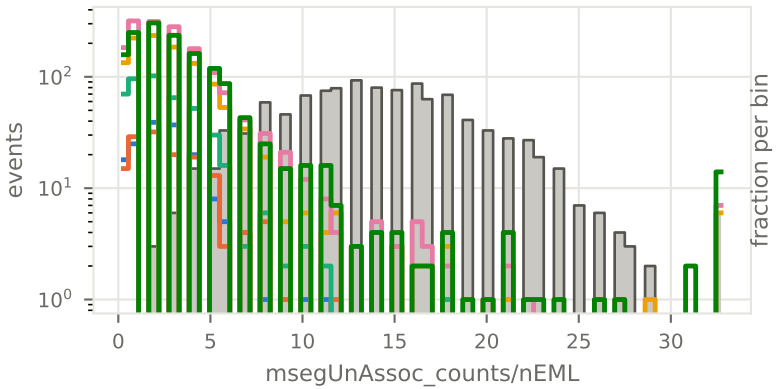
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

# msegUnAssoc_counts/nEML

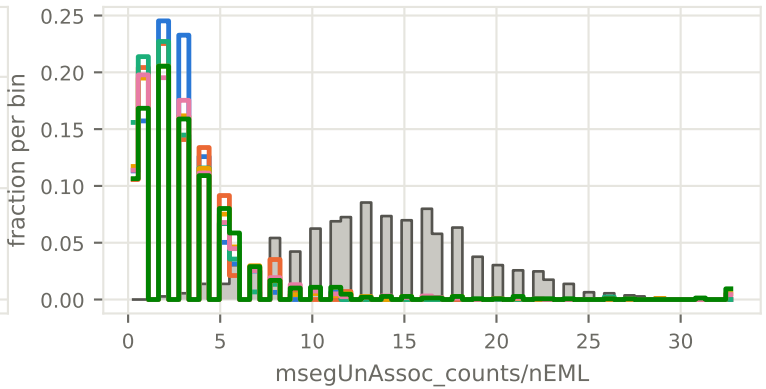
region: BARREL · rank 61 / 98

D_equal 0.858 (noise 0.051) D_pooled 0.843 min 0.821 (mH1000_mS275) max 0.894 (mH200_mS50) bkg n = 1,089 sig n = 4,987

1a · each mass point vs background · raw

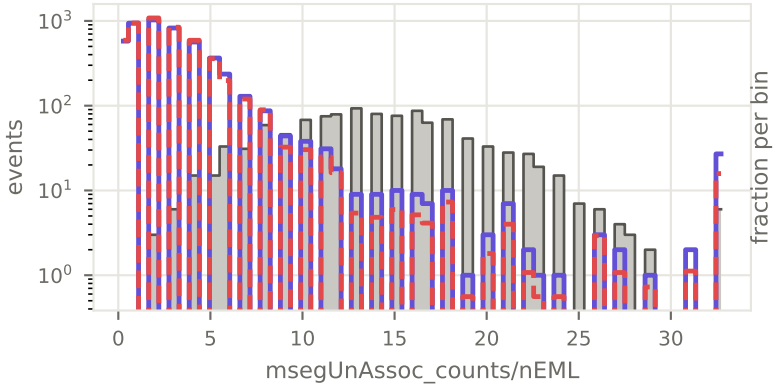



1b · each mass point vs background · normalized

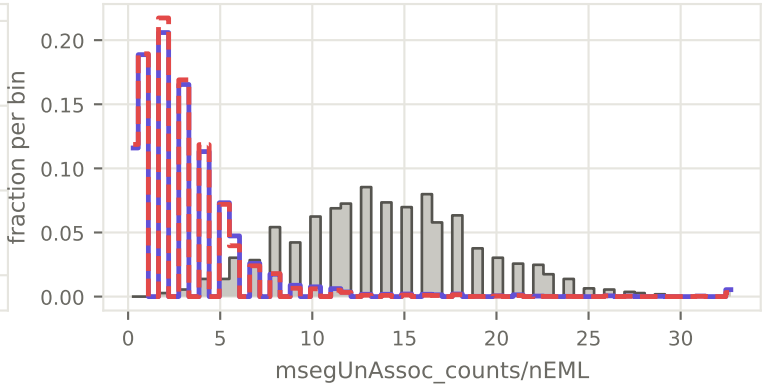



1 · each mass point
background n 1,089
mH125_mS15 D 0.890 n 159
mH125_mS55 D 0.866 n 142
mH200_mS50 D 0.894 n 449
mH400_mS100 D 0.851 n 1,144
mH600_mS150 D 0.840 n 1,608
mH1000_mS275 D 0.821 n 1,485

2a · combined signal vs background · raw

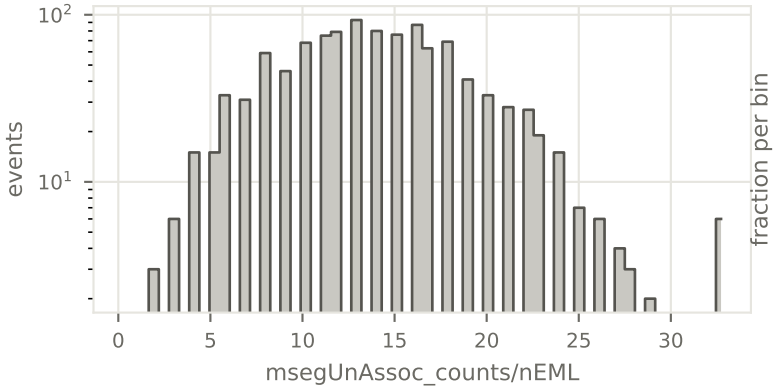



2b · combined signal vs background · normalized

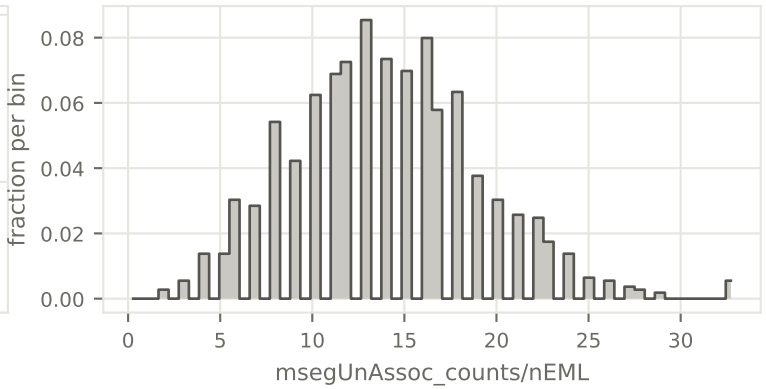



2 · combined signal
background n 1,089
combined, pooled D 0.843 n 4,987
equal weight per mass D 0.858

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0% −999 0.0%
background n 1,089

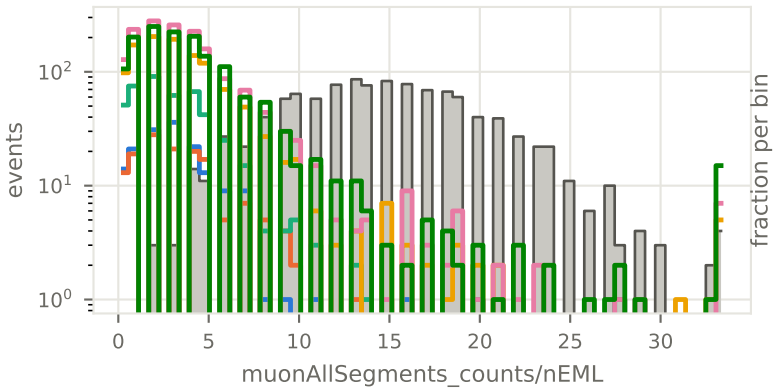
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

# muonAllSegments_counts/nEML

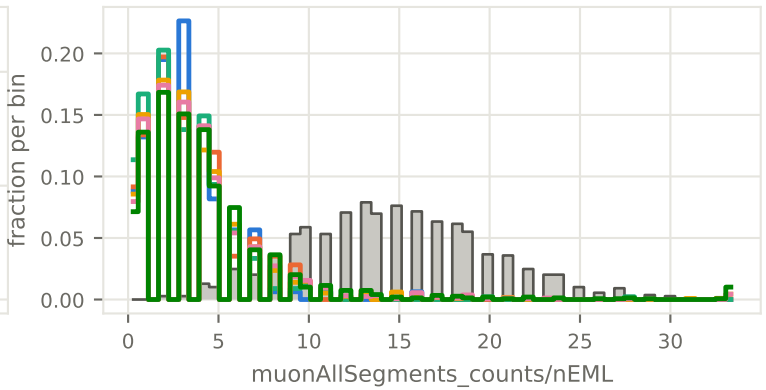
region: BARREL · rank 62 / 98

D_equal 0.848 (noise 0.051)  D_pooled 0.828  min 0.799 (mH1000_mS275)  max 0.901 (mH125_mS15)  bkg n = 1,089  sig n = 4,987

1a · each mass point vs background · raw

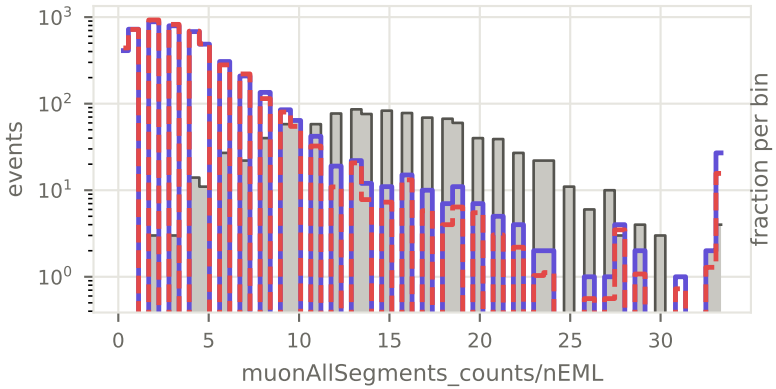



1b · each mass point vs background · normalized

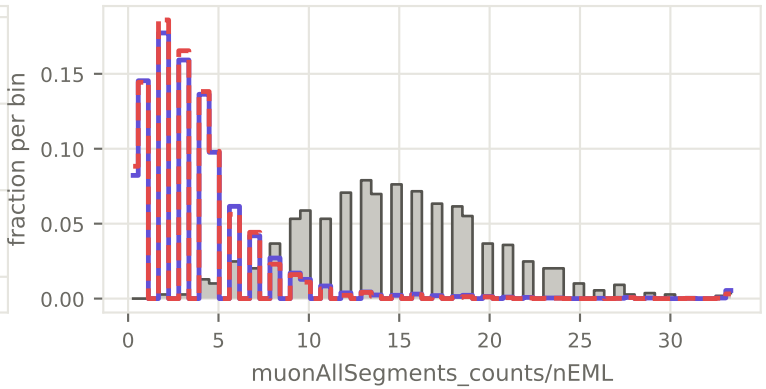



1 · each mass point
background n 1,089
mH125_mS15 D 0.901 n 159
mH125_mS55 D 0.842 n 142
mH200_mS50 D 0.880 n 449
mH400_mS100 D 0.839 n 1,144
mH600_mS150 D 0.825 n 1,608
mH1000_mS275 D 0.799 n 1,485

2a · combined signal vs background · raw

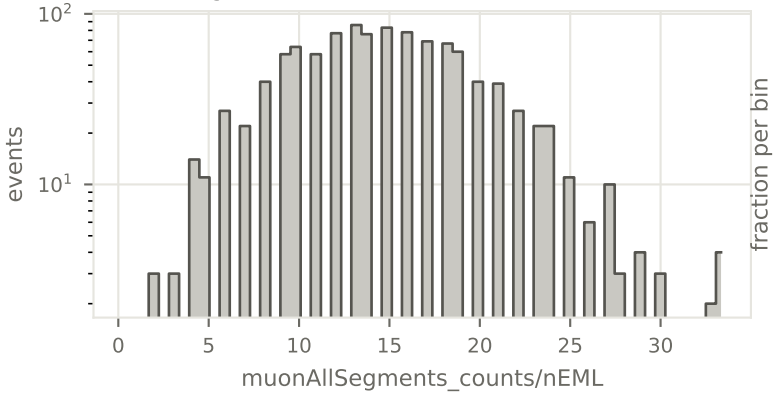



2b · combined signal vs background · normalized

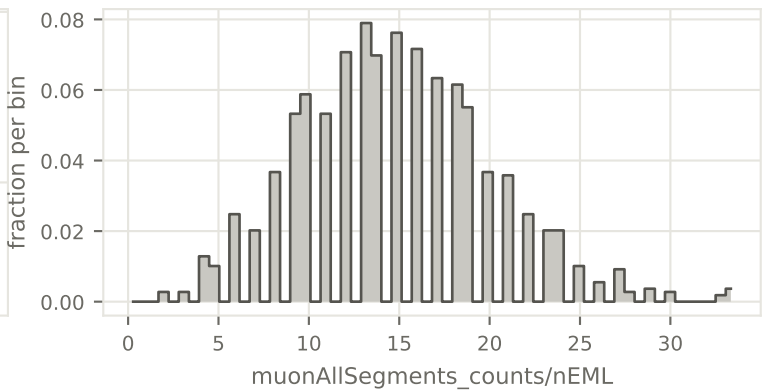



2 · combined signal
background n 1,089
combined, pooled D 0.828 n 4,987
equal weight per mass D 0.848

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0% −999 0.0%
background n 1,089

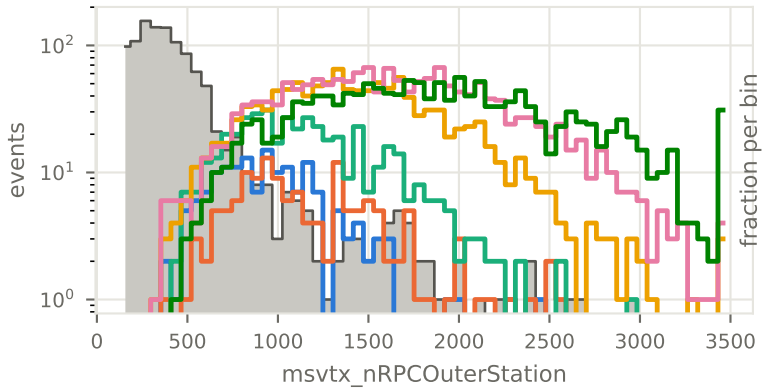
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

# msvtx_nRPCOuterStation

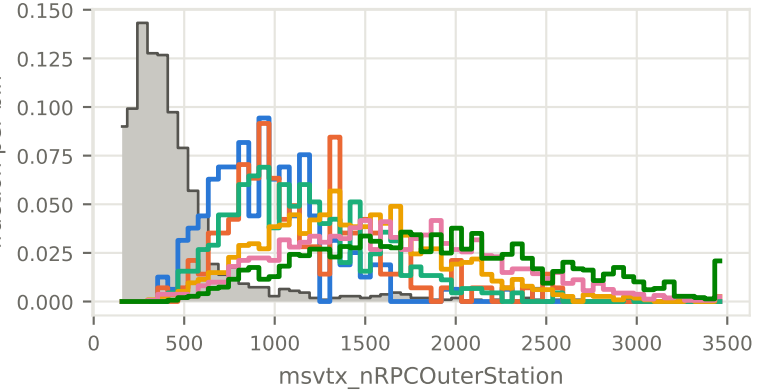
region: BARREL · rank 66 / 98

D_equal 0.815 (noise 0.051) D_pooled 0.843 min 0.740 (mH125_mS15) max 0.881 (mH1000_mS275) bkg n = 1,089 sig n = 4,987

1a · each mass point vs background · raw

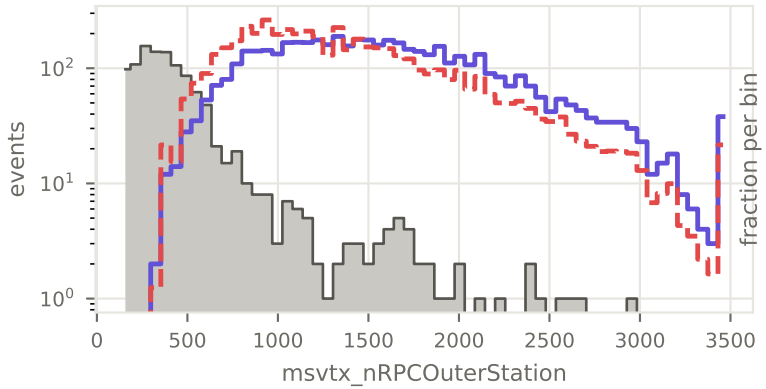



1b · each mass point vs background · normalized

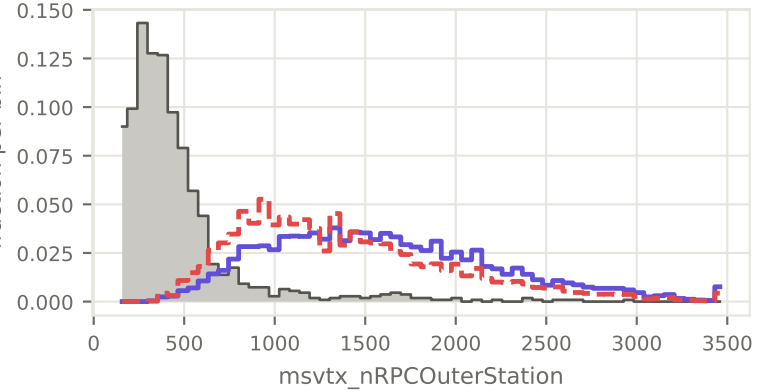



1 · each mass point
background n 1,089
mH125_mS15 D 0.740 n 159
mH125_mS55 D 0.817 n 142
mH200_mS50 D 0.807 n 449
mH400_mS100 D 0.841 n 1,144
mH600_mS150 D 0.859 n 1,608
mH1000_mS275 D 0.881 n 1,485

2a · combined signal vs background · raw

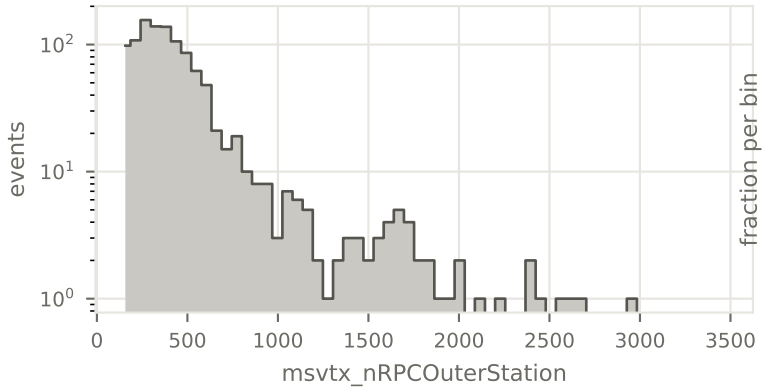



2b · combined signal vs background · normalized

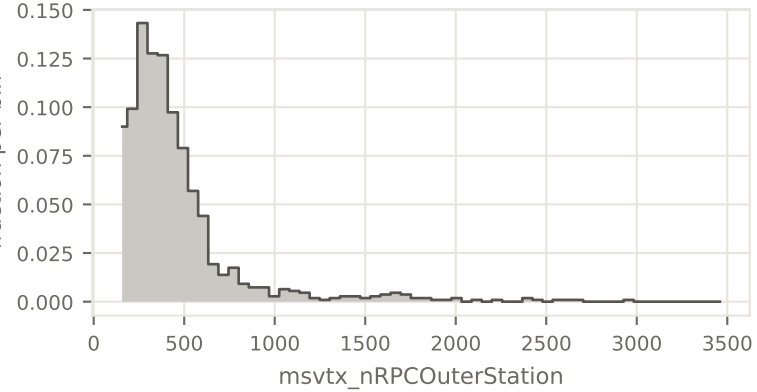



2 · combined signal
background n 1,089
combined, pooled D 0.843 n 4,987
equal weight per mass D 0.815

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0% −999 0.0%
background n 1,089

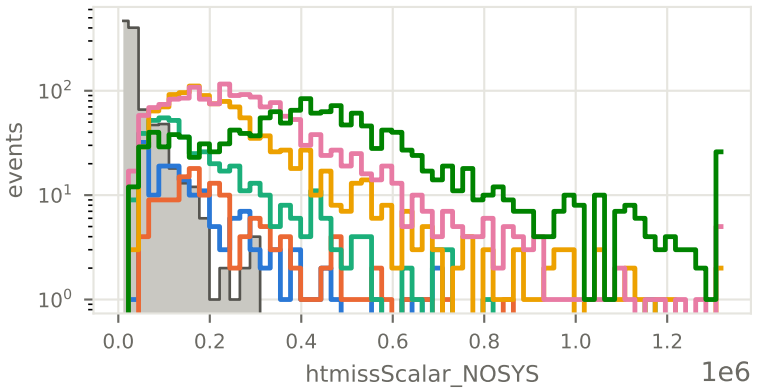
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

# htmissScalar_NOSYS

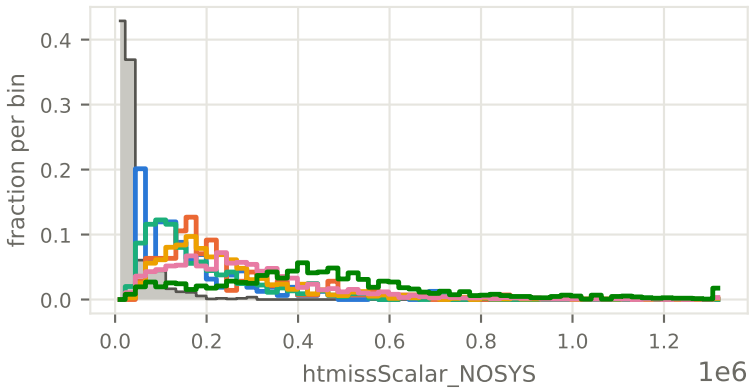
region: BARREL · rank 3 / 98

D_equal 0.798 (noise 0.051) D_pooled 0.818 min 0.798 (mH125_mS15) max 0.872 (mH1000_mS275) bkg n = 1,089 sig n = 4,987

1a · each mass point vs background · raw

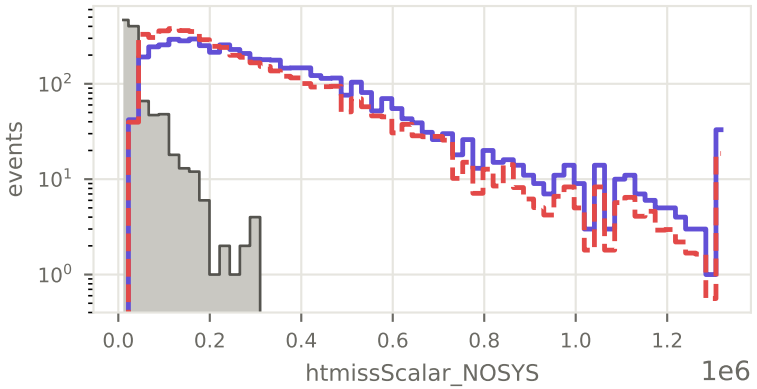



1b · each mass point vs background · normalized

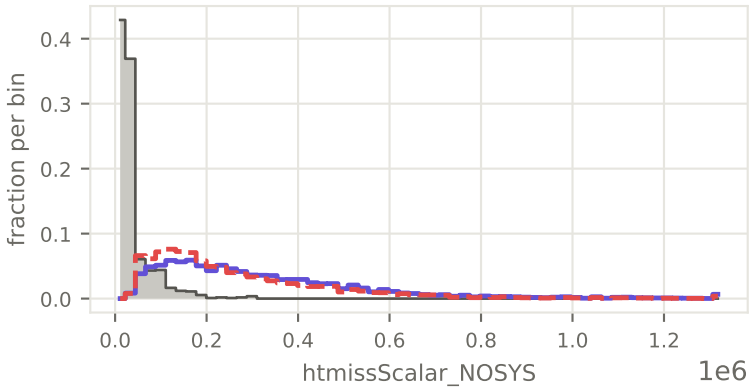



1 · each mass point
background n 1,089
mH125_mS15 D 0.798 n 159
mH125_mS55 D 0.839 n 142
mH200_mS50 D 0.798 n 449
mH400_mS100 D 0.837 n 1,144
mH600_mS150 D 0.820 n 1,608
mH1000_mS275 D 0.872 n 1,485

2a · combined signal vs background · raw

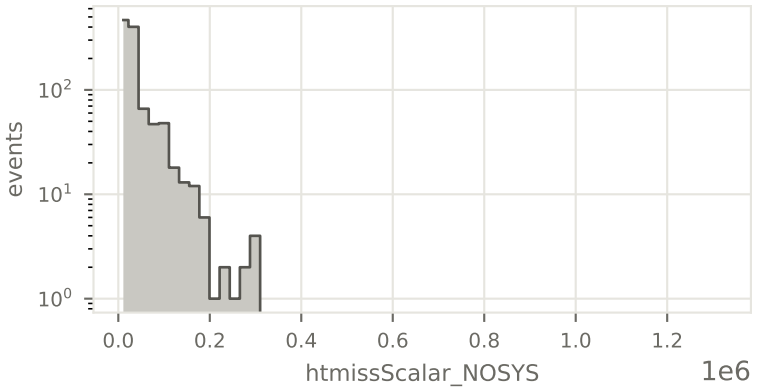



2b · combined signal vs background · normalized

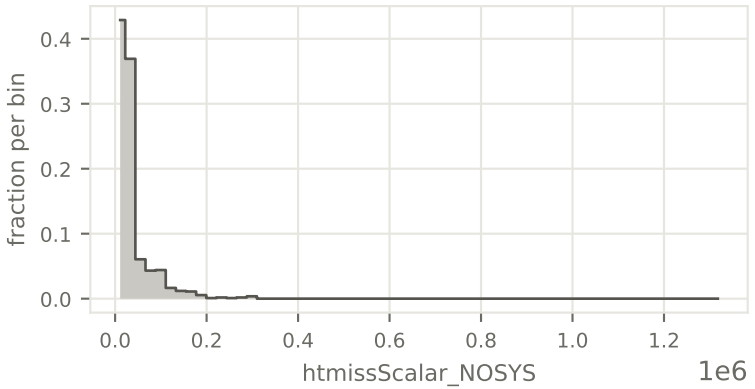



2 · combined signal
background n 1,089
combined, pooled D 0.818 n 4,987
equal weight per mass D 0.798

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0% −999 0.0%
background n 1,089

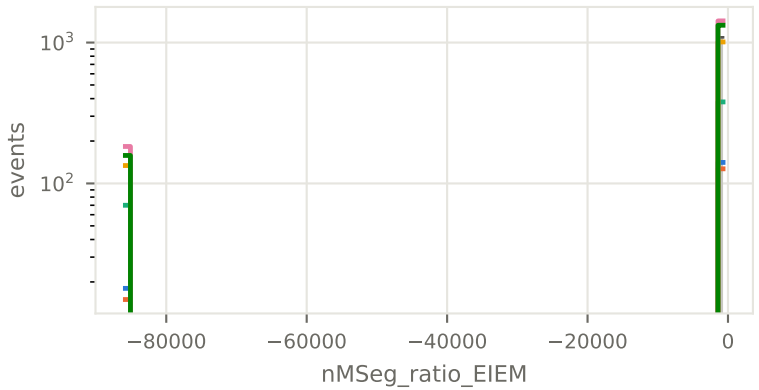
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

# nMSeg_ratio_EIEM

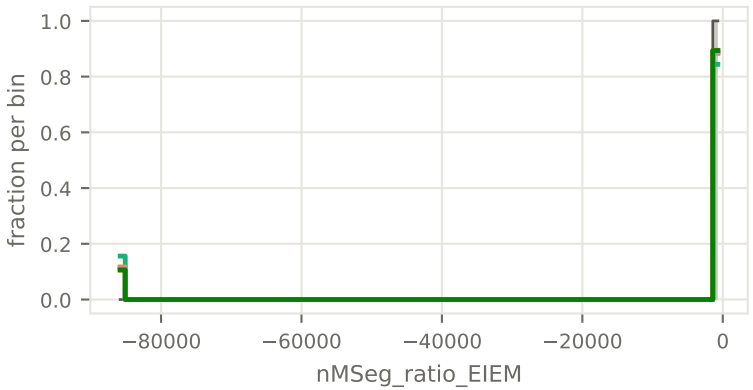
region: BARREL · rank 81 / 98 · in: NN1 endcap

D_equal 0.755 (noise 0.051)  D_pooled 0.746  min 0.727 (mH1000_mS275)  max 0.805 (mH125_mS15)  bkg n = 1,089  sig n = 4,987

1a · each mass point vs background · raw

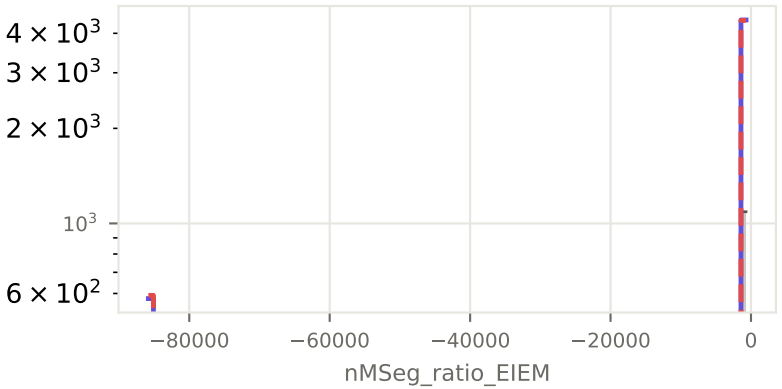



1b · each mass point vs background · normalized

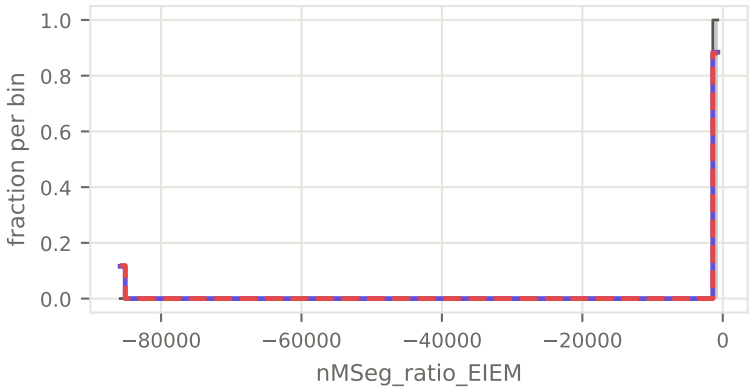



1 · each mass point
background n 1,089
mH125_mS15 D 0.805 n 159
mH125_mS55 D 0.755 n 142
mH200_mS50 D 0.750 n 449
mH400_mS100 D 0.759 n 1,144
mH600_mS150 D 0.752 n 1,608
mH1000_mS275 D 0.727 n 1,485

2a · combined signal vs background · raw

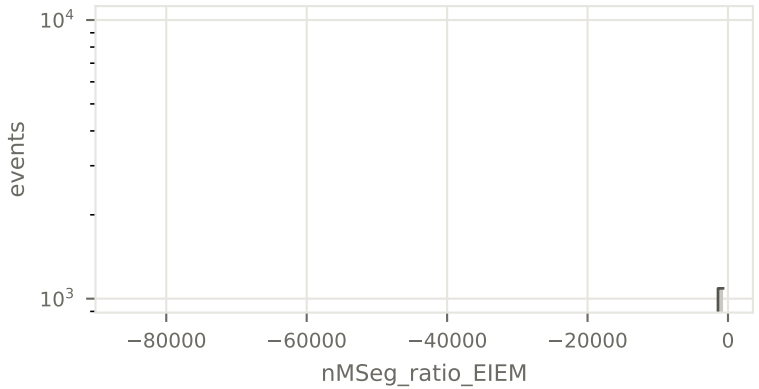



2b · combined signal vs background · normalized

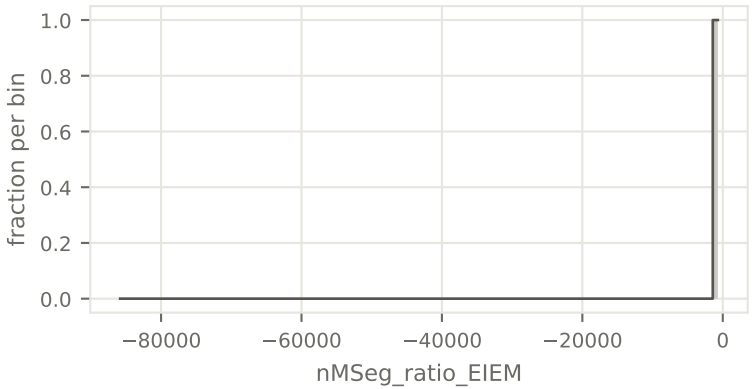



2 · combined signal
background n 1,089
combined, pooled D 0.746 n 4,987
equal weight per mass D 0.755

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0% −999 0.0%
background n 1,089

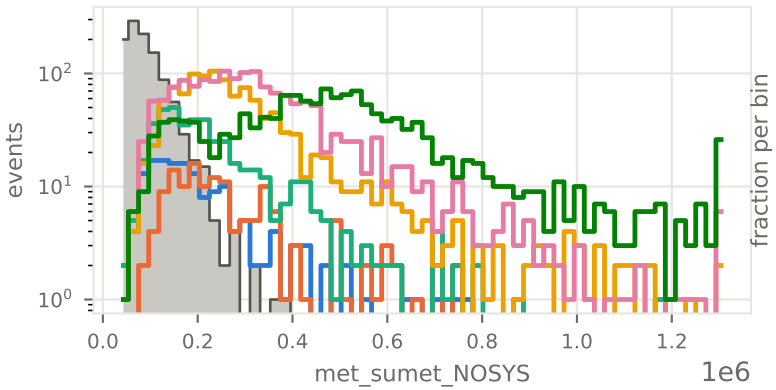
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

# met_sumet_NOSYS

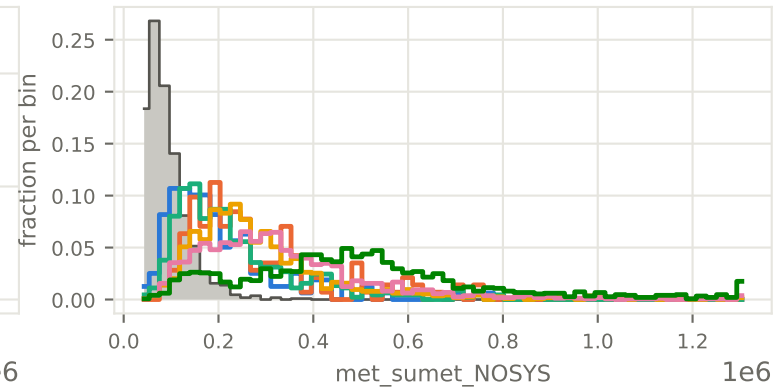
region: BARREL · rank 9 / 98

D_equal 0.737 (noise 0.051) D_pooled 0.783 min 0.588 (mH125_mS15) max 0.852 (mH1000_mS275) bkg n = 1,089 sig n = 4,987

1a · each mass point vs background · raw

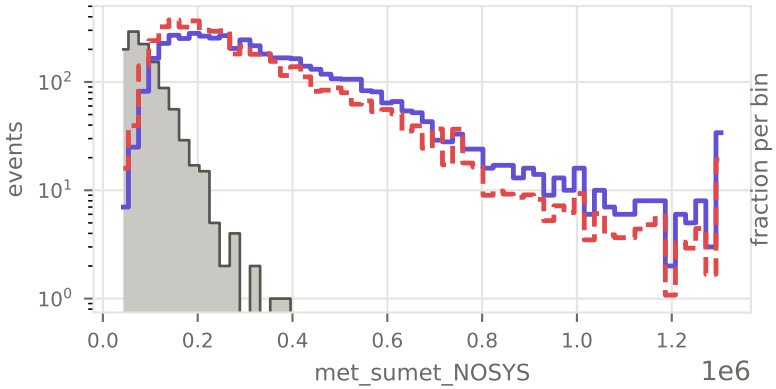



1b · each mass point vs background · normalized

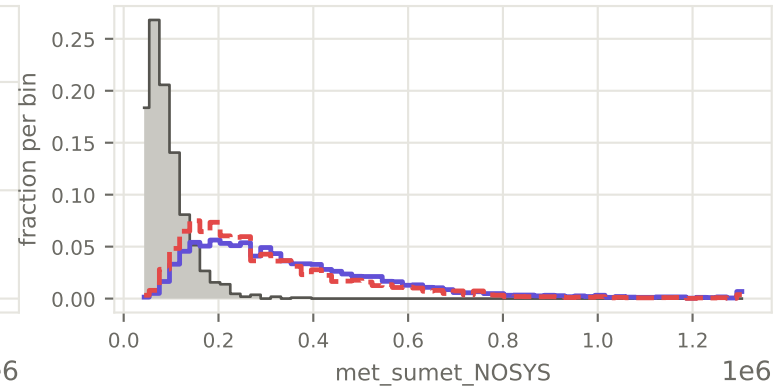



1 · each mass point
background n 1,089
mH125_mS15 D 0.588 n 159
mH125_mS55 D 0.799 n 142
mH200_mS50 D 0.680 n 449
mH400_mS100 D 0.794 n 1,144
mH600_mS150 D 0.798 n 1,608
mH1000_mS275 D 0.852 n 1,485

2a · combined signal vs background · raw

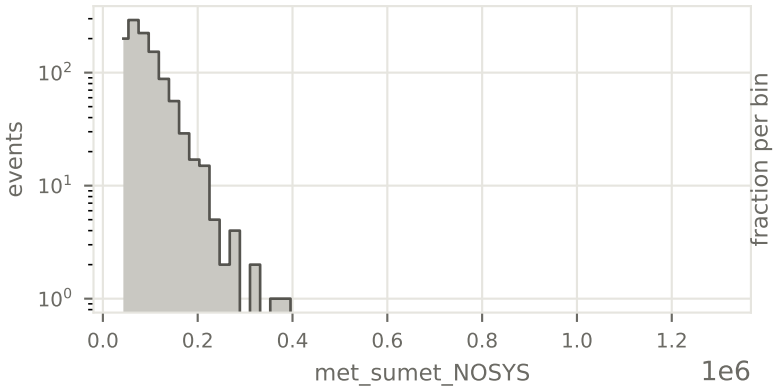



2b · combined signal vs background · normalized

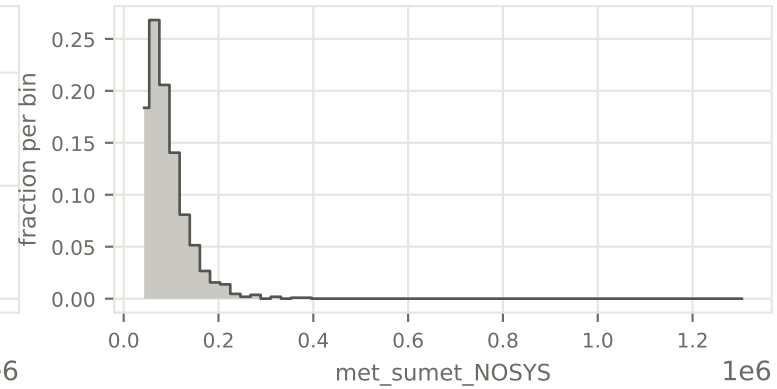



2 · combined signal
background n 1,089
combined, pooled D 0.783 n 4,987
equal weight per mass D 0.737

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0% −999 0.0%
background n 1,089

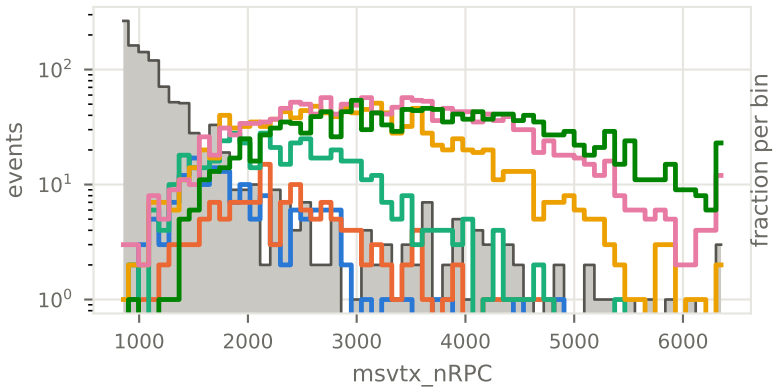
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

# msvtx_nRPC

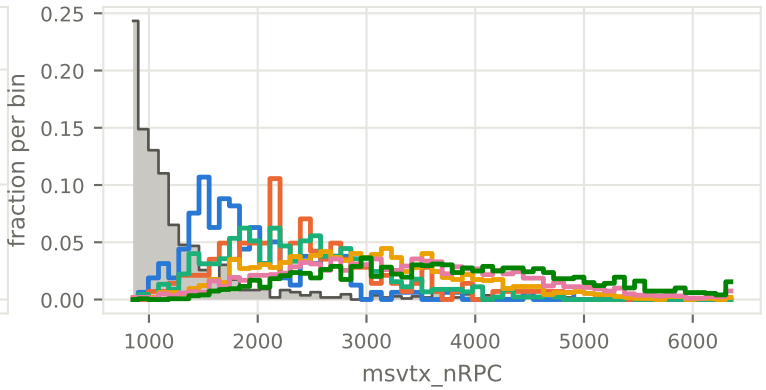
region: BARREL · rank 68 / 98

D_equal 0.725 (noise 0.051) D_pooled 0.787 min 0.647 (mH125_mS15) max 0.850 (mH1000_mS275) bkg n = 1,089 sig n = 4,987

1a · each mass point vs background · raw

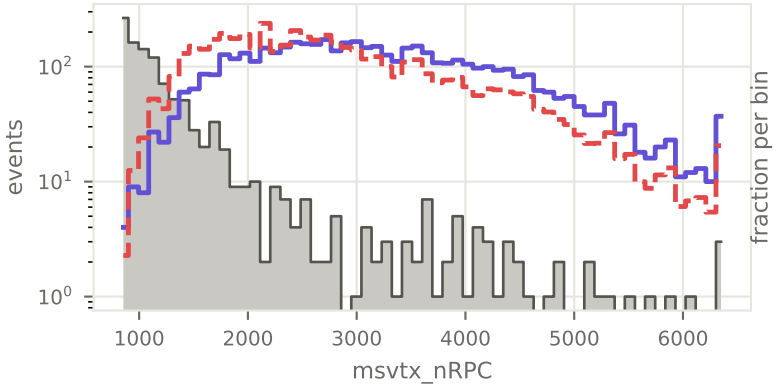



1b · each mass point vs background · normalized

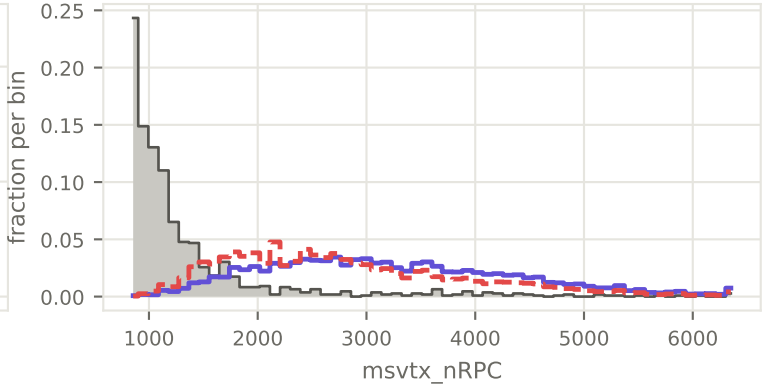



1 · each mass point
background n 1,089
mH125_mS15 D 0.647 n 159
mH125_mS55 D 0.734 n 142
mH200_mS50 D 0.716 n 449
mH400_mS100 D 0.792 n 1,144
mH600_mS150 D 0.809 n 1,608
mH1000_mS275 D 0.850 n 1,485

2a · combined signal vs background · raw

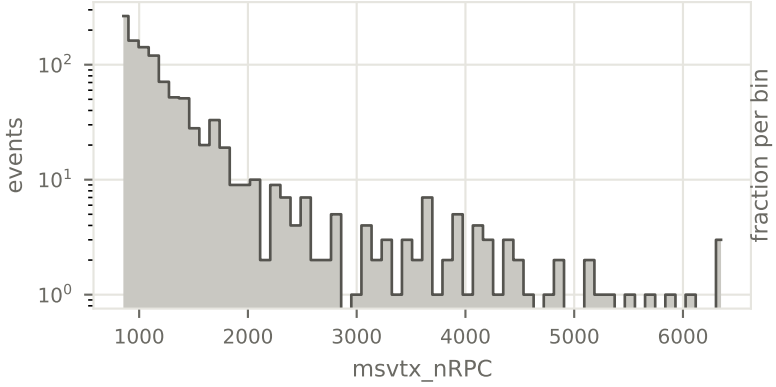



2b · combined signal vs background · normalized

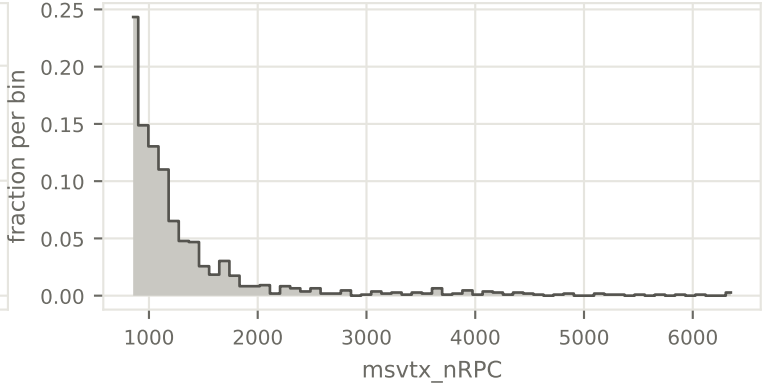



2 · combined signal
background n 1,089
combined, pooled D 0.787 n 4,987
equal weight per mass D 0.725

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0% −999 0.0%
background n 1,089

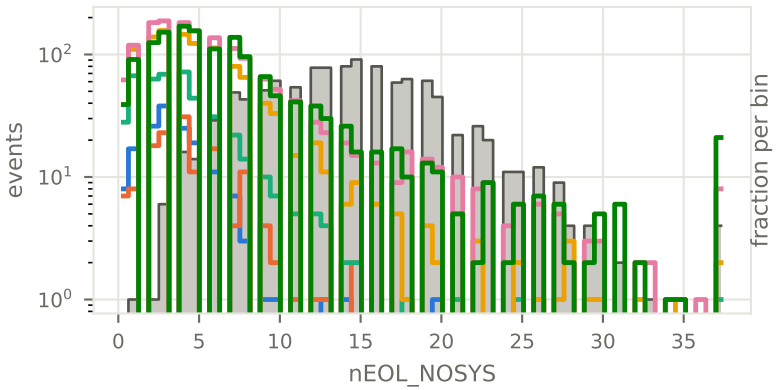
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

# nEOL_NOSYS

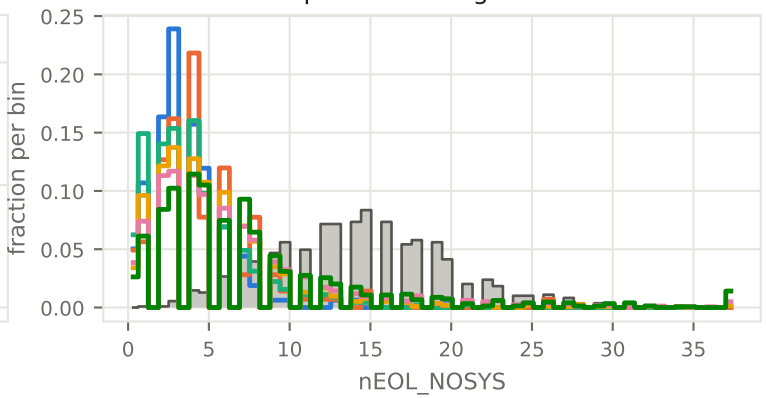
region: BARREL · rank 76 / 98

D_equal 0.710 (noise 0.051) D_pooled 0.651 min 0.580 (mH1000_mS275) max 0.844 (mH125_mS15) bkg n = 1,089 sig n = 4,987

1a · each mass point vs background · raw

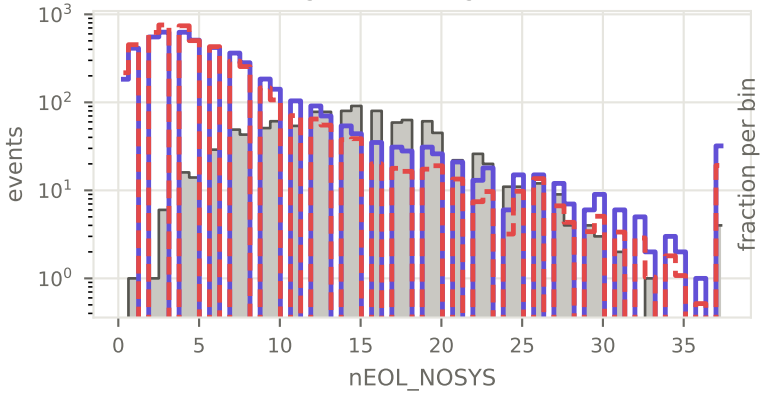



1b · each mass point vs background · normalized

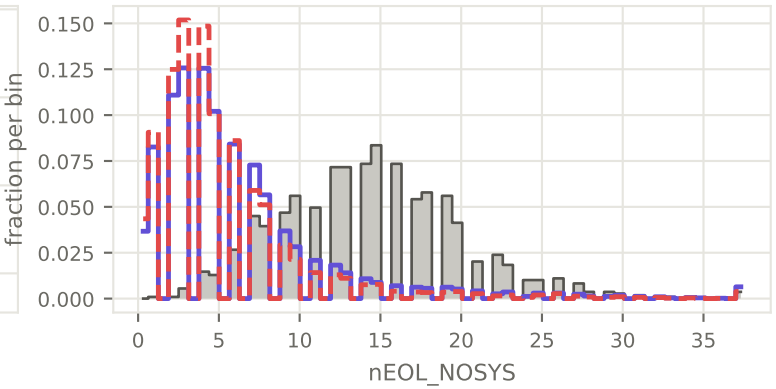



1 · each mass point
background n 1,089
mH125_mS15 D 0.844 n 159
mH125_mS55 D 0.769 n 142
mH200_mS50 D 0.775 n 449
mH400_mS100 D 0.704 n 1,144
mH600_mS150 D 0.620 n 1,608
mH1000_mS275 D 0.580 n 1,485

2a · combined signal vs background · raw

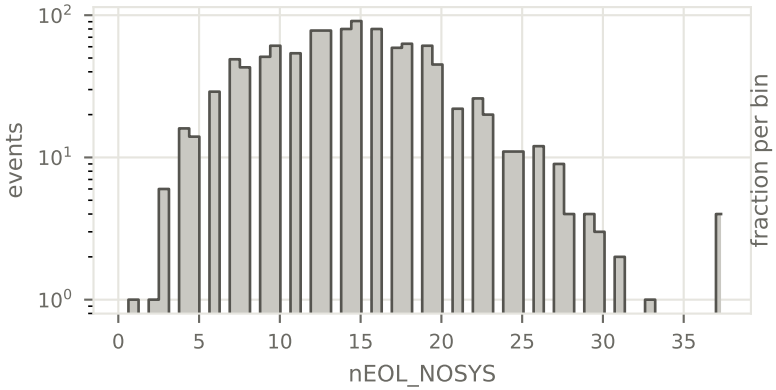



2b · combined signal vs background · normalized

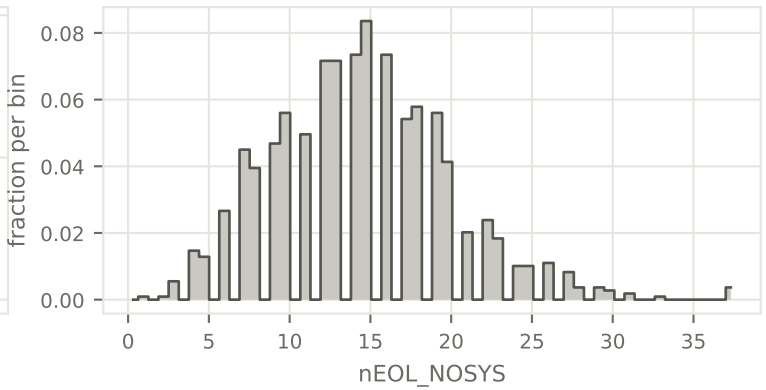



2 · combined signal
background n 1,089
combined, pooled D 0.651 n 4,987
equal weight per mass D 0.710

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0% −999 0.0%
background n 1,089

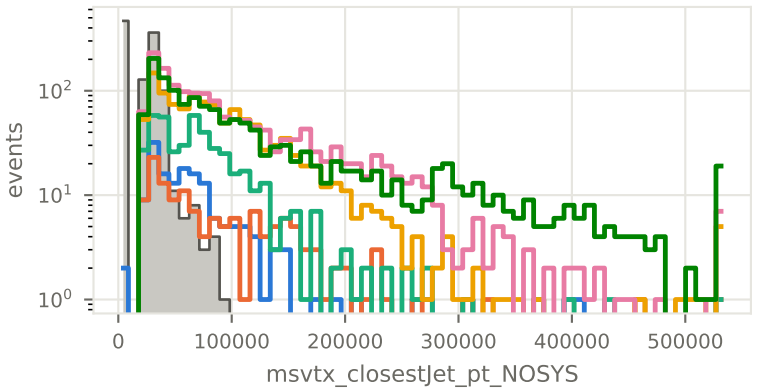
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

# msvtx_closestJet_pt_NOSYS

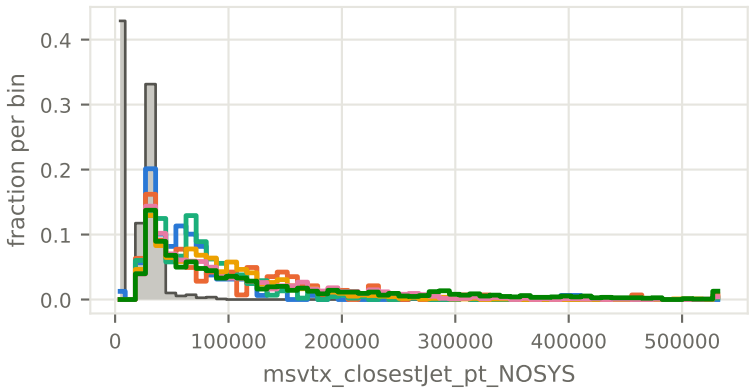
region: BARREL · rank 8 / 98

D_equal 0.707 (noise 0.051)  D_pooled 0.728  min 0.639 (mH125_mS15)  max 0.740 (mH1000_mS275)  bkg n = 1,089  sig n = 4,987

1a · each mass point vs background · raw

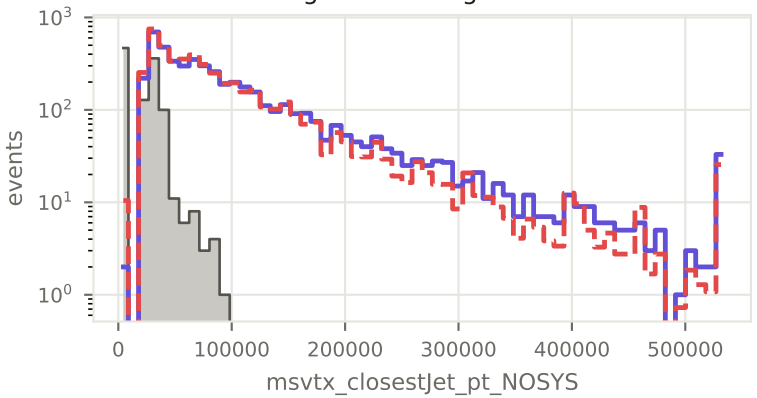



1b · each mass point vs background · normalized

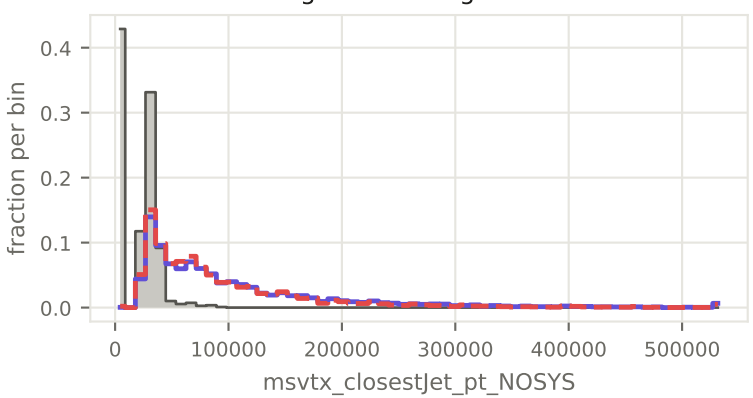



1 · each mass point
background n 1,089
mH125_mS15 D 0.639 n 159
mH125_mS55 D 0.685 n 142
mH200_mS50 D 0.715 n 449
mH400_mS100 D 0.738 n 1,144
mH600_mS150 D 0.727 n 1,608
mH1000_mS275 D 0.740 n 1,485

2a · combined signal vs background · raw

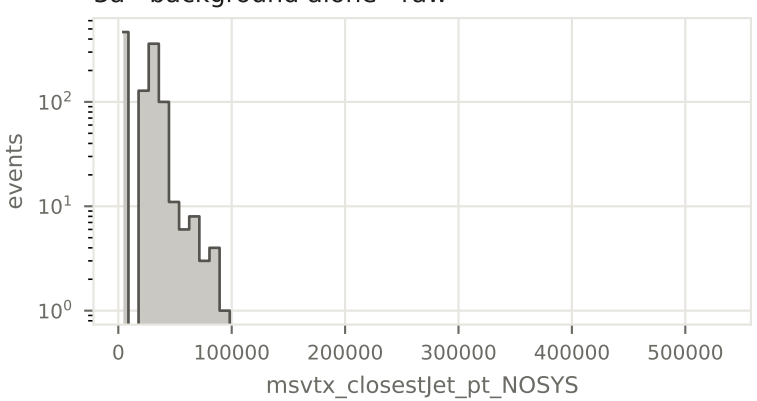



2b · combined signal vs background · normalized

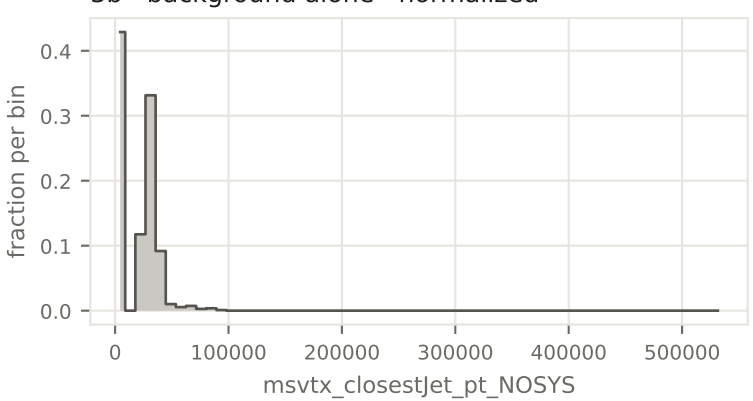



2 · combined signal
background n 1,089
combined, pooled D 0.728 n 4,987
equal weight per mass D 0.707

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0%  −999 0.0%
background n 1,089

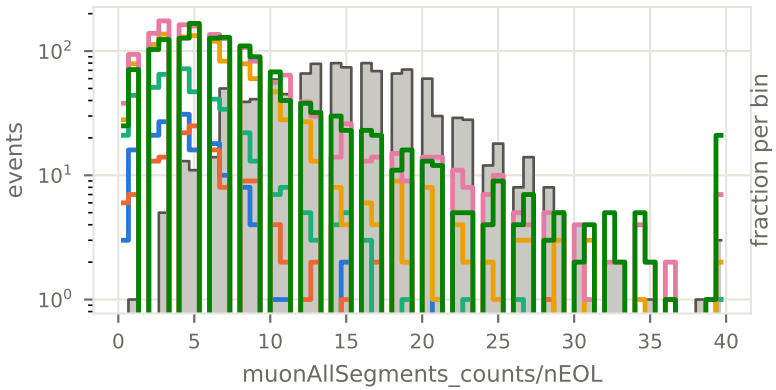
a = raw (log y)  b = normalized · edge bins include under/overflow · NaN and -999 removed

# muonAllSegments_counts/nEOL

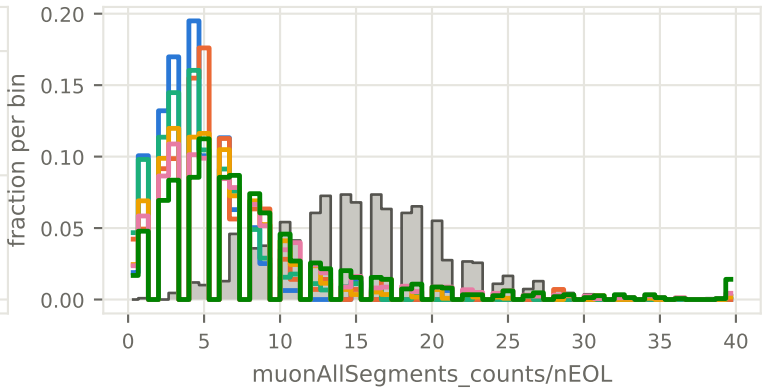
region: BARREL · rank 72 / 98

D_equal 0.692 (noise 0.051)  D_pooled 0.632  min 0.563 (mH1000_mS275)  max 0.821 (mH125_mS15)  bkg n = 1,089  sig n = 4,987

1a · each mass point vs background · raw

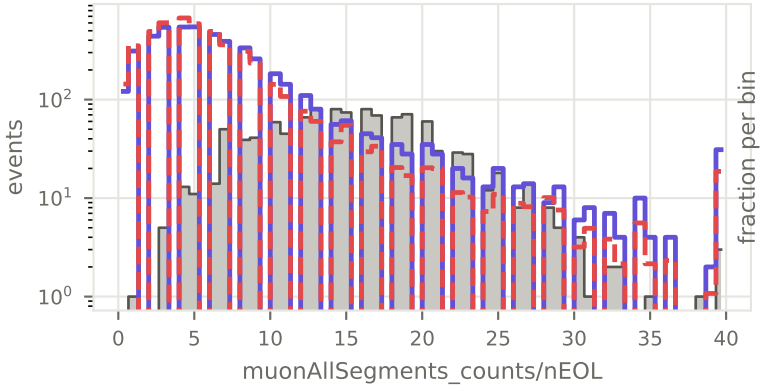



1b · each mass point vs background · normalized

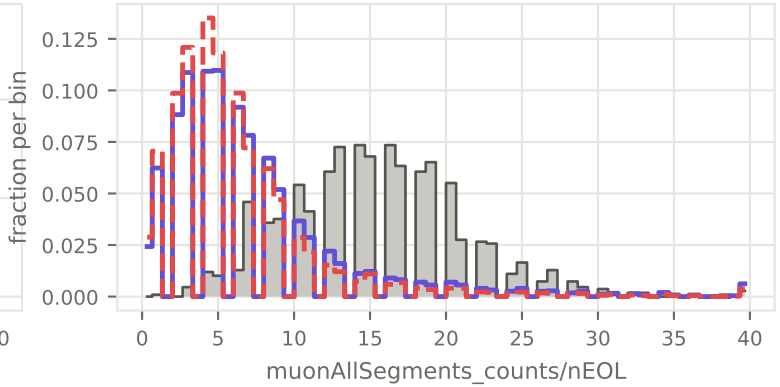



1 · each mass point
- background n 1,089
- mH125_mS15 D 0.821 n 159
- mH125_mS55 D 0.749 n 142
- mH200_mS50 D 0.762 n 449
- mH400_mS100 D 0.681 n 1,144
- mH600_mS150 D 0.599 n 1,608
- mH1000_mS275 D 0.563 n 1,485

2a · combined signal vs background · raw

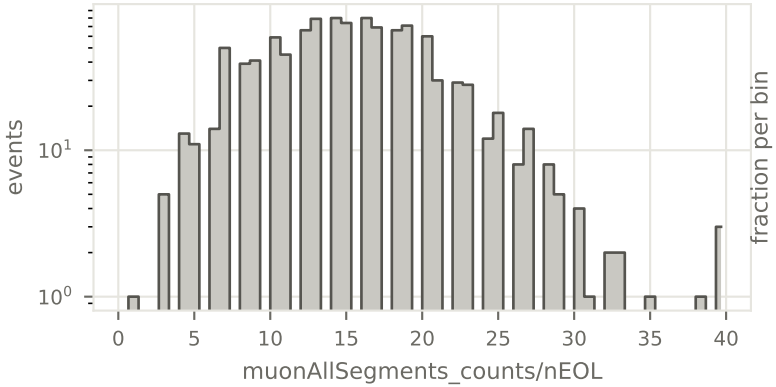



2b · combined signal vs background · normalized

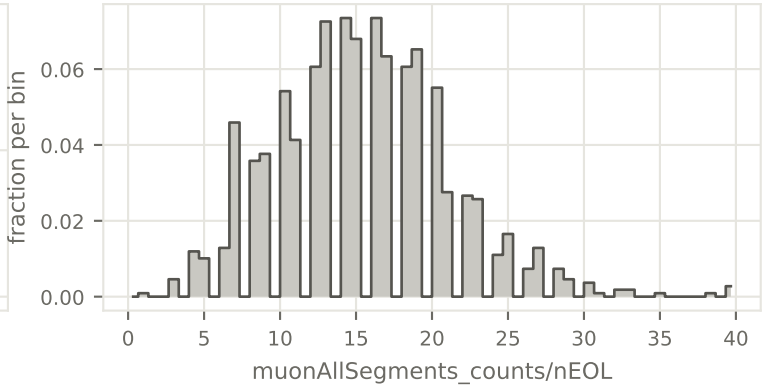



2 · combined signal
- background n 1,089
- combined, pooled D 0.632 n 4,987
- equal weight per mass D 0.692

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0%  −999 0.0%
- background n 1,089

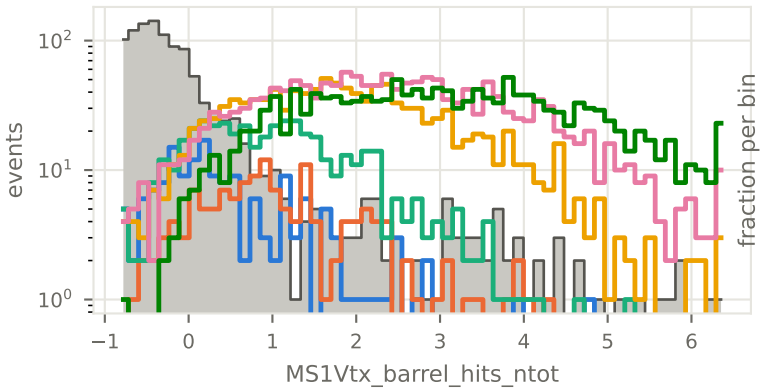
a = raw (log y)  b = normalized · edge bins include under/overflow · NaN and -999 removed

# MS1Vtx_barrel_hits_ntot

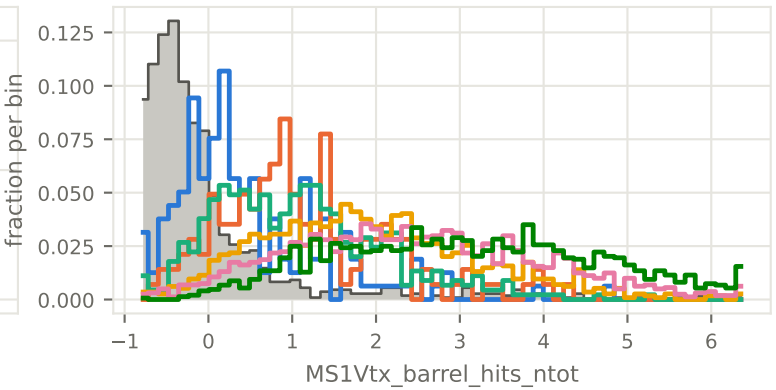
region: BARREL · rank 94 / 98 · in: NN2 barrel

D_equal 0.632 (noise 0.051) D_pooled 0.727 min 0.405 (mH125_mS15) max 0.834 (mH1000_mS275) bkg n = 1,089 sig n = 4,987

1a · each mass point vs background · raw

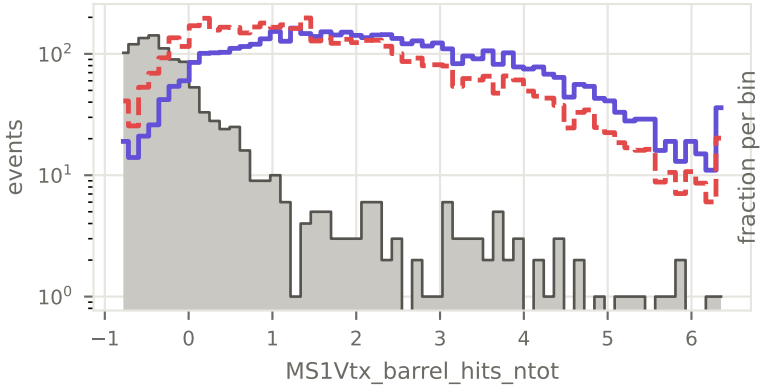



1b · each mass point vs background · normalized

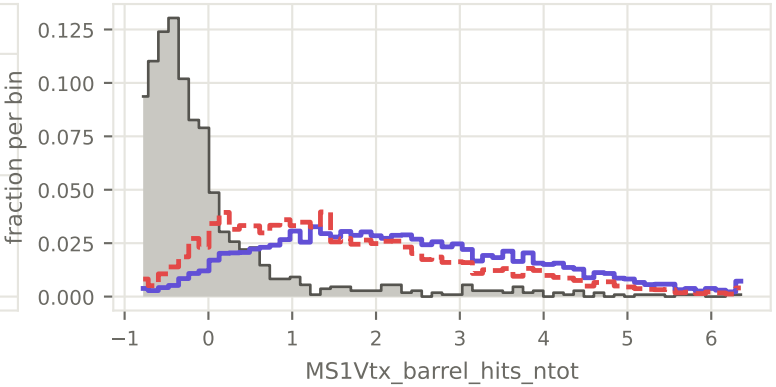



1 · each mass point
background n 1,089
mH125_mS15 D 0.405 n 159
mH125_mS55 D 0.628 n 142
mH200_mS50 D 0.610 n 449
mH400_mS100 D 0.727 n 1,144
mH600_mS150 D 0.766 n 1,608
mH1000_mS275 D 0.834 n 1,485

2a · combined signal vs background · raw

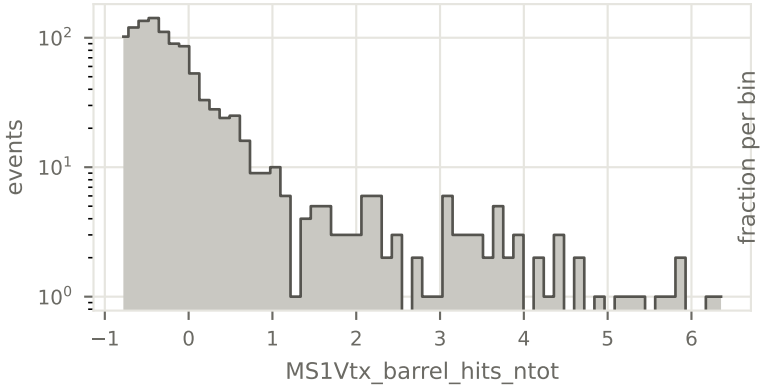



2b · combined signal vs background · normalized

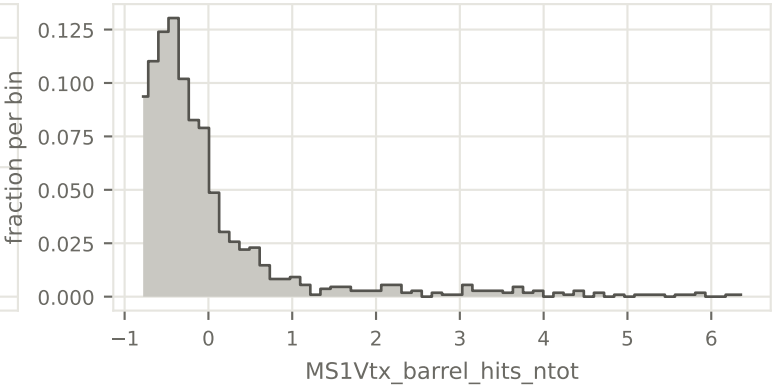



2 · combined signal
background n 1,089
combined, pooled D 0.727 n 4,987
equal weight per mass D 0.632

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0% −999 0.0%
background n 1,089

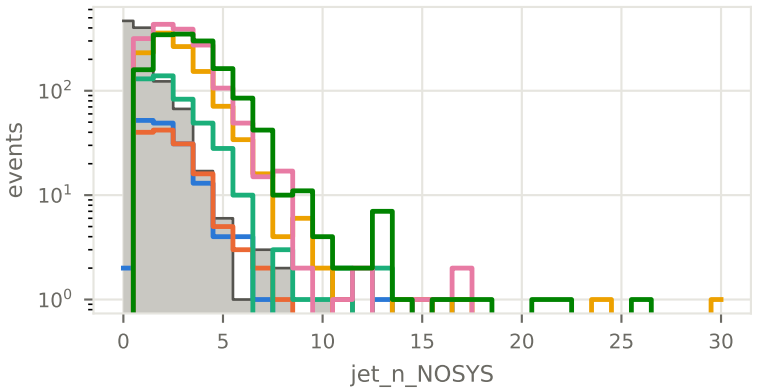
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

# jet_n_NOSYS

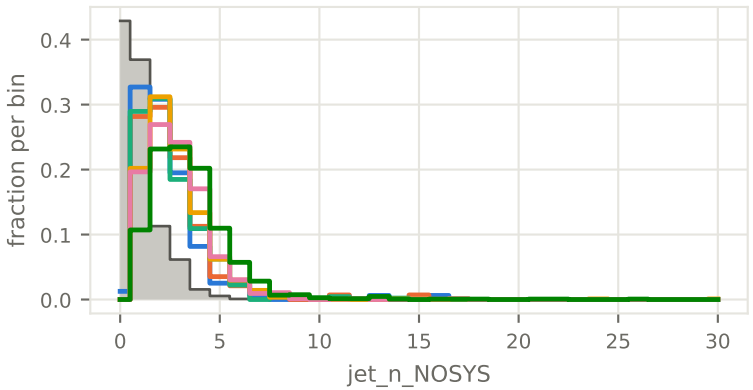
region: BARREL · rank 15 / 98

D_equal 0.562 (noise 0.051) D_pooled 0.611 min 0.458 (mH125_mS15) max 0.691 (mH1000_mS275) bkg n = 1,089 sig n = 4,987

1a · each mass point vs background · raw

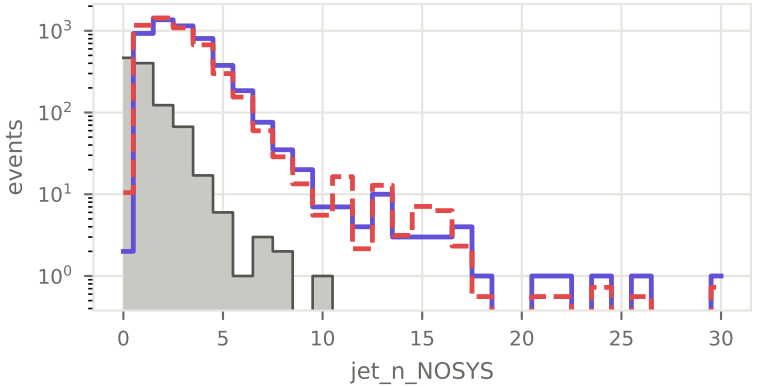



1b · each mass point vs background · normalized

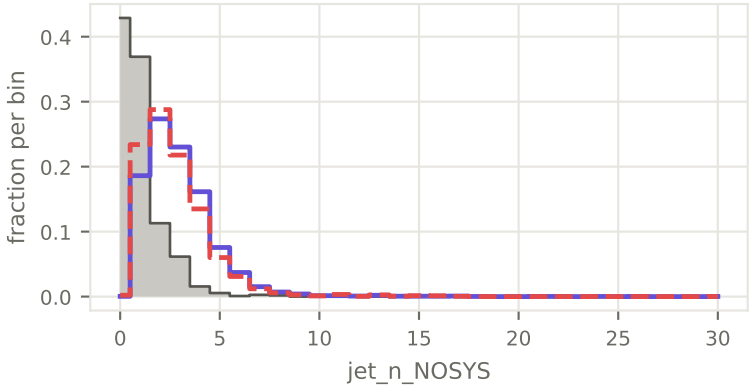



1 · each mass point
background n 1,089
mH125_mS15 D 0.458 n 159
mH125_mS55 D 0.516 n 142
mH200_mS50 D 0.508 n 449
mH400_mS100 D 0.596 n 1,144
mH600_mS150 D 0.601 n 1,608
mH1000_mS275 D 0.691 n 1,485

2a · combined signal vs background · raw

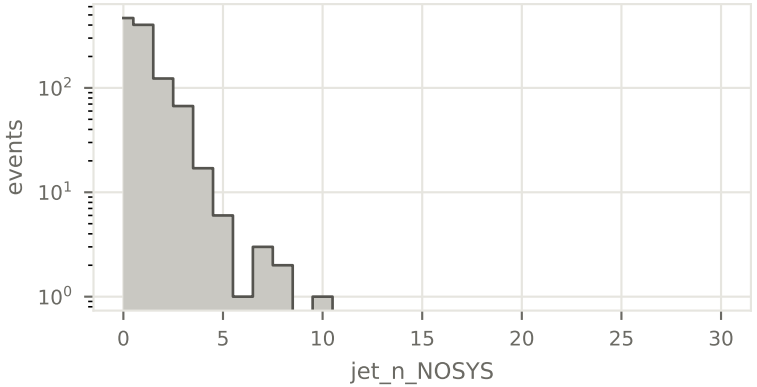



2b · combined signal vs background · normalized

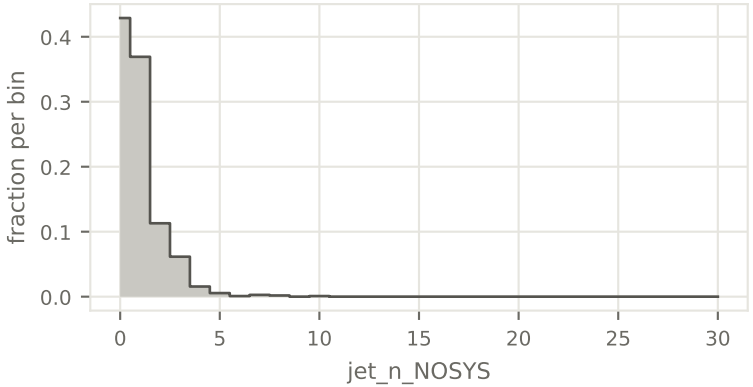



2 · combined signal
background n 1,089
combined, pooled D 0.611 n 4,987
equal weight per mass D 0.562

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0% −999 0.0%
background n 1,089

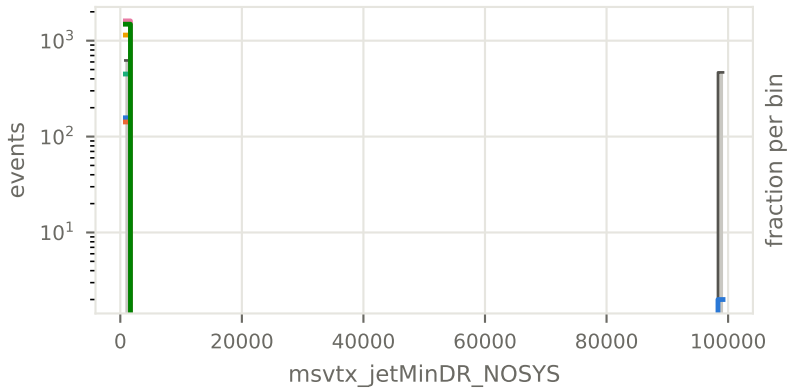
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

# msvtx_jetMinDR_NOSYS

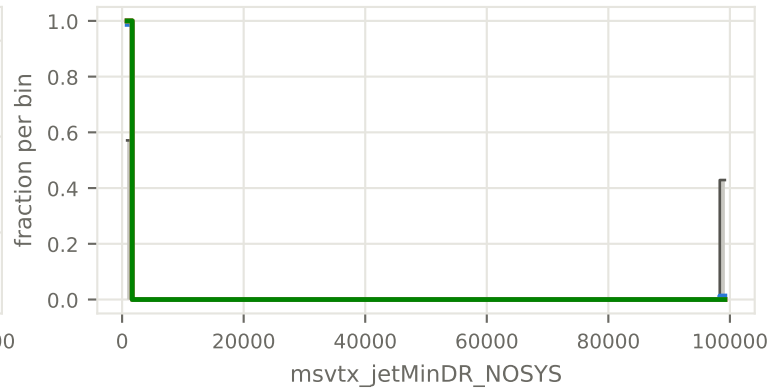
region: BARREL · rank 16 / 98

D_equal 0.555 (noise 0.051) D_pooled 0.583 min 0.510 (mH125_mS55) max 0.622 (mH1000_mS275) bkg n = 1,089 sig n = 4,987

1a · each mass point vs background · raw

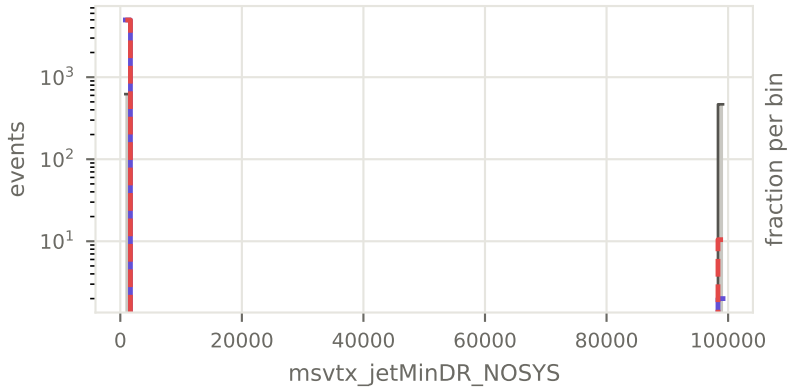



1b · each mass point vs background · normalized

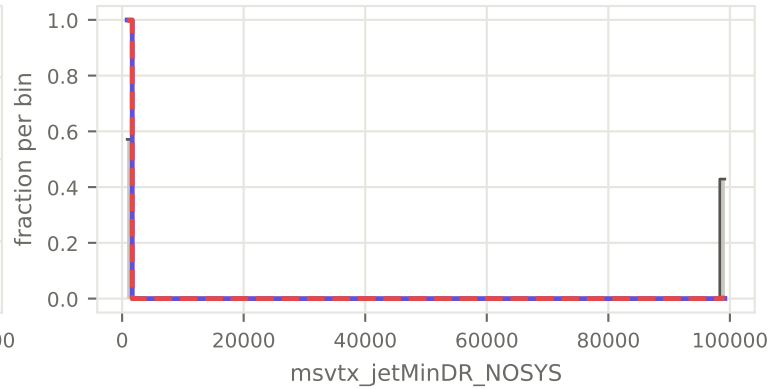



1 · each mass point
background n 1,089
mH125_mS15 D 0.543 n 159
mH125_mS55 D 0.510 n 142
mH200_mS50 D 0.554 n 449
mH400_mS100 D 0.602 n 1,144
mH600_mS150 D 0.575 n 1,608
mH1000_mS275 D 0.622 n 1,485

2a · combined signal vs background · raw

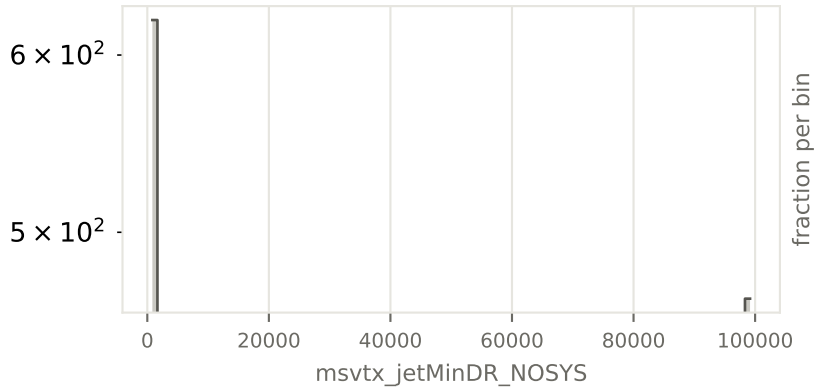



2b · combined signal vs background · normalized

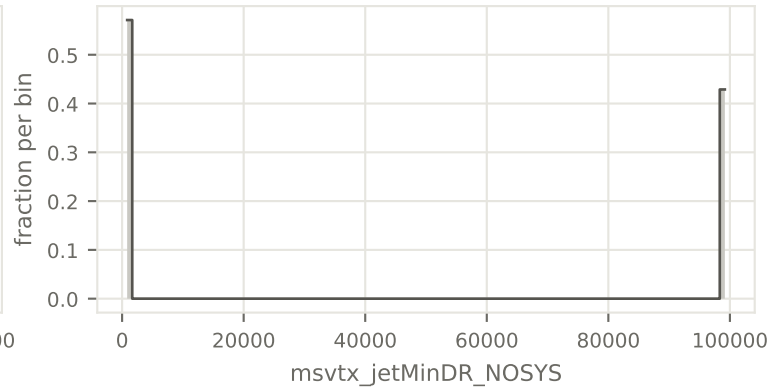



2 · combined signal
background n 1,089
combined, pooled D 0.583 n 4,987
equal weight per mass D 0.555

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0% −999 0.0%
background n 1,089

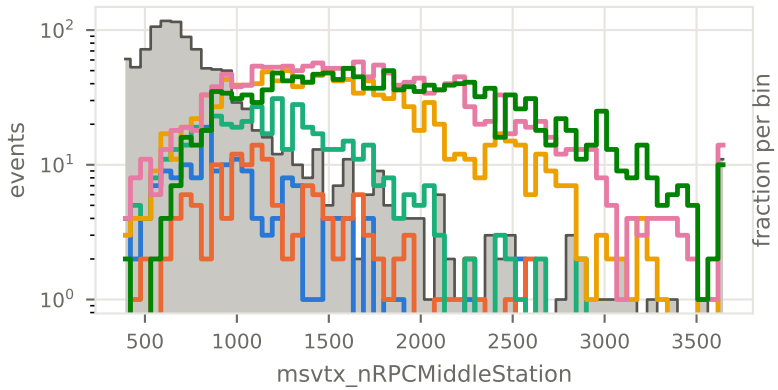
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

# msvtx_nRPCMiddleStation

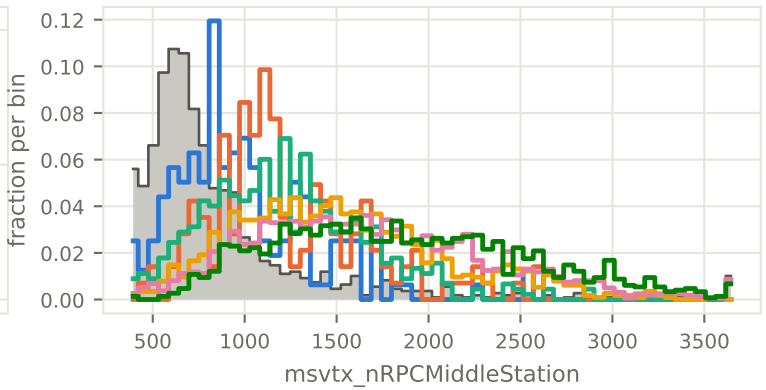
region: BARREL · rank 69 / 98

D_equal 0.520 (noise 0.051) D_pooled 0.611 min 0.314 (mH125_mS15) max 0.691 (mH1000_mS275) bkg n = 1,089 sig n = 4,987

1a · each mass point vs background · raw

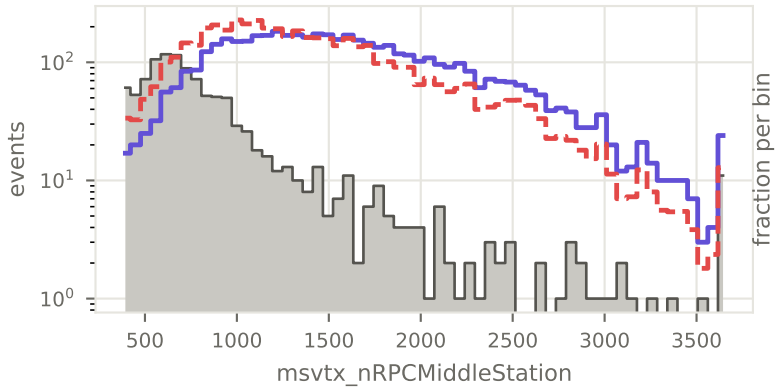



1b · each mass point vs background · normalized

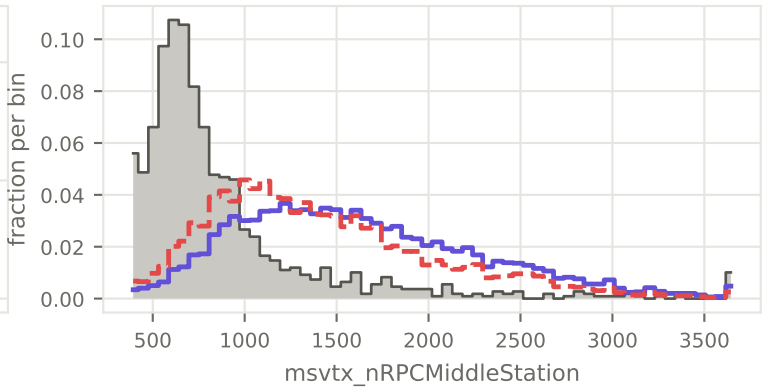



1 · each mass point
background n 1,089
mH125_mS15 D 0.314 n 159
mH125_mS55 D 0.528 n 142
mH200_mS50 D 0.469 n 449
mH400_mS100 D 0.605 n 1,144
mH600_mS150 D 0.643 n 1,608
mH1000_mS275 D 0.691 n 1,485

2a · combined signal vs background · raw

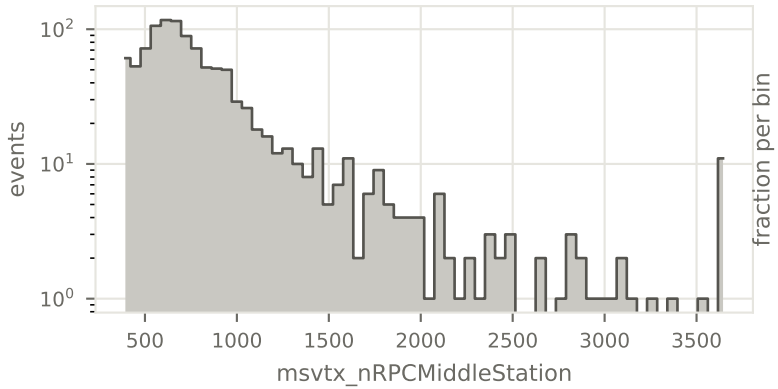



2b · combined signal vs background · normalized

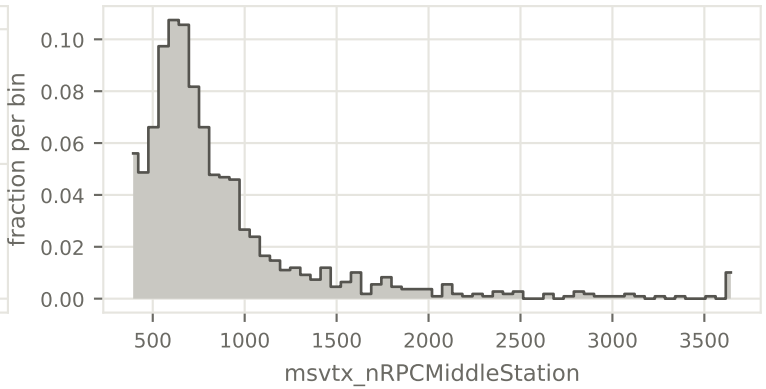



2 · combined signal
background n 1,089
combined, pooled D 0.611 n 4,987
equal weight per mass D 0.520

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0% −999 0.0%
background n 1,089

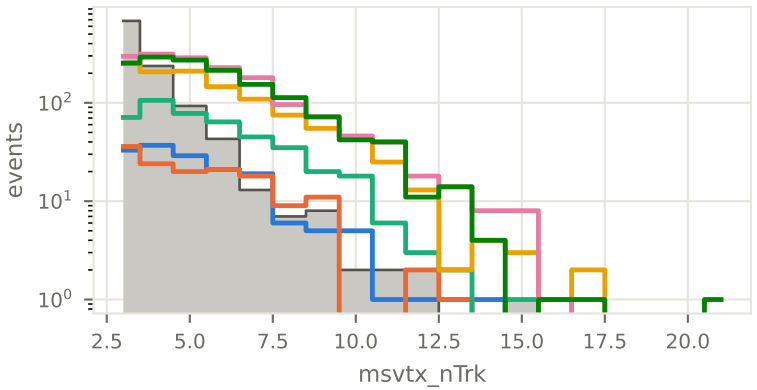
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

# msvtx_nTrk

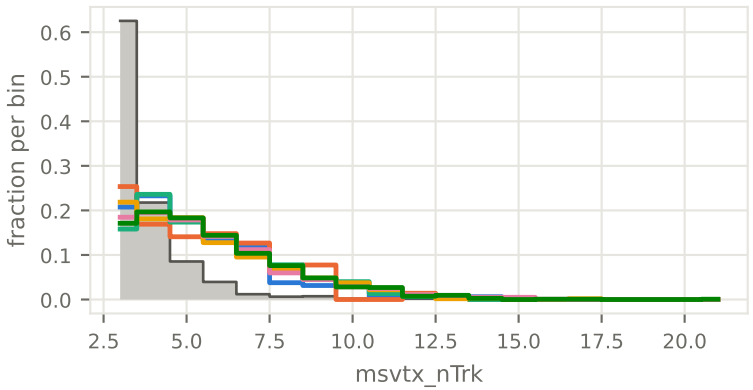
region: BARREL · rank 12 / 98 · in: NN2 barrel, NN2 endcap

D_equal 0.442 (noise 0.051)  D_pooled 0.458  min 0.418 (mH125_mS15)  max 0.475 (mH1000_mS275)  bkg n = 1,089  sig n = 4,987

1a · each mass point vs background · raw

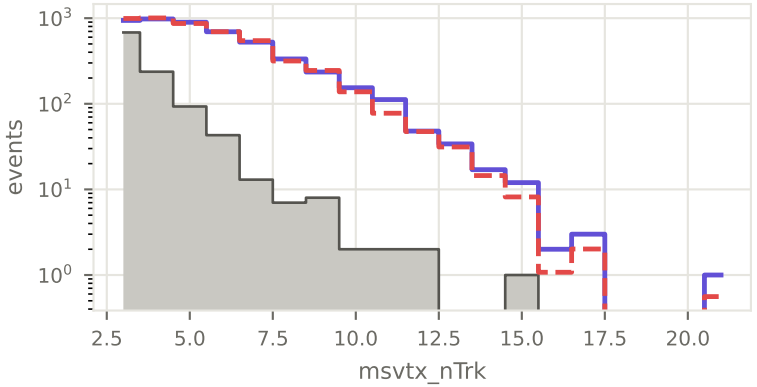



1b · each mass point vs background · normalized

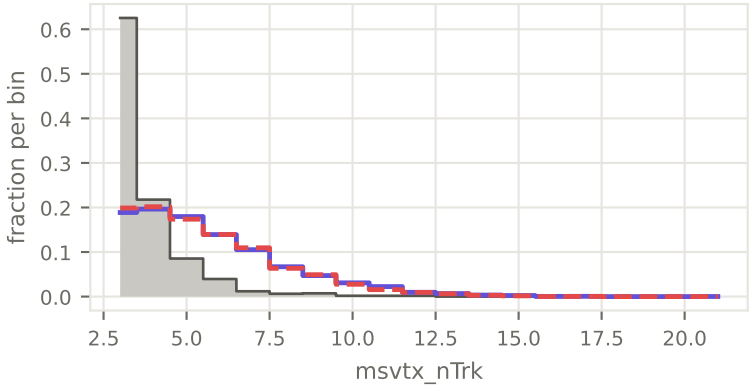



1 · each mass point
background n 1,089
mH125_mS15 D 0.418 n 159
mH125_mS55 D 0.420 n 142
mH200_mS50 D 0.467 n 449
mH400_mS100 D 0.443 n 1,144
mH600_mS150 D 0.464 n 1,608
mH1000_mS275 D 0.475 n 1,485

2a · combined signal vs background · raw

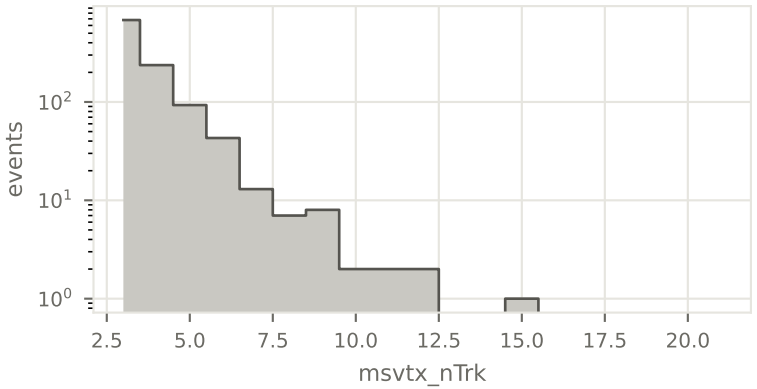



2b · combined signal vs background · normalized

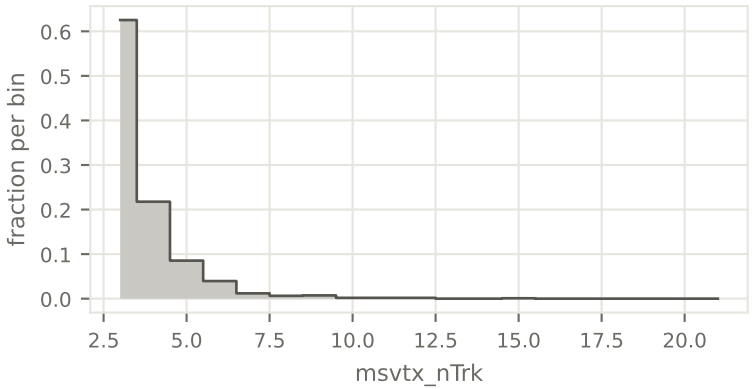



2 · combined signal
background n 1,089
combined, pooled D 0.458 n 4,987
equal weight per mass D 0.442

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0% −999 0.0%
background n 1,089

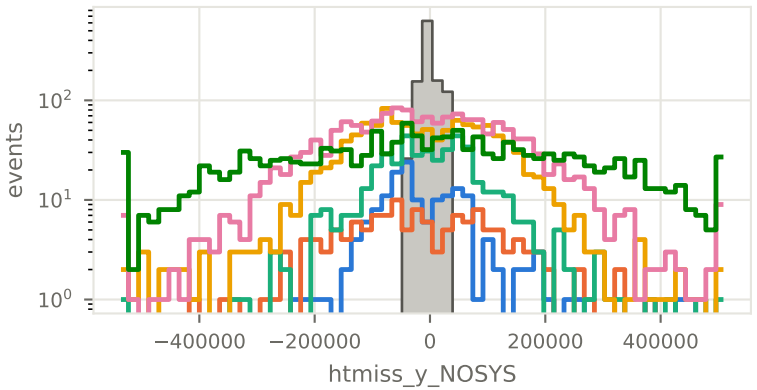
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

# htmiss_y_NOSYS

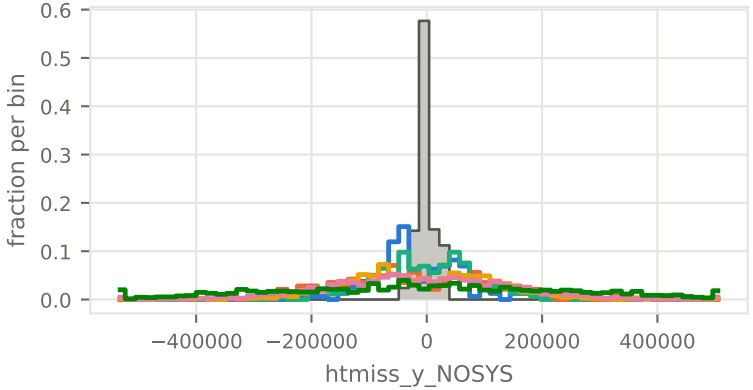
region: BARREL · rank 19 / 98

D_equal 0.436 (noise 0.051) D_pooled 0.437 min 0.374 (mH200_mS50) max 0.470 (mH125_mS15) bkg n = 1,089 sig n = 4,987

1a · each mass point vs background · raw

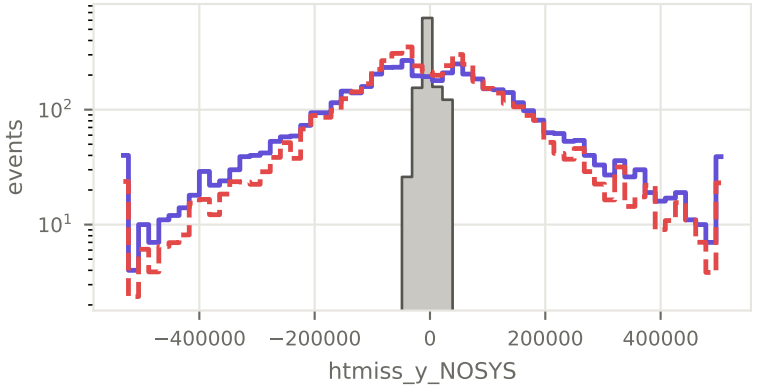



1b · each mass point vs background · normalized

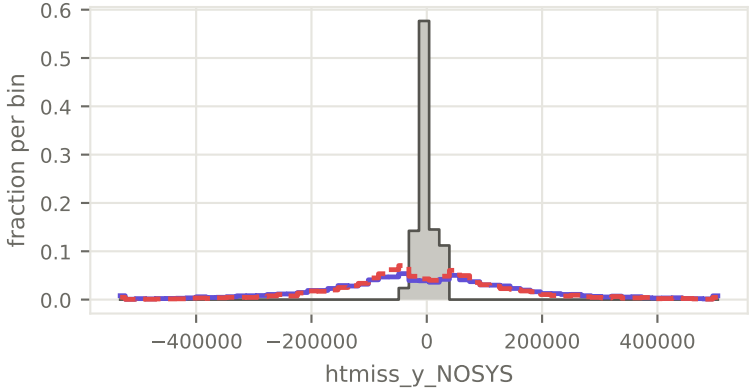



1 · each mass point
background n 1,089
mH125_mS15 D 0.470 n 159
mH125_mS55 D 0.468 n 142
mH200_mS50 D 0.374 n 449
mH400_mS100 D 0.439 n 1,144
mH600_mS150 D 0.447 n 1,608
mH1000_mS275 D 0.444 n 1,485

2a · combined signal vs background · raw

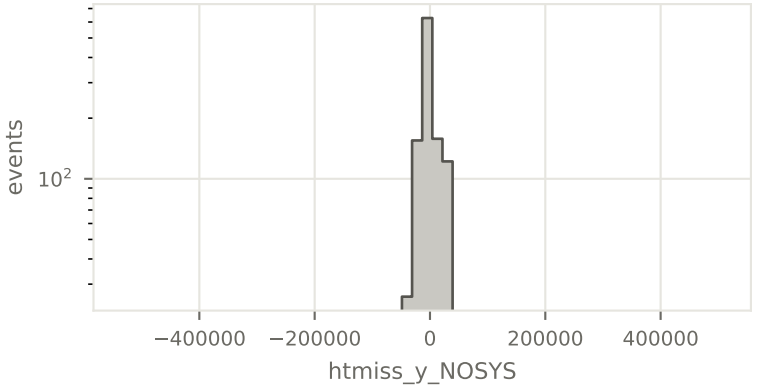



2b · combined signal vs background · normalized

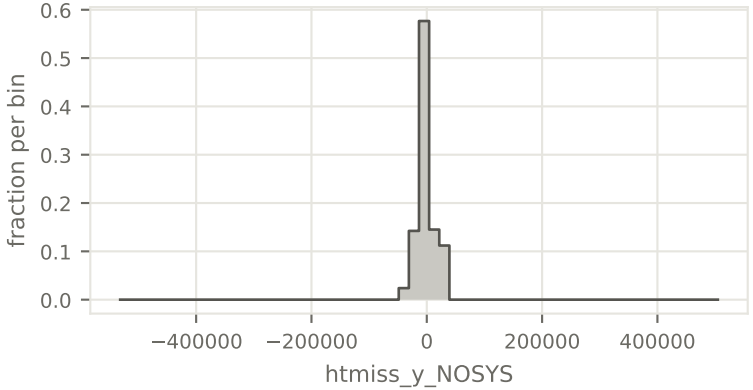



2 · combined signal
background n 1,089
combined, pooled D 0.437 n 4,987
equal weight per mass D 0.436

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0% −999 0.0%
background n 1,089

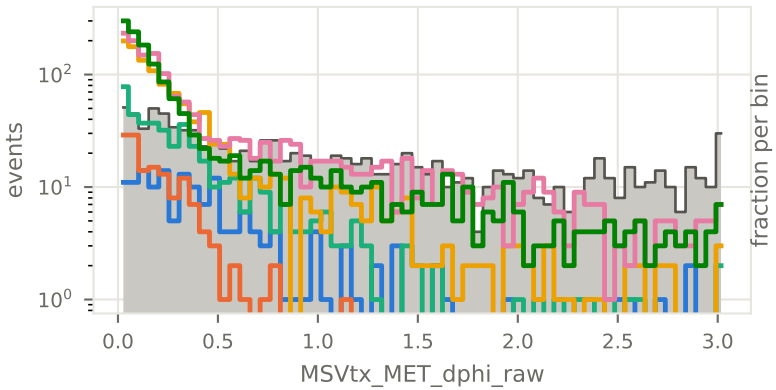
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

# MSVtx_MET_dphi_raw

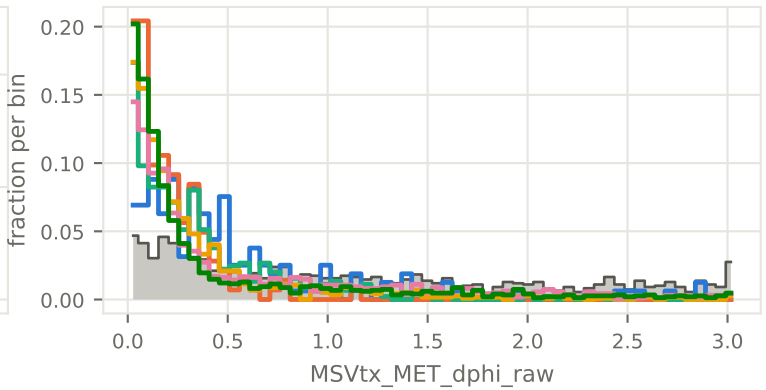
region: BARREL · rank 28 / 98 · in: NN2 barrel, NN2 endcap

D_equal 0.429 (noise 0.051) D_pooled 0.407 min 0.332 (mH600_mS150) max 0.620 (mH125_mS55) bkg n = 1,089 sig n = 4,987

1a · each mass point vs background · raw

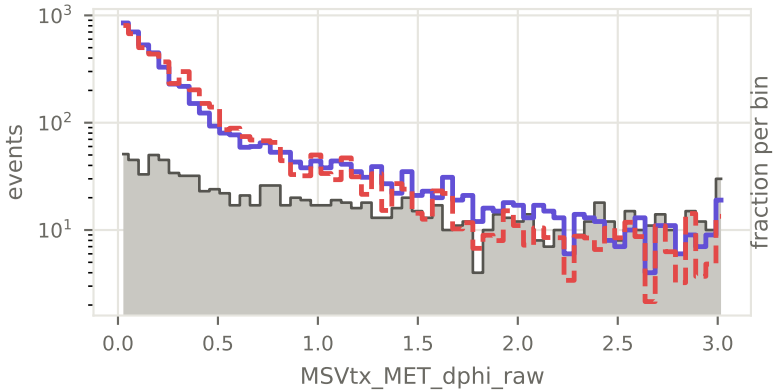



1b · each mass point vs background · normalized

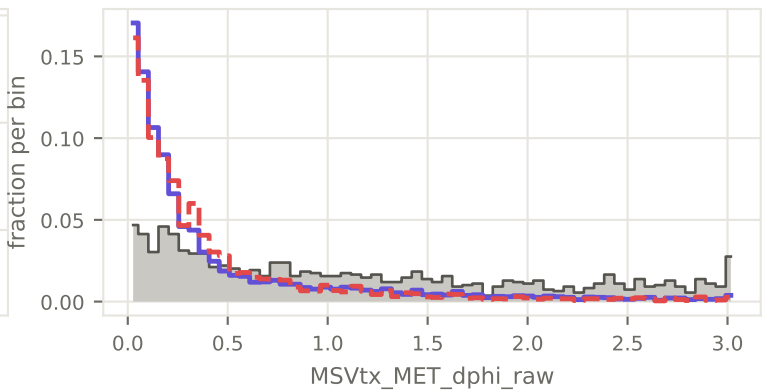



1 · each mass point
background n 1,089
mH125_mS15 D 0.386 n 159
mH125_mS55 D 0.620 n 142
mH200_mS50 D 0.433 n 449
mH400_mS100 D 0.482 n 1,144
mH600_mS150 D 0.332 n 1,608
mH1000_mS275 D 0.439 n 1,485

2a · combined signal vs background · raw

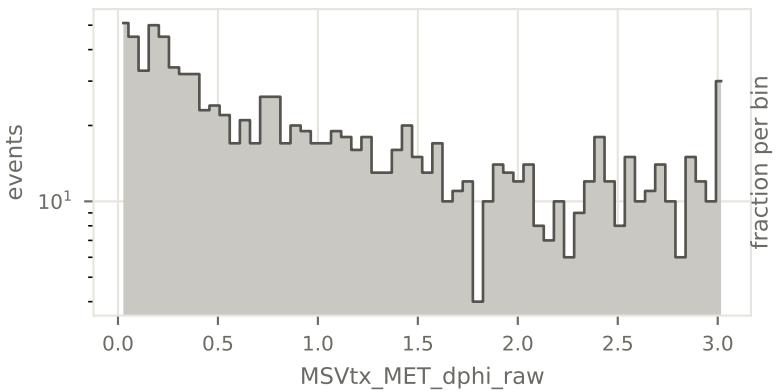



2b · combined signal vs background · normalized

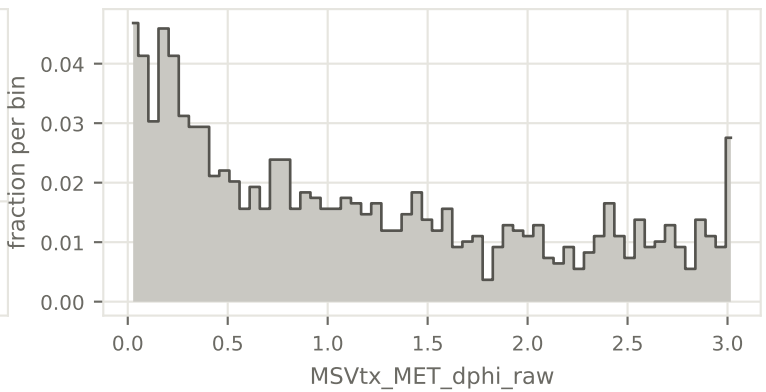



2 · combined signal
background n 1,089
combined, pooled D 0.407 n 4,987
equal weight per mass D 0.429

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0% −999 0.0%
background n 1,089

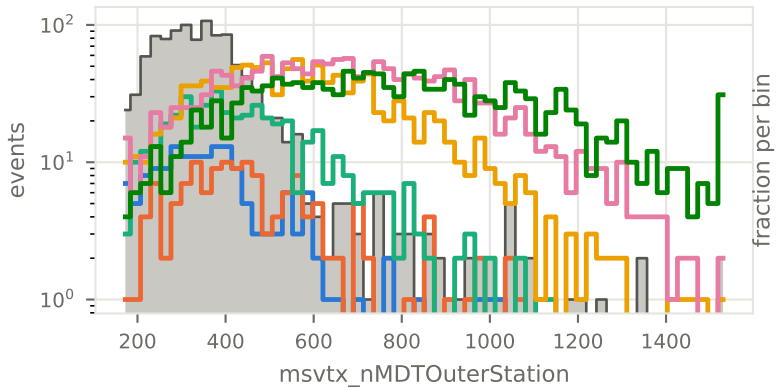
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

# msvtx_nMDTOuterStation

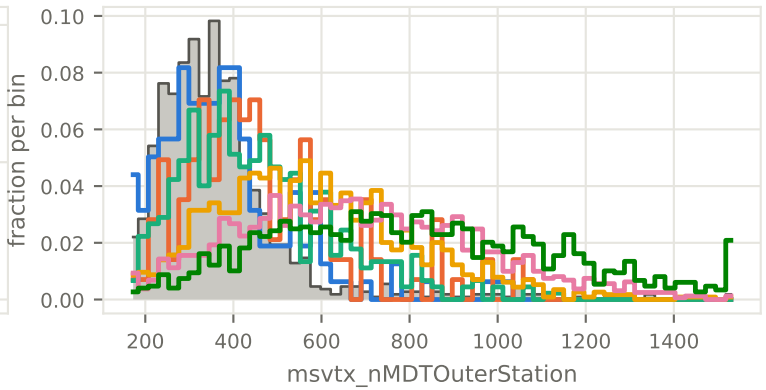
region: BARREL · rank 14 / 98

D_equal 0.414 (noise 0.051)  D_pooled 0.565  min 0.080 (mH125_mS15)  max 0.710 (mH1000_mS275)  bkg n = 1,089  sig n = 4,987

1a · each mass point vs background · raw

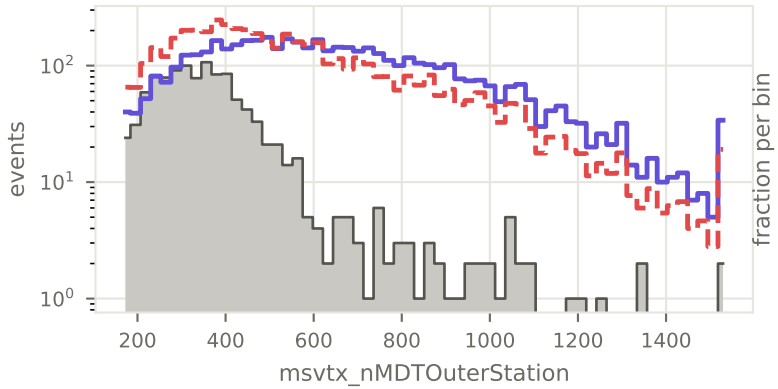



1b · each mass point vs background · normalized

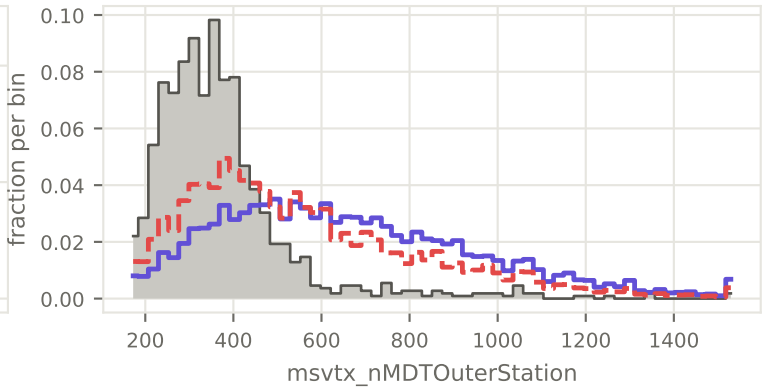



1 · each mass point
background n 1,089
mH125_mS15 D 0.080 n 159
mH125_mS55 D 0.334 n 142
mH200_mS50 D 0.297 n 449
mH400_mS100 D 0.525 n 1,144
mH600_mS150 D 0.629 n 1,608
mH1000_mS275 D 0.710 n 1,485

2a · combined signal vs background · raw

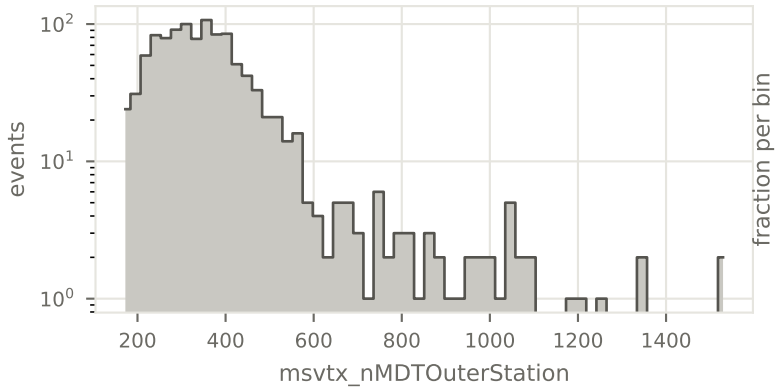



2b · combined signal vs background · normalized

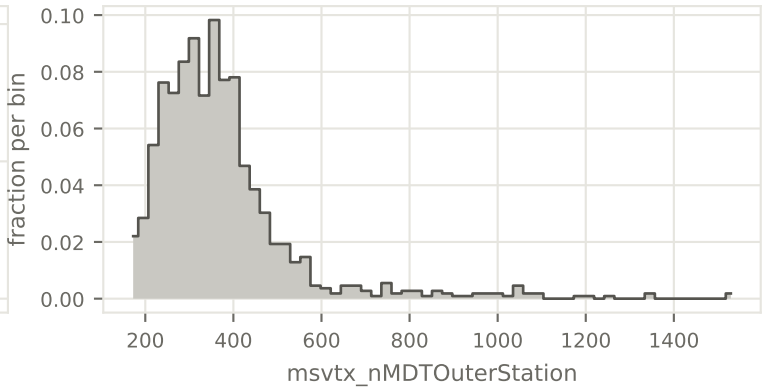



2 · combined signal
background n 1,089
combined, pooled D 0.565 n 4,987
equal weight per mass D 0.414

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0%  −999 0.0%
background n 1,089

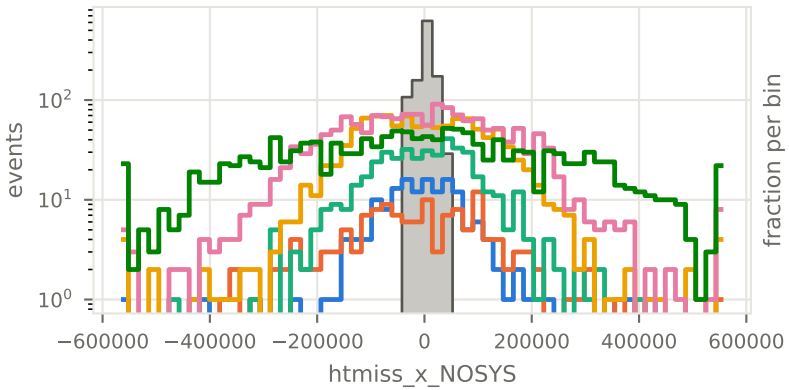
a = raw (log y)  b = normalized · edge bins include under/overflow · NaN and -999 removed

# htmiss_x_NOSYS

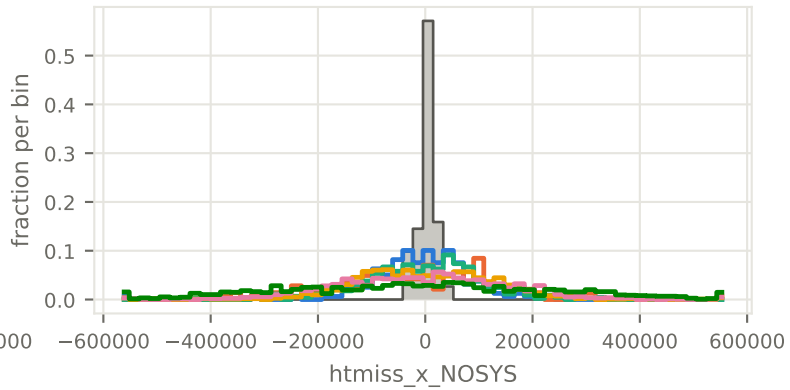
region: BARREL · rank 25 / 98

D_equal 0.409 (noise 0.051) D_pooled 0.421 min 0.363 (mH125_mS15) max 0.458 (mH1000_mS275) bkg n = 1,089 sig n = 4,987

1a · each mass point vs background · raw

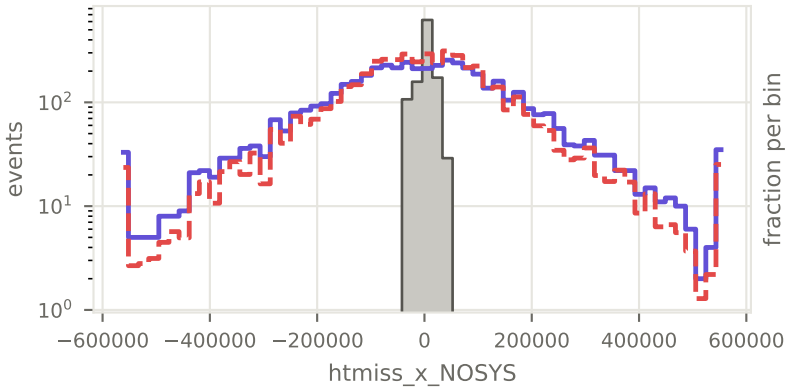



1b · each mass point vs background · normalized

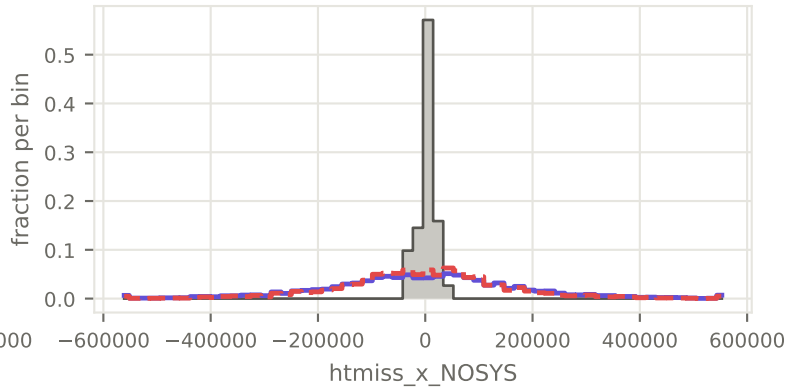



1 · each mass point
background n 1,089
mH125_mS15 D 0.363 n 159
mH125_mS55 D 0.446 n 142
mH200_mS50 D 0.386 n 449
mH400_mS100 D 0.407 n 1,144
mH600_mS150 D 0.418 n 1,608
mH1000_mS275 D 0.458 n 1,485

2a · combined signal vs background · raw

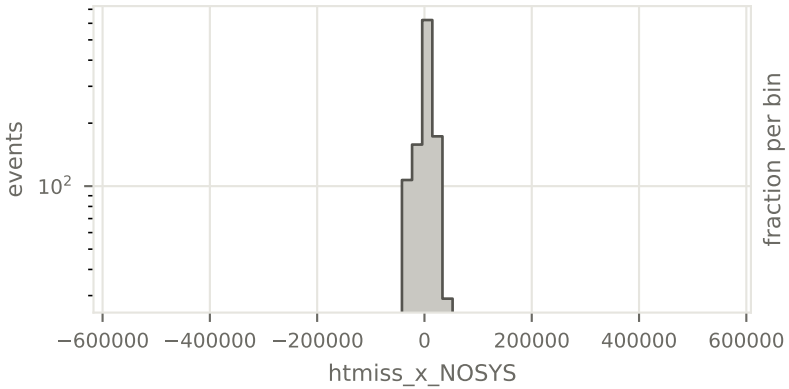



2b · combined signal vs background · normalized

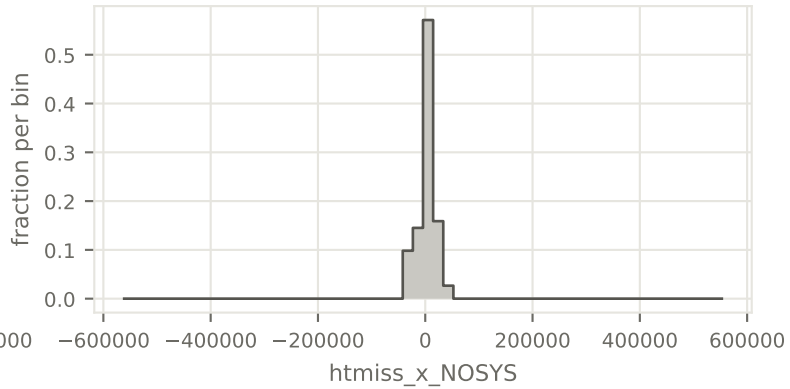



2 · combined signal
background n 1,089
combined, pooled D 0.421 n 4,987
equal weight per mass D 0.409

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0% −999 0.0%
background n 1,089

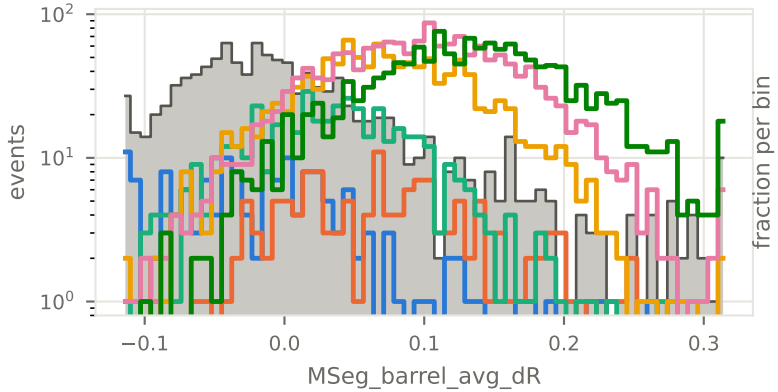
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

# MSeg_barrel_avg_dR

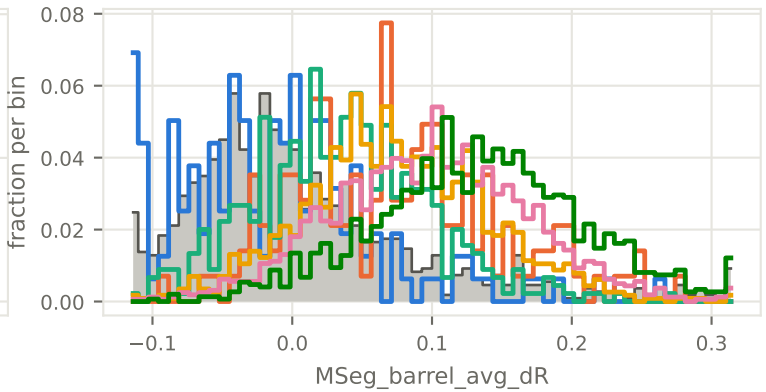
region: BARREL · rank 97 / 98 · in: NN2 barrel

D_equal 0.366 (noise 0.051) D_pooled 0.497 min 0.101 (mH125_mS15) max 0.645 (mH1000_mS275) bkg n = 1,089 sig n = 4,987

1a · each mass point vs background · raw

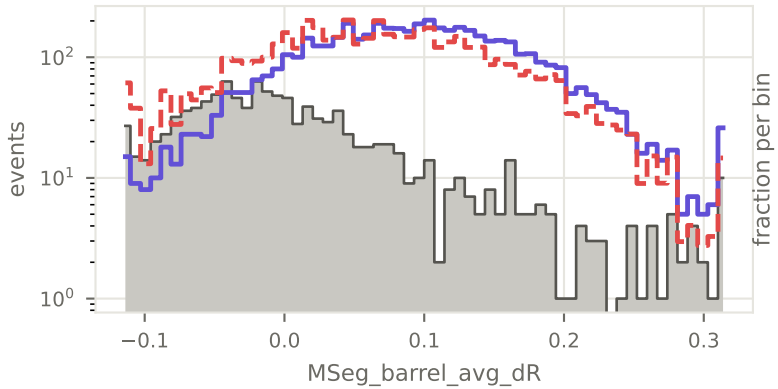



1b · each mass point vs background · normalized

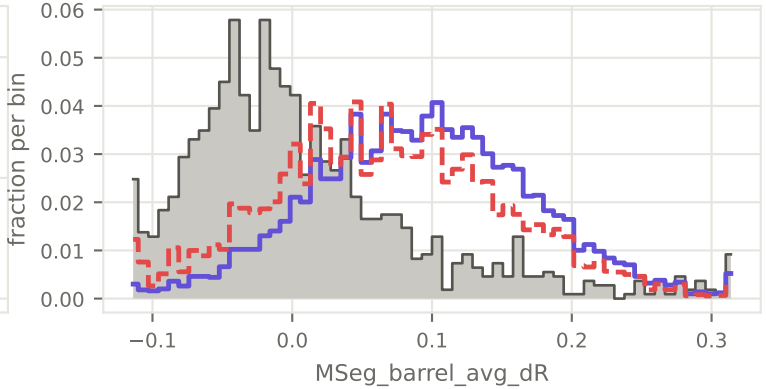



1 · each mass point
background n 1,089
mH125_mS15 D 0.101 n 159
mH125_mS55 D 0.431 n 142
mH200_mS50 D 0.291 n 449
mH400_mS100 D 0.471 n 1,144
mH600_mS150 D 0.540 n 1,608
mH1000_mS275 D 0.645 n 1,485

2a · combined signal vs background · raw

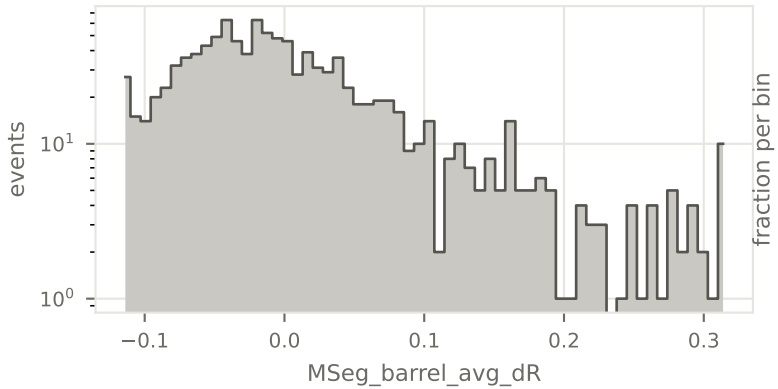



2b · combined signal vs background · normalized

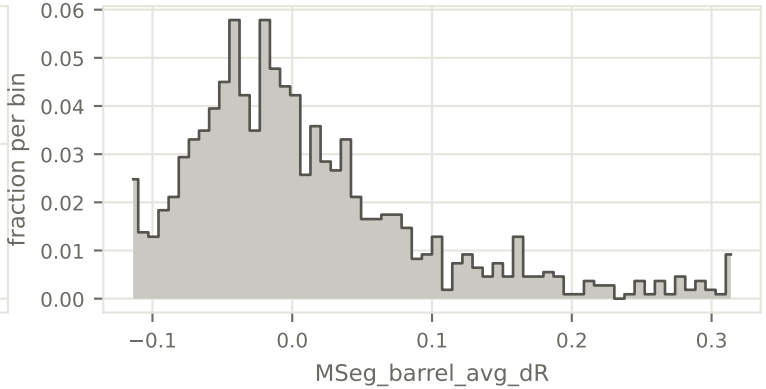



2 · combined signal
background n 1,089
combined, pooled D 0.497 n 4,987
equal weight per mass D 0.366

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0% −999 0.0%
background n 1,089

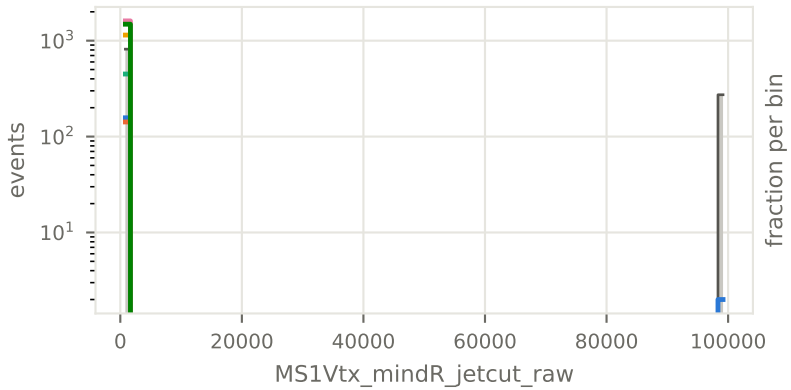
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

# MS1Vtx_mindR_jetcut_raw

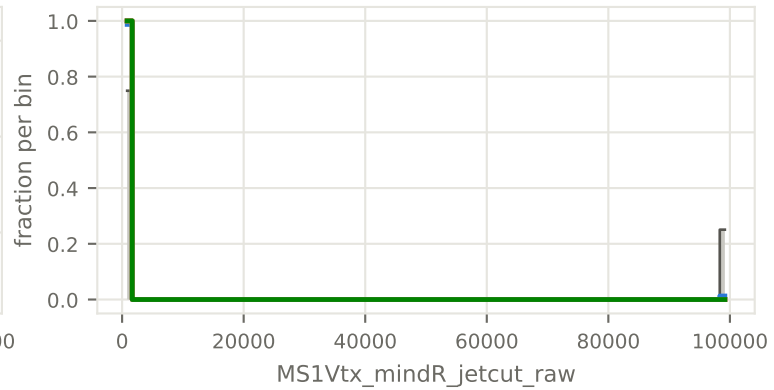
region: BARREL · rank 46 / 98 · in: NN1 barrel, NN1 endcap

D_equal 0.365 (noise 0.051) D_pooled 0.391 min 0.307 (mH125_mS55) max 0.432 (mH1000_mS275) bkg n = 1,089 sig n = 4,987

1a · each mass point vs background · raw

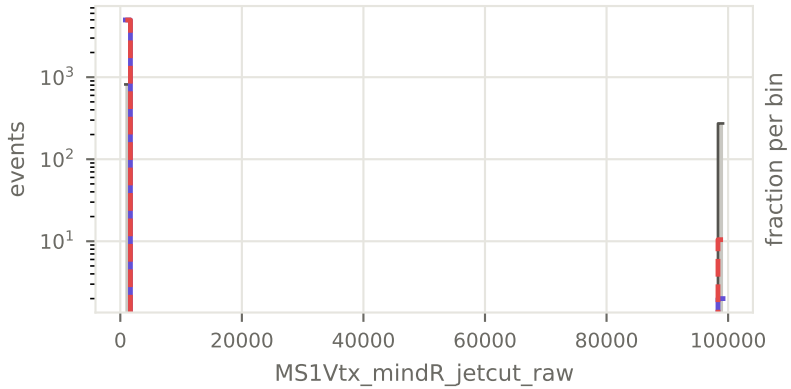



1b · each mass point vs background · normalized

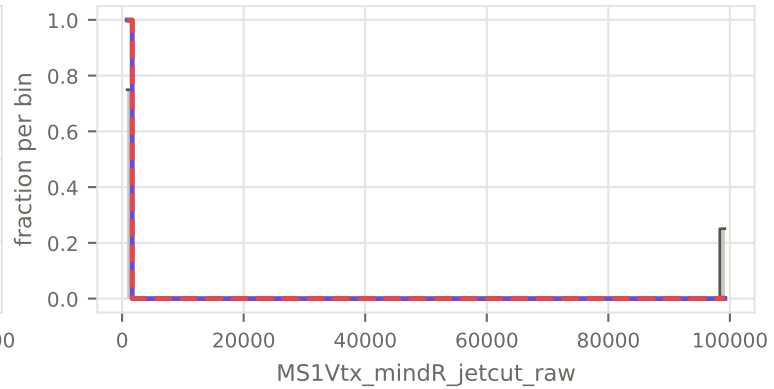



1 · each mass point
background n 1,089
mH125_mS15 D 0.341 n 159
mH125_mS55 D 0.307 n 142
mH200_mS50 D 0.353 n 449
mH400_mS100 D 0.398 n 1,144
mH600_mS150 D 0.385 n 1,608
mH1000_mS275 D 0.432 n 1,485

2a · combined signal vs background · raw

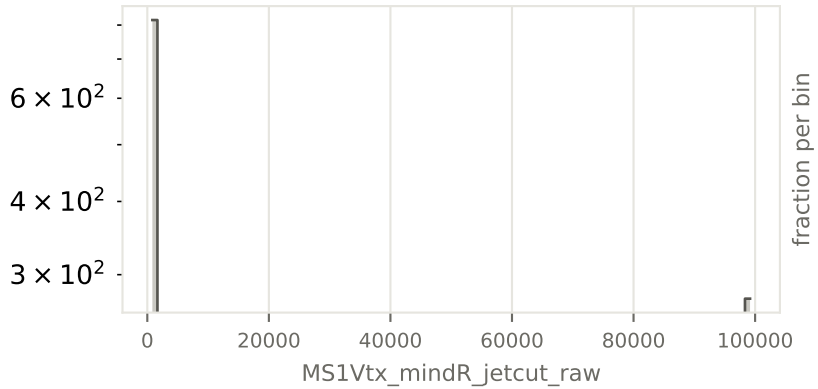



2b · combined signal vs background · normalized

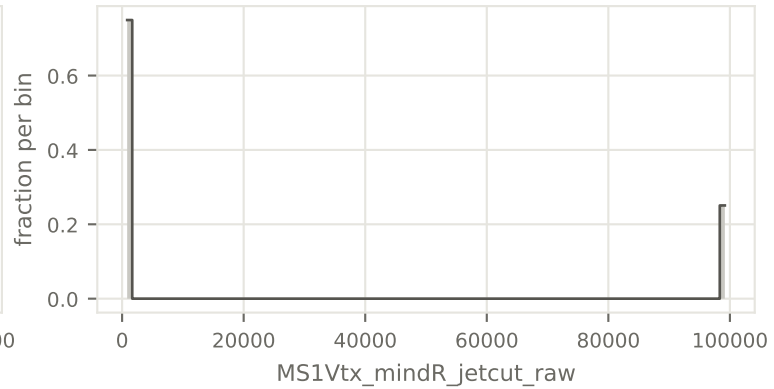



2 · combined signal
background n 1,089
combined, pooled D 0.391 n 4,987
equal weight per mass D 0.365

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0% −999 0.0%
background n 1,089

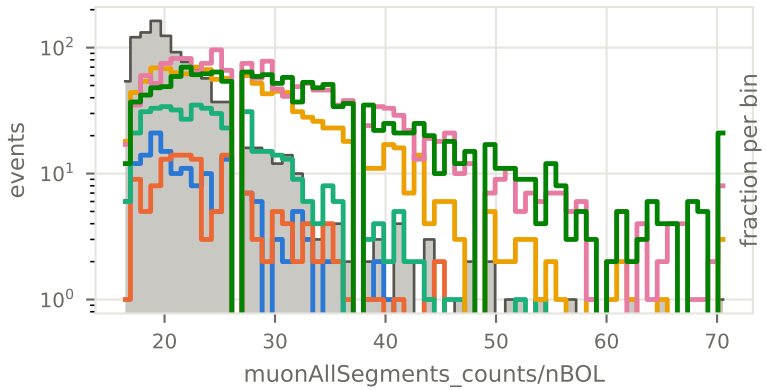
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

# muonAllSegments_counts/nBOL

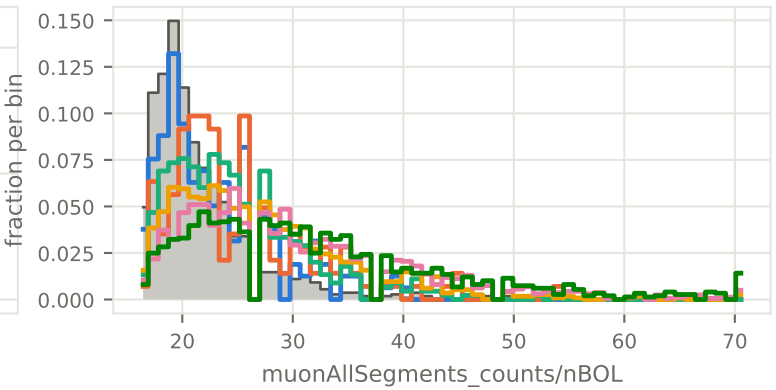
region: BARREL · rank 51 / 98

D_equal 0.333 (noise 0.051) D_pooled 0.425 min 0.151 (mH125_mS15) max 0.517 (mH1000_mS275) bkg n = 1,089 sig n = 4,987

1a · each mass point vs background · raw

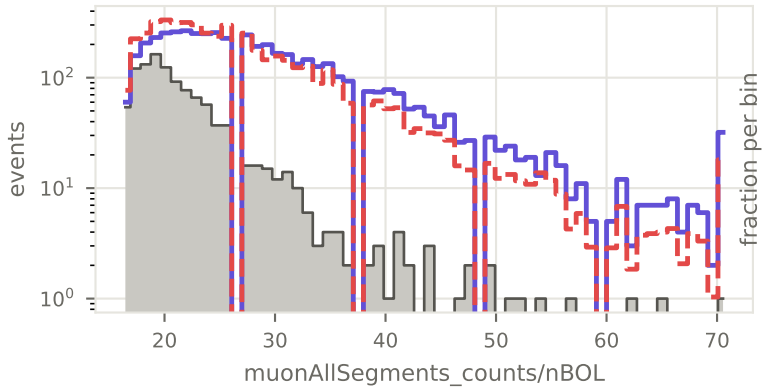



1b · each mass point vs background · normalized

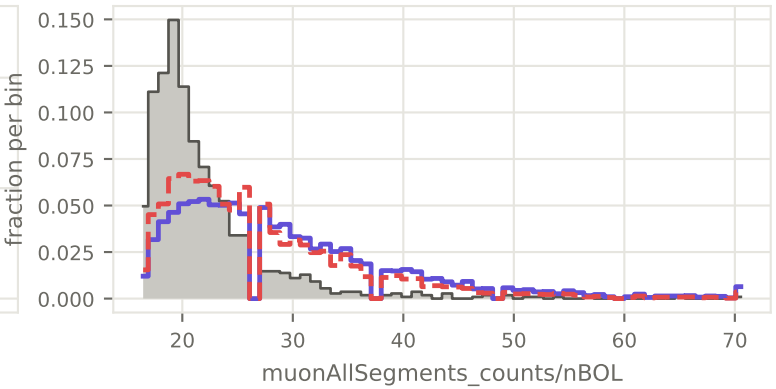



1 · each mass point
background n 1,089
mH125_mS15 D 0.151 n 159
mH125_mS55 D 0.292 n 142
mH200_mS50 D 0.291 n 449
mH400_mS100 D 0.370 n 1,144
mH600_mS150 D 0.477 n 1,608
mH1000_mS275 D 0.517 n 1,485

2a · combined signal vs background · raw

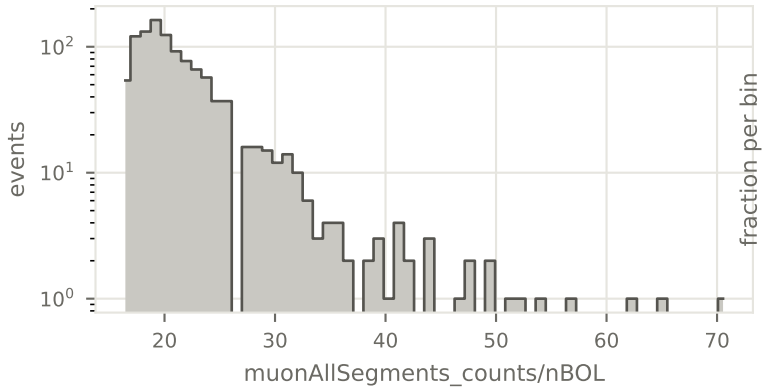



2b · combined signal vs background · normalized

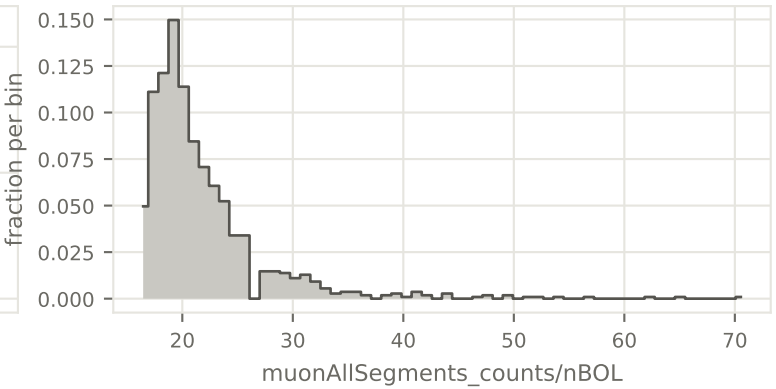



2 · combined signal
background n 1,089
combined, pooled D 0.425 n 4,987
equal weight per mass D 0.333

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0% −999 0.0%
background n 1,089

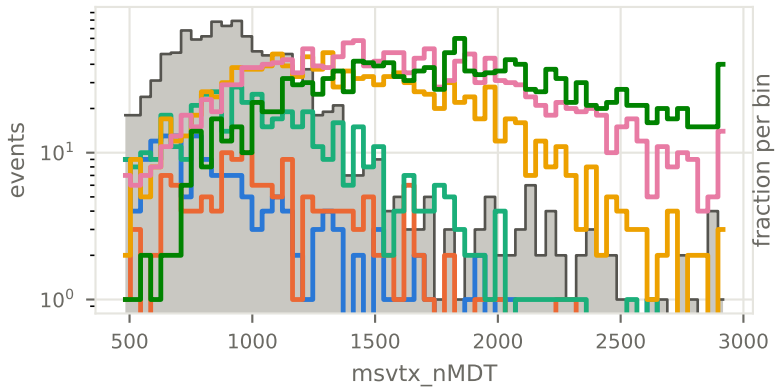
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

# msvtx_nMDT

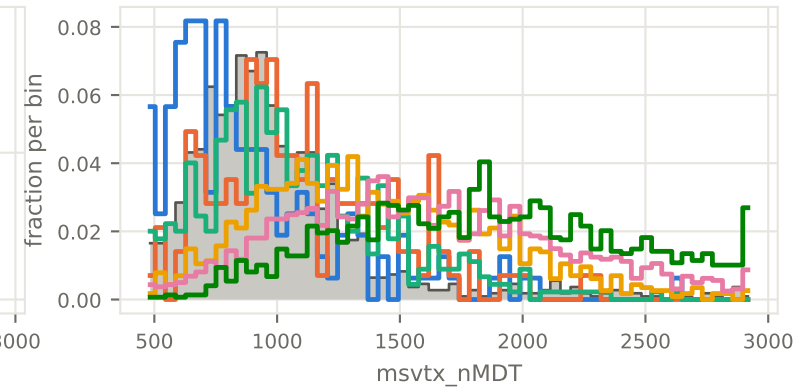
region: BARREL · rank 10 / 98

D_equal 0.331 (noise 0.051) D_pooled 0.500 min 0.161 (mH200_mS50) max 0.687 (mH1000_mS275) bkg n = 1,089 sig n = 4,987

1a · each mass point vs background · raw

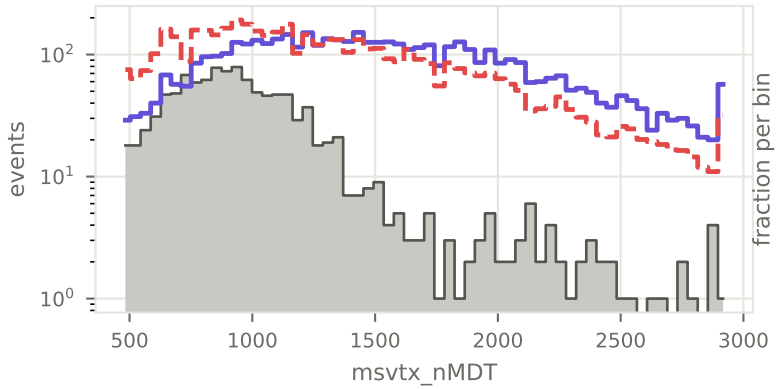



1b · each mass point vs background · normalized

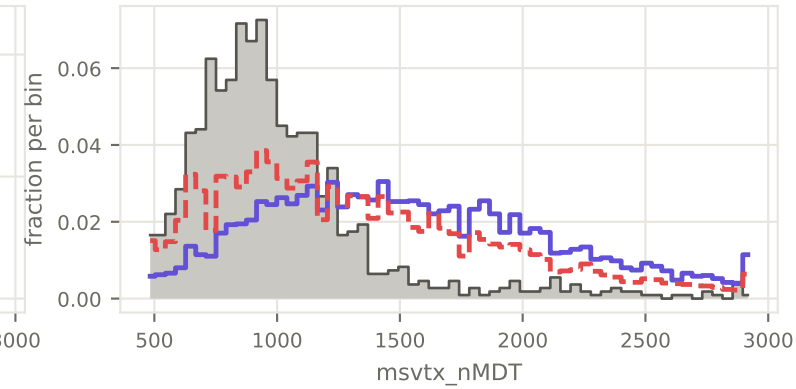



1 · each mass point
background n 1,089
mH125_mS15 D 0.224 n 159
mH125_mS55 D 0.177 n 142
mH200_mS50 D 0.161 n 449
mH400_mS100 D 0.439 n 1,144
mH600_mS150 D 0.555 n 1,608
mH1000_mS275 D 0.687 n 1,485

2a · combined signal vs background · raw

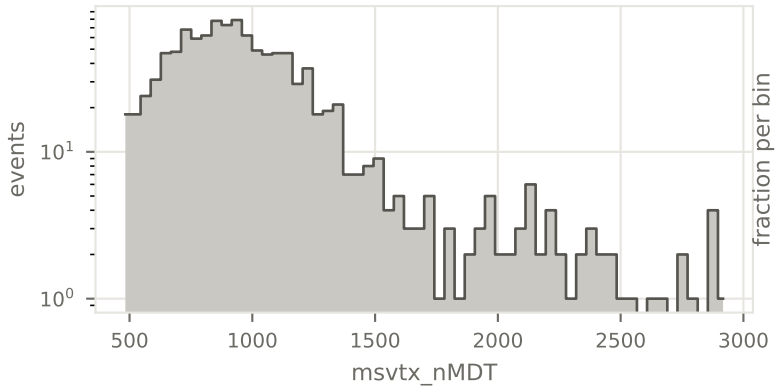



2b · combined signal vs background · normalized

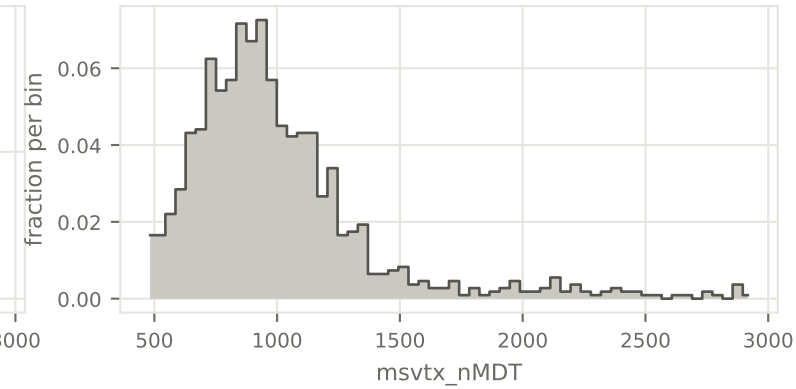



2 · combined signal
background n 1,089
combined, pooled D 0.500 n 4,987
equal weight per mass D 0.331

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0% −999 0.0%
background n 1,089

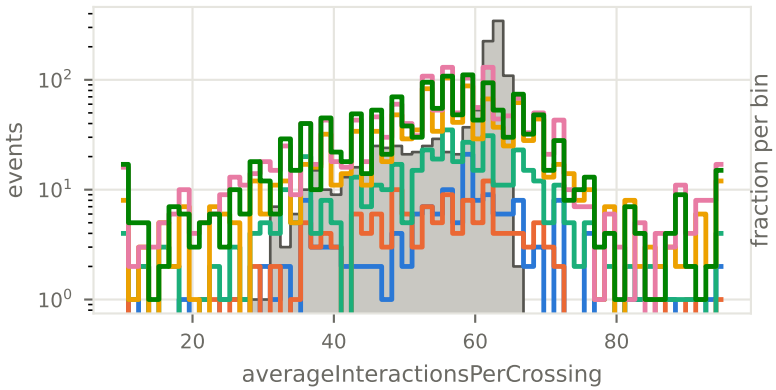
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

# averageInteractionsPerCrossing

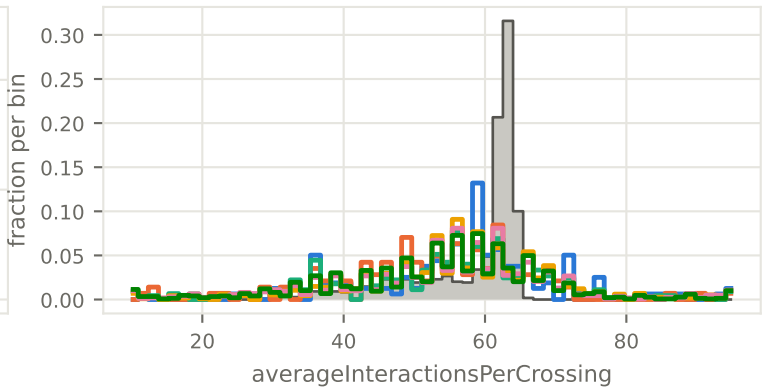
region: BARREL · rank 30 / 98

D_equal 0.326 (noise 0.051) D_pooled 0.324 min 0.307 (mH125_mS15) max 0.367 (mH125_mS55) bkg n = 1,089 sig n = 4,987

1a · each mass point vs background · raw

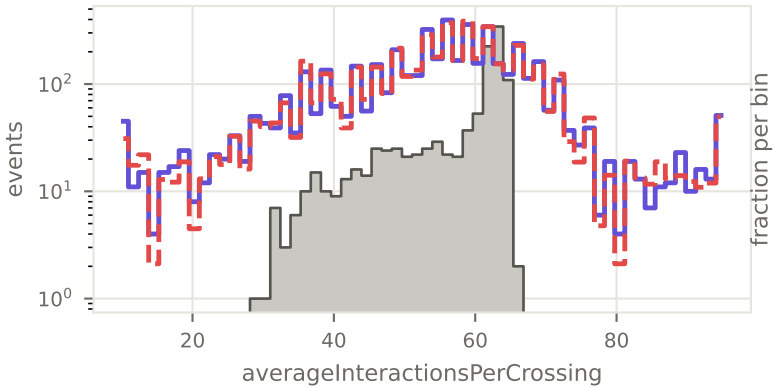



1b · each mass point vs background · normalized

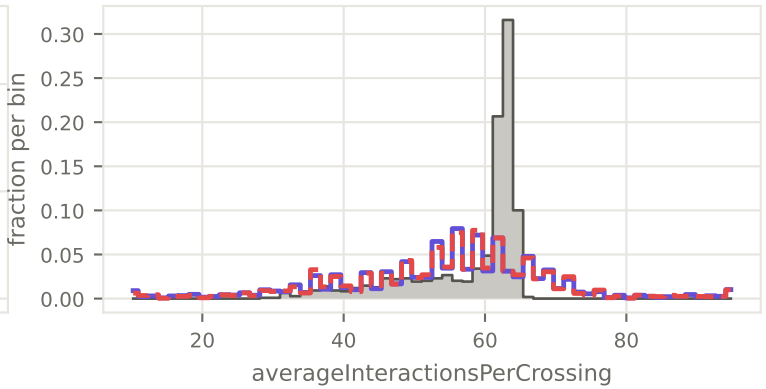



1 · each mass point
background n 1,089
mH125_mS15 D 0.307 n 159
mH125_mS55 D 0.367 n 142
mH200_mS50 D 0.315 n 449
mH400_mS100 D 0.324 n 1,144
mH600_mS150 D 0.316 n 1,608
mH1000_mS275 D 0.339 n 1,485

2a · combined signal vs background · raw

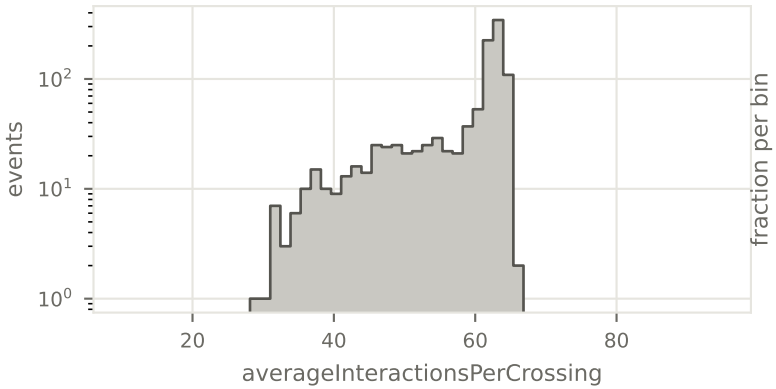



2b · combined signal vs background · normalized

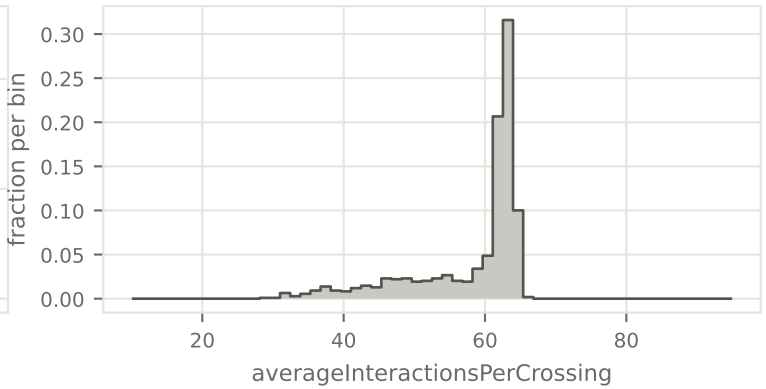



2 · combined signal
background n 1,089
combined, pooled D 0.324 n 4,987
equal weight per mass D 0.326

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0% −999 0.0%
background n 1,089

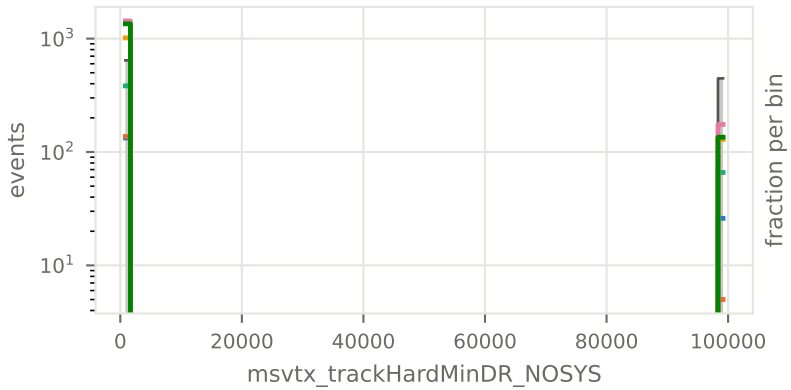
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

# msvtx_trackHardMinDR_NOSYS

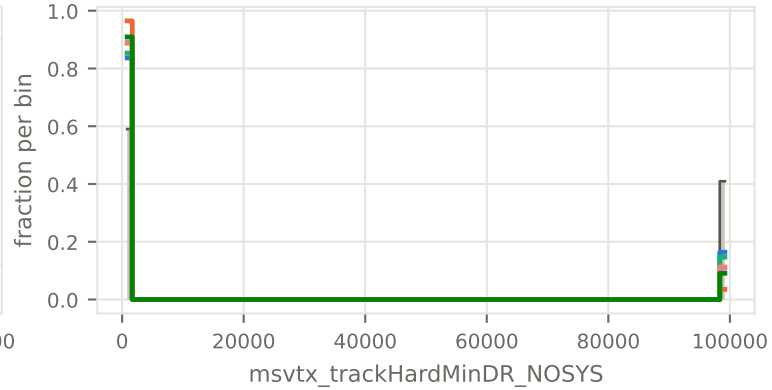
region: BARREL · rank 53 / 98

D_equal 0.320 (noise 0.051) D_pooled 0.346 min 0.273 (mH125_mS15) max 0.393 (mH1000_mS275) bkg n = 1,089 sig n = 4,987

1a · each mass point vs background · raw

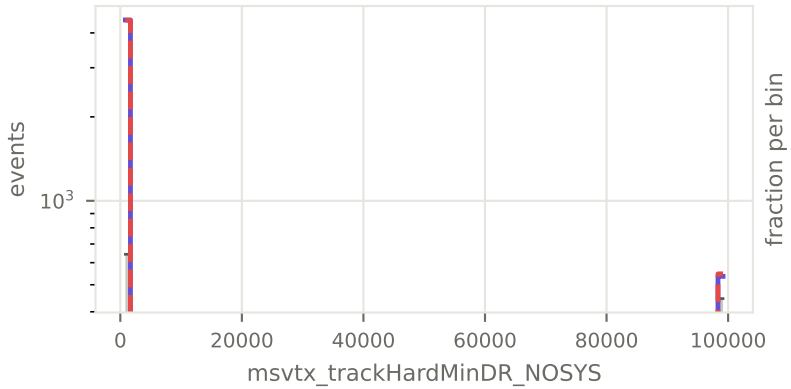



1b · each mass point vs background · normalized

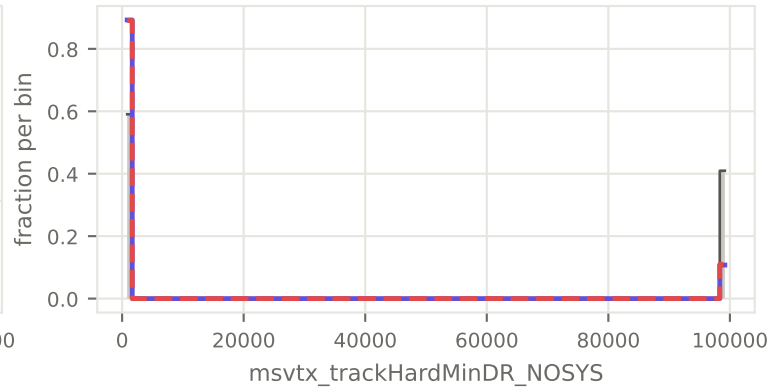



1 · each mass point
background n 1,089
mH125_mS15 D 0.273 n 159
mH125_mS55 D 0.378 n 142
mH200_mS50 D 0.291 n 449
mH400_mS100 D 0.327 n 1,144
mH600_mS150 D 0.349 n 1,608
mH1000_mS275 D 0.393 n 1,485

2a · combined signal vs background · raw

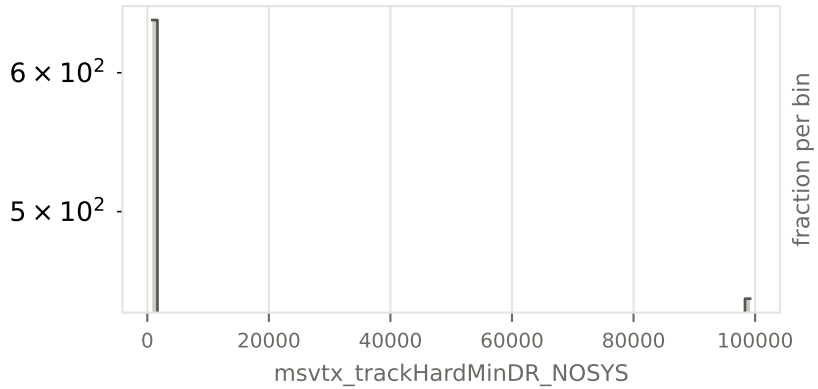



2b · combined signal vs background · normalized

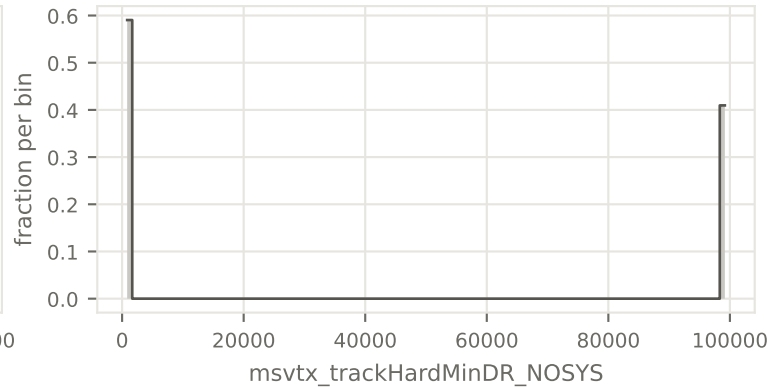



2 · combined signal
background n 1,089
combined, pooled D 0.346 n 4,987
equal weight per mass D 0.320

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0% −999 0.0%
background n 1,089

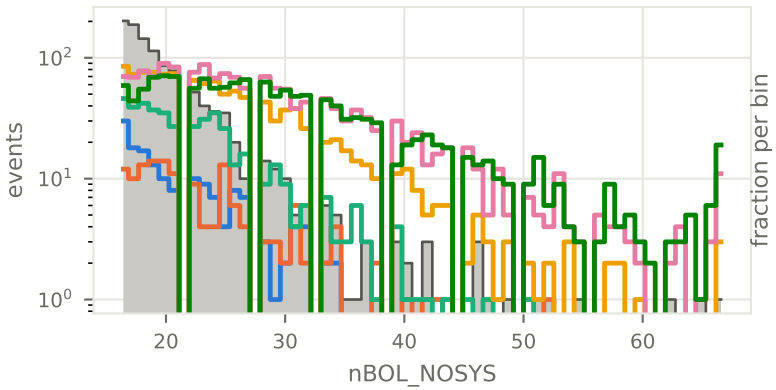
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

# nBOL_NOSYS

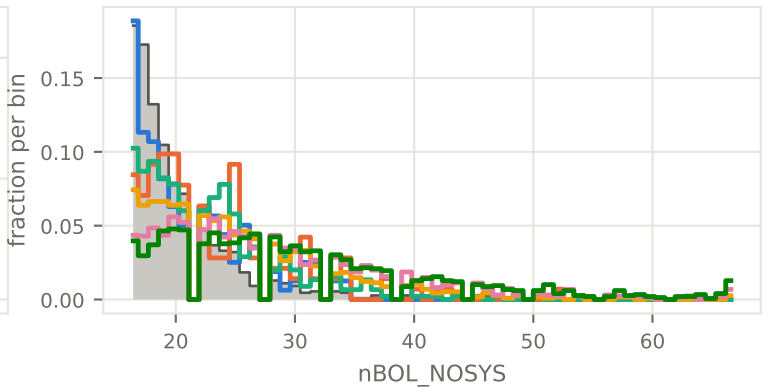
region: BARREL · rank 49 / 98 · in: NN1 barrel

D_equal 0.319 (noise 0.051) D_pooled 0.408 min 0.142 (mH125_mS15) max 0.508 (mH1000_mS275) bkg n = 1,089 sig n = 4,987

1a · each mass point vs background · raw

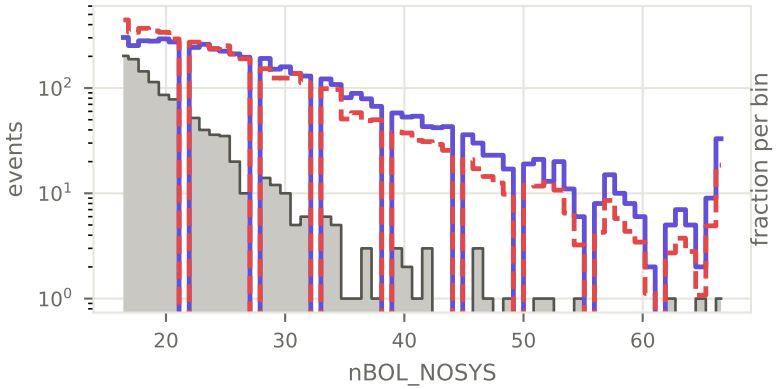



1b · each mass point vs background · normalized

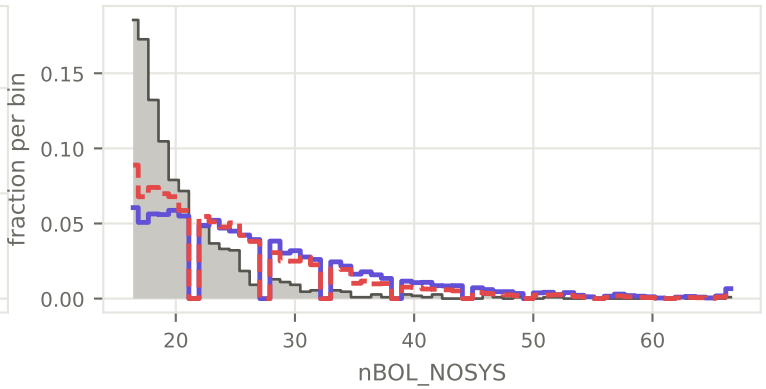



1 · each mass point
background n 1,089
mH125_mS15 D 0.142 n 159
mH125_mS55 D 0.250 n 142
mH200_mS50 D 0.242 n 449
mH400_mS100 D 0.346 n 1,144
mH600_mS150 D 0.459 n 1,608
mH1000_mS275 D 0.508 n 1,485

2a · combined signal vs background · raw

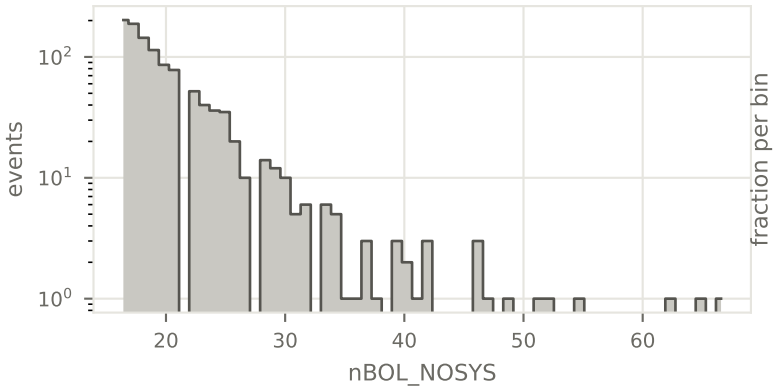



2b · combined signal vs background · normalized

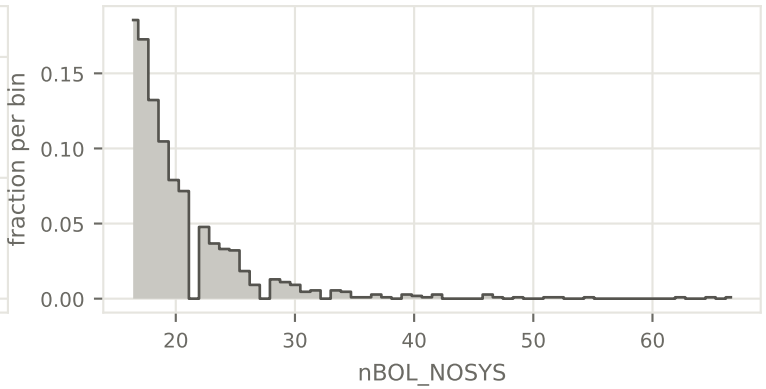



2 · combined signal
background n 1,089
combined, pooled D 0.408 n 4,987
equal weight per mass D 0.319

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0% −999 0.0%
background n 1,089

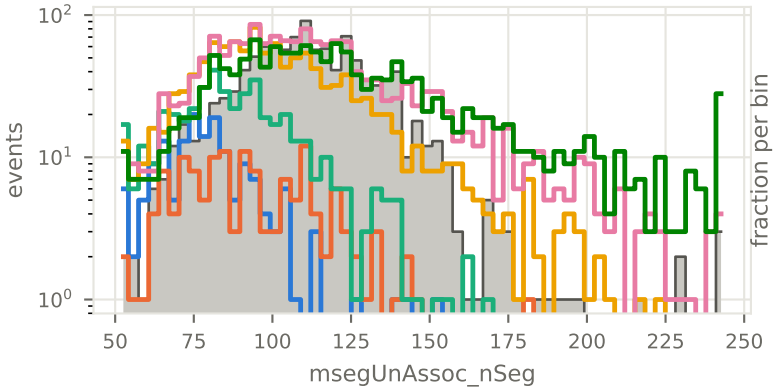
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

# msegUnAssoc_nSeg

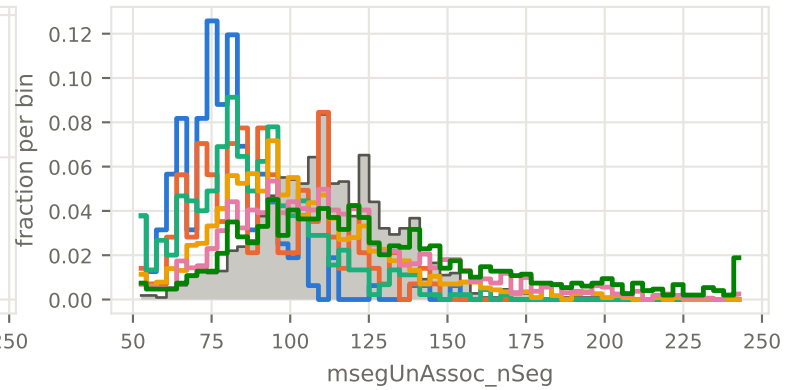
region: BARREL · rank 60 / 98

D_equal 0.314 (noise 0.051)  D_pooled 0.175  min 0.113 (mH600_mS150)  max 0.652 (mH125_mS15)  bkg n = 1,089  sig n = 4,987

1a · each mass point vs background · raw

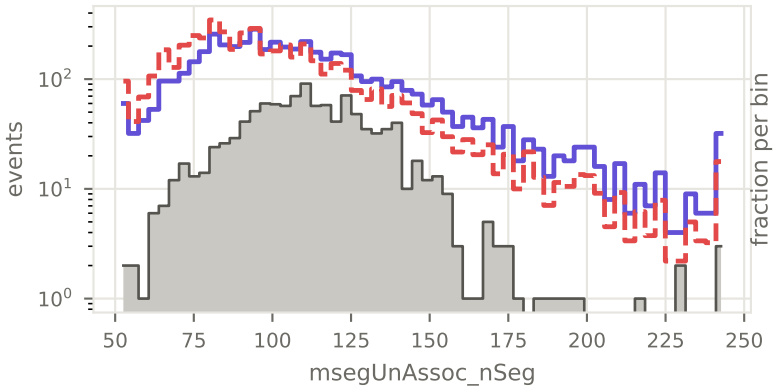



1b · each mass point vs background · normalized

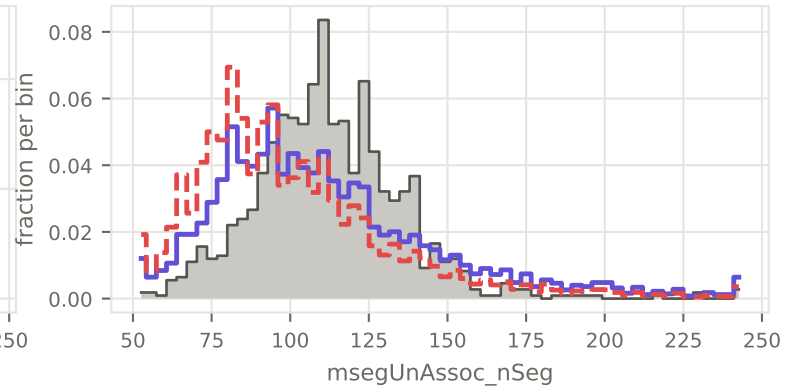



1 · each mass point
background n 1,089
mH125_mS15 D 0.652 n 159
mH125_mS55 D 0.391 n 142
mH200_mS50 D 0.472 n 449
mH400_mS100 D 0.239 n 1,144
mH600_mS150 D 0.113 n 1,608
mH1000_mS275 D 0.208 n 1,485

2a · combined signal vs background · raw

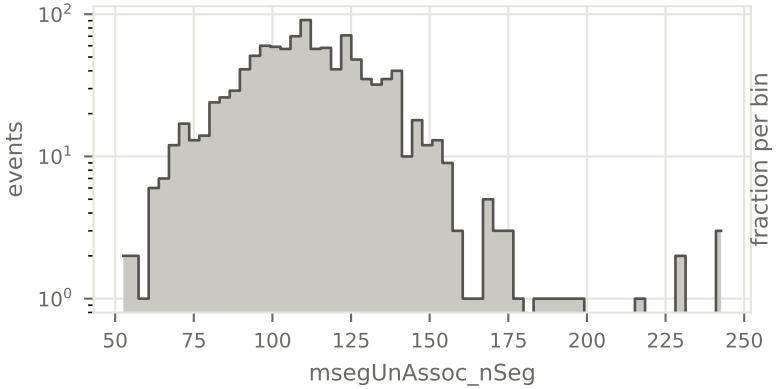



2b · combined signal vs background · normalized

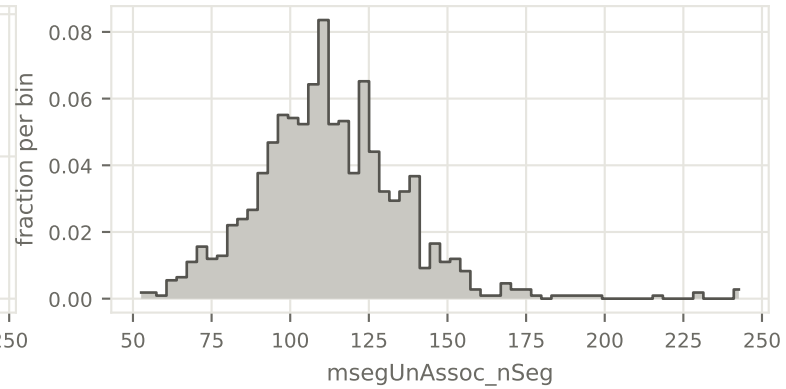



2 · combined signal
background n 1,089
combined, pooled D 0.175 n 4,987
equal weight per mass D 0.314

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0% −999 0.0%
background n 1,089

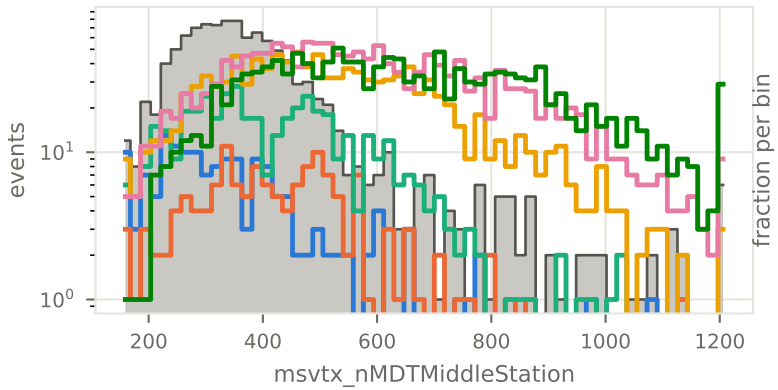
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

# msvtx_nMDTMiddleStation

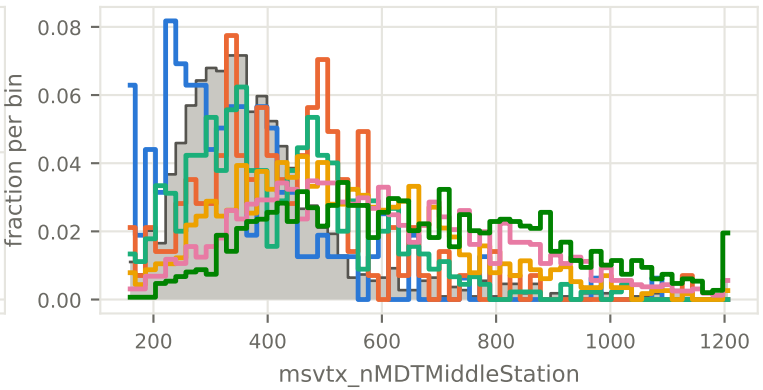
region: BARREL · rank 11 / 98

D_equal 0.313 (noise 0.051) D_pooled 0.444 min 0.196 (mH125_mS15) max 0.573 (mH1000_mS275) bkg n = 1,089 sig n = 4,987

1a · each mass point vs background · raw

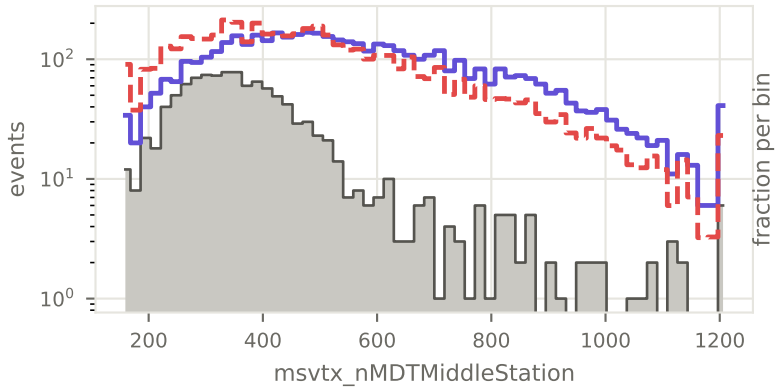



1b · each mass point vs background · normalized

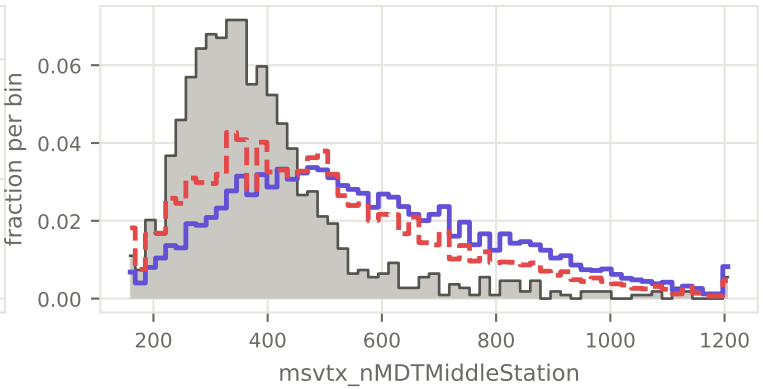



1 · each mass point
background n 1,089
mH125_mS15 D 0.196 n 159
mH125_mS55 D 0.242 n 142
mH200_mS50 D 0.216 n 449
mH400_mS100 D 0.405 n 1,144
mH600_mS150 D 0.488 n 1,608
mH1000_mS275 D 0.573 n 1,485

2a · combined signal vs background · raw

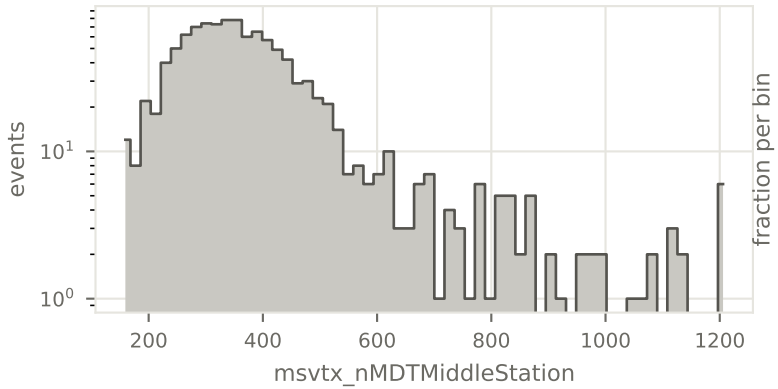



2b · combined signal vs background · normalized

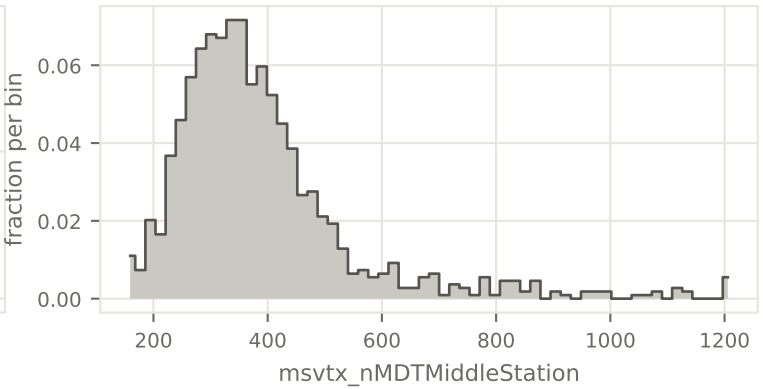



2 · combined signal
background n 1,089
combined, pooled D 0.444 n 4,987
equal weight per mass D 0.313

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0% −999 0.0%
background n 1,089

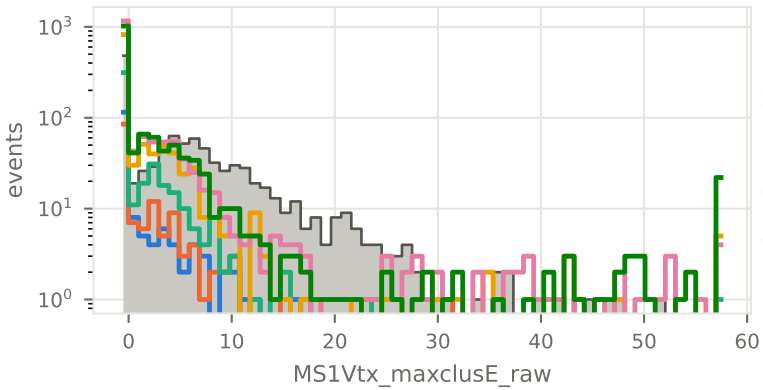
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

# MS1Vtx_maxclusE_raw

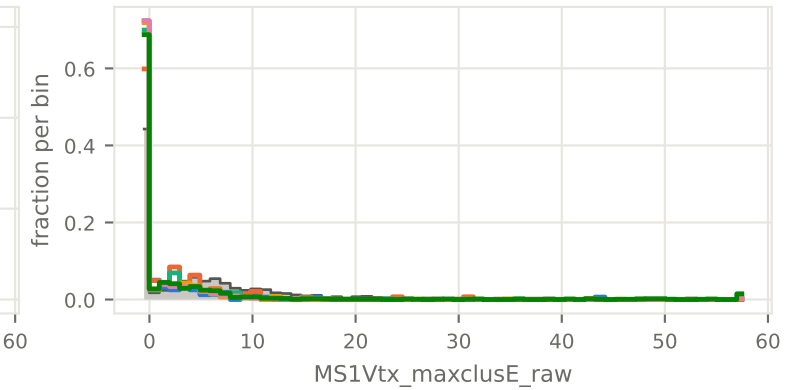
region: BARREL · rank 29 / 98 · in: NN1 barrel, NN1 endcap

D_equal 0.312 (noise 0.051) D_pooled 0.314 min 0.275 (mH125_mS55) max 0.332 (mH125_mS15) bkg n = 1,089 sig n = 4,987

1a · each mass point vs background · raw

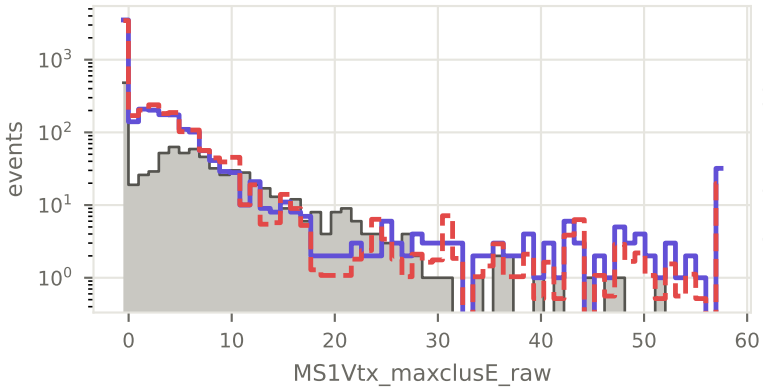



1b · each mass point vs background · normalized

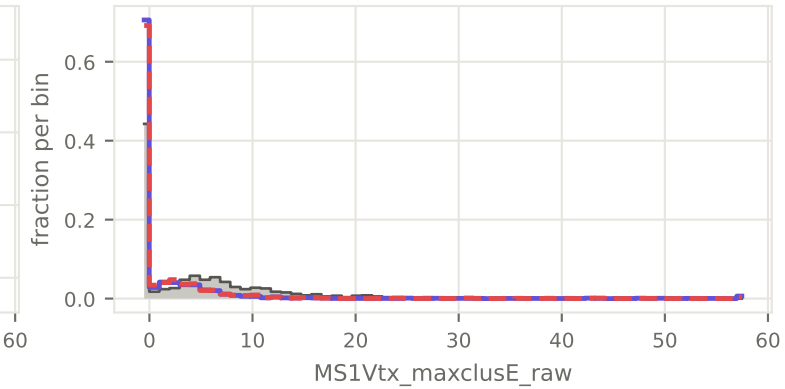



1 · each mass point
background n 1,089
mH125_mS15 D 0.332 n 159
mH125_mS55 D 0.275 n 142
mH200_mS50 D 0.331 n 449
mH400_mS100 D 0.328 n 1,144
mH600_mS150 D 0.319 n 1,608
mH1000_mS275 D 0.295 n 1,485

2a · combined signal vs background · raw

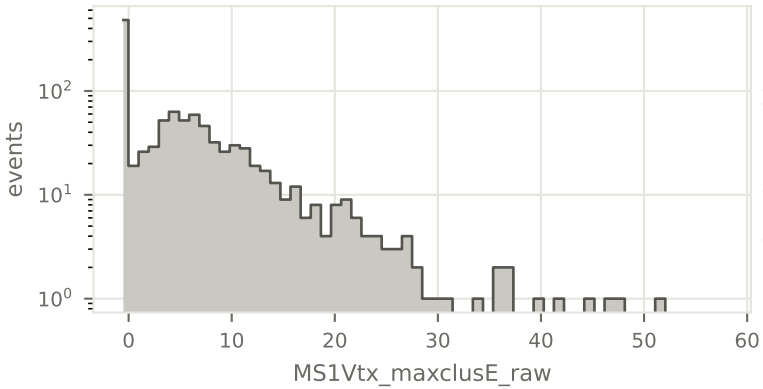



2b · combined signal vs background · normalized

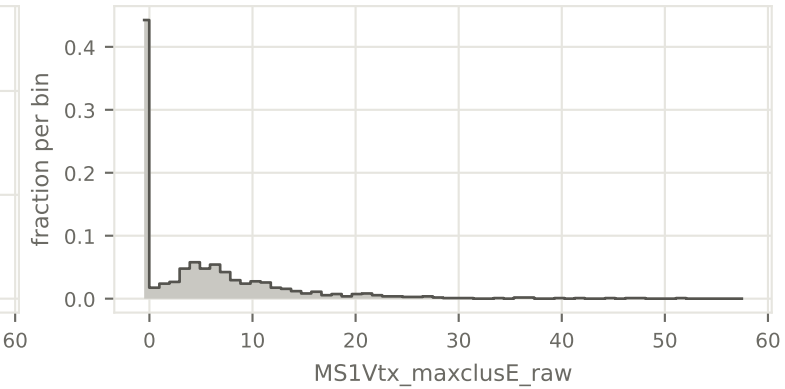



2 · combined signal
background n 1,089
combined, pooled D 0.314 n 4,987
equal weight per mass D 0.312

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0% −999 0.0%
background n 1,089

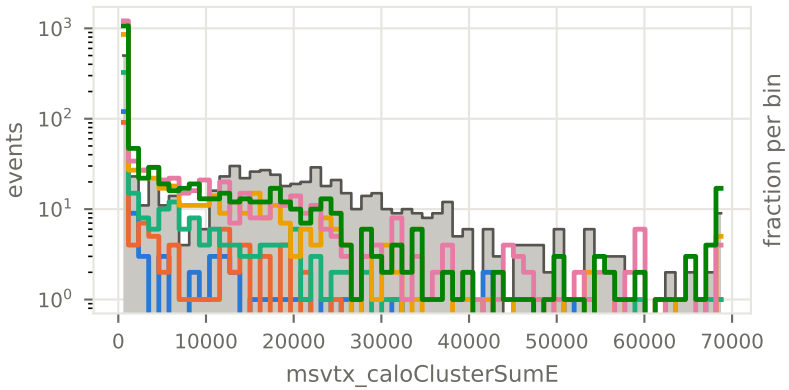
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

# msvtx_caloClusterSumE

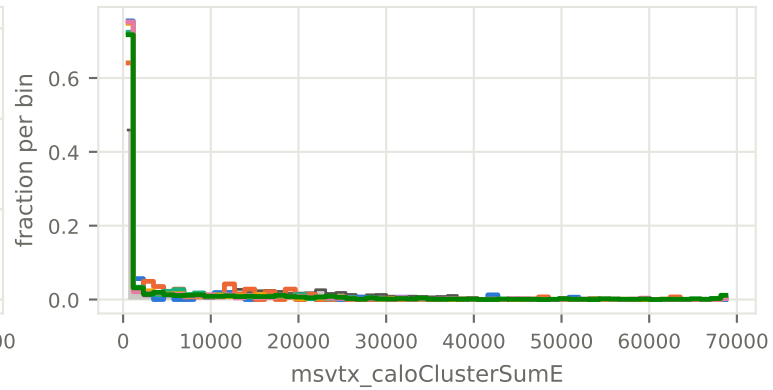
region: BARREL · rank 26 / 98 · in: NN1 barrel, NN1 endcap

D_equal 0.310 (noise 0.051) D_pooled 0.312 min 0.282 (mH125_mS55) max 0.343 (mH125_mS15) bkg n = 1,089 sig n = 4,987

1a · each mass point vs background · raw

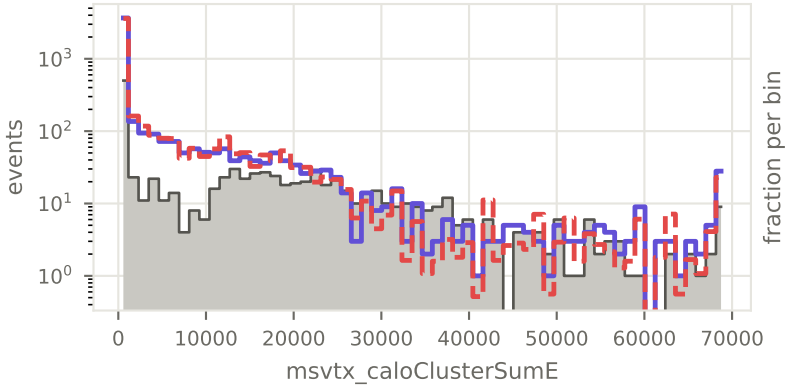



1b · each mass point vs background · normalized

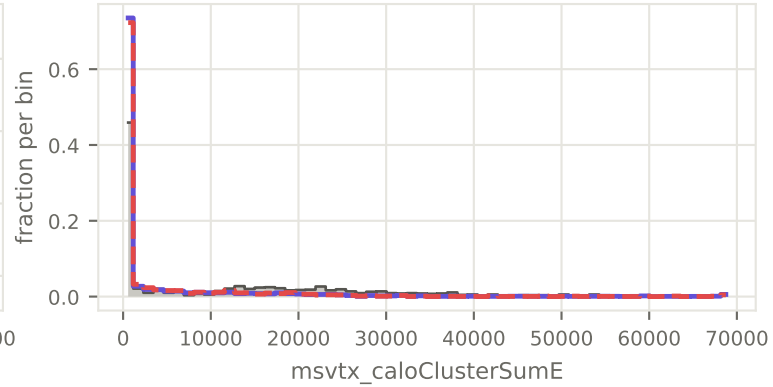



1 · each mass point
background n 1,089
mH125_mS15 D 0.343 n 159
mH125_mS55 D 0.282 n 142
mH200_mS50 D 0.330 n 449
mH400_mS100 D 0.327 n 1,144
mH600_mS150 D 0.318 n 1,608
mH1000_mS275 D 0.292 n 1,485

2a · combined signal vs background · raw

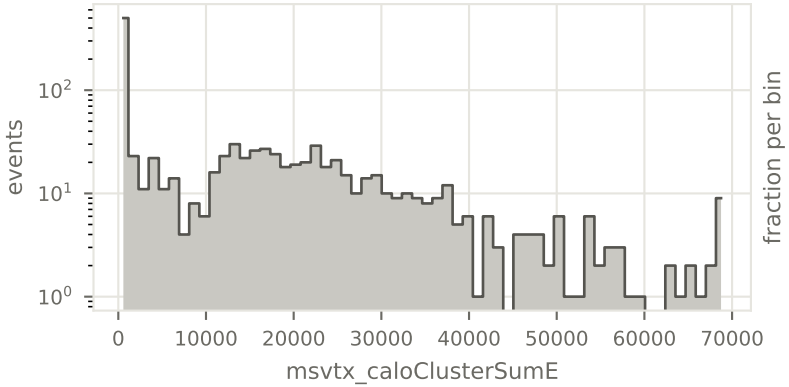



2b · combined signal vs background · normalized

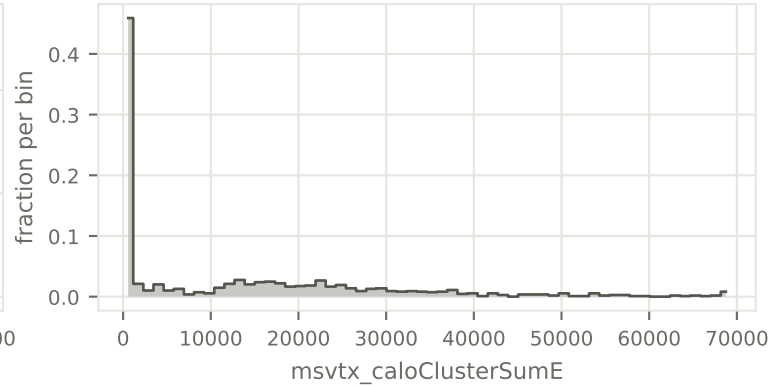



2 · combined signal
background n 1,089
combined, pooled D 0.312 n 4,987
equal weight per mass D 0.310

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0% −999 0.0%
background n 1,089

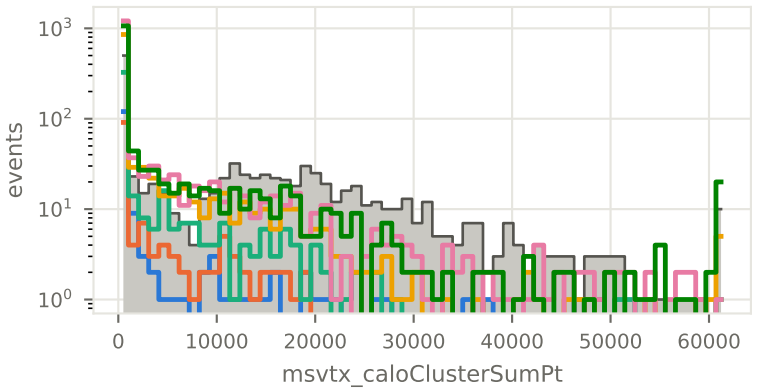
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

# msvtx_caloClusterSumPt

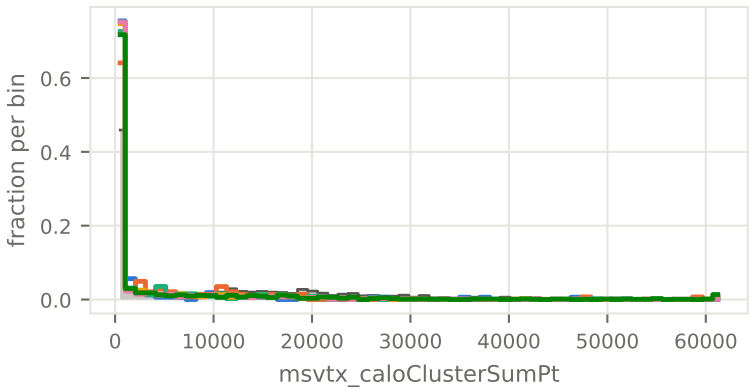
region: BARREL · rank 23 / 98

D_equal 0.308 (noise 0.051) D_pooled 0.308 min 0.284 (mH125_mS55) max 0.346 (mH125_mS15) bkg n = 1,089 sig n = 4,987

1a · each mass point vs background · raw

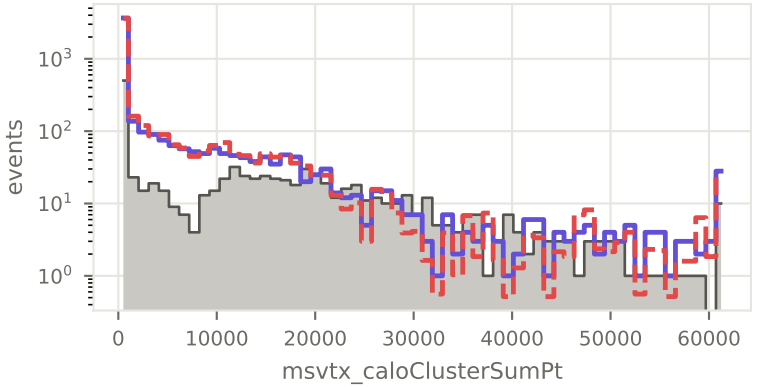



1b · each mass point vs background · normalized

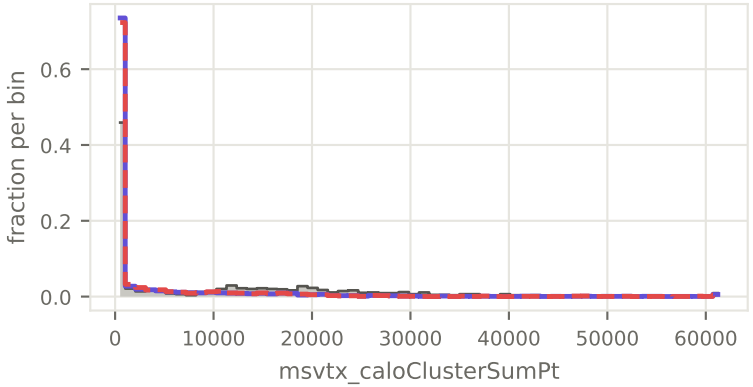



1 · each mass point
background n 1,089
mH125_mS15 D 0.346 n 159
mH125_mS55 D 0.284 n 142
mH200_mS50 D 0.329 n 449
mH400_mS100 D 0.325 n 1,144
mH600_mS150 D 0.312 n 1,608
mH1000_mS275 D 0.286 n 1,485

2a · combined signal vs background · raw

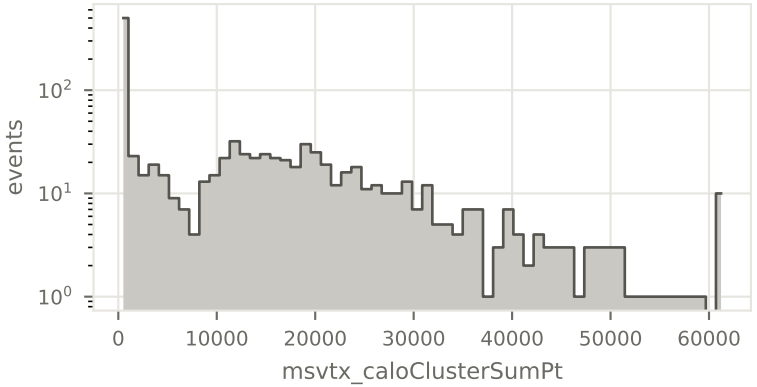



2b · combined signal vs background · normalized

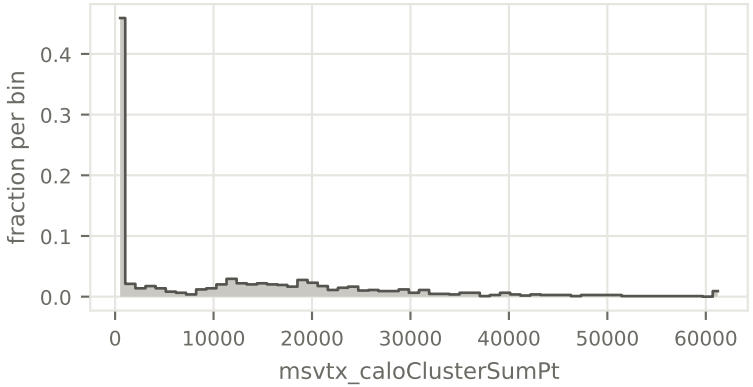



2 · combined signal
background n 1,089
combined, pooled D 0.308 n 4,987
equal weight per mass D 0.308

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0% −999 0.0%
background n 1,089

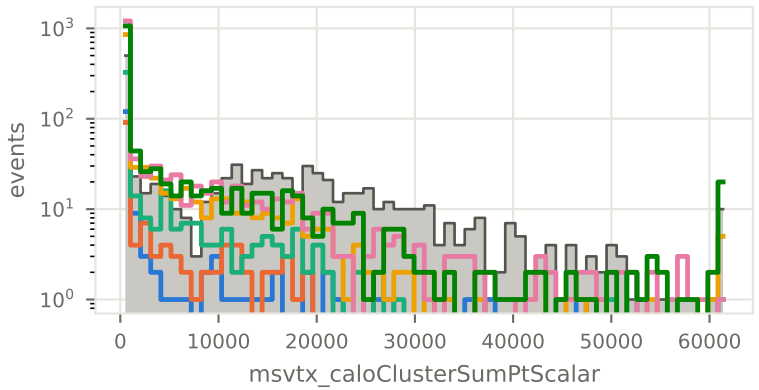
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

# msvtx_caloClusterSumPtScalar

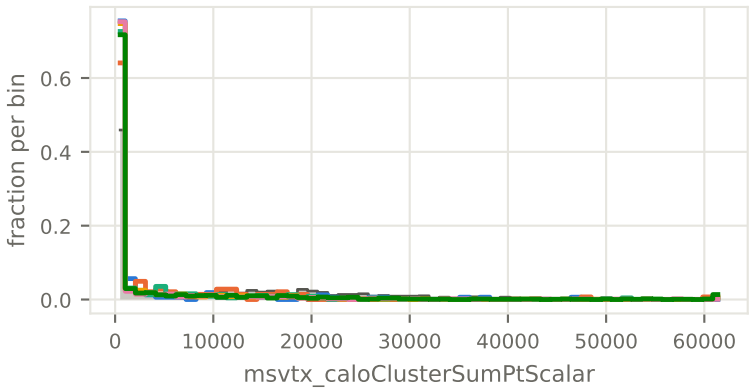
region: BARREL · rank 22 / 98 · in: NN1 barrel

D_equal 0.307 (noise 0.051) D_pooled 0.308 min 0.284 (mH125_mS55) max 0.347 (mH125_mS15) bkg n = 1,089 sig n = 4,987

1a · each mass point vs background · raw

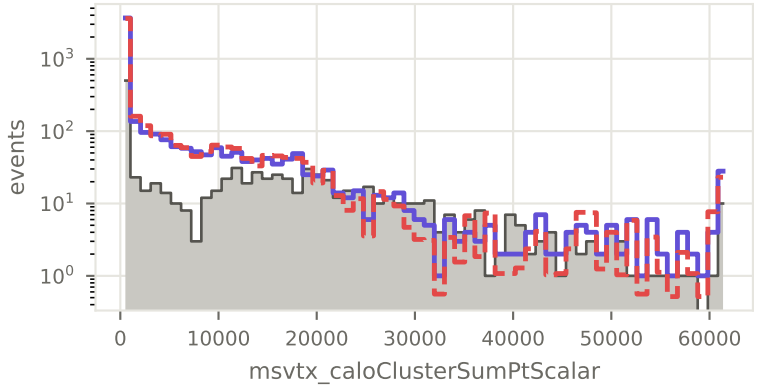



1b · each mass point vs background · normalized

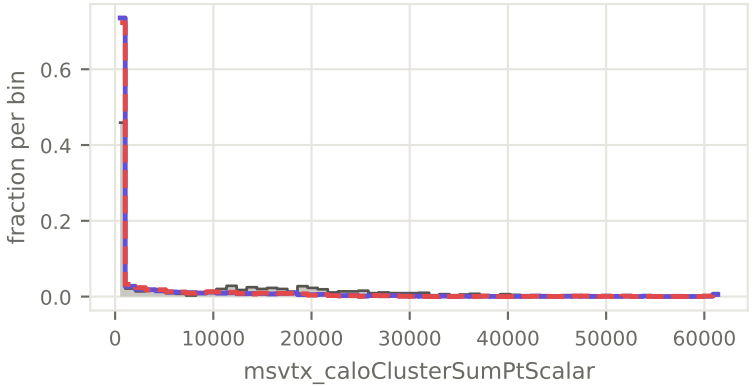



1 · each mass point
background n 1,089
mH125_mS15 D 0.347 n 159
mH125_mS55 D 0.284 n 142
mH200_mS50 D 0.329 n 449
mH400_mS100 D 0.326 n 1,144
mH600_mS150 D 0.313 n 1,608
mH1000_mS275 D 0.286 n 1,485

2a · combined signal vs background · raw

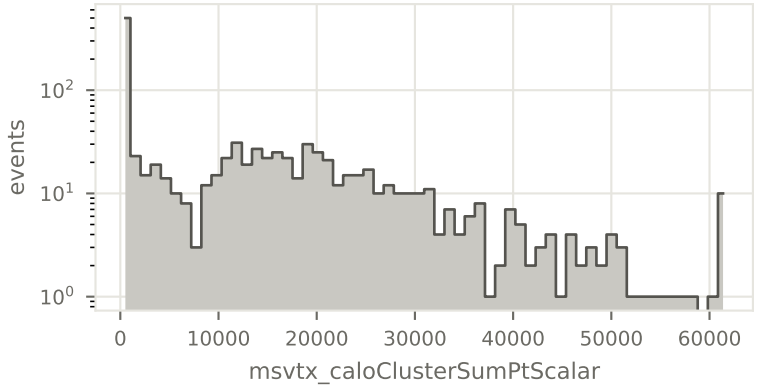



2b · combined signal vs background · normalized

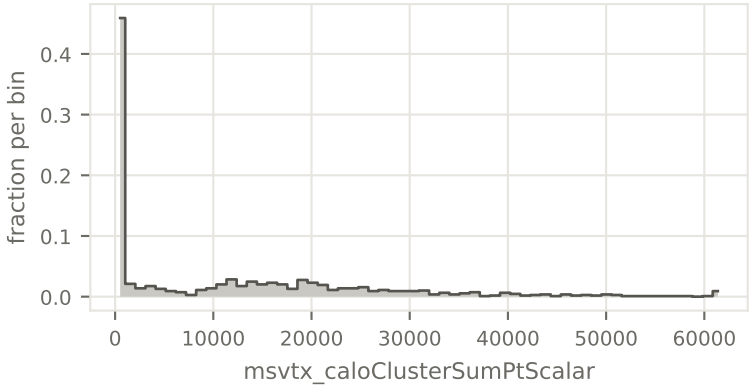



2 · combined signal
background n 1,089
combined, pooled D 0.308 n 4,987
equal weight per mass D 0.307

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0% −999 0.0%
background n 1,089

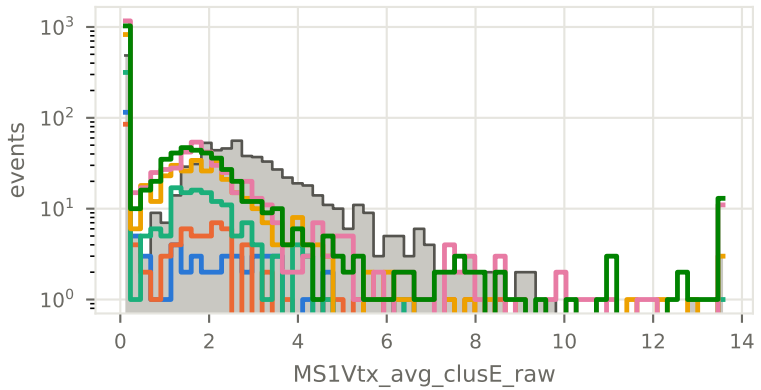
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

# MS1Vtx_avg_clusE_raw

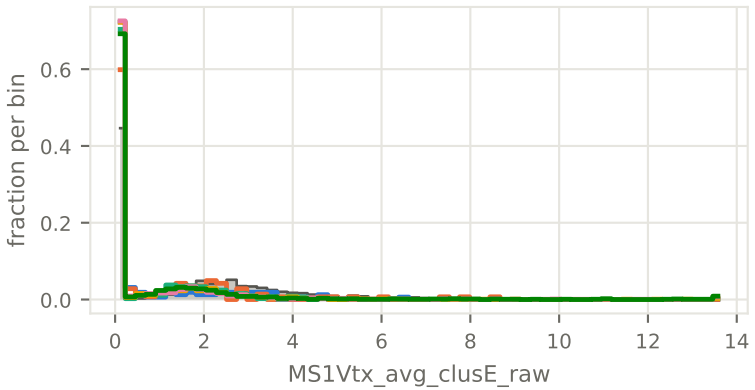
region: BARREL · rank 35 / 98 · in: NN1 barrel, NN1 endcap

D_equal 0.304 (noise 0.051) D_pooled 0.310 min 0.252 (mH125_mS55) max 0.333 (mH125_mS15) bkg n = 1,089 sig n = 4,987

1a · each mass point vs background · raw

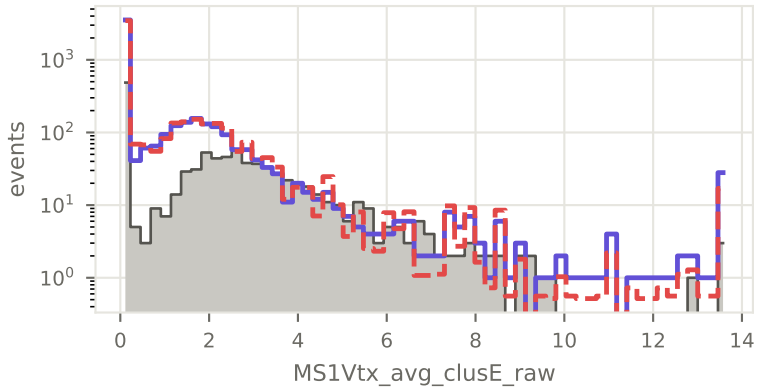



1b · each mass point vs background · normalized

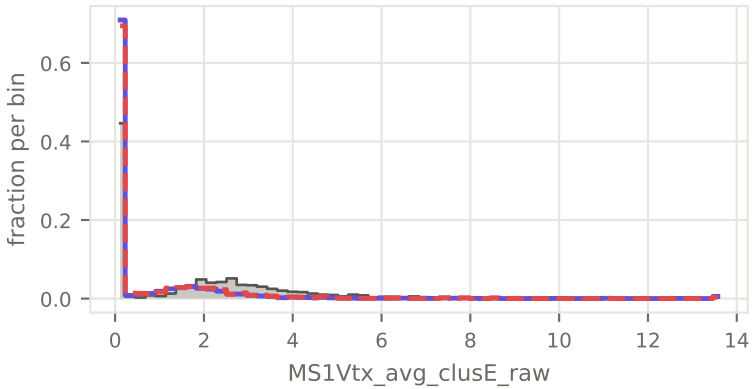



1 · each mass point
background n 1,089
mH125_mS15 D 0.333 n 159
mH125_mS55 D 0.252 n 142
mH200_mS50 D 0.315 n 449
mH400_mS100 D 0.323 n 1,144
mH600_mS150 D 0.319 n 1,608
mH1000_mS275 D 0.302 n 1,485

2a · combined signal vs background · raw

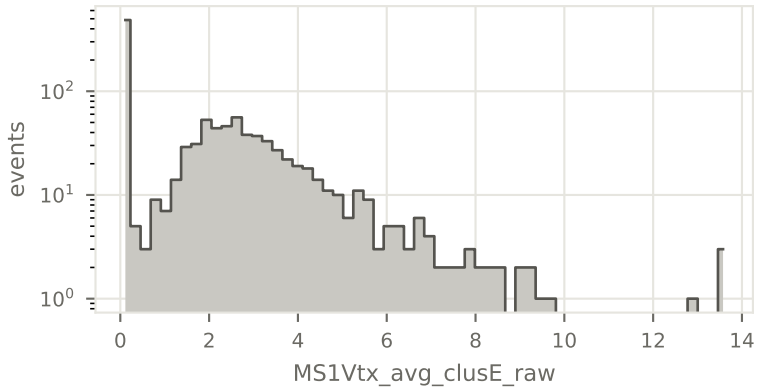



2b · combined signal vs background · normalized

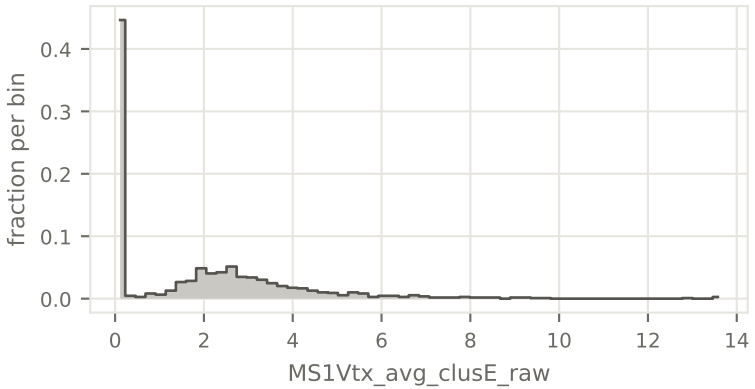



2 · combined signal
background n 1,089
combined, pooled D 0.310 n 4,987
equal weight per mass D 0.304

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0% −999 0.0%
background n 1,089

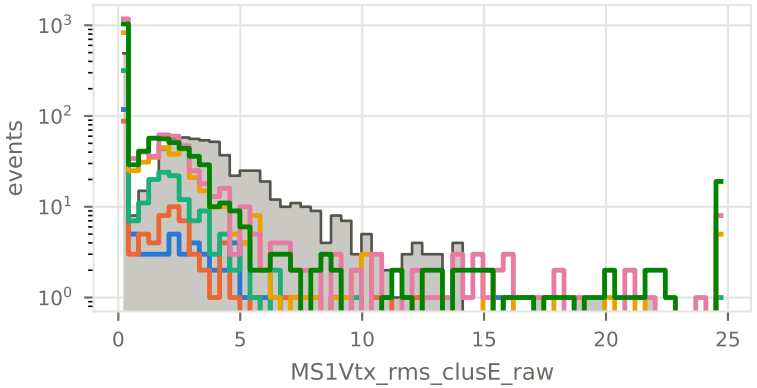
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

# MS1Vtx_rms_clusE_raw

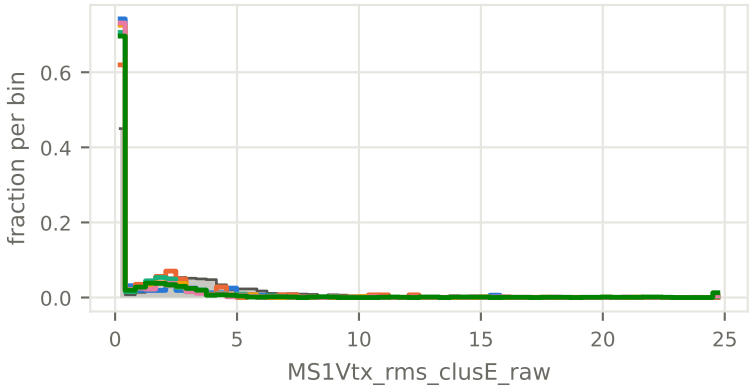
region: BARREL · rank 32 / 98 · in: NN1 barrel, NN1 endcap

D_equal 0.299 (noise 0.051) D_pooled 0.309 min 0.261 (mH125_mS55) max 0.331 (mH125_mS15) bkg n = 1,089 sig n = 4,987

1a · each mass point vs background · raw

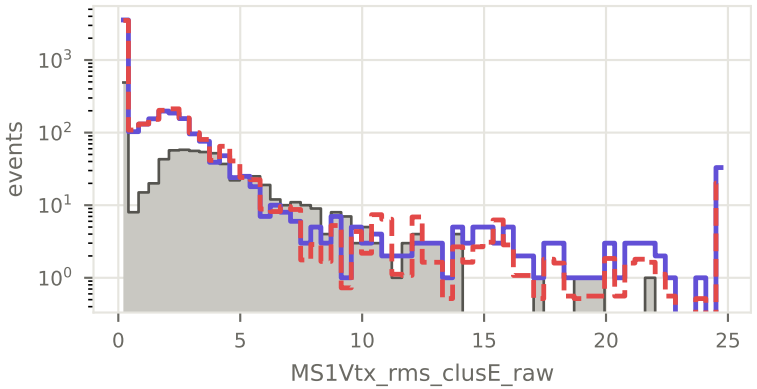



1b · each mass point vs background · normalized

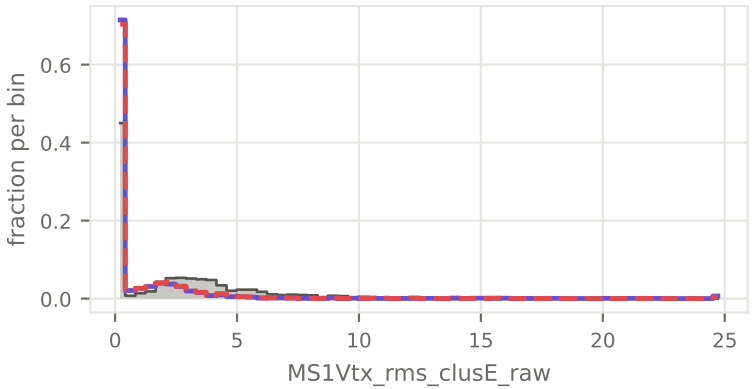



1 · each mass point
background n 1,089
mH125_mS15 D 0.331 n 159
mH125_mS55 D 0.261 n 142
mH200_mS50 D 0.324 n 449
mH400_mS100 D 0.322 n 1,144
mH600_mS150 D 0.317 n 1,608
mH1000_mS275 D 0.298 n 1,485

2a · combined signal vs background · raw

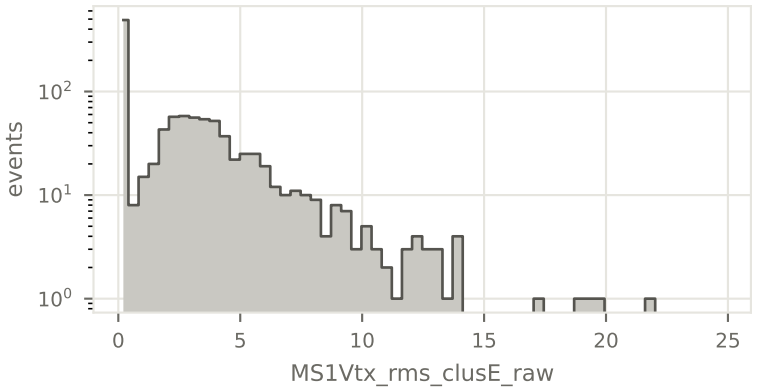



2b · combined signal vs background · normalized

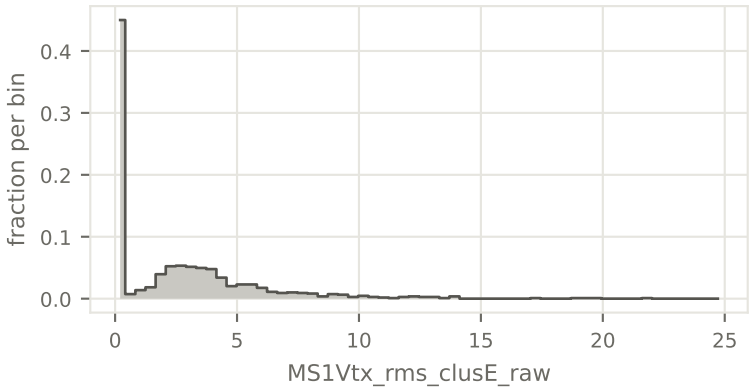



2 · combined signal
background n 1,089
combined, pooled D 0.309 n 4,987
equal weight per mass D 0.299

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0% −999 0.0%
background n 1,089

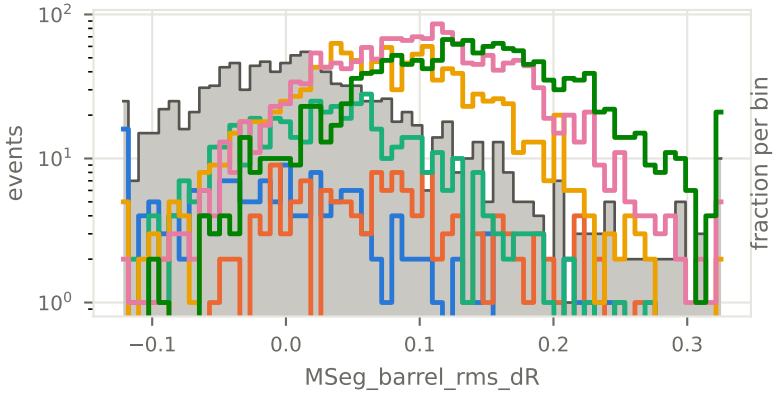
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

# MSeg_barrel_rms_dR

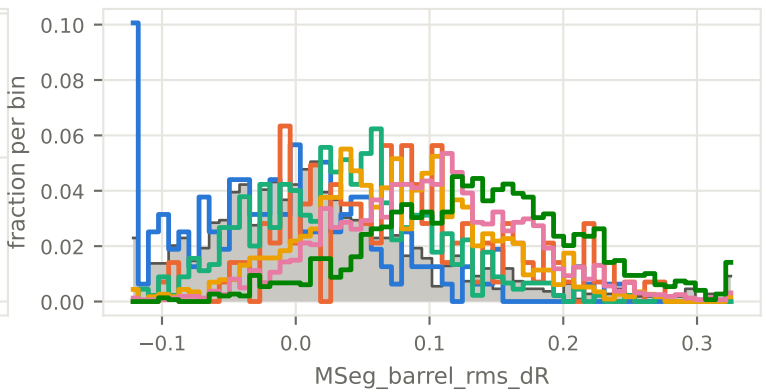
region: BARREL · rank 98 / 98 · in: NN2 barrel

D_equal 0.297 (noise 0.051) D_pooled 0.414 min 0.133 (mH125_mS15) max 0.584 (mH1000_mS275) bkg n = 1,089 sig n = 4,987

1a · each mass point vs background · raw

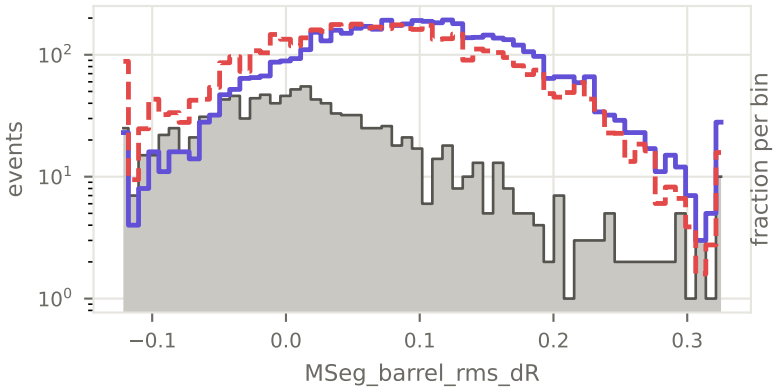



1b · each mass point vs background · normalized

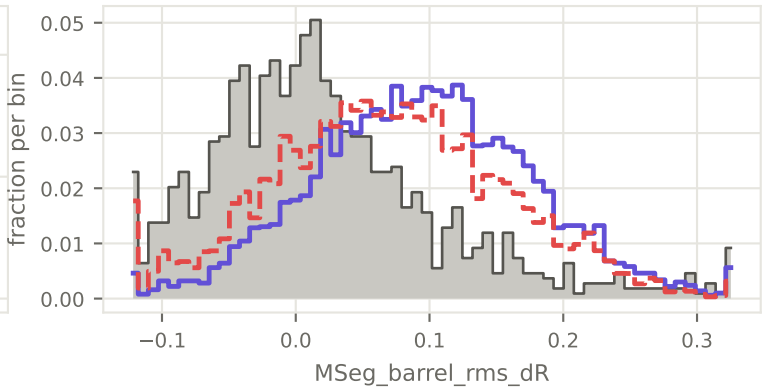



1 · each mass point
background n 1,089
mH125_mS15 D 0.133 n 159
mH125_mS55 D 0.339 n 142
mH200_mS50 D 0.172 n 449
mH400_mS100 D 0.377 n 1,144
mH600_mS150 D 0.457 n 1,608
mH1000_mS275 D 0.584 n 1,485

2a · combined signal vs background · raw

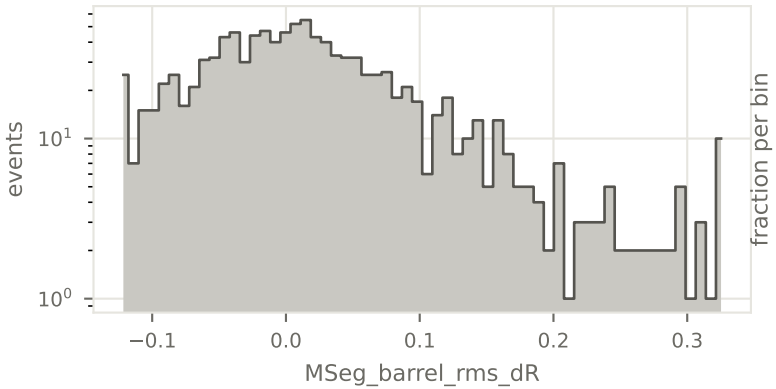



2b · combined signal vs background · normalized

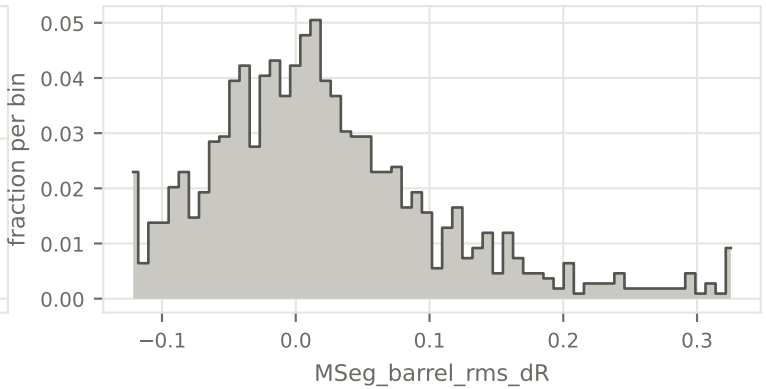



2 · combined signal
background n 1,089
combined, pooled D 0.414 n 4,987
equal weight per mass D 0.297

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0% −999 0.0%
background n 1,089

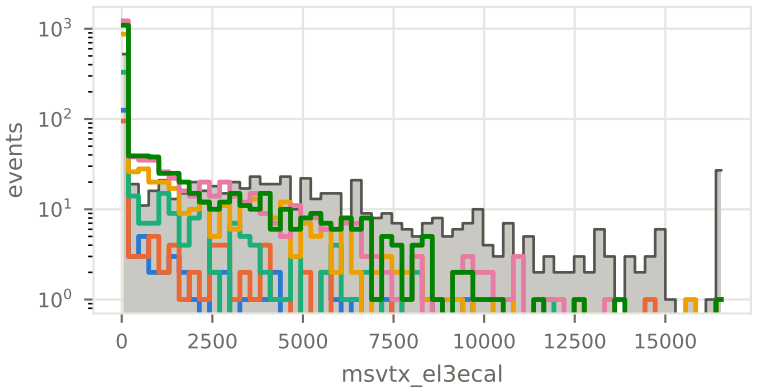
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

# msvtx_el3ecal

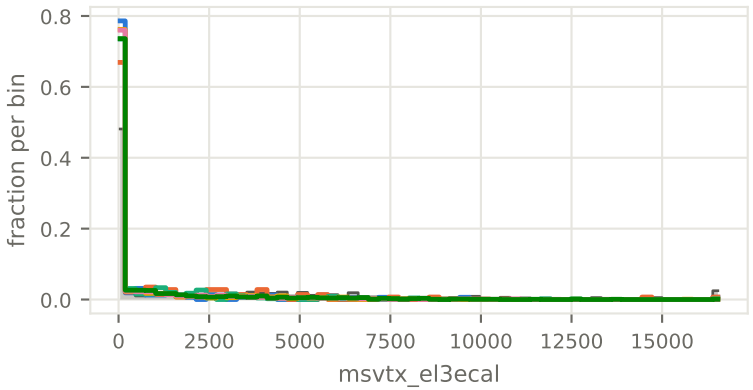
region: BARREL · rank 34 / 98 · in: NN1 barrel, NN1 endcap

D_equal 0.293 (noise 0.051) D_pooled 0.300 min 0.251 (mH125_mS55) max 0.331 (mH125_mS15) bkg n = 1,089 sig n = 4,987

1a · each mass point vs background · raw

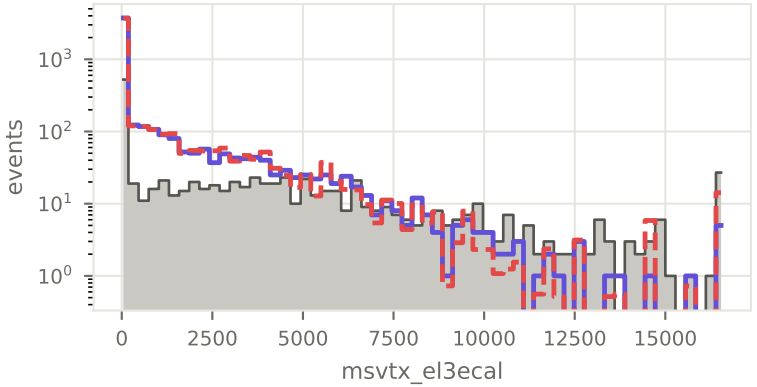



1b · each mass point vs background · normalized

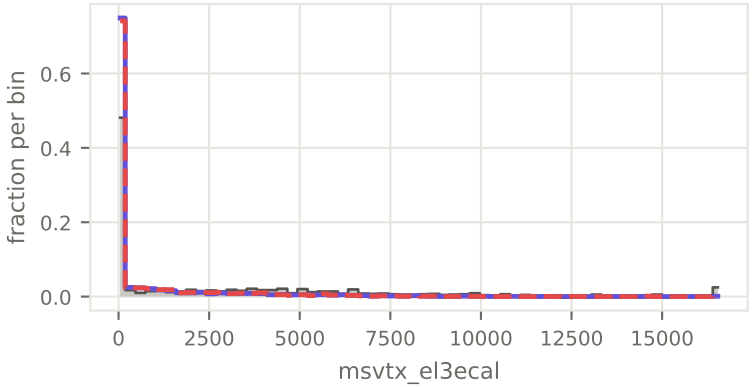



1 · each mass point
background n 1,089
mH125_mS15 D 0.331 n 159
mH125_mS55 D 0.251 n 142
mH200_mS50 D 0.305 n 449
mH400_mS100 D 0.308 n 1,144
mH600_mS150 D 0.307 n 1,608
mH1000_mS275 D 0.297 n 1,485

2a · combined signal vs background · raw

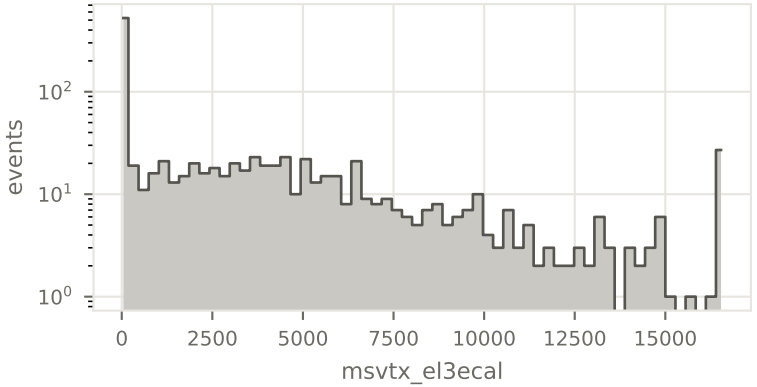



2b · combined signal vs background · normalized

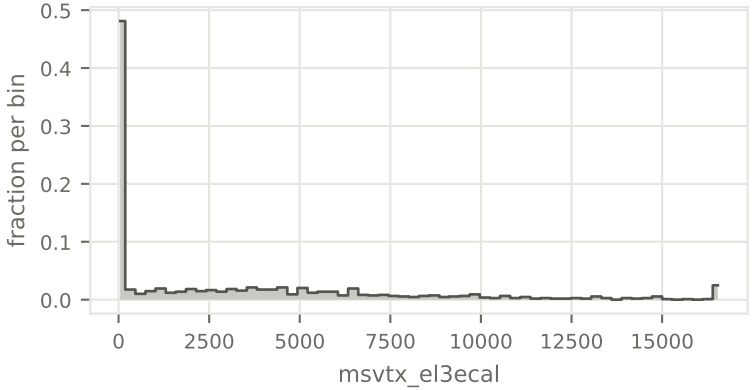



2 · combined signal
background n 1,089
combined, pooled D 0.300 n 4,987
equal weight per mass D 0.293

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0% −999 0.0%
background n 1,089

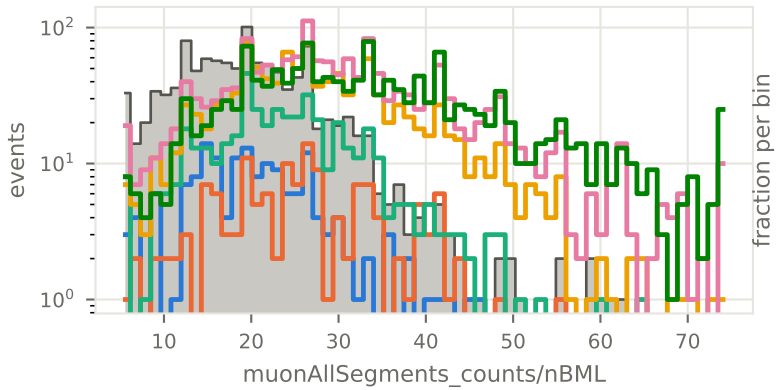
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

# muonAllSegments_counts/nBML

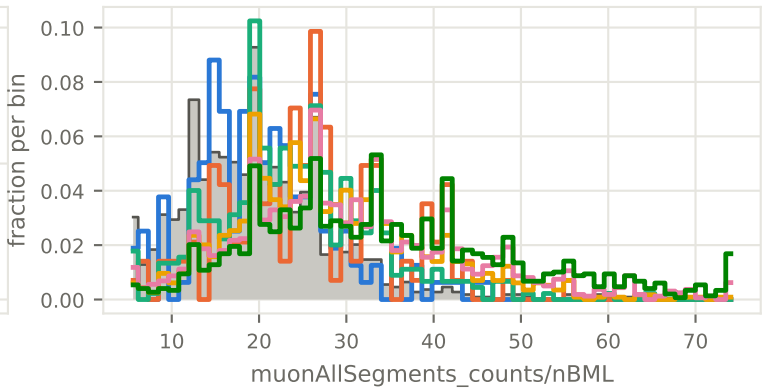
region: BARREL · rank 83 / 98

D_equal 0.292 (noise 0.051)  D_pooled 0.371  min 0.097 (mH125_mS15)  max 0.477 (mH1000_mS275)  bkg n = 1,089  sig n = 4,987

1a · each mass point vs background · raw

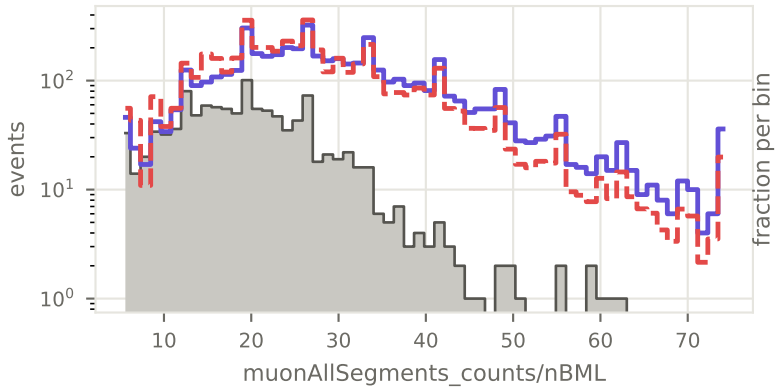



1b · each mass point vs background · normalized

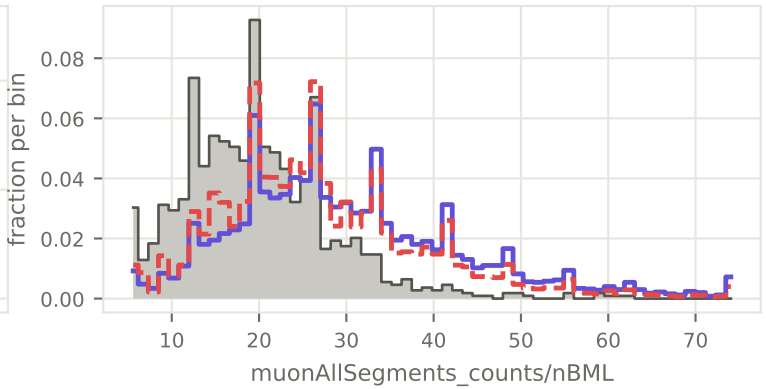



1 · each mass point
background n 1,089
mH125_mS15 D 0.097 n 159
mH125_mS55 D 0.323 n 142
mH200_mS50 D 0.226 n 449
mH400_mS100 D 0.351 n 1,144
mH600_mS150 D 0.396 n 1,608
mH1000_mS275 D 0.477 n 1,485

2a · combined signal vs background · raw

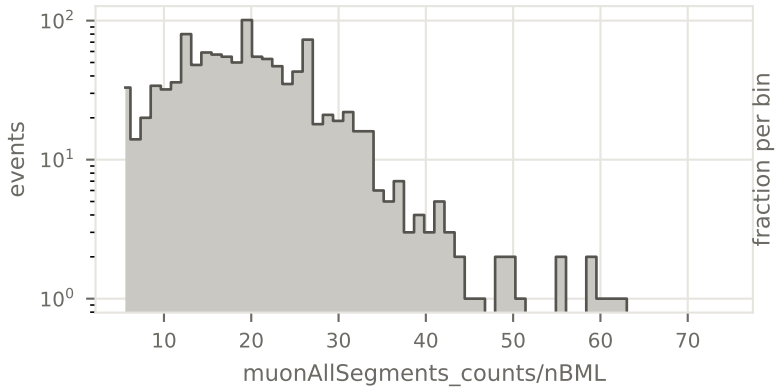



2b · combined signal vs background · normalized

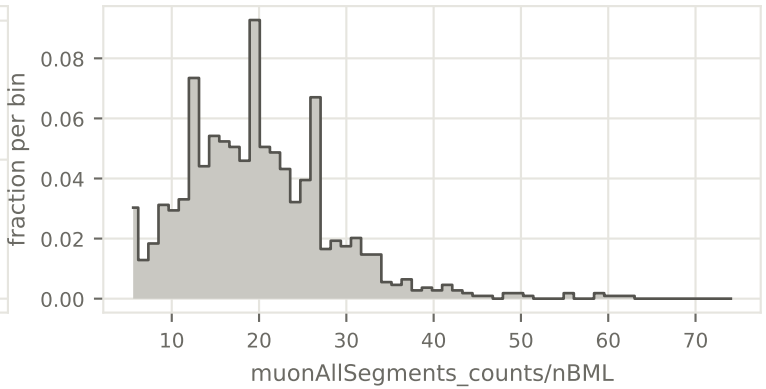



2 · combined signal
background n 1,089
combined, pooled D 0.371 n 4,987
equal weight per mass D 0.292

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0% −999 0.0%
background n 1,089

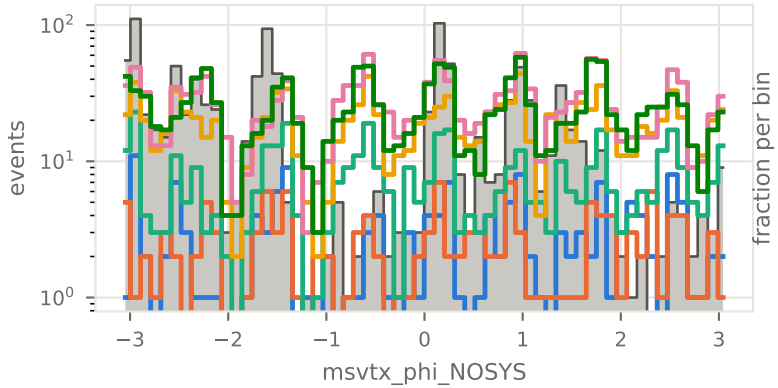
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

# msvtx_phi_NOSYS

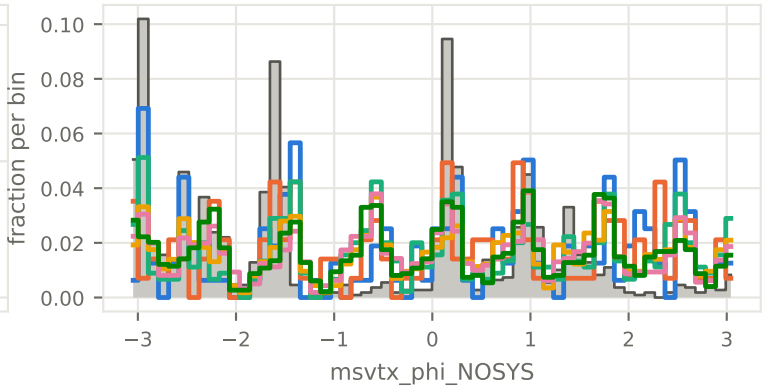
region: BARREL · rank 78 / 98

D_equal 0.285 (noise 0.051) D_pooled 0.285 min 0.279 (mH125_mS55) max 0.292 (mH200_mS50) bkg n = 1,089 sig n = 4,987

1a · each mass point vs background · raw

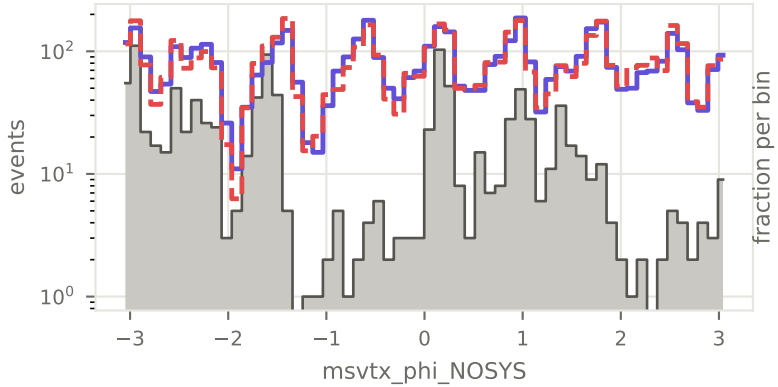



1b · each mass point vs background · normalized

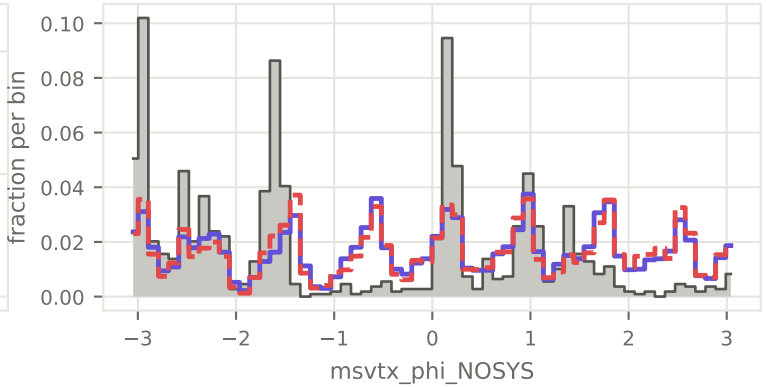



1 · each mass point
background n 1,089
mH125_mS15 D 0.290 n 159
mH125_mS55 D 0.279 n 142
mH200_mS50 D 0.292 n 449
mH400_mS100 D 0.283 n 1,144
mH600_mS150 D 0.290 n 1,608
mH1000_mS275 D 0.280 n 1,485

2a · combined signal vs background · raw

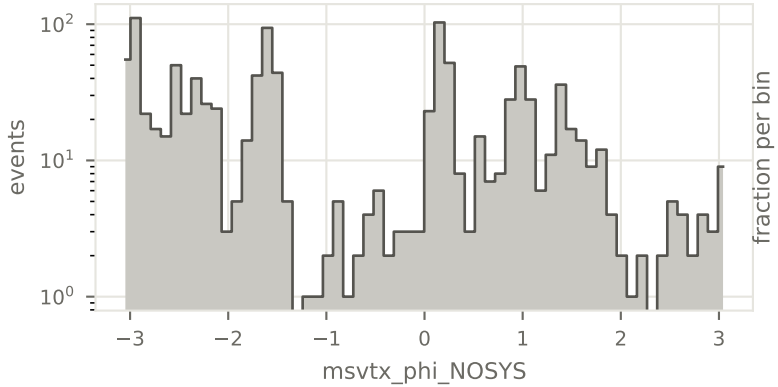



2b · combined signal vs background · normalized

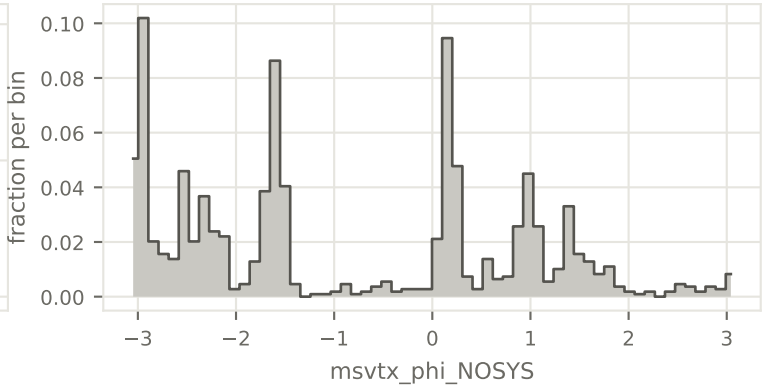



2 · combined signal
background n 1,089
combined, pooled D 0.285 n 4,987
equal weight per mass D 0.285

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0% −999 0.0%
background n 1,089

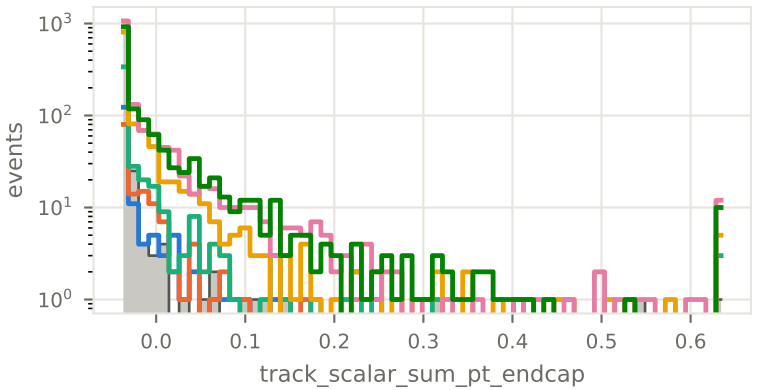
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

# track_scalar_sum_pt_endcap

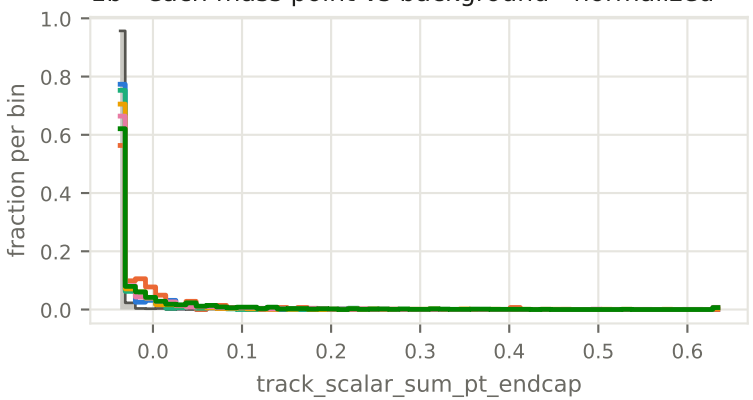
region: BARREL · rank 48 / 98 · in: NN1 endcap

D_equal 0.284 (noise 0.051) D_pooled 0.302 min 0.195 (mH125_mS15) max 0.406 (mH125_mS55) bkg n = 1,089 sig n = 4,987

1a · each mass point vs background · raw

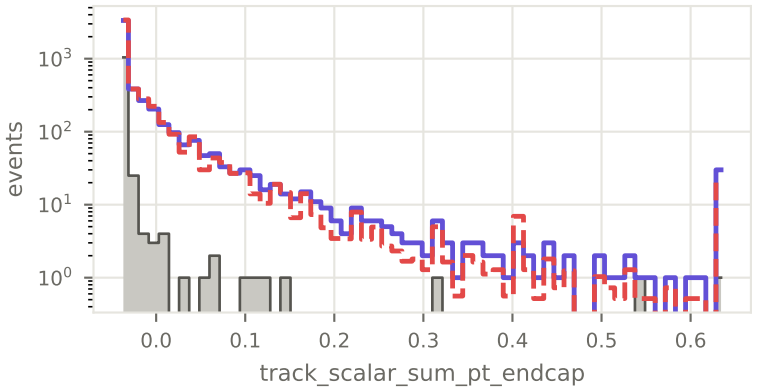



1b · each mass point vs background · normalized

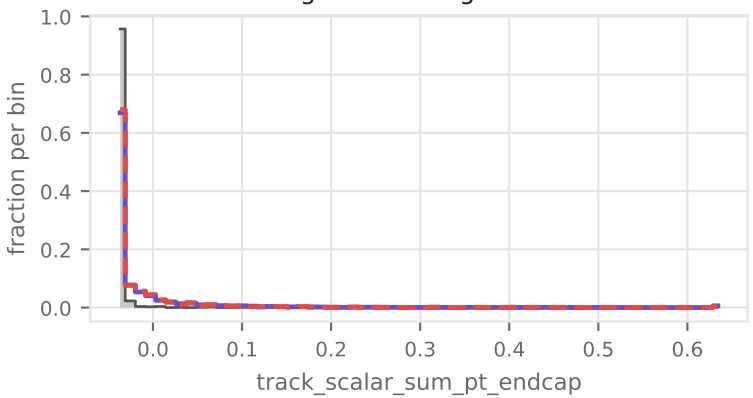



1 · each mass point
background n 1,089
mH125_mS15 D 0.195 n 159
mH125_mS55 D 0.406 n 142
mH200_mS50 D 0.223 n 449
mH400_mS100 D 0.270 n 1,144
mH600_mS150 D 0.310 n 1,608
mH1000_mS275 D 0.354 n 1,485

2a · combined signal vs background · raw

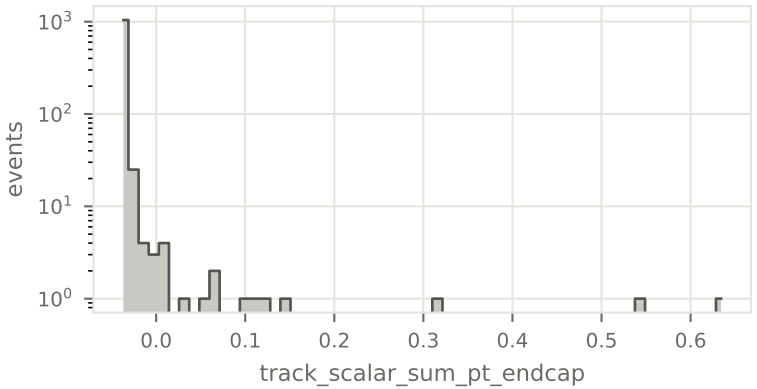



2b · combined signal vs background · normalized

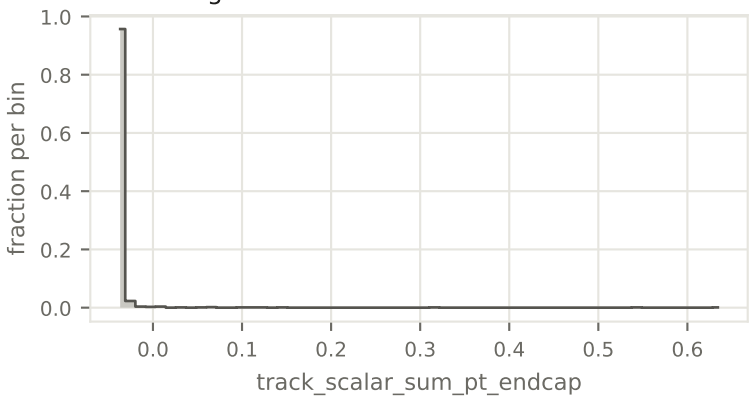



2 · combined signal
background n 1,089
combined, pooled D 0.302 n 4,987
equal weight per mass D 0.284

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0% −999 0.0%
background n 1,089

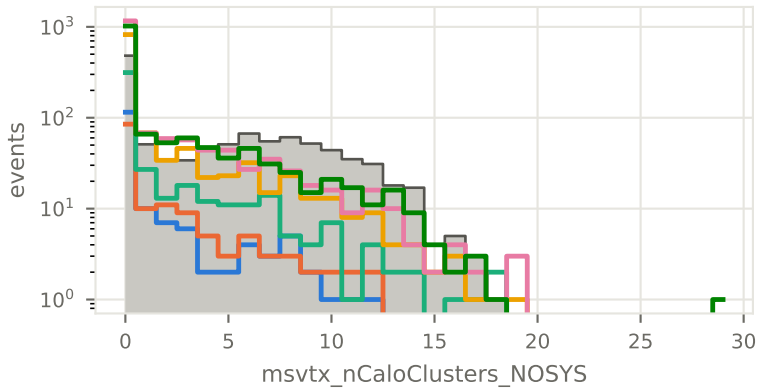
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

# msvtx_nCaloClusters_NOSYS

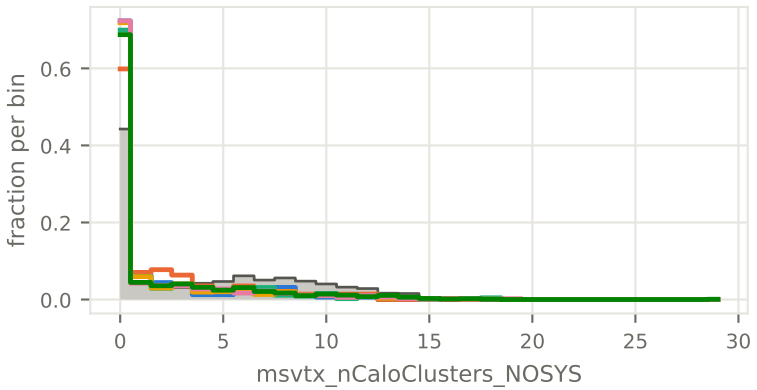
region: BARREL · rank 33 / 98

D_equal 0.281 (noise 0.051) D_pooled 0.279 min 0.256 (mH1000_mS275) max 0.316 (mH125_mS15) bkg n = 1,089 sig n = 4,987

1a · each mass point vs background · raw

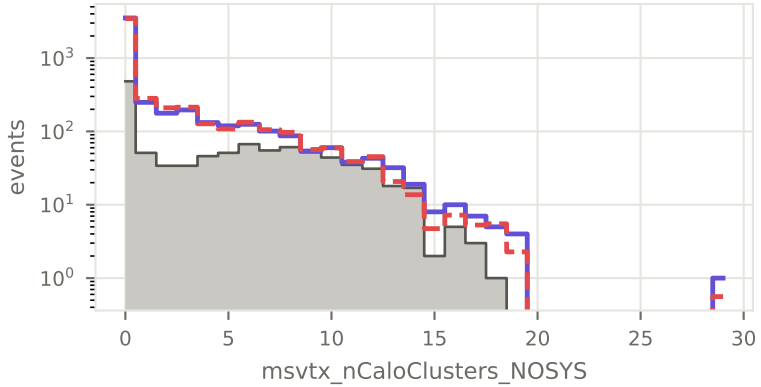



1b · each mass point vs background · normalized

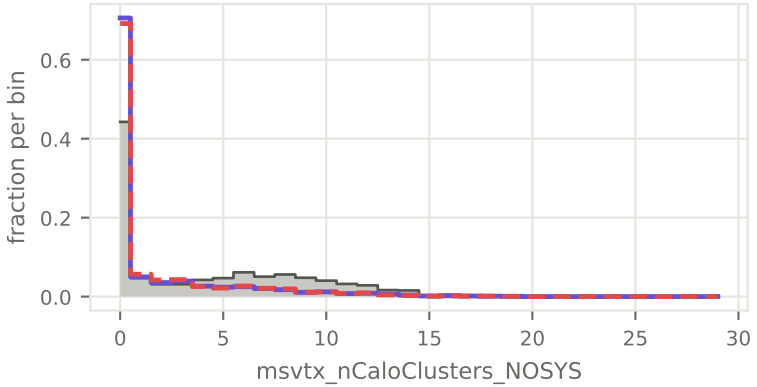



1 · each mass point
background n 1,089
mH125_mS15 D 0.316 n 159
mH125_mS55 D 0.258 n 142
mH200_mS50 D 0.277 n 449
mH400_mS100 D 0.296 n 1,144
mH600_mS150 D 0.286 n 1,608
mH1000_mS275 D 0.256 n 1,485

2a · combined signal vs background · raw

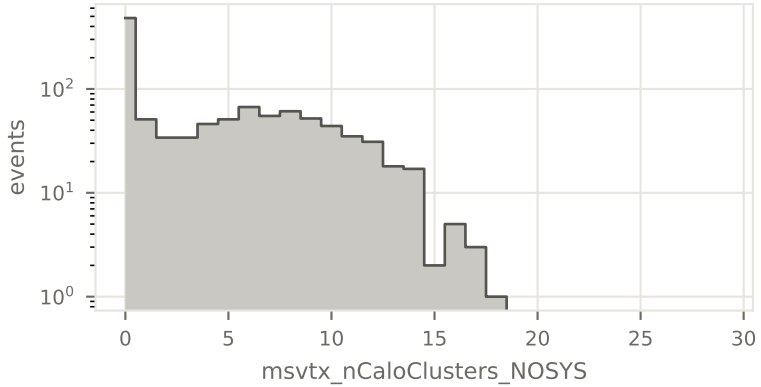



2b · combined signal vs background · normalized

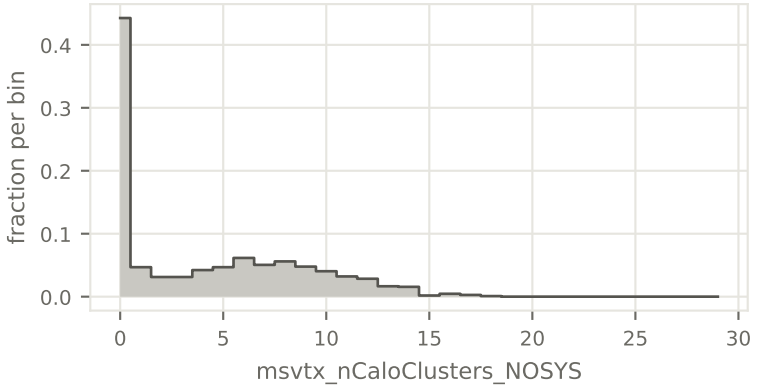



2 · combined signal
background n 1,089
combined, pooled D 0.279 n 4,987
equal weight per mass D 0.281

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0% −999 0.0%
background n 1,089

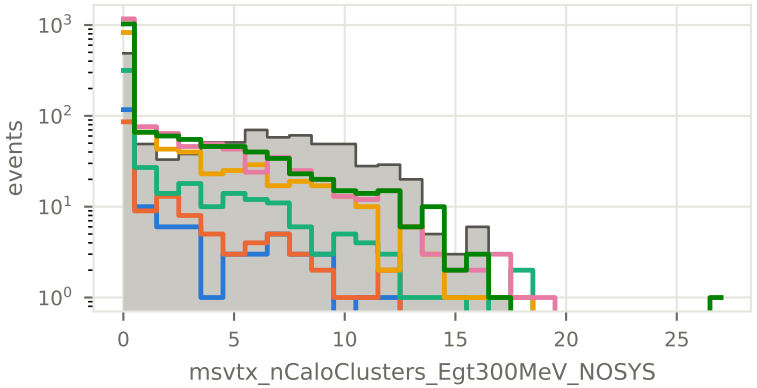
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

# msvtx_nCaloClusters_Egt300MeV_NOSYS

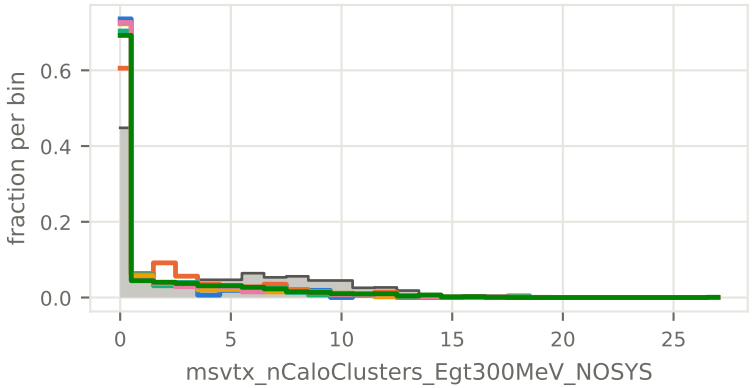
region: BARREL · rank 31 / 98

D_equal 0.281 (noise 0.051) D_pooled 0.278 min 0.256 (mH1000_mS275) max 0.316 (mH125_mS15) bkg n = 1,089 sig n = 4,987

1a · each mass point vs background · raw

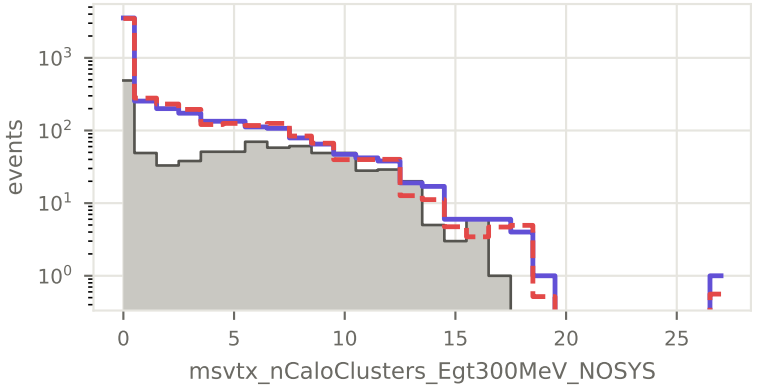



1b · each mass point vs background · normalized

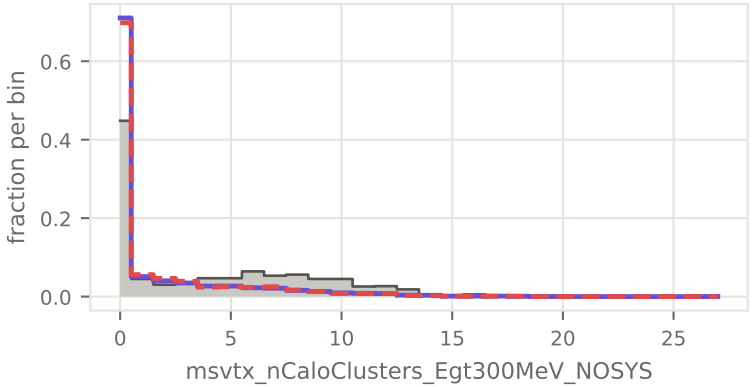



1 · each mass point
background n 1,089
mH125_mS15 D 0.316 n 159
mH125_mS55 D 0.259 n 142
mH200_mS50 D 0.277 n 449
mH400_mS100 D 0.295 n 1,144
mH600_mS150 D 0.290 n 1,608
mH1000_mS275 D 0.256 n 1,485

2a · combined signal vs background · raw

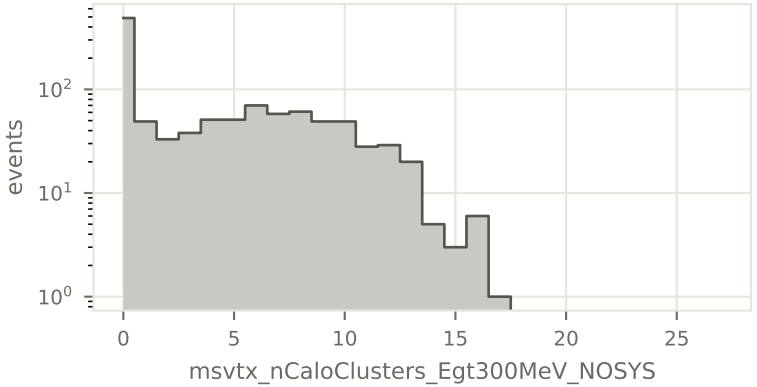



2b · combined signal vs background · normalized

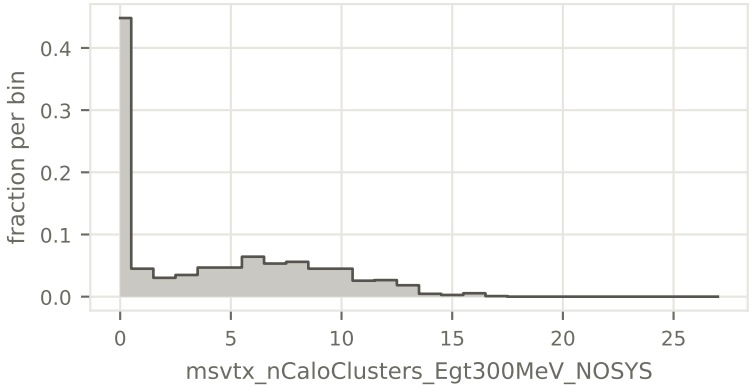



2 · combined signal
background n 1,089
combined, pooled D 0.278 n 4,987
equal weight per mass D 0.281

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0% −999 0.0%
background n 1,089

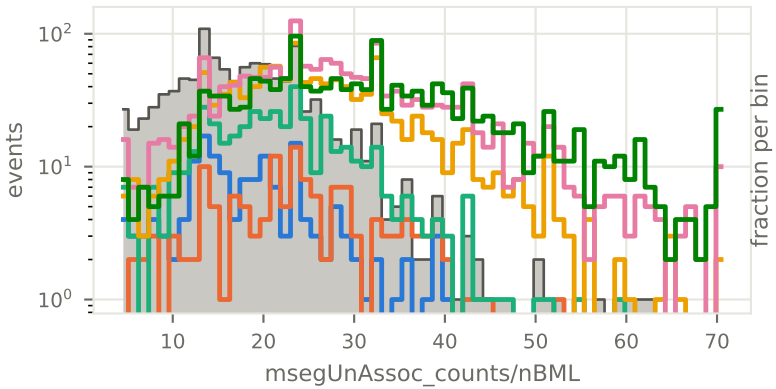
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

# msegUnAssoc_counts/nBML

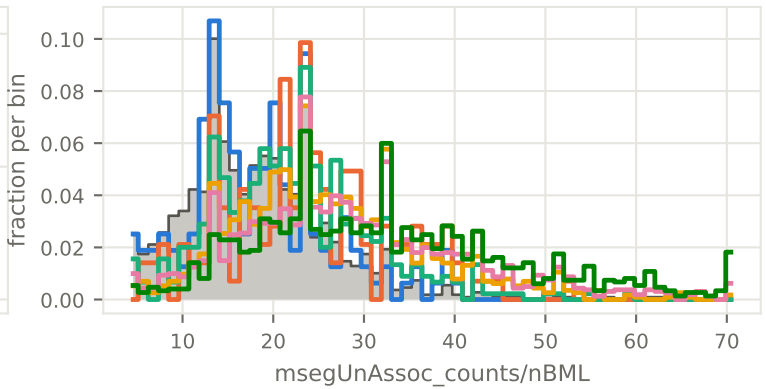
region: BARREL · rank 82 / 98 · in: NN1 barrel

D_equal 0.277 (noise 0.051) D_pooled 0.368 min 0.053 (mH125_mS15) max 0.473 (mH1000_mS275) bkg n = 1,089 sig n = 4,987

1a · each mass point vs background · raw

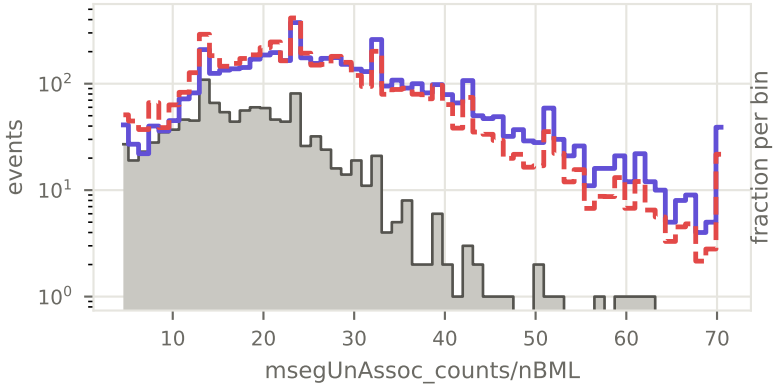



1b · each mass point vs background · normalized

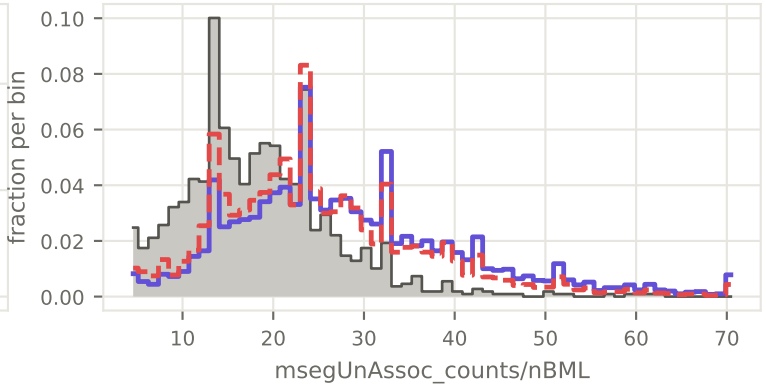



1 · each mass point
background n 1,089
mH125_mS15 D 0.053 n 159
mH125_mS55 D 0.312 n 142
mH200_mS50 D 0.202 n 449
mH400_mS100 D 0.331 n 1,144
mH600_mS150 D 0.405 n 1,608
mH1000_mS275 D 0.473 n 1,485

2a · combined signal vs background · raw

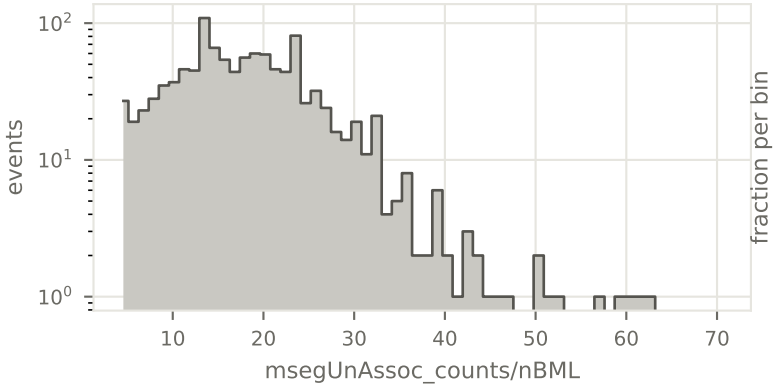



2b · combined signal vs background · normalized

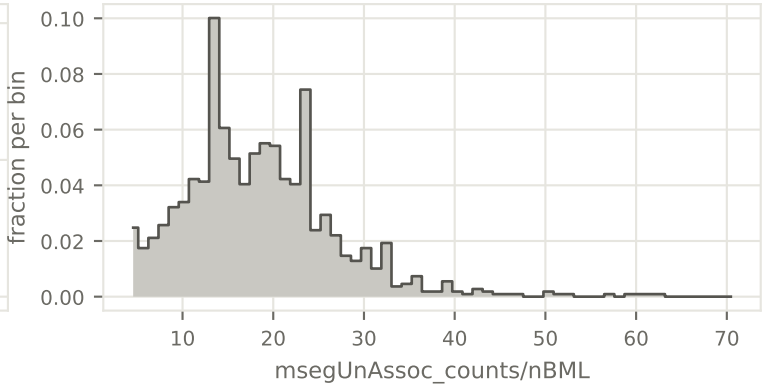



2 · combined signal
background n 1,089
combined, pooled D 0.368 n 4,987
equal weight per mass D 0.277

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0% −999 0.0%
background n 1,089

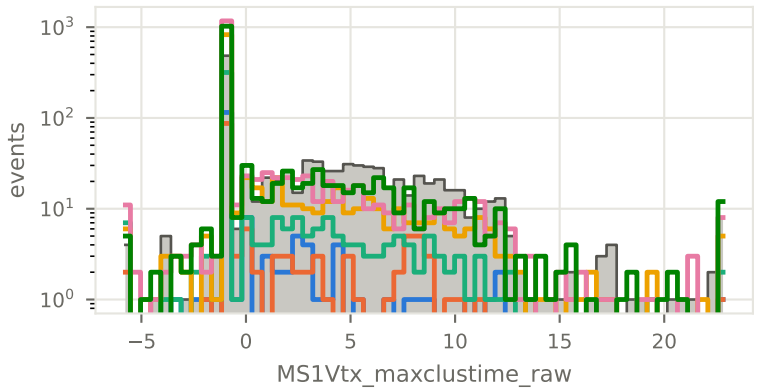
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

# MS1Vtx_maxclustime_raw

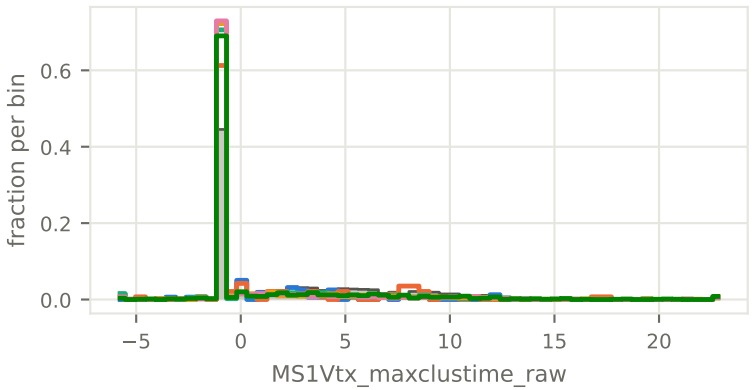
region: BARREL · rank 37 / 98 · in: NN2 barrel, NN2 endcap

D_equal 0.274 (noise 0.051)  D_pooled 0.278  min 0.221 (mH125_mS55)  max 0.322 (mH125_mS15)  bkg n = 1,089  sig n = 4,987

1a · each mass point vs background · raw

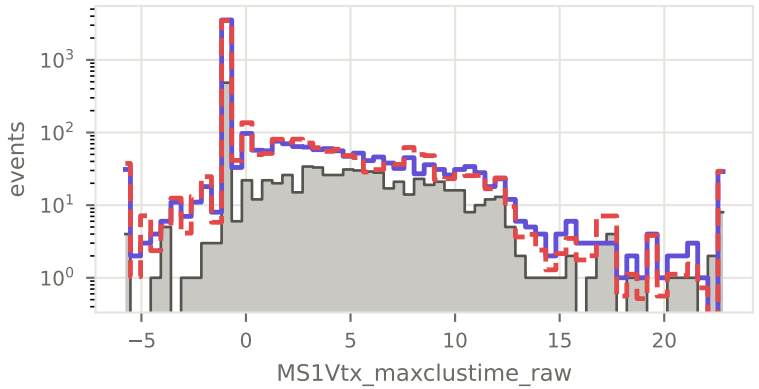



1b · each mass point vs background · normalized

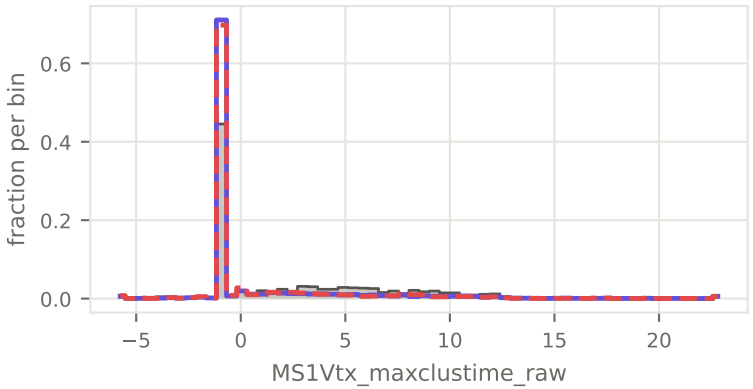



1 · each mass point
background n 1,089
mH125_mS15 D 0.322 n 159
mH125_mS55 D 0.221 n 142
mH200_mS50 D 0.283 n 449
mH400_mS100 D 0.288 n 1,144
mH600_mS150 D 0.291 n 1,608
mH1000_mS275 D 0.255 n 1,485

2a · combined signal vs background · raw

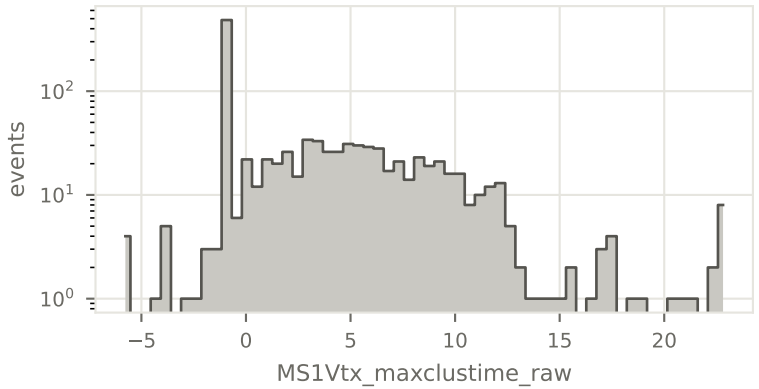



2b · combined signal vs background · normalized

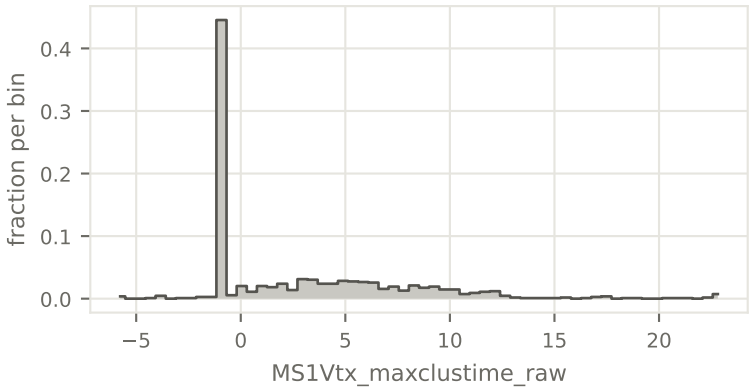



2 · combined signal
background n 1,089
combined, pooled D 0.278 n 4,987
equal weight per mass D 0.274

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0%  −999 0.0%
background n 1,089

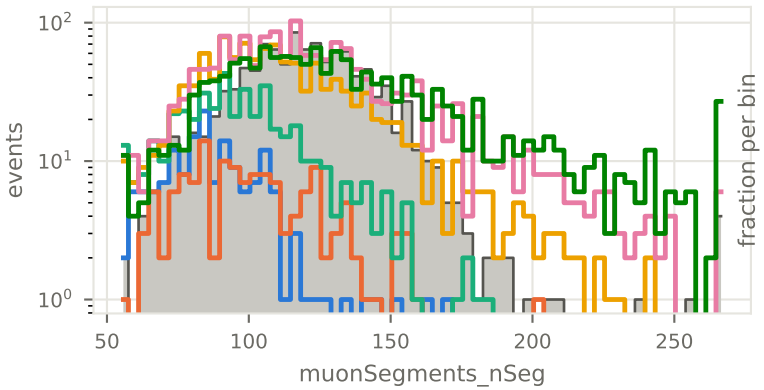
a = raw (log y)  b = normalized · edge bins include under/overflow · NaN and -999 removed

# muonSegments_nSeg

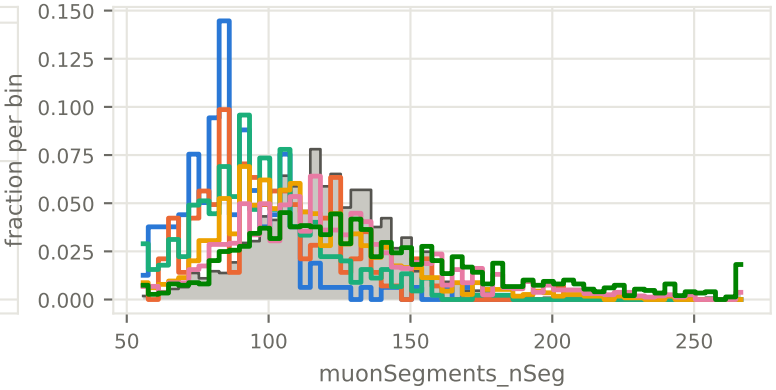
region: BARREL · rank 58 / 98

D_equal 0.274 (noise 0.051)  D_pooled 0.147  min 0.110 (mH600_mS150)  max 0.594 (mH125_mS15)  bkg n = 1,089  sig n = 4,987

1a · each mass point vs background · raw

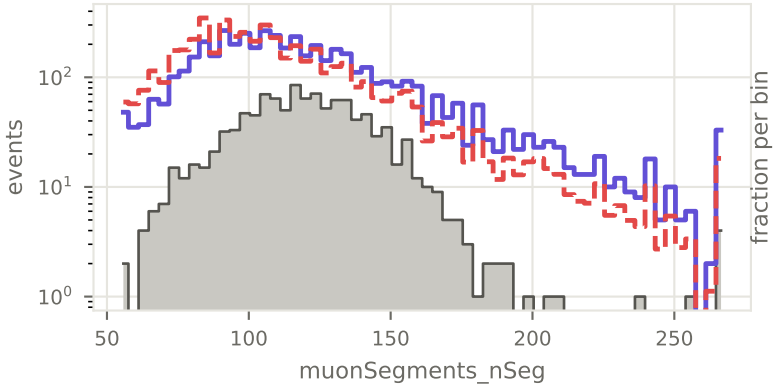



1b · each mass point vs background · normalized

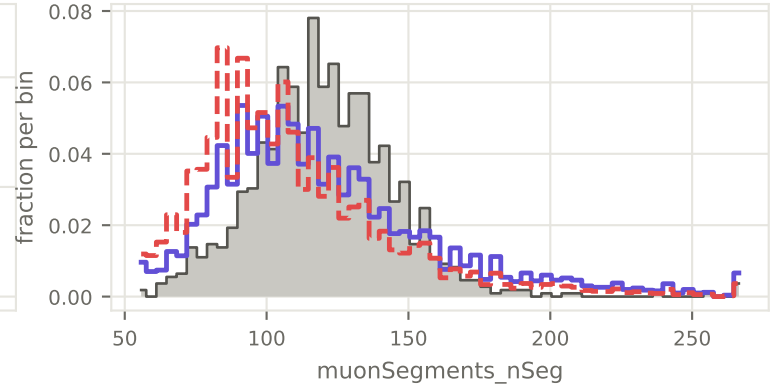



1 · each mass point
background n 1,089
mH125_mS15 D 0.594 n 159
mH125_mS55 D 0.353 n 142
mH200_mS50 D 0.425 n 449
mH400_mS100 D 0.210 n 1,144
mH600_mS150 D 0.110 n 1,608
mH1000_mS275 D 0.213 n 1,485

2a · combined signal vs background · raw

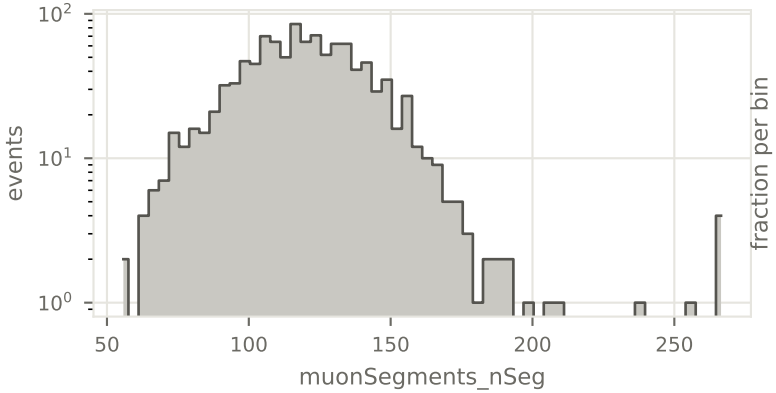



2b · combined signal vs background · normalized

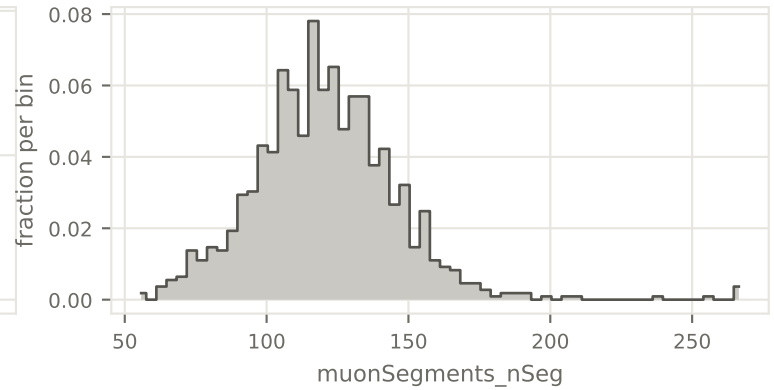



2 · combined signal
background n 1,089
combined, pooled D 0.147 n 4,987
equal weight per mass D 0.274

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0% −999 0.0%
background n 1,089

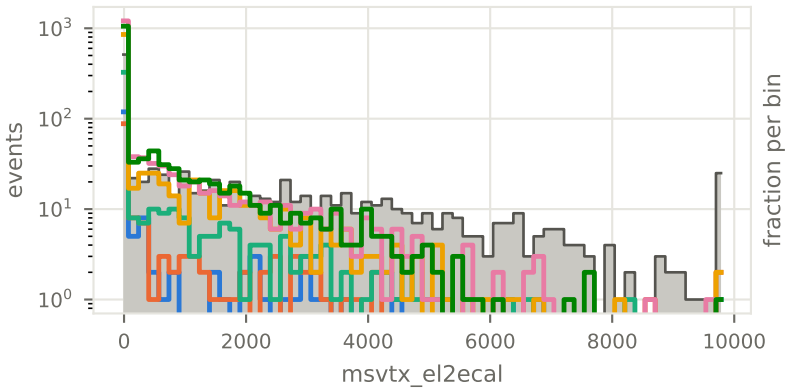
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

# msvtx_el2ecal

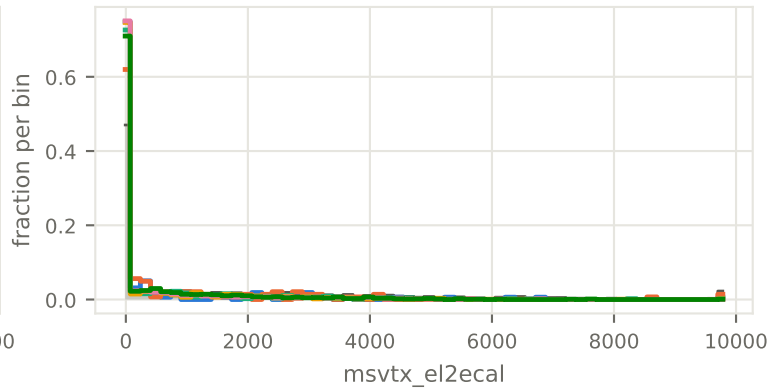
region: BARREL · rank 38 / 98 · in: NN1 barrel, NN1 endcap

D_equal 0.267 (noise 0.051)  D_pooled 0.270  min 0.218 (mH125_mS55)  max 0.321 (mH125_mS15)  bkg n = 1,089  sig n = 4,987

1a · each mass point vs background · raw

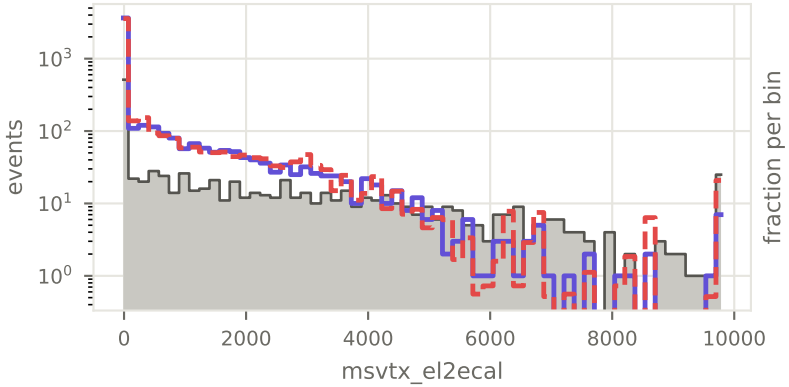



1b · each mass point vs background · normalized

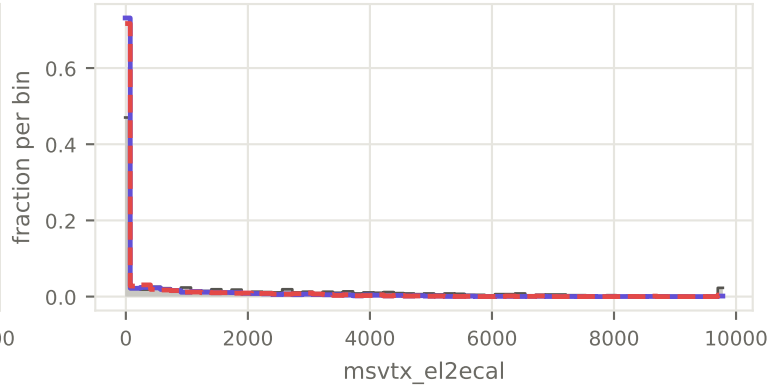



1 · each mass point
background n 1,089
mH125_mS15 D 0.321 n 159
mH125_mS55 D 0.218 n 142
mH200_mS50 D 0.258 n 449
mH400_mS100 D 0.277 n 1,144
mH600_mS150 D 0.290 n 1,608
mH1000_mS275 D 0.257 n 1,485

2a · combined signal vs background · raw

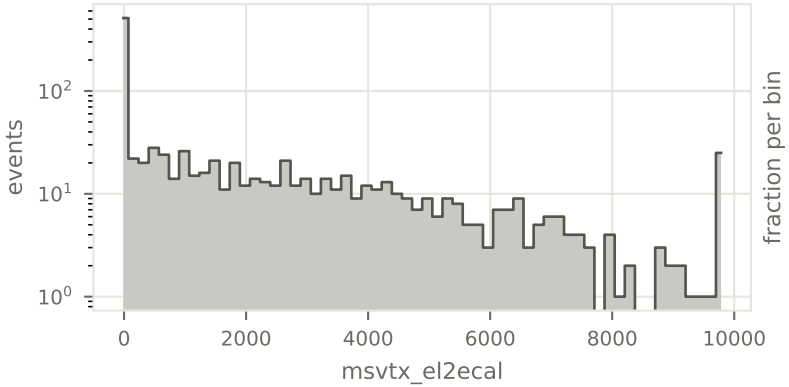



2b · combined signal vs background · normalized

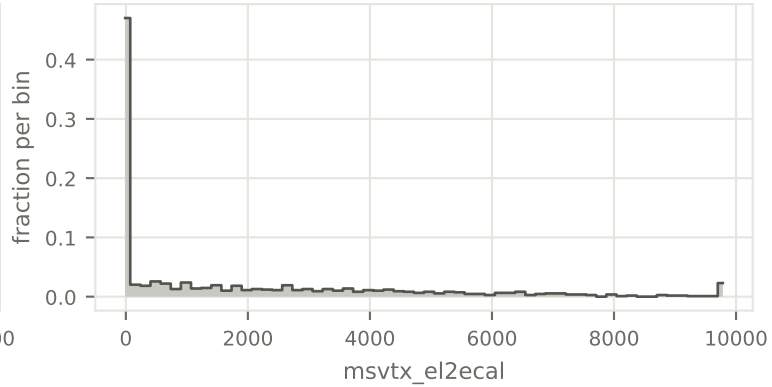



2 · combined signal
background n 1,089
combined, pooled D 0.270 n 4,987
equal weight per mass D 0.267

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0% −999 0.0%
background n 1,089

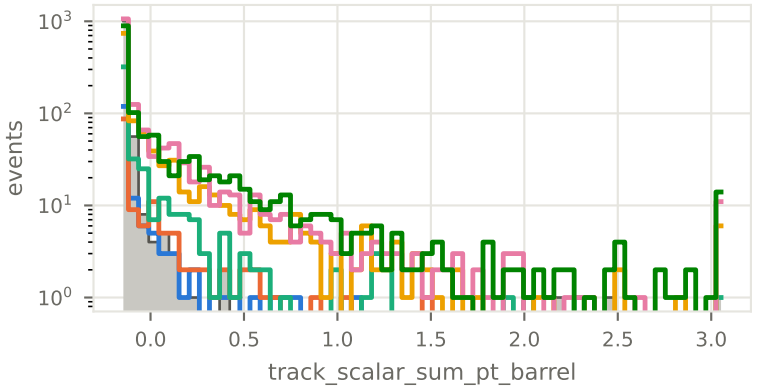
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

# track_scalar_sum_pt_barrel

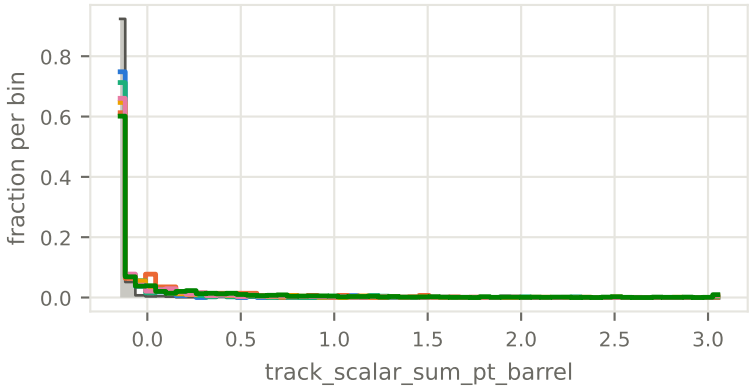
region: BARREL · rank 57 / 98 · in: NN1 barrel

D_equal 0.265 (noise 0.051)  D_pooled 0.281  min 0.183 (mH125_mS15)  max 0.327 (mH1000_mS275)  bkg n = 1,089  sig n = 4,987

1a · each mass point vs background · raw

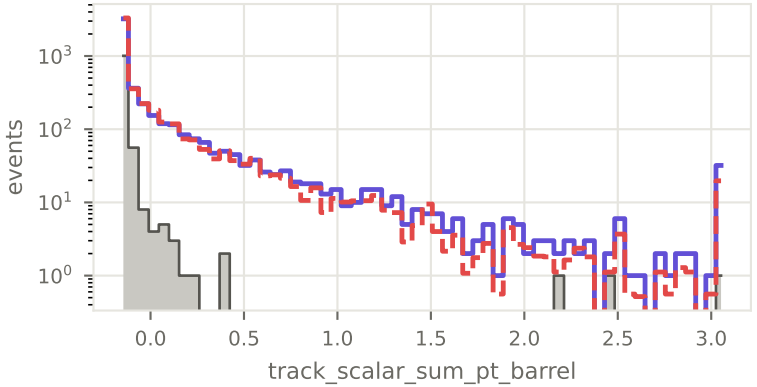



1b · each mass point vs background · normalized

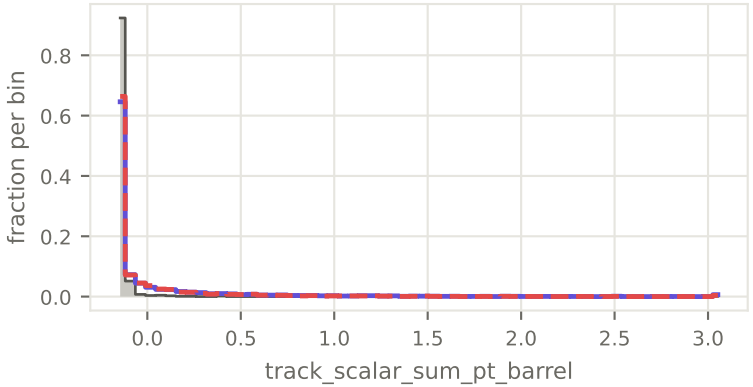



1 · each mass point
background n 1,089
mH125_mS15 D 0.183 n 159
mH125_mS55 D 0.323 n 142
mH200_mS50 D 0.230 n 449
mH400_mS100 D 0.283 n 1,144
mH600_mS150 D 0.267 n 1,608
mH1000_mS275 D 0.327 n 1,485

2a · combined signal vs background · raw

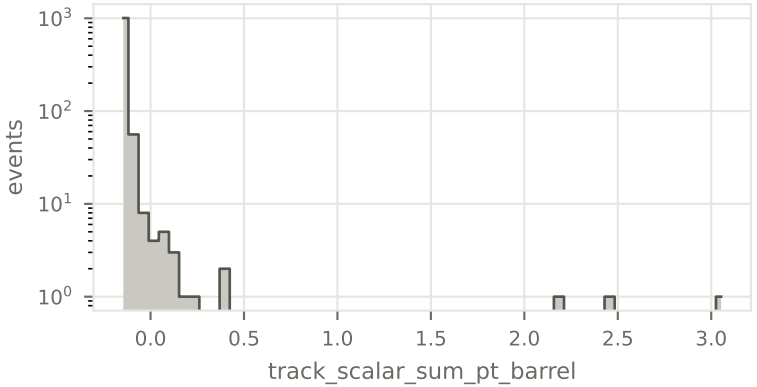



2b · combined signal vs background · normalized

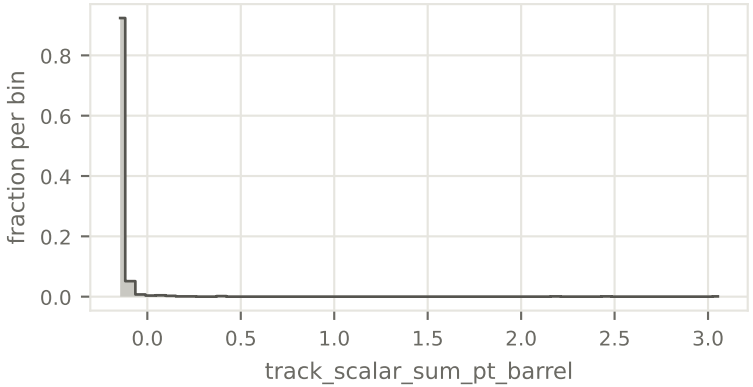



2 · combined signal
background n 1,089
combined, pooled D 0.281 n 4,987
equal weight per mass D 0.265

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0% −999 0.0%
background n 1,089

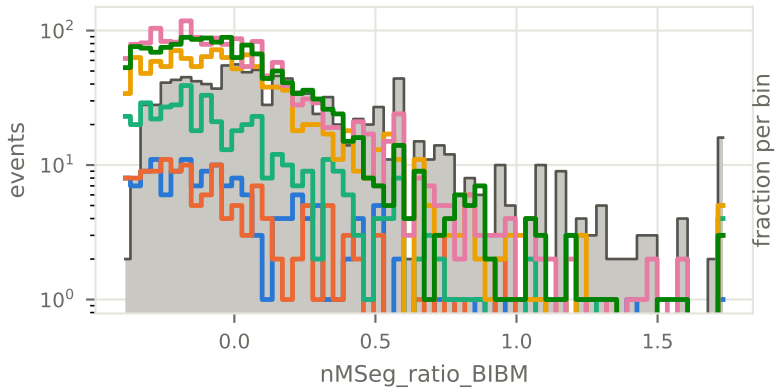
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

# nMSeg_ratio_BIBM

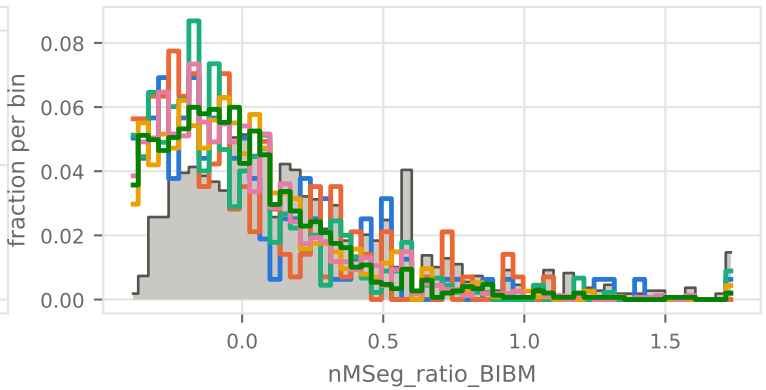
region: BARREL · rank 41 / 98 · in: NN1 barrel

D_equal 0.263 (noise 0.051) D_pooled 0.251 min 0.242 (mH400_mS100) max 0.313 (mH125_mS55) bkg n = 1,089 sig n = 4,987

1a · each mass point vs background · raw

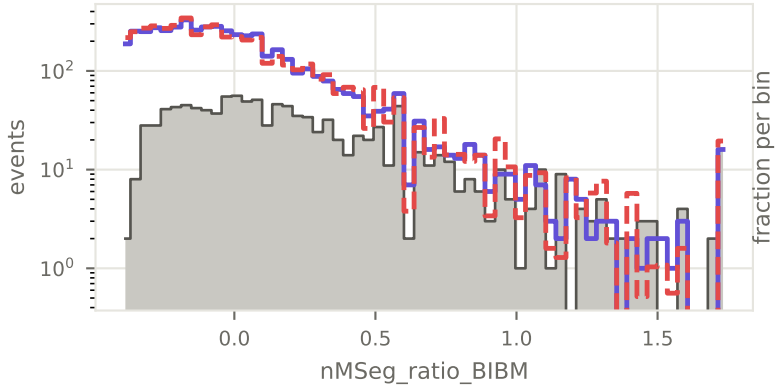



1b · each mass point vs background · normalized

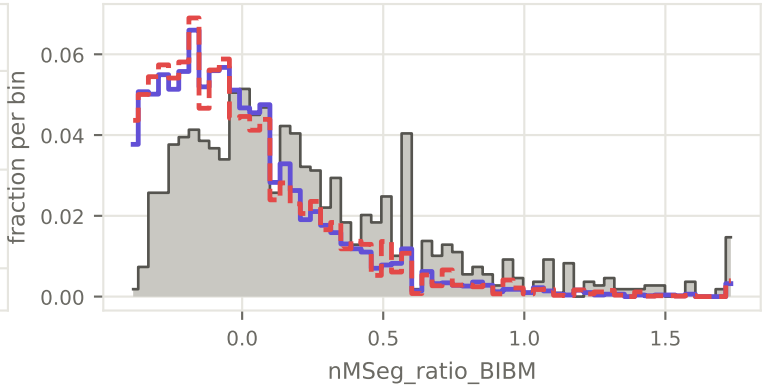



1 · each mass point
background n 1,089
mH125_mS15 D 0.267 n 159
mH125_mS55 D 0.313 n 142
mH200_mS50 D 0.293 n 449
mH400_mS100 D 0.242 n 1,144
mH600_mS150 D 0.263 n 1,608
mH1000_mS275 D 0.246 n 1,485

2a · combined signal vs background · raw

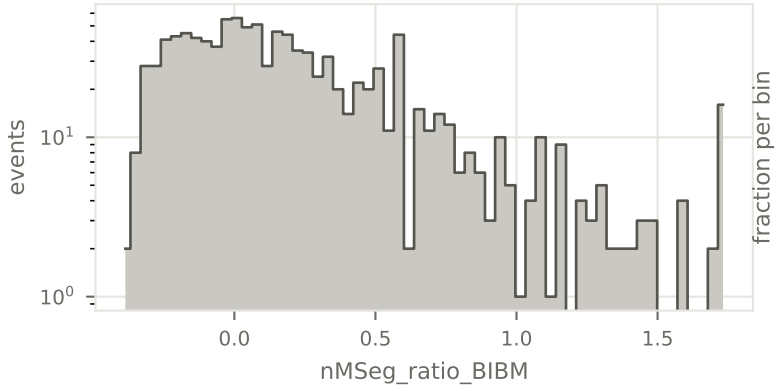



2b · combined signal vs background · normalized

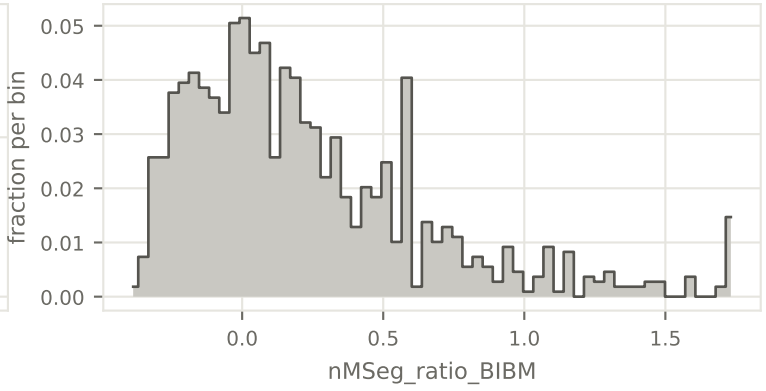



2 · combined signal
background n 1,089
combined, pooled D 0.251 n 4,987
equal weight per mass D 0.263

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0% −999 0.0%
background n 1,089

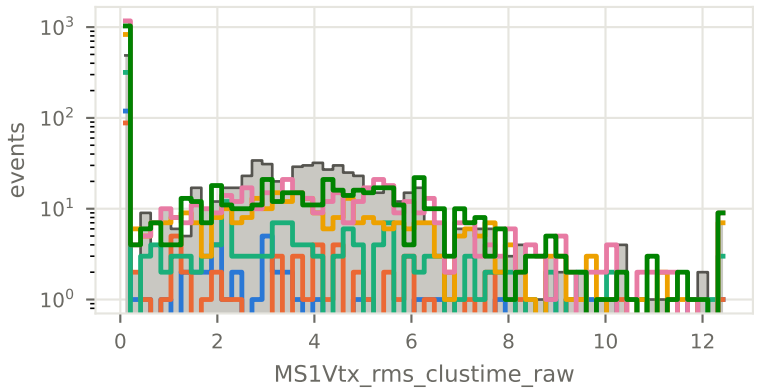
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

# MS1Vtx_rms_clustime_raw

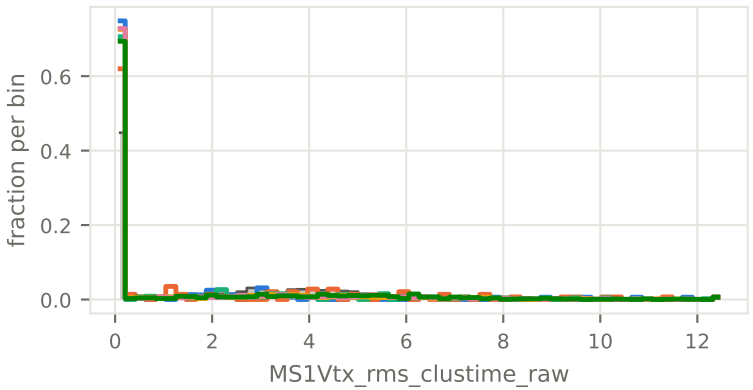
region: BARREL · rank 40 / 98 · in: NN2 barrel, NN2 endcap

D_equal 0.262 (noise 0.051)  D_pooled 0.268  min 0.218 (mH125_mS55)  max 0.317 (mH125_mS15)  bkg n = 1,089  sig n = 4,987

1a · each mass point vs background · raw

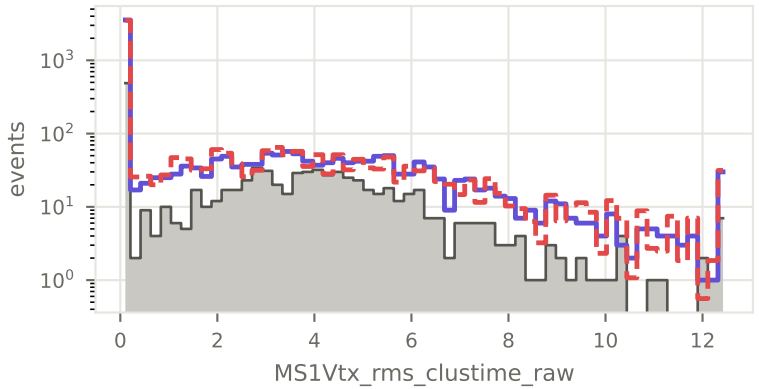



1b · each mass point vs background · normalized

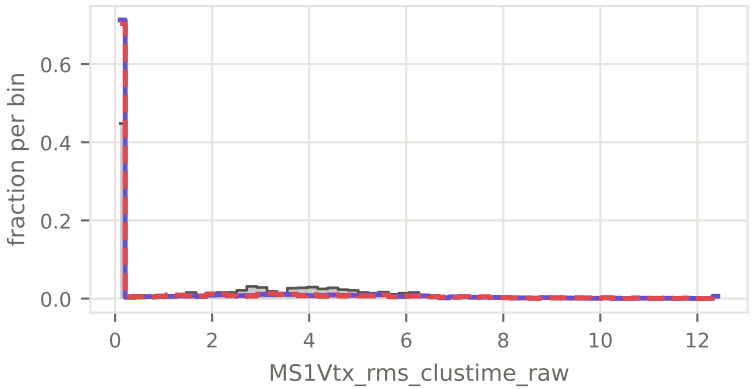



1 · each mass point
- background n 1,089
- mH125_mS15 D 0.317 n 159
- mH125_mS55 D 0.218 n 142
- mH200_mS50 D 0.261 n 449
- mH400_mS100 D 0.282 n 1,144
- mH600_mS150 D 0.284 n 1,608
- mH1000_mS275 D 0.249 n 1,485

2a · combined signal vs background · raw

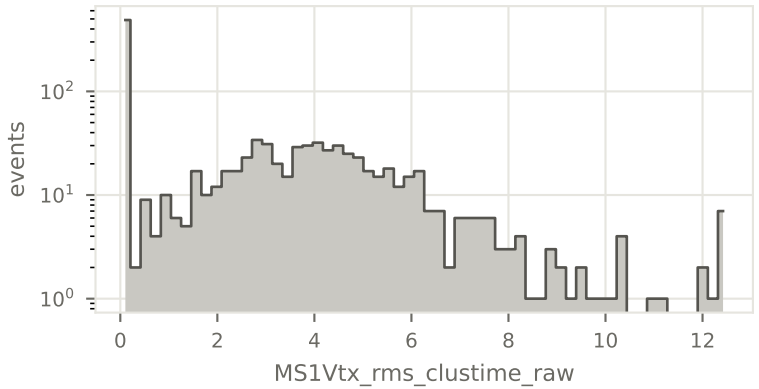



2b · combined signal vs background · normalized

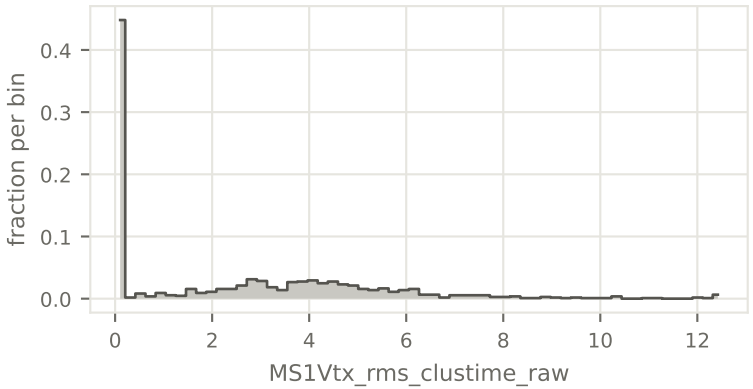



2 · combined signal
- background n 1,089
- combined, pooled D 0.268 n 4,987
- equal weight per mass D 0.262

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0%  −999 0.0%
- background n 1,089

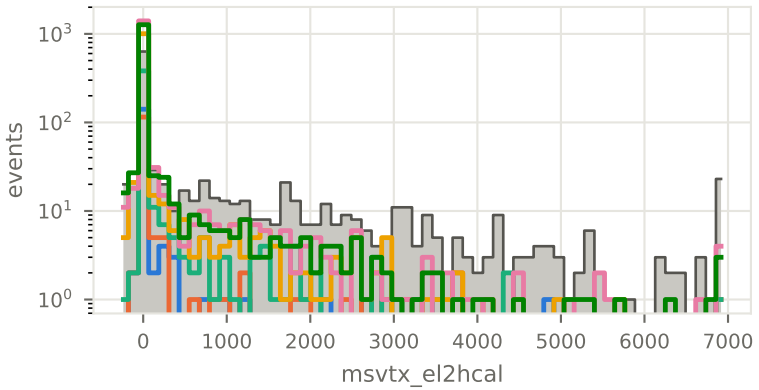
a = raw (log y)  b = normalized · edge bins include under/overflow · NaN and -999 removed

# msvtx_el2hcal

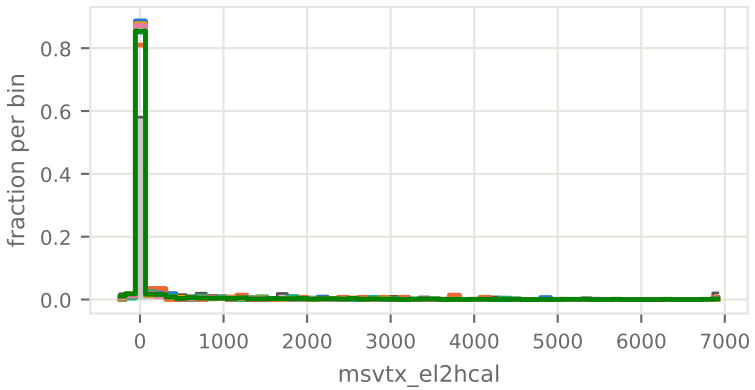
region: BARREL · rank 45 / 98 · in: NN1 barrel, NN1 endcap

D_equal 0.259 (noise 0.051)  D_pooled 0.269  min 0.239 (mH125_mS55)  max 0.293 (mH125_mS15)  bkg n = 1,089  sig n = 4,987

1a · each mass point vs background · raw

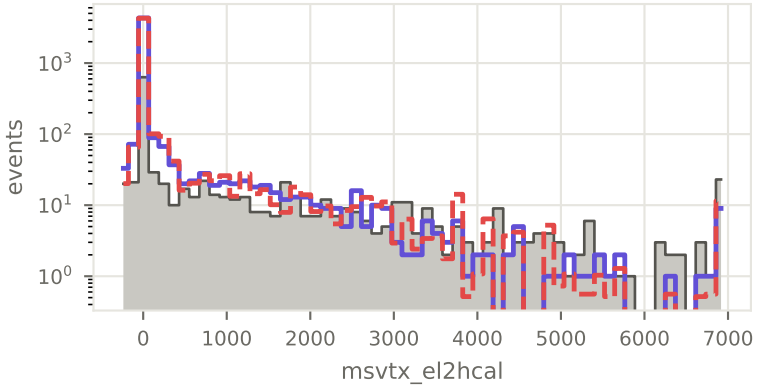



1b · each mass point vs background · normalized

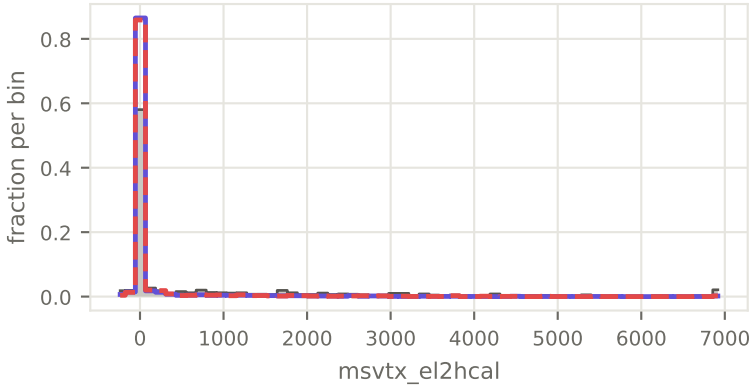



1 · each mass point
background n 1,089
mH125_mS15 D 0.293 n 159
mH125_mS55 D 0.239 n 142
mH200_mS50 D 0.246 n 449
mH400_mS100 D 0.283 n 1,144
mH600_mS150 D 0.277 n 1,608
mH1000_mS275 D 0.267 n 1,485

2a · combined signal vs background · raw

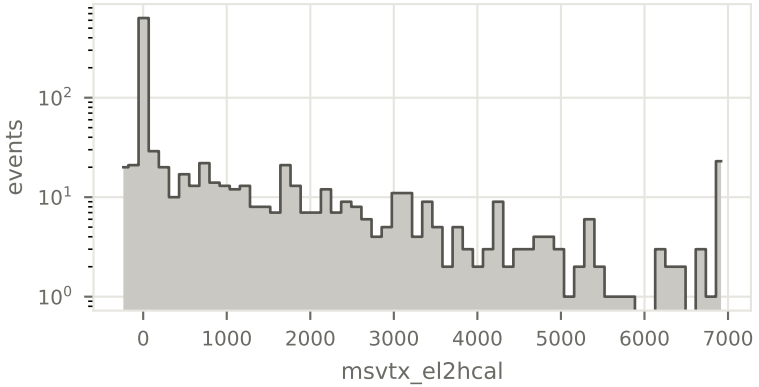



2b · combined signal vs background · normalized

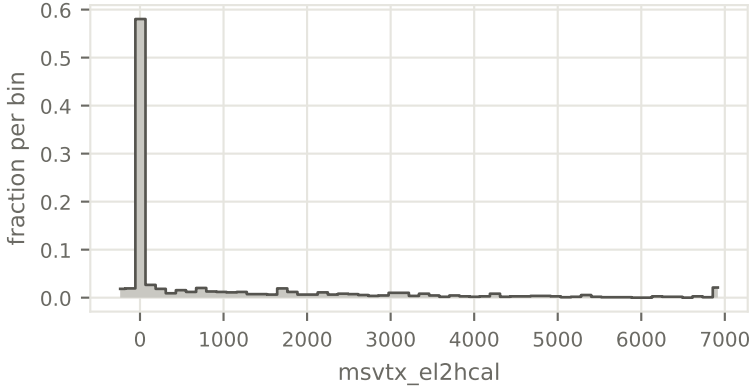



2 · combined signal
background n 1,089
combined, pooled D 0.269 n 4,987
equal weight per mass D 0.259

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0% −999 0.0%
background n 1,089

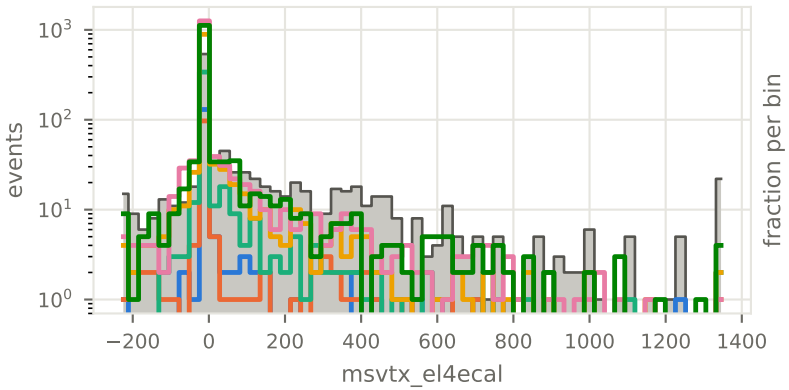
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

# msvtx_el4ecal

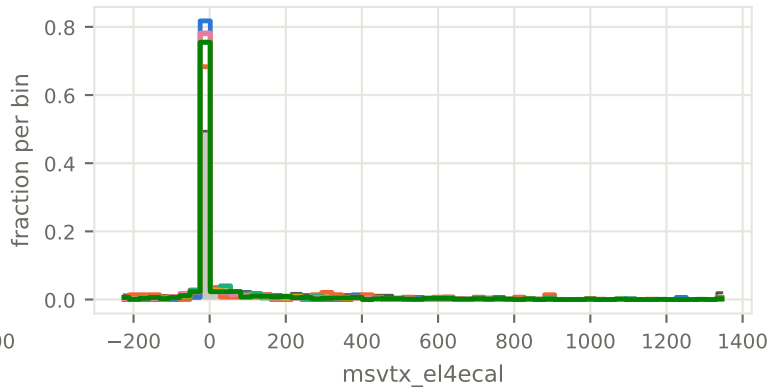
region: BARREL · rank 39 / 98 · in: NN1 barrel, NN1 endcap

D_equal 0.243 (noise 0.051)  D_pooled 0.251  min 0.201 (mH125_mS55)  max 0.272 (mH125_mS15)  bkg n = 1,089  sig n = 4,987

1a · each mass point vs background · raw

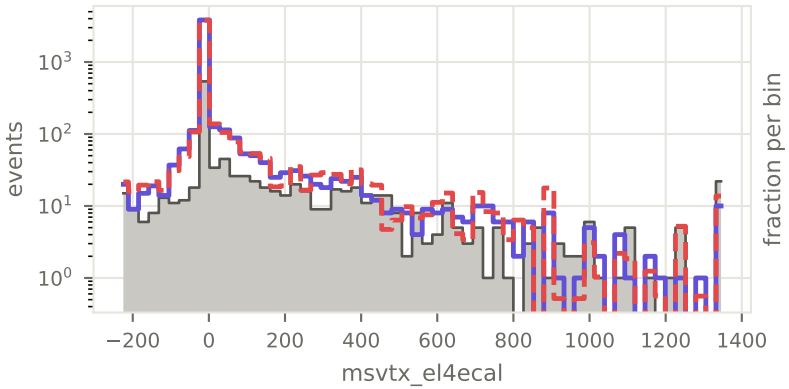



1b · each mass point vs background · normalized

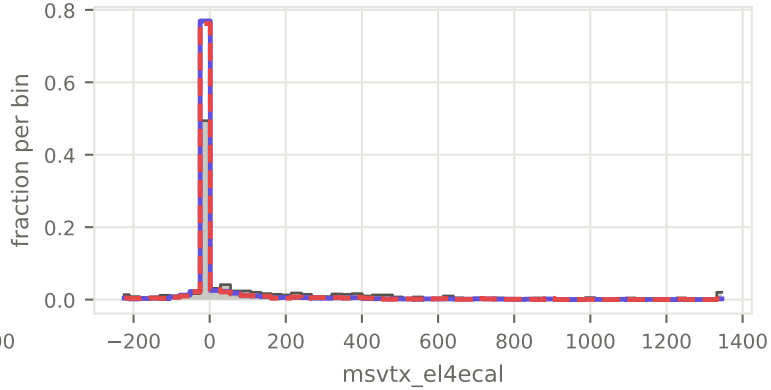



1 · each mass point
background n 1,089
mH125_mS15 D 0.272 n 159
mH125_mS55 D 0.201 n 142
mH200_mS50 D 0.222 n 449
mH400_mS100 D 0.266 n 1,144
mH600_mS150 D 0.266 n 1,608
mH1000_mS275 D 0.237 n 1,485

2a · combined signal vs background · raw

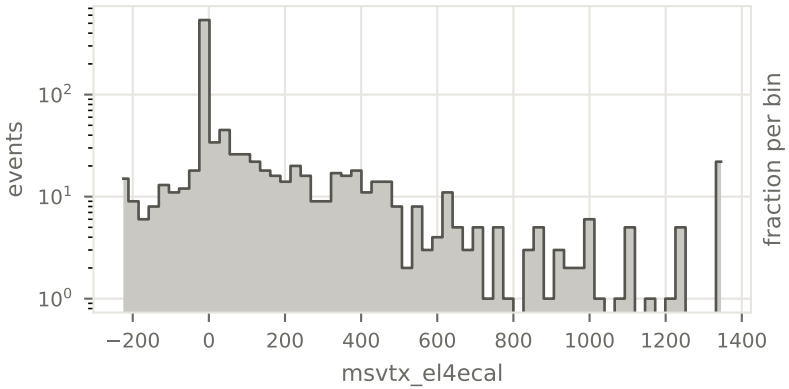



2b · combined signal vs background · normalized

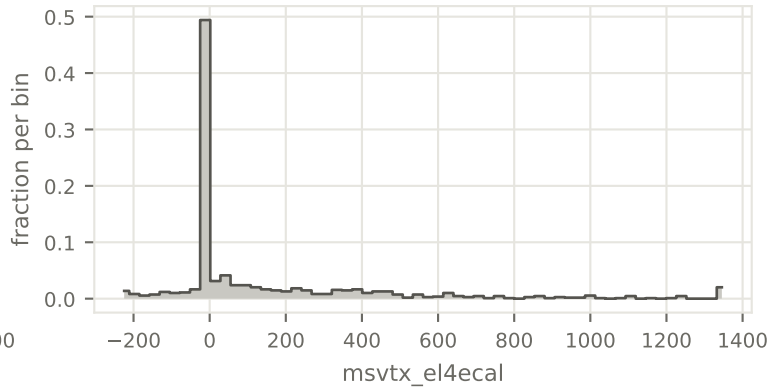



2 · combined signal
background n 1,089
combined, pooled D 0.251 n 4,987
equal weight per mass D 0.243

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0%  −999 0.0%
background n 1,089

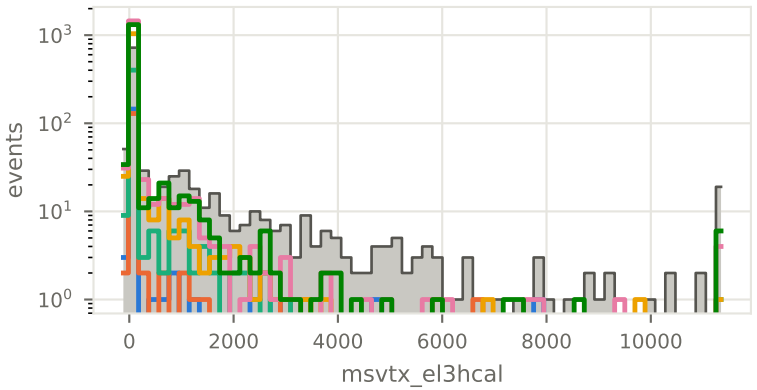
a = raw (log y)  b = normalized · edge bins include under/overflow · NaN and -999 removed

# msvtx_el3hcal

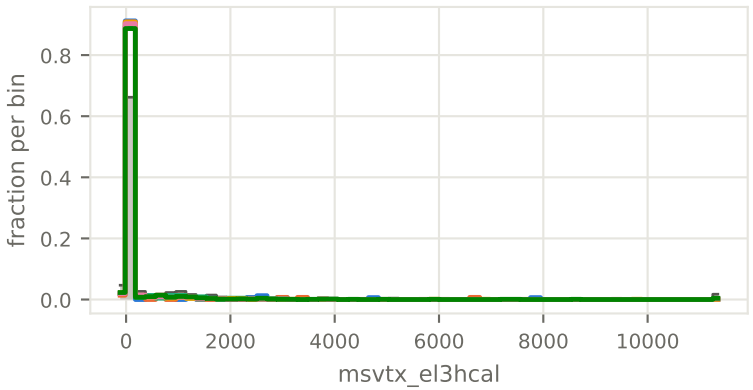
region: BARREL · rank 59 / 98 · in: NN1 barrel, NN1 endcap

D_equal 0.240 (noise 0.051) D_pooled 0.236 min 0.224 (mH1000_mS275) max 0.265 (mH125_mS15) bkg n = 1,089 sig n = 4,987

1a · each mass point vs background · raw

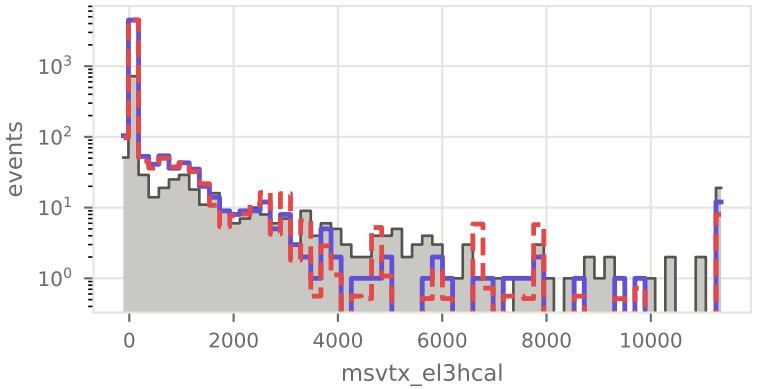



1b · each mass point vs background · normalized

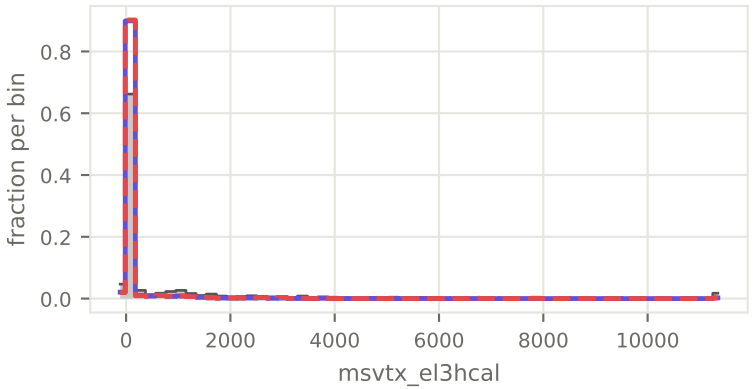



1 · each mass point
background n 1,089
mH125_mS15 D 0.265 n 159
mH125_mS55 D 0.235 n 142
mH200_mS50 D 0.236 n 449
mH400_mS100 D 0.245 n 1,144
mH600_mS150 D 0.239 n 1,608
mH1000_mS275 D 0.224 n 1,485

2a · combined signal vs background · raw

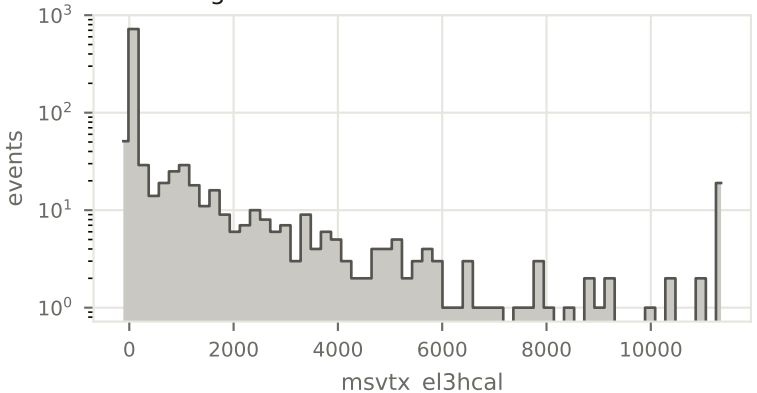



2b · combined signal vs background · normalized

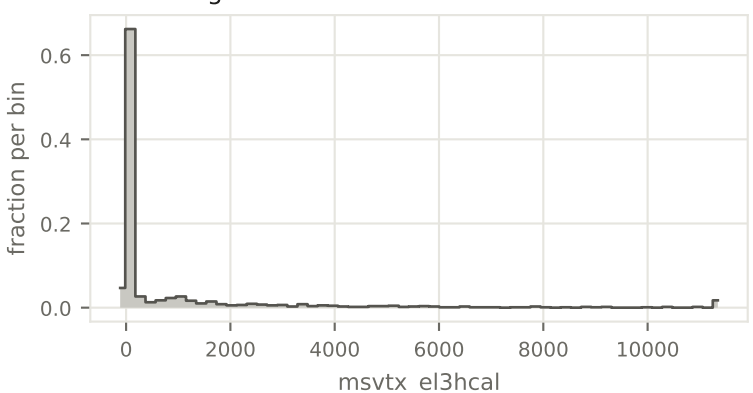



2 · combined signal
background n 1,089
combined, pooled D 0.236 n 4,987
equal weight per mass D 0.240

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0% −999 0.0%
background n 1,089

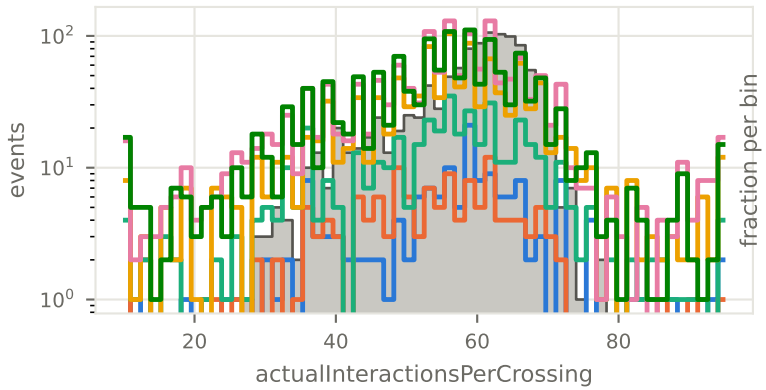
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

# actualInteractionsPerCrossing

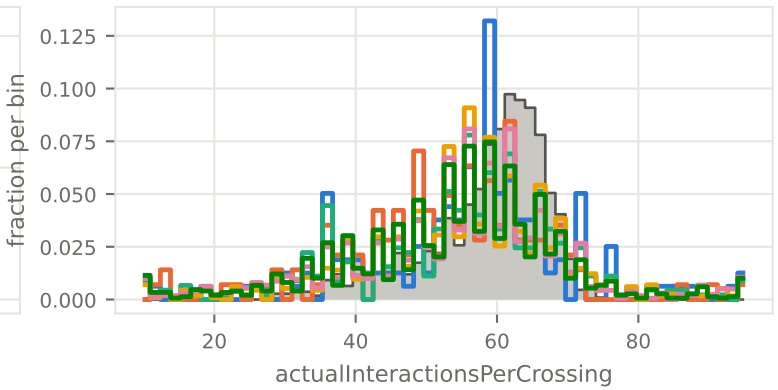
region: BARREL · rank 47 / 98

D_equal 0.237 (noise 0.051) D_pooled 0.244 min 0.187 (mH125_mS15) max 0.297 (mH125_mS55) bkg n = 1,089 sig n = 4,987

1a · each mass point vs background · raw

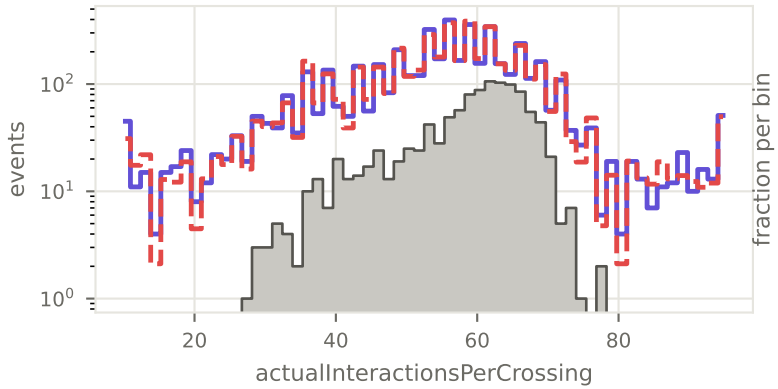



1b · each mass point vs background · normalized

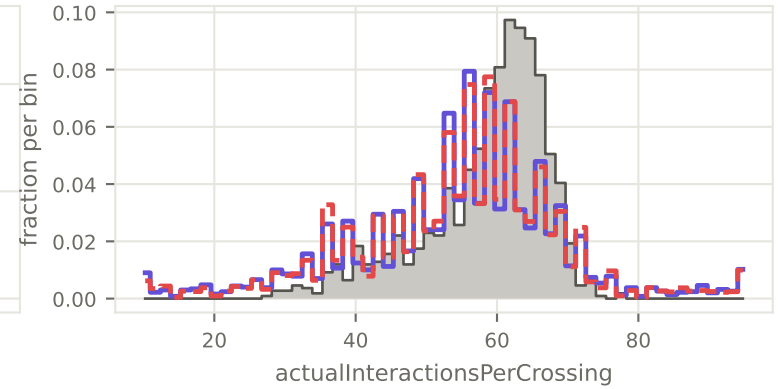



1 · each mass point
background n 1,089
mH125_mS15 D 0.187 n 159
mH125_mS55 D 0.297 n 142
mH200_mS50 D 0.244 n 449
mH400_mS100 D 0.239 n 1,144
mH600_mS150 D 0.240 n 1,608
mH1000_mS275 D 0.257 n 1,485

2a · combined signal vs background · raw

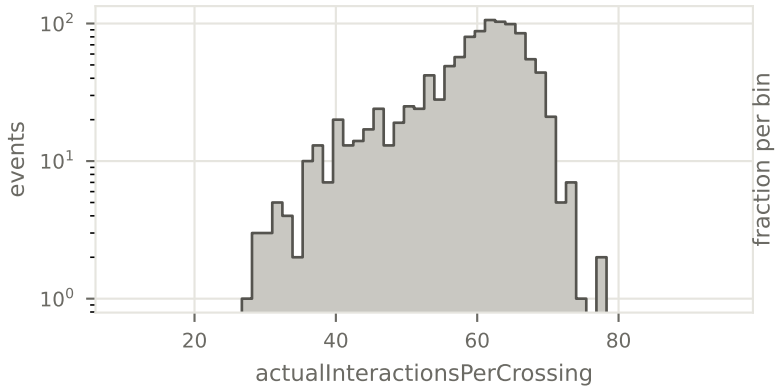



2b · combined signal vs background · normalized

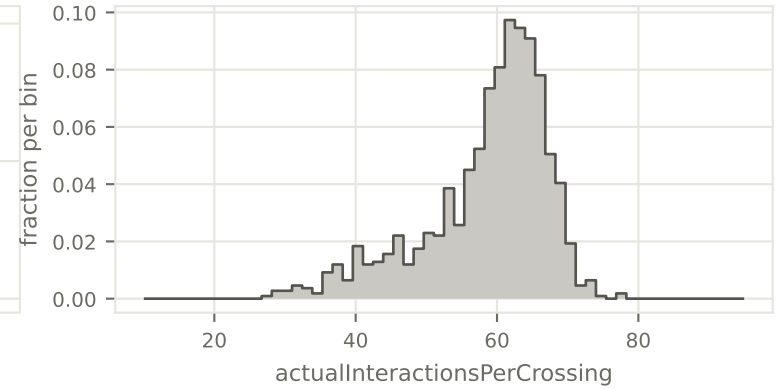



2 · combined signal
background n 1,089
combined, pooled D 0.244 n 4,987
equal weight per mass D 0.237

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0% −999 0.0%
background n 1,089

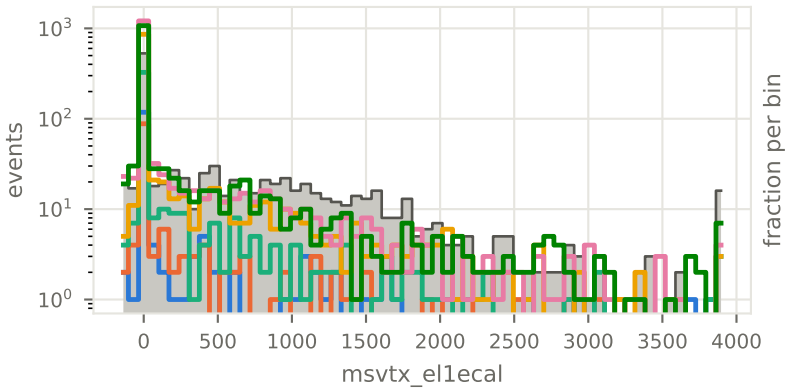
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

# msvtx_el1ecal

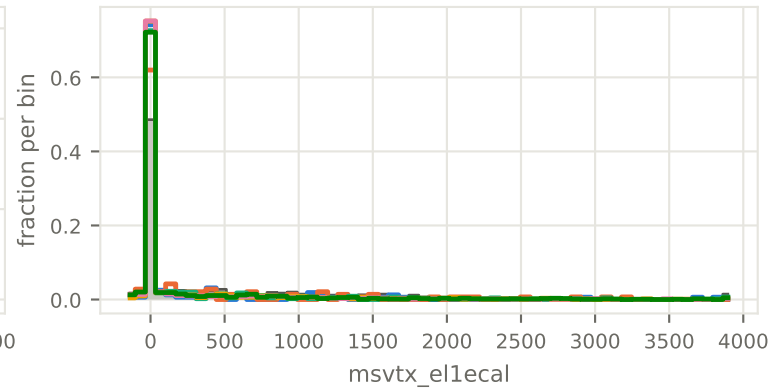
region: BARREL · rank 42 / 98 · in: NN1 barrel, NN1 endcap

D_equal 0.235 (noise 0.051) D_pooled 0.249 min 0.187 (mH125_mS55) max 0.267 (mH600_mS150) bkg n = 1,089 sig n = 4,987

1a · each mass point vs background · raw

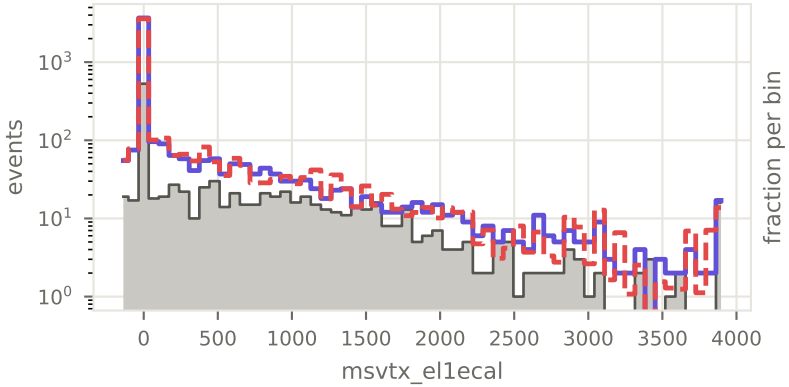



1b · each mass point vs background · normalized

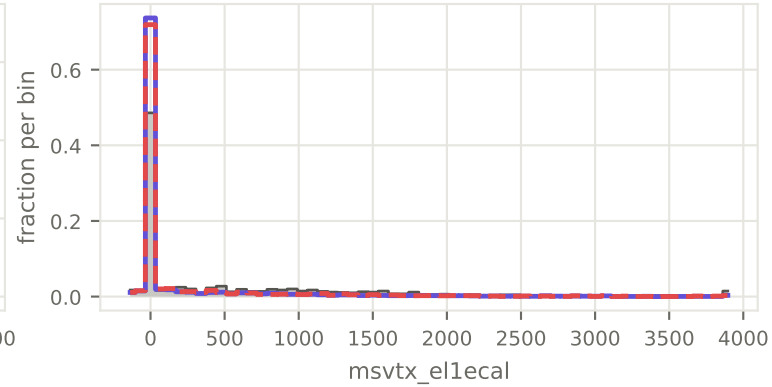



1 · each mass point
background n 1,089
mH125_mS15 D 0.257 n 159
mH125_mS55 D 0.187 n 142
mH200_mS50 D 0.239 n 449
mH400_mS100 D 0.253 n 1,144
mH600_mS150 D 0.267 n 1,608
mH1000_mS275 D 0.241 n 1,485

2a · combined signal vs background · raw

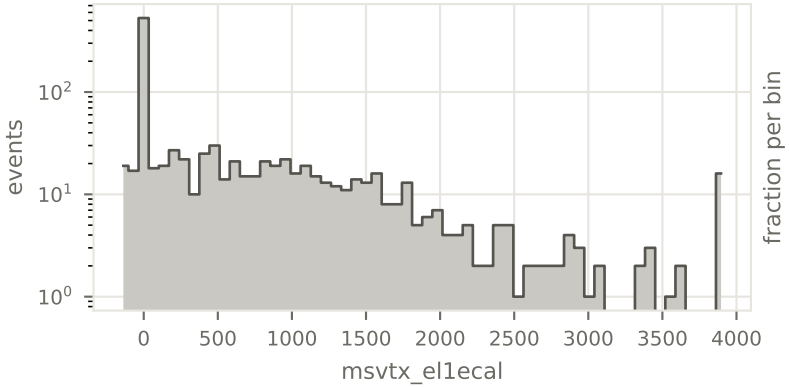



2b · combined signal vs background · normalized

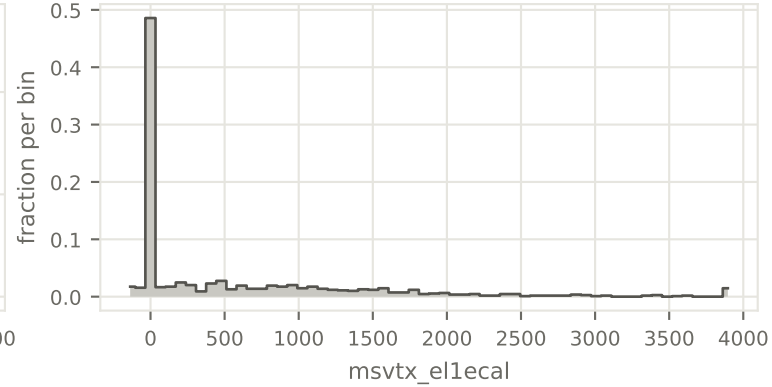



2 · combined signal
background n 1,089
combined, pooled D 0.249 n 4,987
equal weight per mass D 0.235

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0% −999 0.0%
background n 1,089

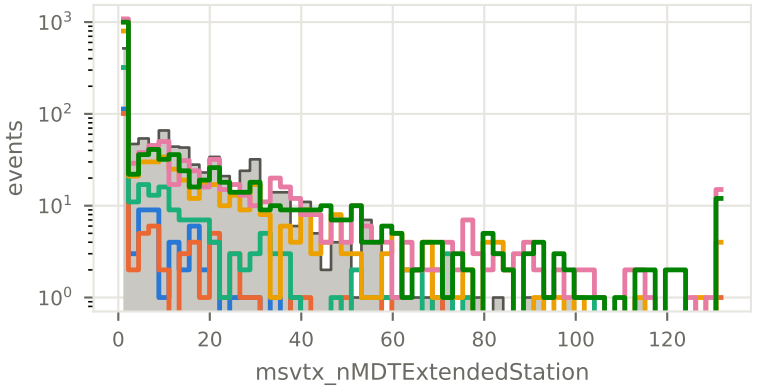
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

# msvtx_nMDTExtendedStation

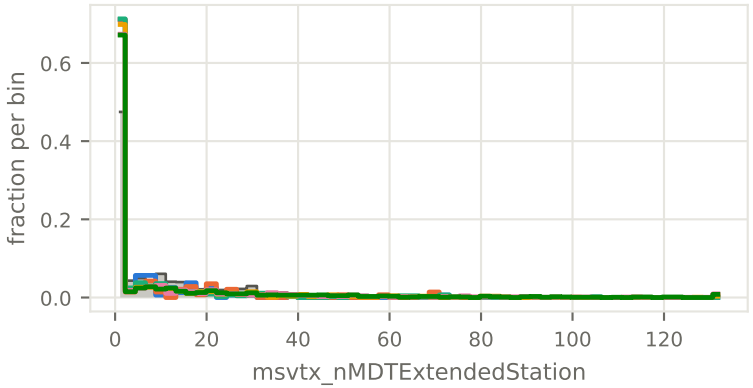
region: BARREL · rank 52 / 98

D_equal 0.222 (noise 0.051) D_pooled 0.212 min 0.203 (mH600_mS150) max 0.246 (mH125_mS15) bkg n = 1,089 sig n = 4,987

1a · each mass point vs background · raw

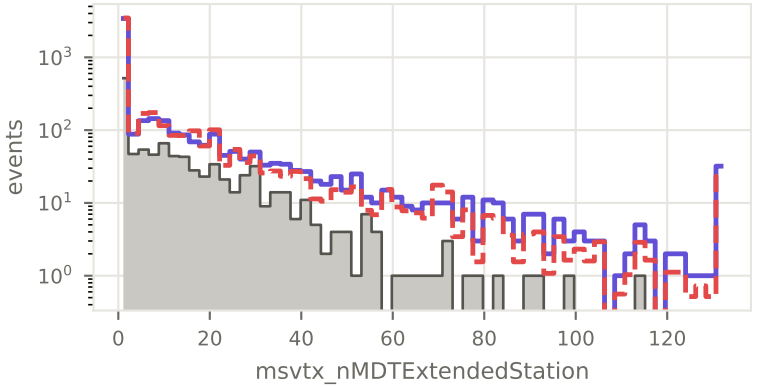



1b · each mass point vs background · normalized

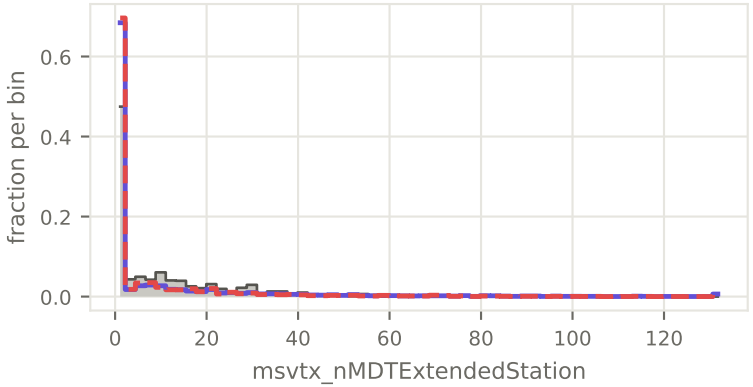



1 · each mass point
background n 1,089
mH125_mS15 D 0.246 n 159
mH125_mS55 D 0.237 n 142
mH200_mS50 D 0.239 n 449
mH400_mS100 D 0.224 n 1,144
mH600_mS150 D 0.203 n 1,608
mH1000_mS275 D 0.204 n 1,485

2a · combined signal vs background · raw

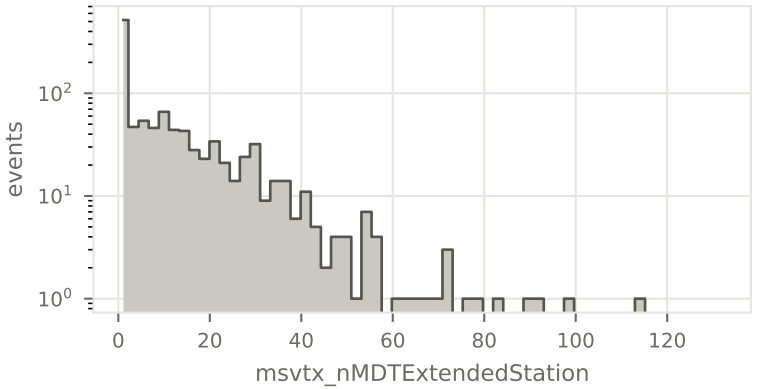



2b · combined signal vs background · normalized

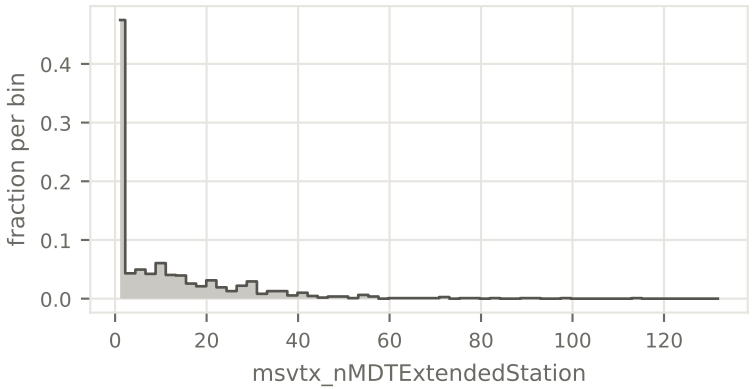



2 · combined signal
background n 1,089
combined, pooled D 0.212 n 4,987
equal weight per mass D 0.222

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0% −999 0.0%
background n 1,089

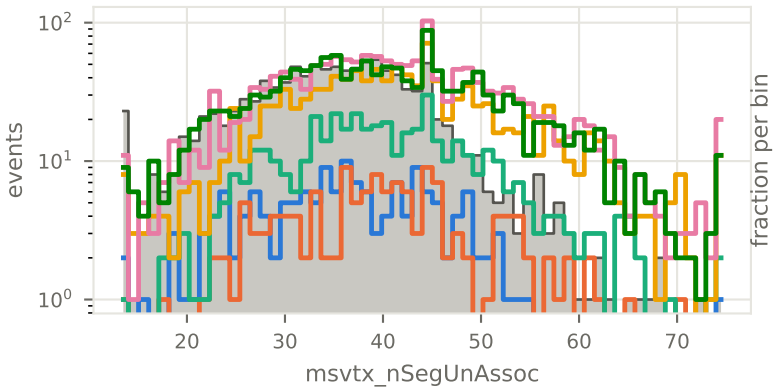
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

# msvtx_nSegUnAssoc

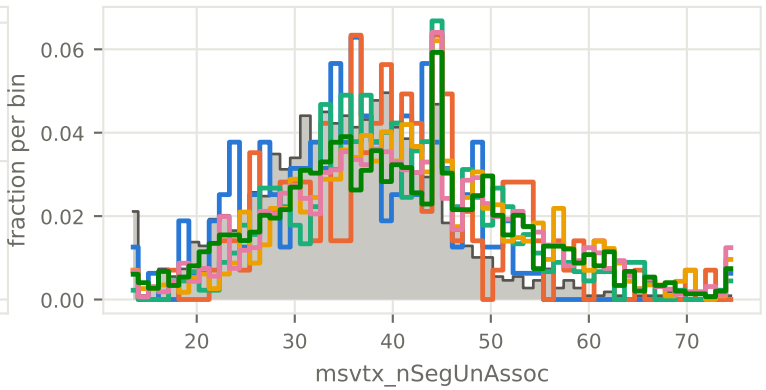
region: BARREL · rank 56 / 98

D_equal 0.193 (noise 0.051)  D_pooled 0.223  min 0.075 (mH125_mS15)  max 0.251 (mH400_mS100)  bkg n = 1,089  sig n = 4,987

1a · each mass point vs background · raw

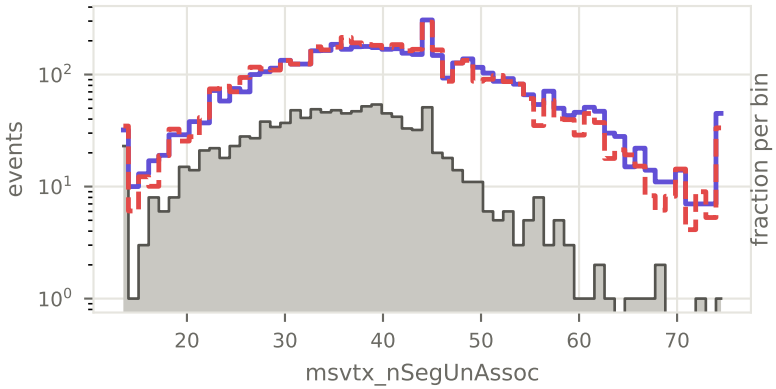



1b · each mass point vs background · normalized

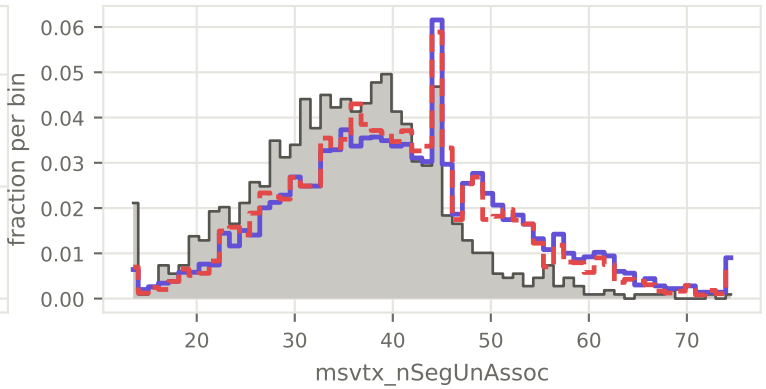



1 · each mass point
background n 1,089
mH125_mS15 D 0.075 n 159
mH125_mS55 D 0.210 n 142
mH200_mS50 D 0.211 n 449
mH400_mS100 D 0.251 n 1,144
mH600_mS150 D 0.242 n 1,608
mH1000_mS275 D 0.221 n 1,485

2a · combined signal vs background · raw

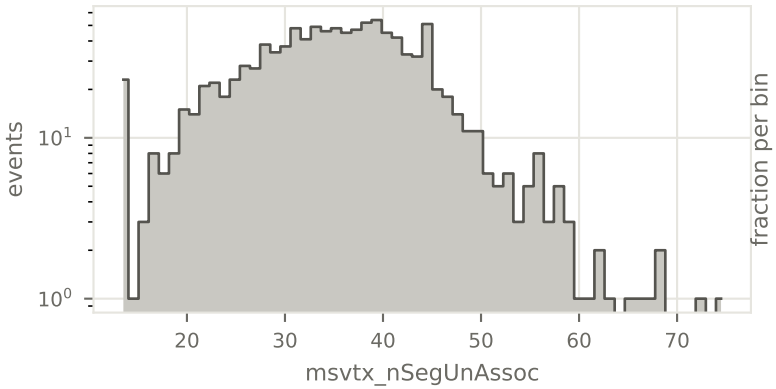



2b · combined signal vs background · normalized

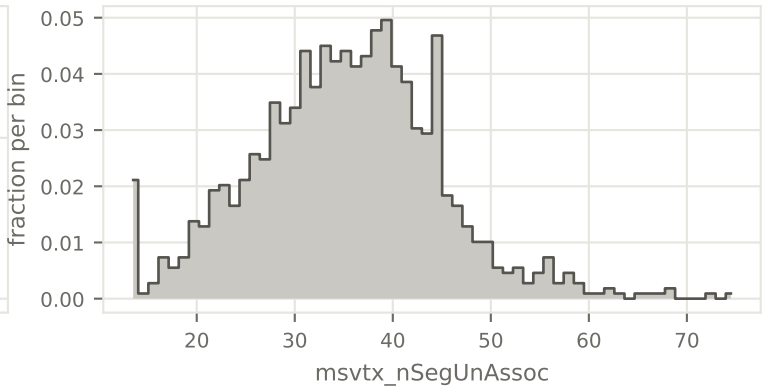



2 · combined signal
background n 1,089
combined, pooled D 0.223 n 4,987
equal weight per mass D 0.193

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0% −999 0.0%
background n 1,089

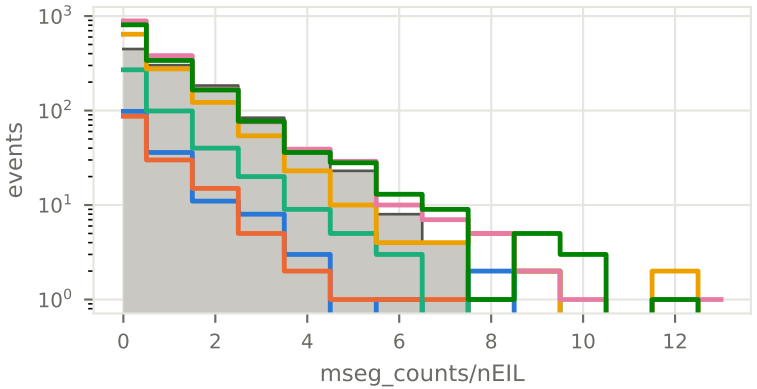
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

# mseg_counts/nEIL

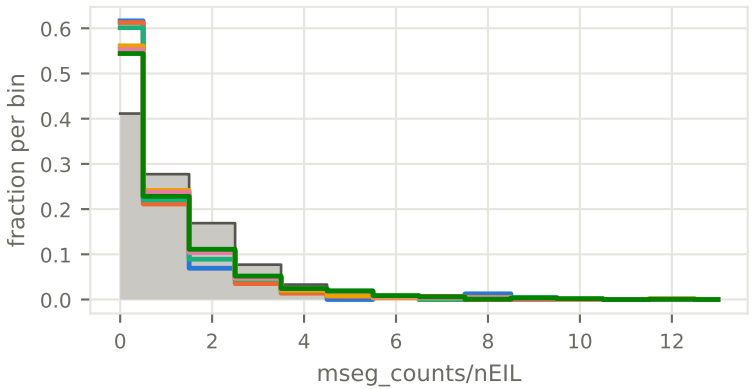
region: BARREL · rank 80 / 98

D_equal 0.170 (noise 0.051) D_pooled 0.149 min 0.133 (mH1000_mS275) max 0.205 (mH125_mS15) bkg n = 1,089 sig n = 4,987

1a · each mass point vs background · raw

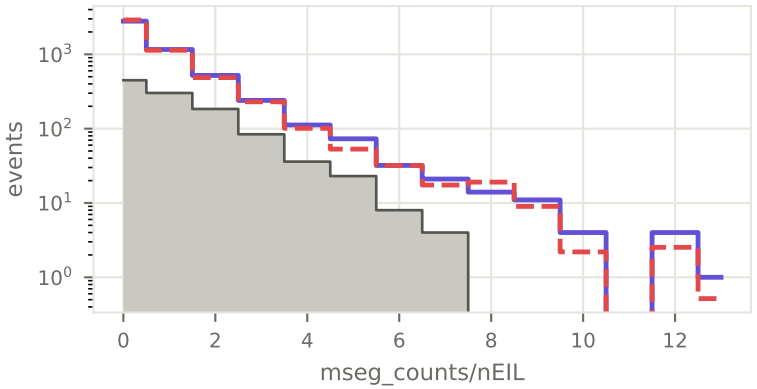



1b · each mass point vs background · normalized

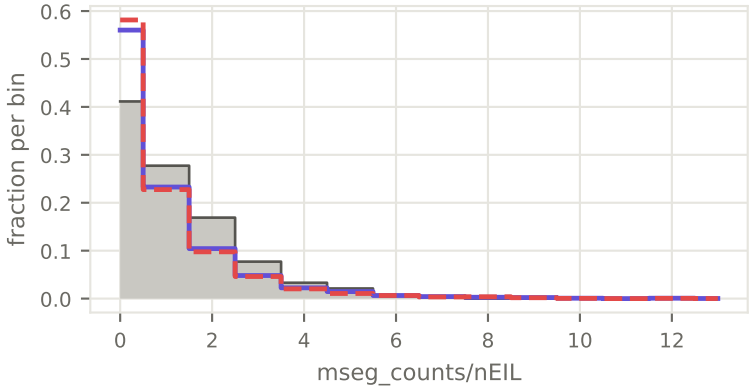



1 · each mass point
background n 1,089
mH125_mS15 D 0.205 n 159
mH125_mS55 D 0.201 n 142
mH200_mS50 D 0.190 n 449
mH400_mS100 D 0.150 n 1,144
mH600_mS150 D 0.141 n 1,608
mH1000_mS275 D 0.133 n 1,485

2a · combined signal vs background · raw

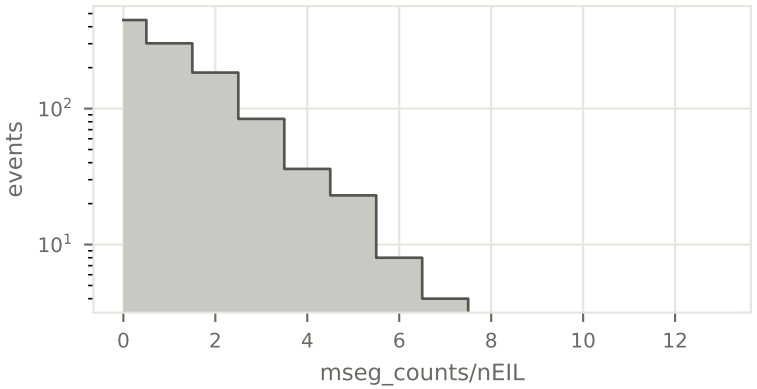



2b · combined signal vs background · normalized

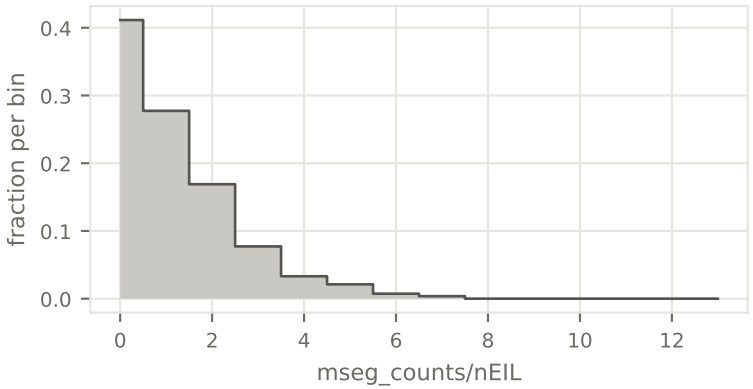



2 · combined signal
background n 1,089
combined, pooled D 0.149 n 4,987
equal weight per mass D 0.170

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0% −999 0.0%
background n 1,089

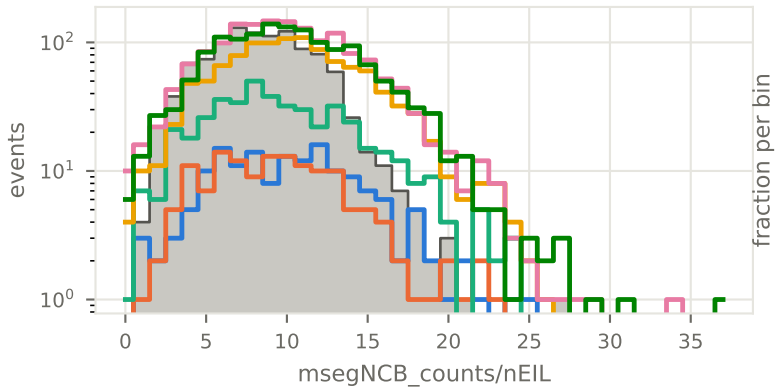
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

# msegNCB_counts/nEIL

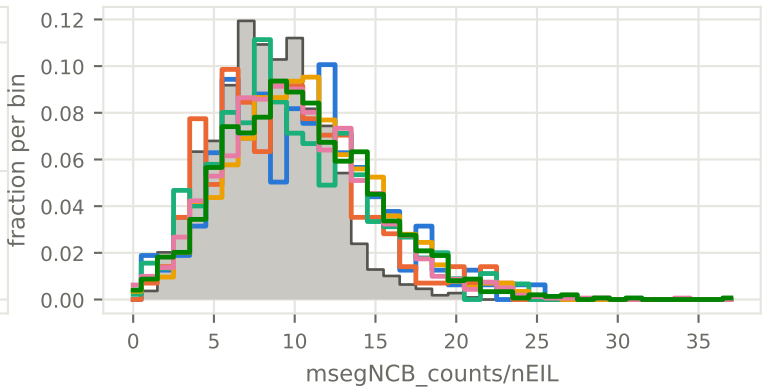
region: BARREL · rank 63 / 98

D_equal 0.168 (noise 0.051) D_pooled 0.176 min 0.121 (mH125_mS55) max 0.204 (mH400_mS100) bkg n = 1,089 sig n = 4,987

1a · each mass point vs background · raw

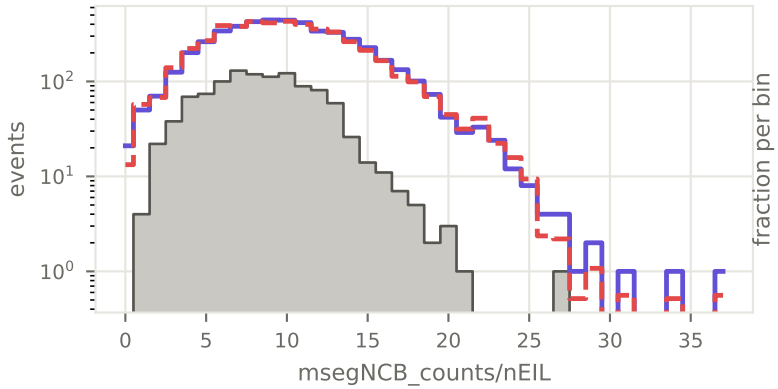



1b · each mass point vs background · normalized

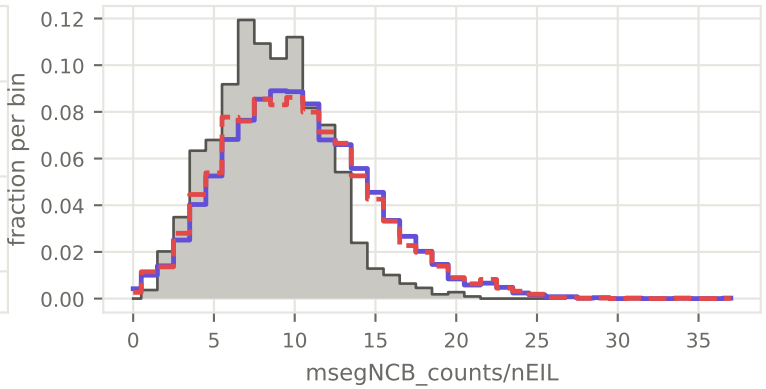



1 · each mass point
background n 1,089
mH125_mS15 D 0.203 n 159
mH125_mS55 D 0.121 n 142
mH200_mS50 D 0.167 n 449
mH400_mS100 D 0.204 n 1,144
mH600_mS150 D 0.170 n 1,608
mH1000_mS275 D 0.182 n 1,485

2a · combined signal vs background · raw

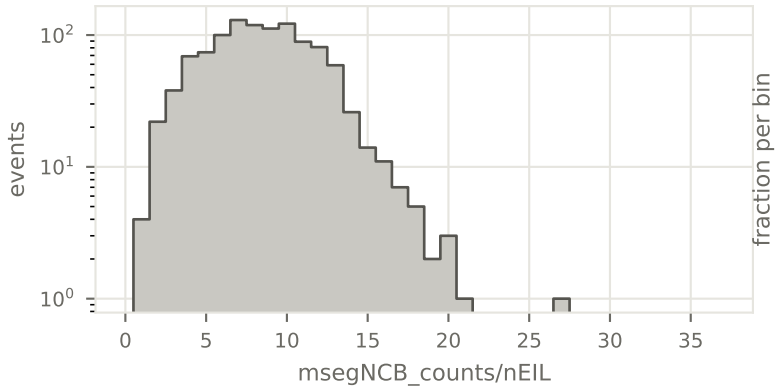



2b · combined signal vs background · normalized

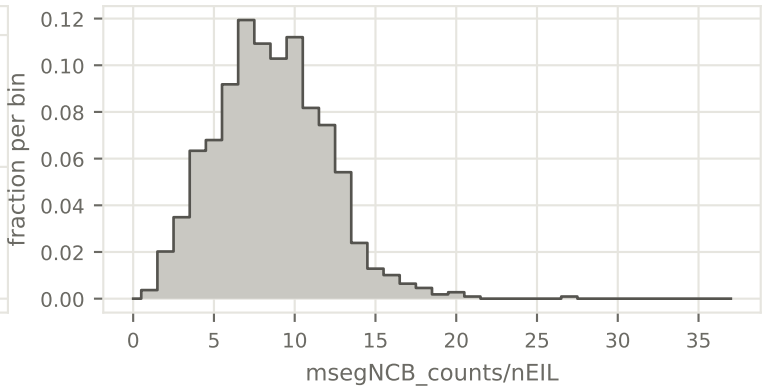



2 · combined signal
background n 1,089
combined, pooled D 0.176 n 4,987
equal weight per mass D 0.168

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0% −999 0.0%
background n 1,089

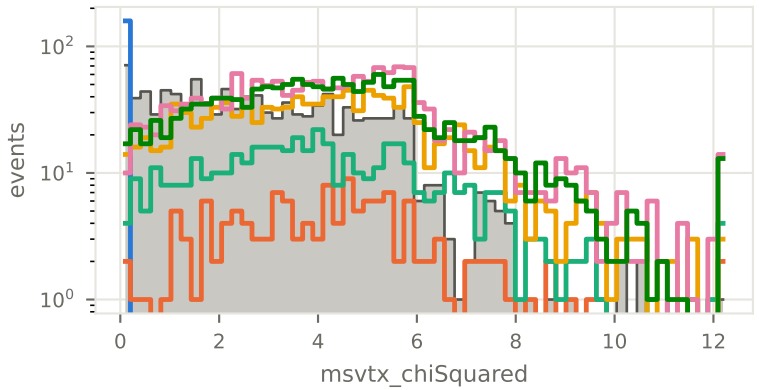
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

# msvtx_chiSquared

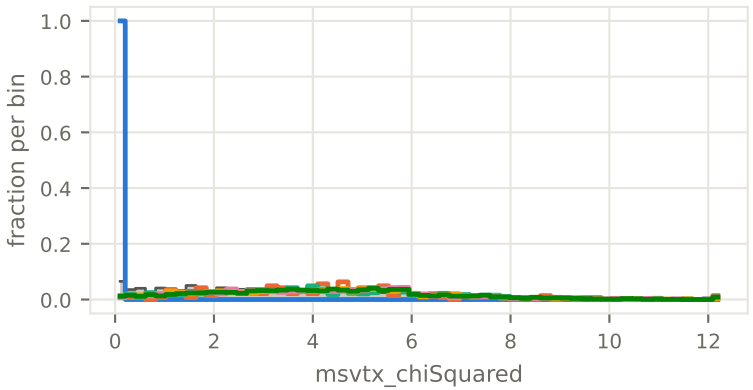
region: BARREL · rank 17 / 98

D_equal 0.167 (noise 0.051) D_pooled 0.220 min 0.234 (mH400_mS100) max 1.000 (mH125_mS15) bkg n = 1,089 sig n = 4,987

1a · each mass point vs background · raw

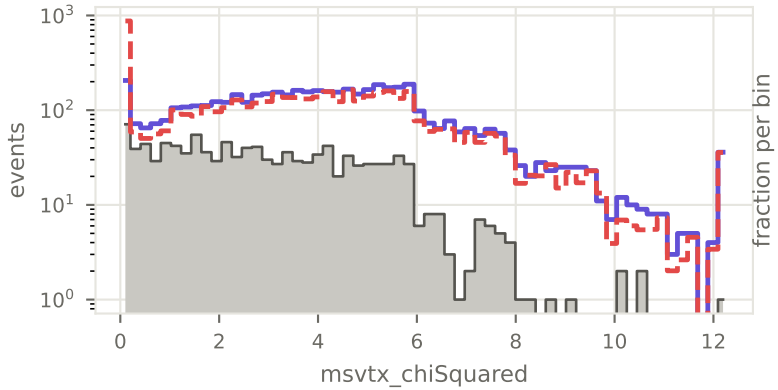



1b · each mass point vs background · normalized

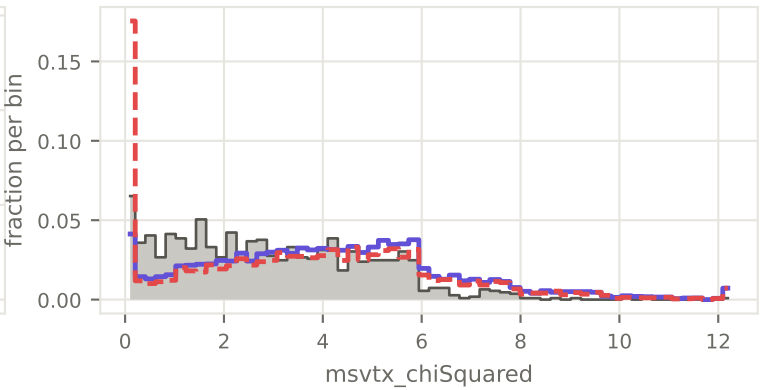



1 · each mass point
background n 1,089
mH125_mS15 D 1.000 n 159
mH125_mS55 D 0.278 n 142
mH200_mS50 D 0.236 n 449
mH400_mS100 D 0.234 n 1,144
mH600_mS150 D 0.242 n 1,608
mH1000_mS275 D 0.253 n 1,485

2a · combined signal vs background · raw

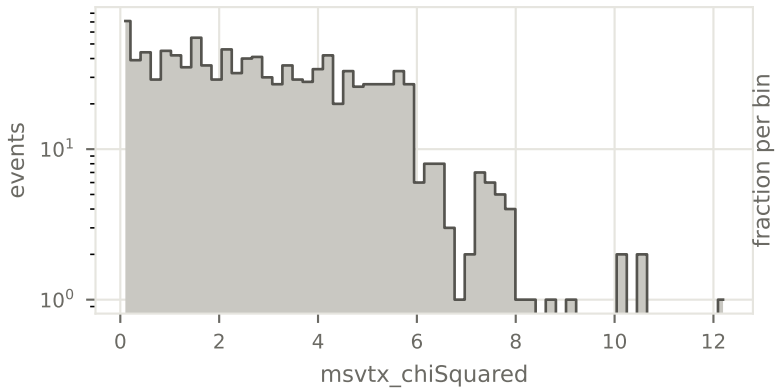



2b · combined signal vs background · normalized

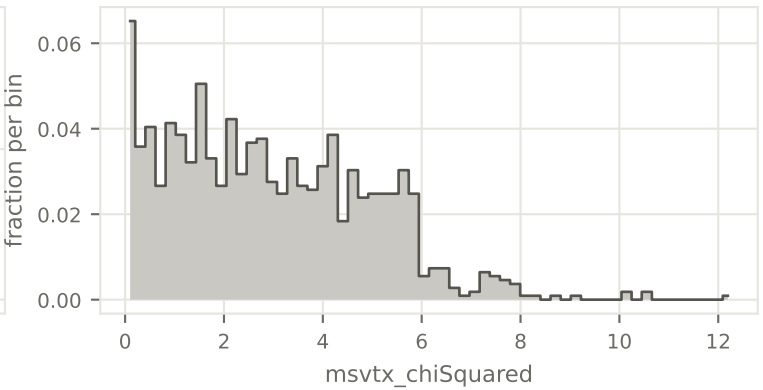



2 · combined signal
background n 1,089
combined, pooled D 0.220 n 4,987
equal weight per mass D 0.167

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0% −999 0.0%
background n 1,089

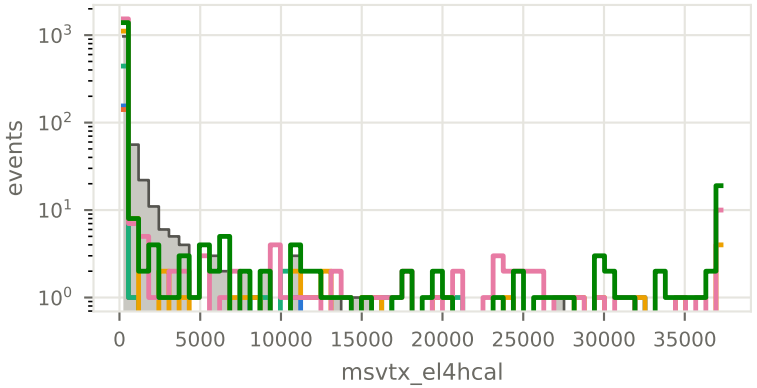
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

# msvtx_el4hcal

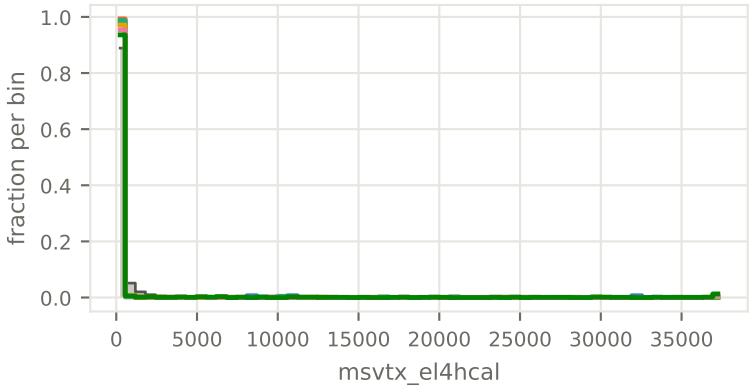
region: BARREL · rank 79 / 98 · in: NN1 barrel, NN1 endcap

D_equal 0.166 (noise 0.051)  D_pooled 0.159  min 0.139 (mH1000_mS275)  max 0.202 (mH125_mS55)  bkg n = 1,089  sig n = 4,987

1a · each mass point vs background · raw

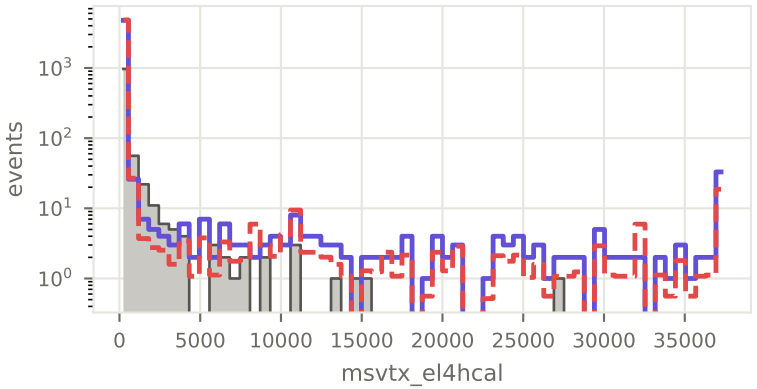



1b · each mass point vs background · normalized

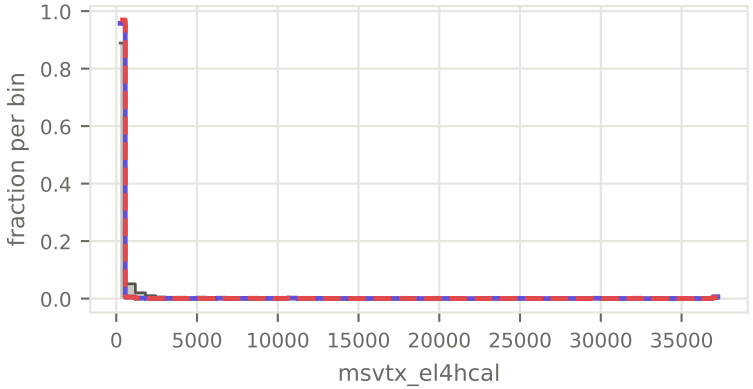



1 · each mass point
background n 1,089
mH125_mS15 D 0.151 n 159
mH125_mS55 D 0.202 n 142
mH200_mS50 D 0.181 n 449
mH400_mS100 D 0.172 n 1,144
mH600_mS150 D 0.160 n 1,608
mH1000_mS275 D 0.139 n 1,485

2a · combined signal vs background · raw

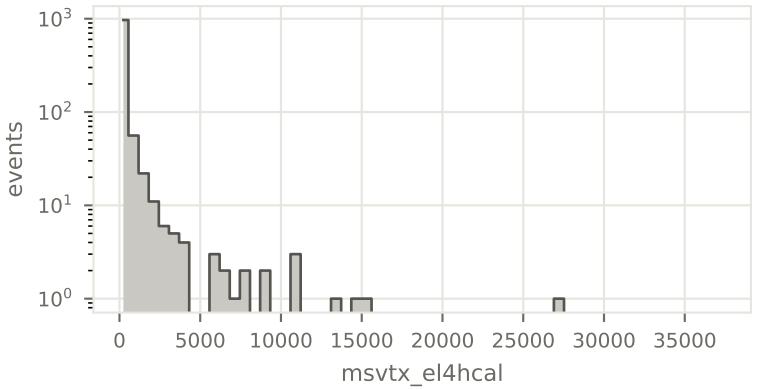



2b · combined signal vs background · normalized

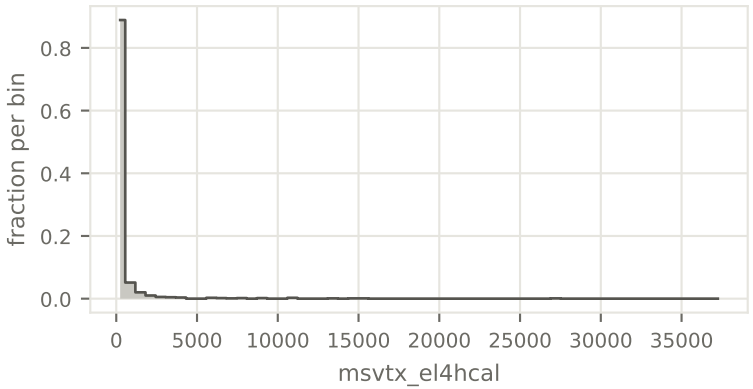



2 · combined signal
background n 1,089
combined, pooled D 0.159 n 4,987
equal weight per mass D 0.166

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0%  −999 0.0%
background n 1,089

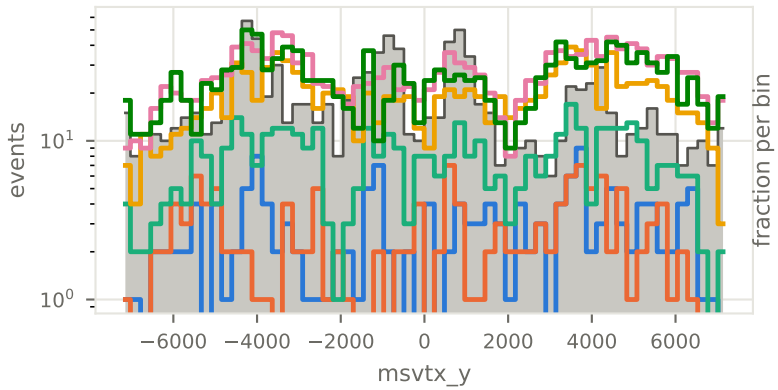
a = raw (log y)  b = normalized · edge bins include under/overflow · NaN and -999 removed

# msvtx_y

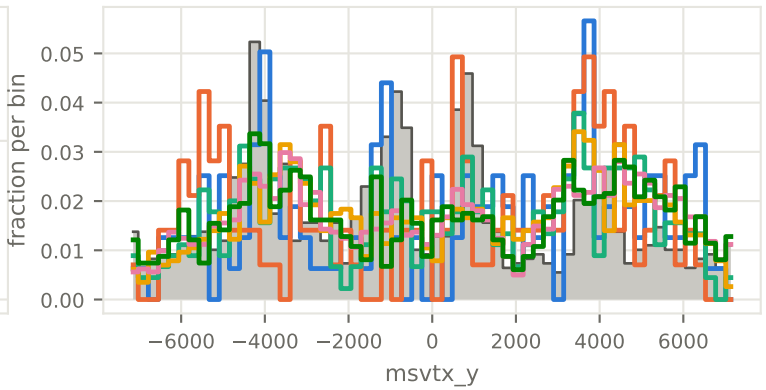
region: BARREL · rank 71 / 98

D_equal 0.166 (noise 0.051) D_pooled 0.160 min 0.135 (mH200_mS50) max 0.203 (mH125_mS15) bkg n = 1,089 sig n = 4,987

1a · each mass point vs background · raw

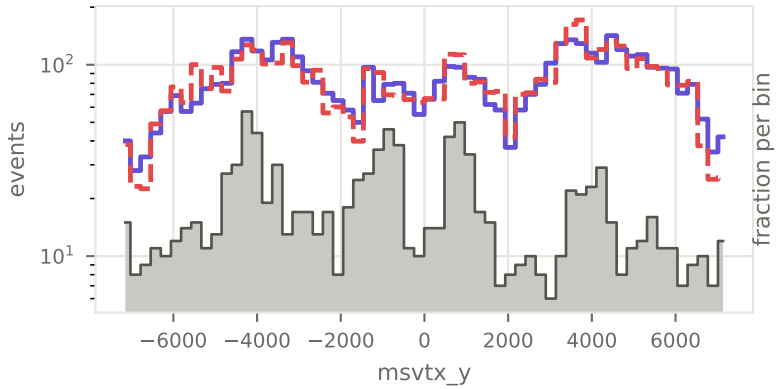



1b · each mass point vs background · normalized

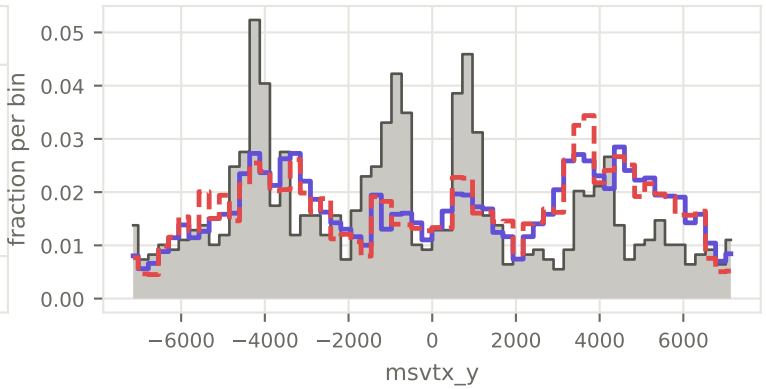



1 · each mass point
background n 1,089
mH125_mS15 D 0.203 n 159
mH125_mS55 D 0.190 n 142
mH200_mS50 D 0.135 n 449
mH400_mS100 D 0.170 n 1,144
mH600_mS150 D 0.163 n 1,608
mH1000_mS275 D 0.150 n 1,485

2a · combined signal vs background · raw

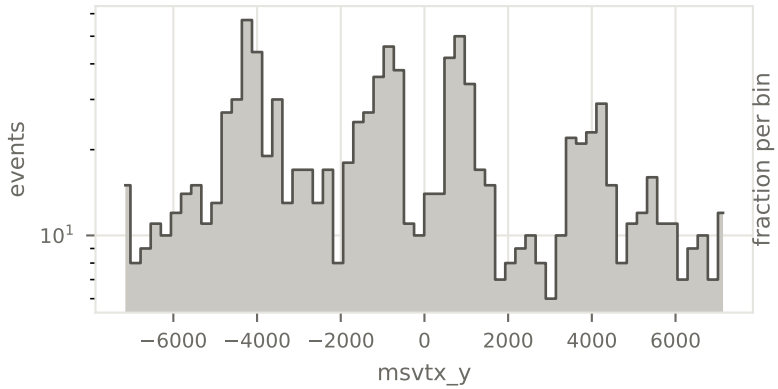



2b · combined signal vs background · normalized

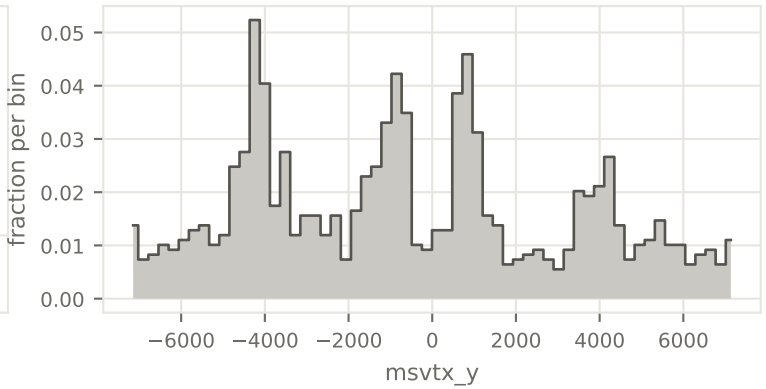



2 · combined signal
background n 1,089
combined, pooled D 0.160 n 4,987
equal weight per mass D 0.166

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0% −999 0.0%
background n 1,089

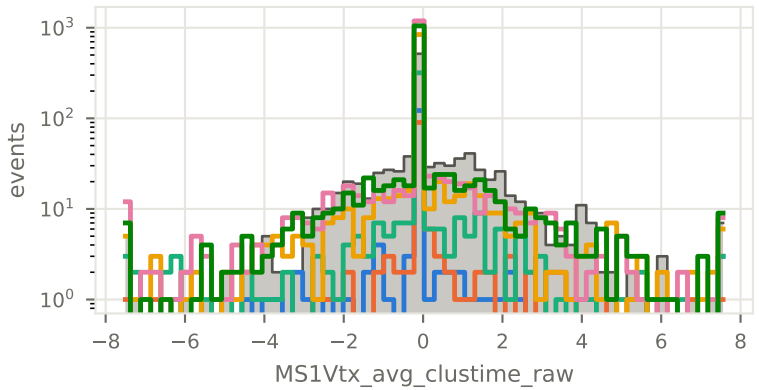
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

# MS1Vtx_avg_clustime_raw

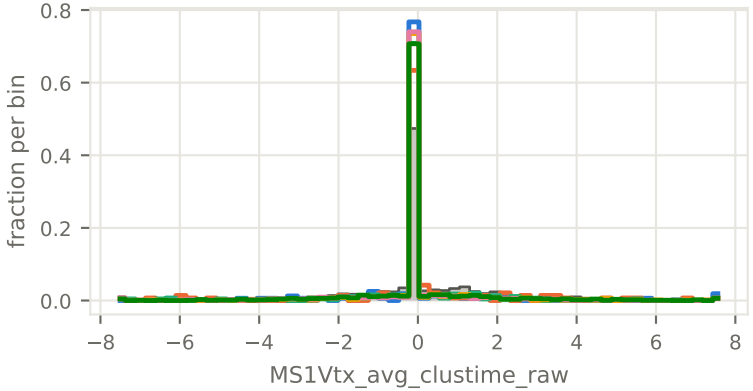
region: BARREL · rank 55 / 98 · in: NN2 barrel, NN2 endcap

D_equal 0.157 (noise 0.051)  D_pooled 0.164  min 0.108 (mH125_mS55)  max 0.199 (mH125_mS15)  bkg n = 1,089  sig n = 4,987

1a · each mass point vs background · raw

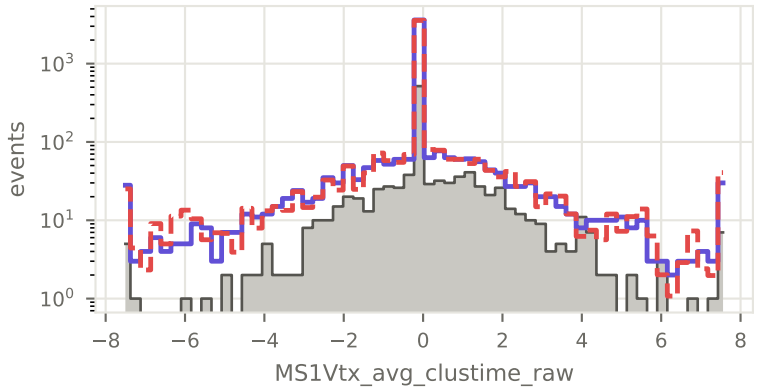



1b · each mass point vs background · normalized

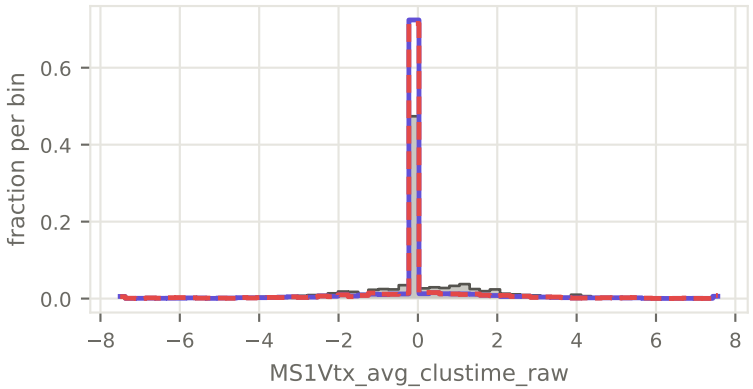



1 · each mass point
- background n 1,089
- mH125_mS15 D 0.199 n 159
- mH125_mS55 D 0.108 n 142
- mH200_mS50 D 0.163 n 449
- mH400_mS100 D 0.165 n 1,144
- mH600_mS150 D 0.177 n 1,608
- mH1000_mS275 D 0.153 n 1,485

2a · combined signal vs background · raw

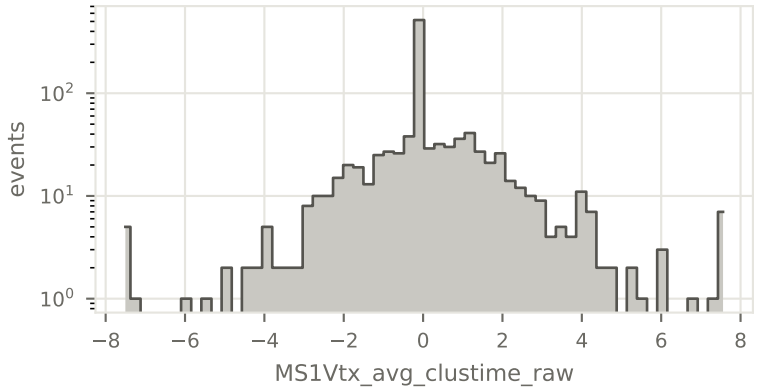



2b · combined signal vs background · normalized

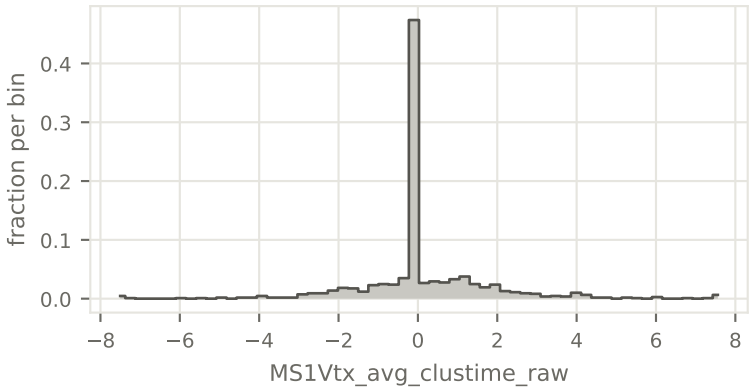



2 · combined signal
- background n 1,089
- combined, pooled D 0.164 n 4,987
- equal weight per mass D 0.157

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0%  −999 0.0%
- background n 1,089

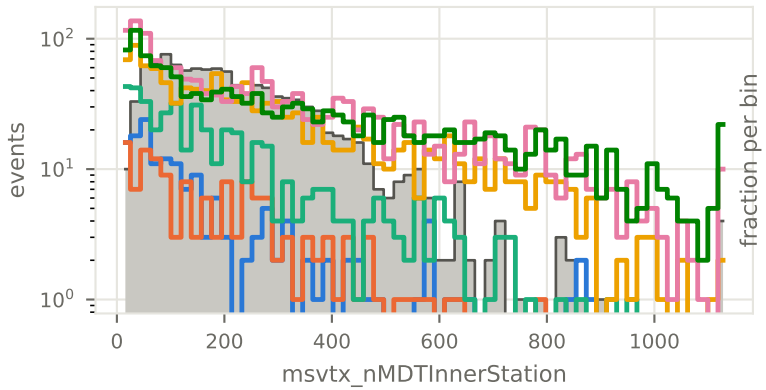
a = raw (log y)  b = normalized · edge bins include under/overflow · NaN and -999 removed

# msvtx_nMDTInnerStation

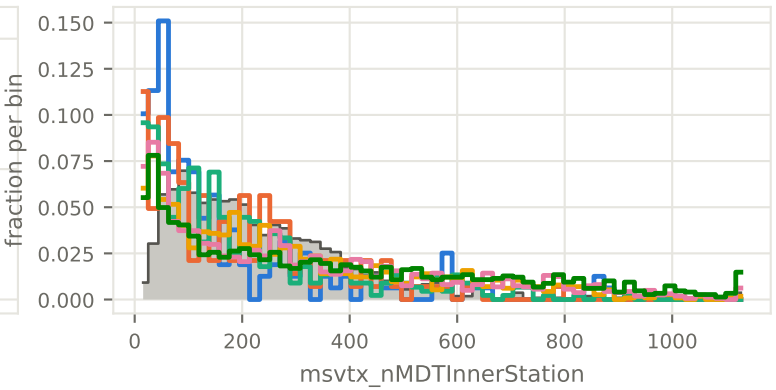
region: BARREL · rank 44 / 98

D_equal 0.154 (noise 0.051) D_pooled 0.161 min 0.139 (mH400_mS100) max 0.304 (mH125_mS15) bkg n = 1,089 sig n = 4,987

1a · each mass point vs background · raw

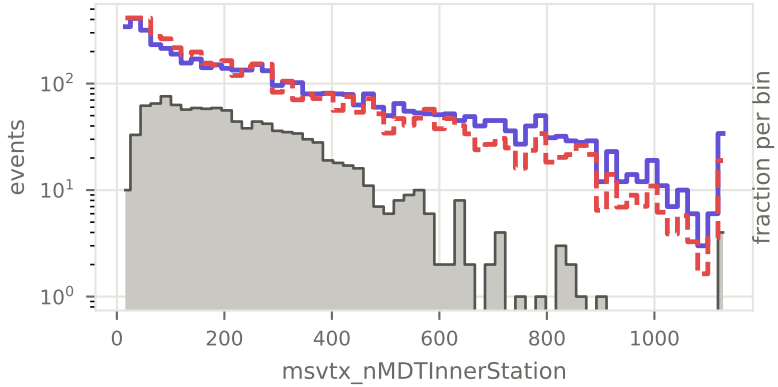



1b · each mass point vs background · normalized

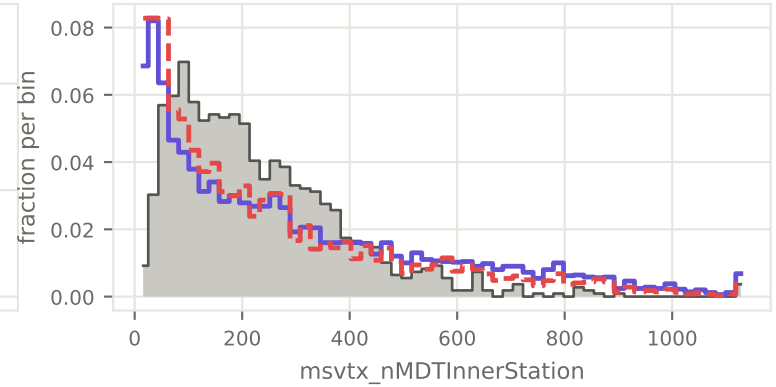



1 · each mass point
background n 1,089
mH125_mS15 D 0.304 n 159
mH125_mS55 D 0.201 n 142
mH200_mS50 D 0.169 n 449
mH400_mS100 D 0.139 n 1,144
mH600_mS150 D 0.179 n 1,608
mH1000_mS275 D 0.245 n 1,485

2a · combined signal vs background · raw

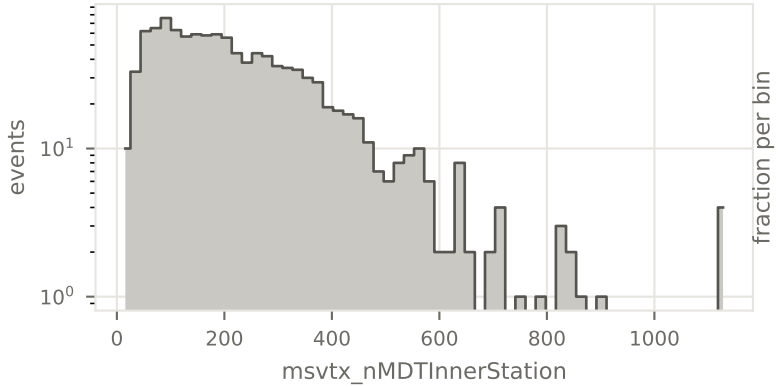



2b · combined signal vs background · normalized

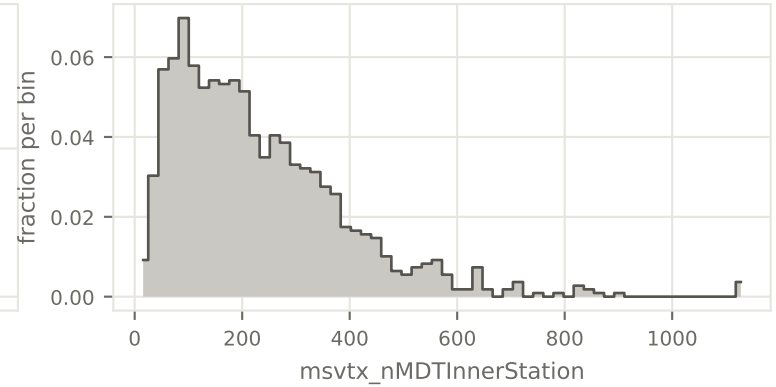



2 · combined signal
background n 1,089
combined, pooled D 0.161 n 4,987
equal weight per mass D 0.154

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0% −999 0.0%
background n 1,089

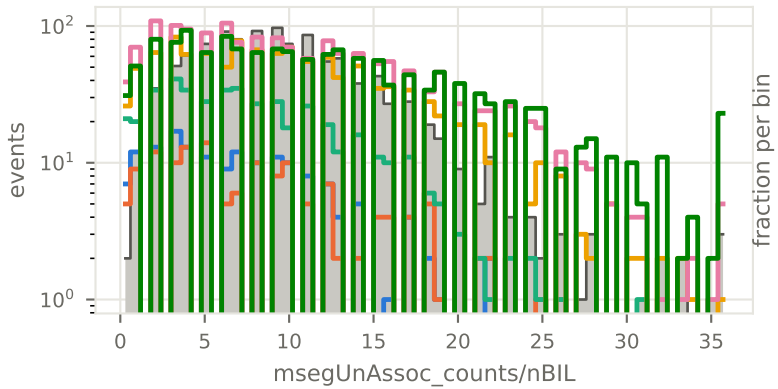
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

# msegUnAssoc_counts/nBIL

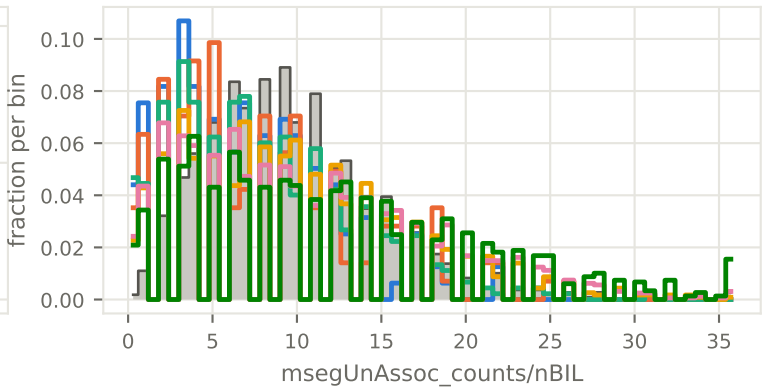
region: BARREL · rank 27 / 98 · in: NN1 barrel

D_equal 0.152 (noise 0.051)  D_pooled 0.111  min 0.102 (mH400_mS100)  max 0.243 (mH125_mS15)  bkg n = 1,089  sig n = 4,987

1a · each mass point vs background · raw

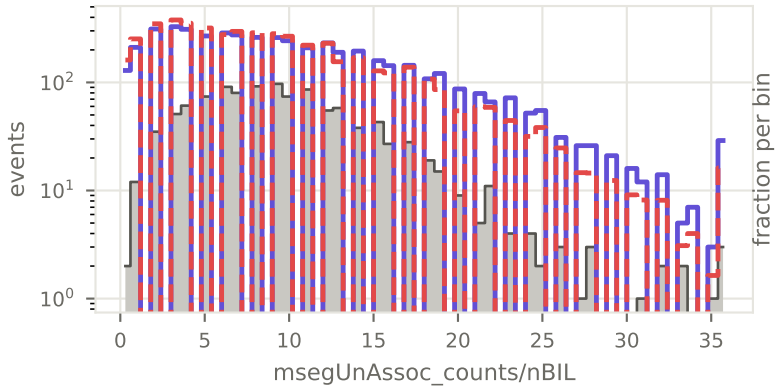



1b · each mass point vs background · normalized

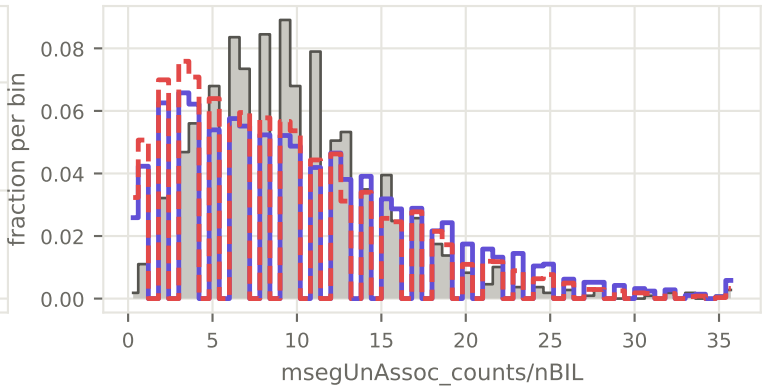



1 · each mass point
background n 1,089
mH125_mS15 D 0.243 n 159
mH125_mS55 D 0.228 n 142
mH200_mS50 D 0.186 n 449
mH400_mS100 D 0.102 n 1,144
mH600_mS150 D 0.110 n 1,608
mH1000_mS275 D 0.171 n 1,485

2a · combined signal vs background · raw

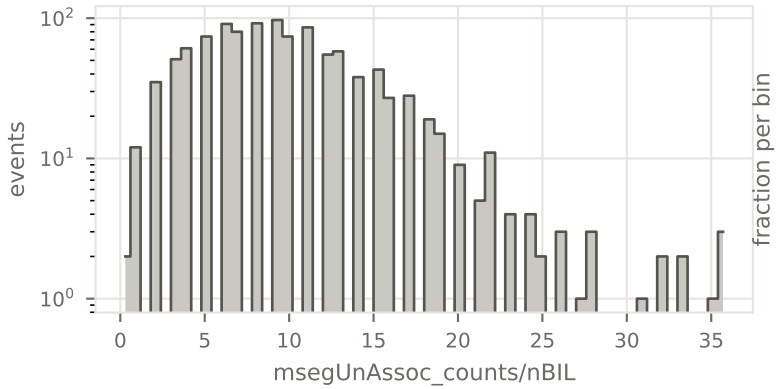



2b · combined signal vs background · normalized

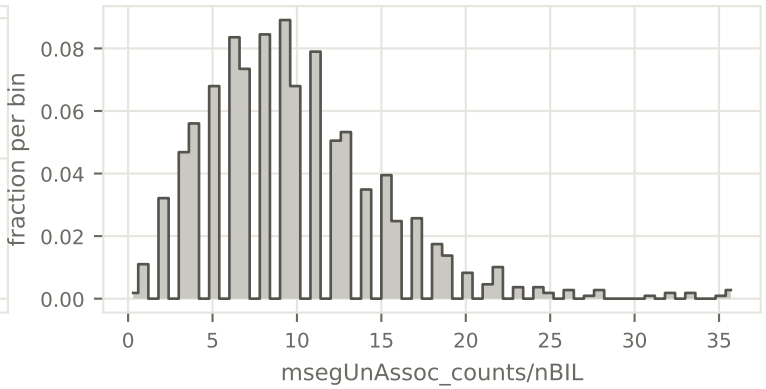



2 · combined signal
background n 1,089
combined, pooled D 0.111 n 4,987
equal weight per mass D 0.152

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0% −999 0.0%
background n 1,089

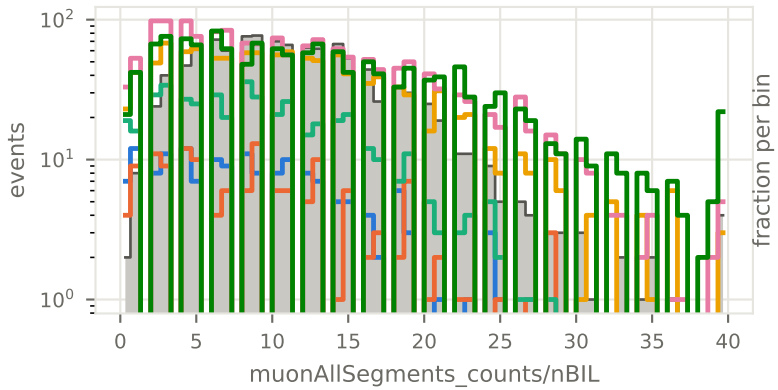
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

# muonAllSegments_counts/nBIL

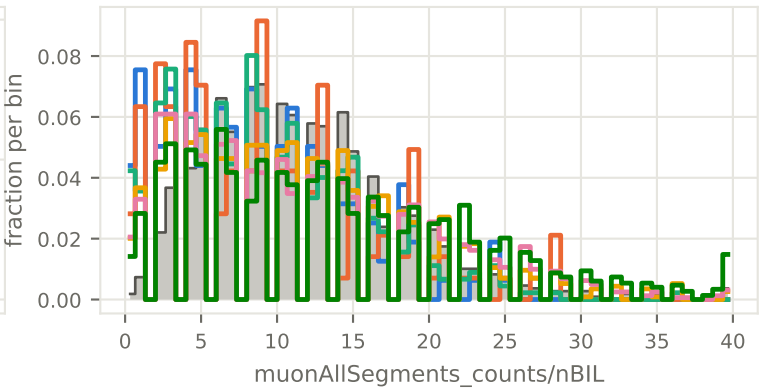
region: BARREL · rank 36 / 98

D_equal 0.146 (noise 0.051)  D_pooled 0.113  min 0.100 (mH400_mS100)  max 0.220 (mH125_mS55)  bkg n = 1,089  sig n = 4,987

1a · each mass point vs background · raw

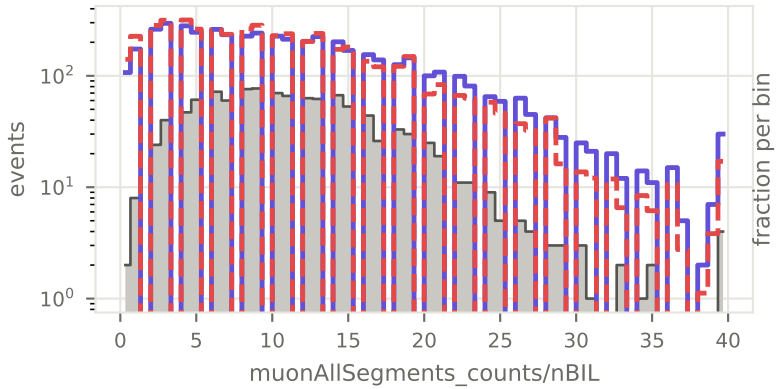



1b · each mass point vs background · normalized

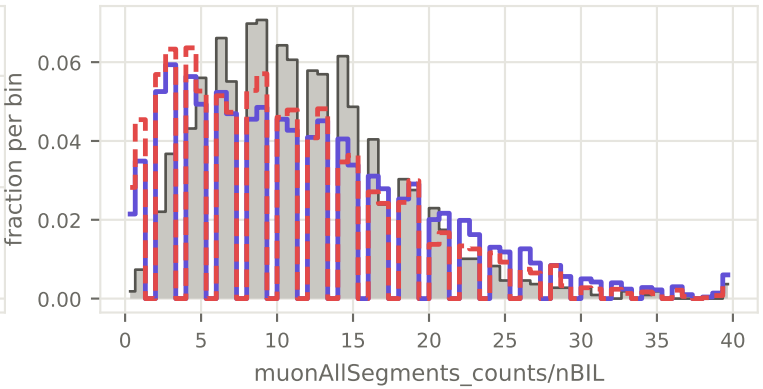



1 · each mass point
background n 1,089
mH125_mS15 D 0.203 n 159
mH125_mS55 D 0.220 n 142
mH200_mS50 D 0.167 n 449
mH400_mS100 D 0.100 n 1,144
mH600_mS150 D 0.125 n 1,608
mH1000_mS275 D 0.150 n 1,485

2a · combined signal vs background · raw

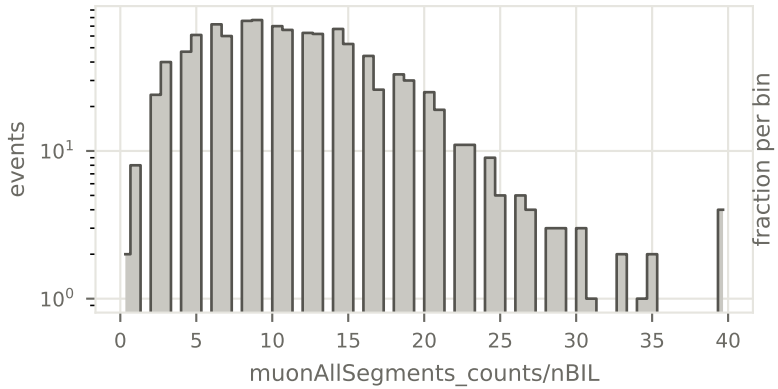



2b · combined signal vs background · normalized

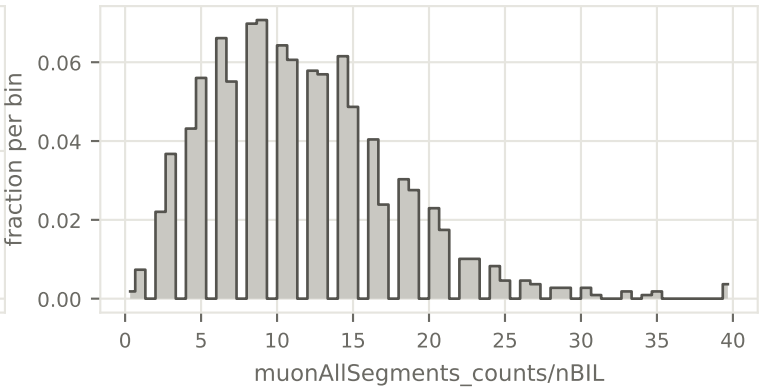



2 · combined signal
background n 1,089
combined, pooled D 0.113 n 4,987
equal weight per mass D 0.146

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0% −999 0.0%
background n 1,089

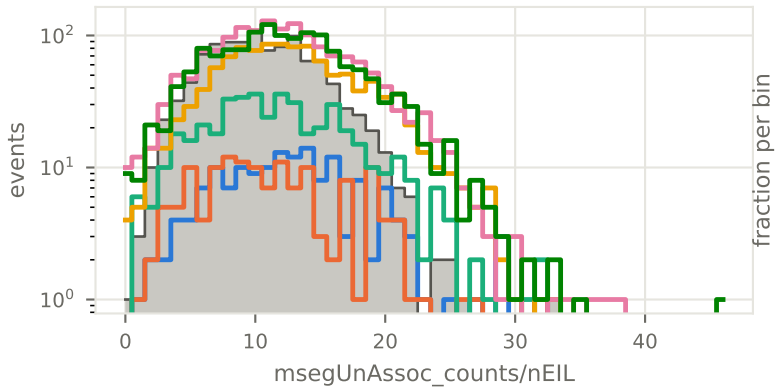
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

# msegUnAssoc_counts/nEIL

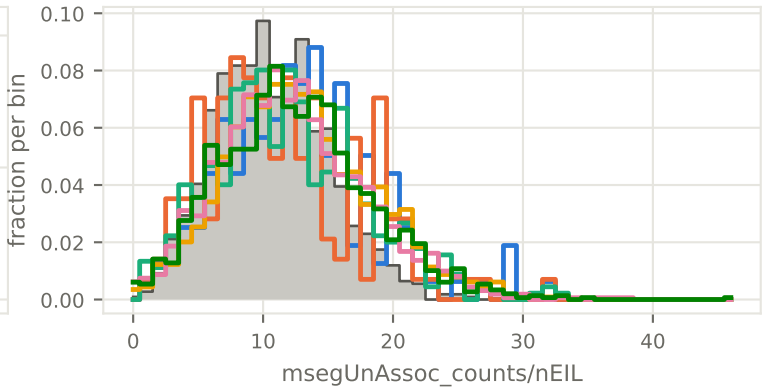
region: BARREL · rank 74 / 98

D_equal 0.136 (noise 0.051)  D_pooled 0.141  min 0.128 (mH600_mS150)  max 0.174 (mH125_mS15)  bkg n = 1,089  sig n = 4,987

1a · each mass point vs background · raw

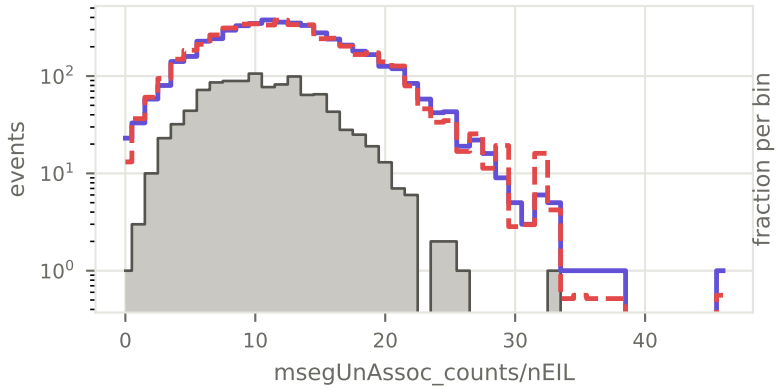



1b · each mass point vs background · normalized

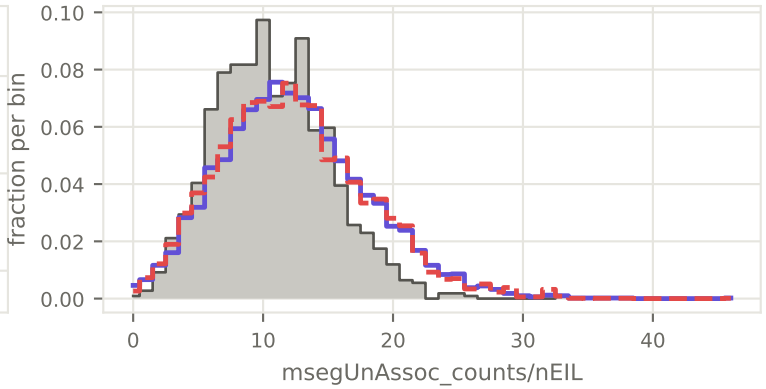



1 · each mass point
background n 1,089
mH125_mS15 D 0.174 n 159
mH125_mS55 D 0.130 n 142
mH200_mS50 D 0.139 n 449
mH400_mS100 D 0.164 n 1,144
mH600_mS150 D 0.128 n 1,608
mH1000_mS275 D 0.155 n 1,485

2a · combined signal vs background · raw

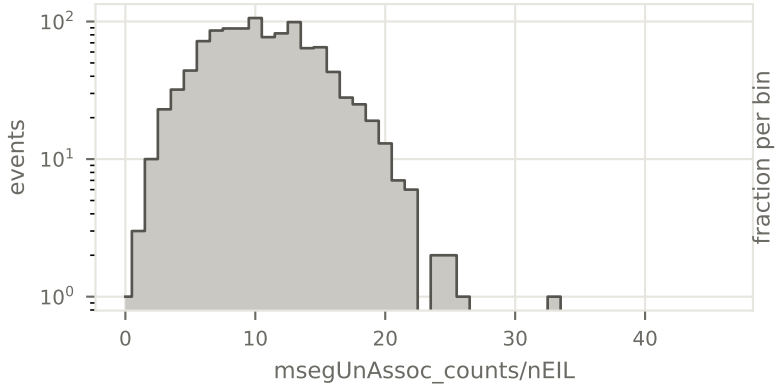



2b · combined signal vs background · normalized

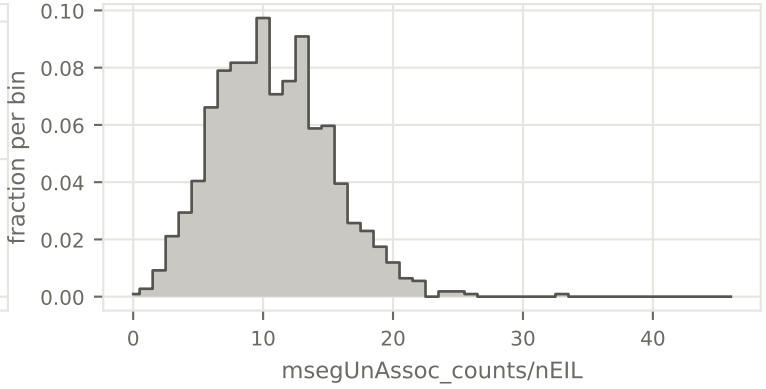



2 · combined signal
background n 1,089
combined, pooled D 0.141 n 4,987
equal weight per mass D 0.136

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0%  −999 0.0%
background n 1,089

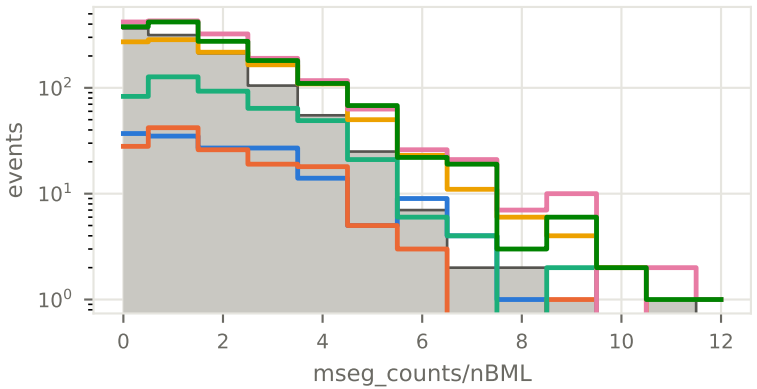
a = raw (log y)  b = normalized · edge bins include under/overflow · NaN and -999 removed

# mseg_counts/nBML

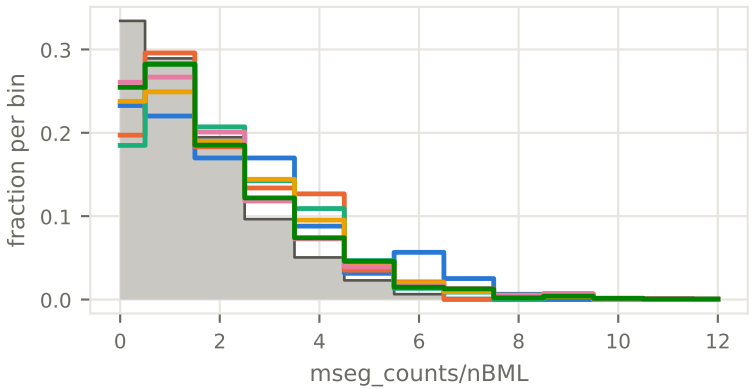
region: BARREL · rank 86 / 98

D_equal 0.135 (noise 0.051) D_pooled 0.113 min 0.096 (mH600_mS150) max 0.196 (mH125_mS15) bkg n = 1,089 sig n = 4,987

1a · each mass point vs background · raw

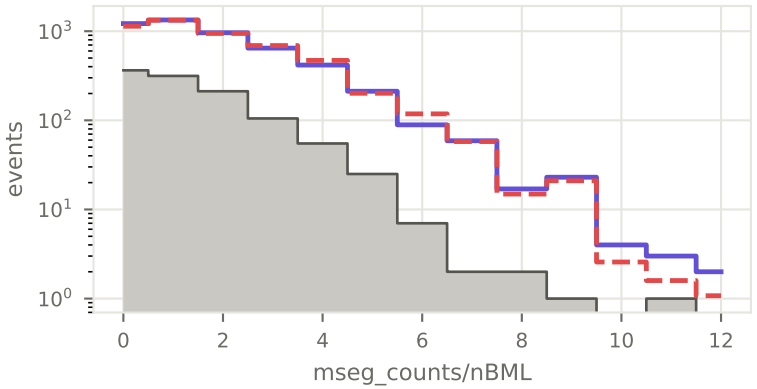



1b · each mass point vs background · normalized

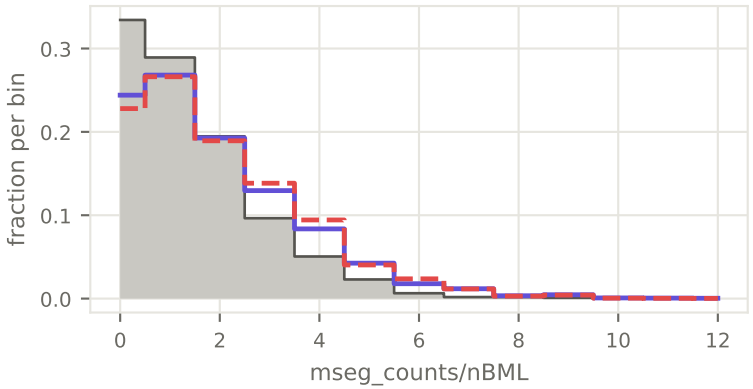



1 · each mass point
background n 1,089
mH125_mS15 D 0.196 n 159
mH125_mS55 D 0.142 n 142
mH200_mS50 D 0.156 n 449
mH400_mS100 D 0.142 n 1,144
mH600_mS150 D 0.096 n 1,608
mH1000_mS275 D 0.096 n 1,485

2a · combined signal vs background · raw

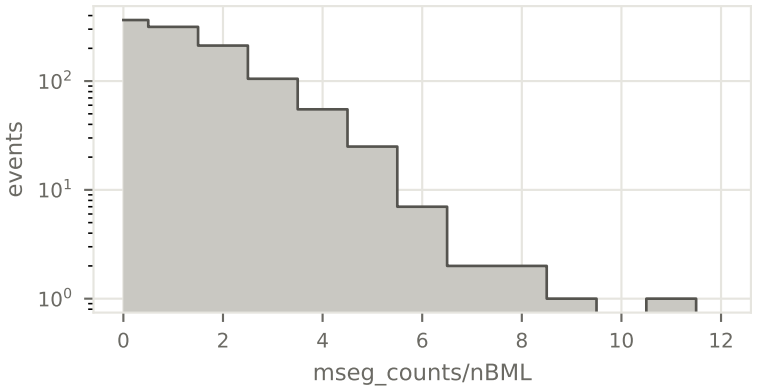



2b · combined signal vs background · normalized

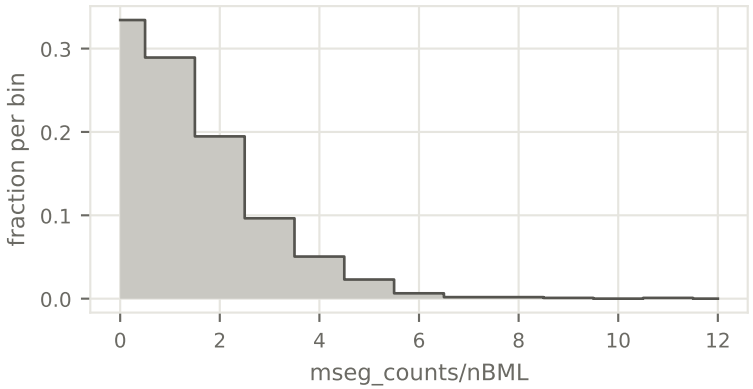



2 · combined signal
background n 1,089
combined, pooled D 0.113 n 4,987
equal weight per mass D 0.135

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0% −999 0.0%
background n 1,089

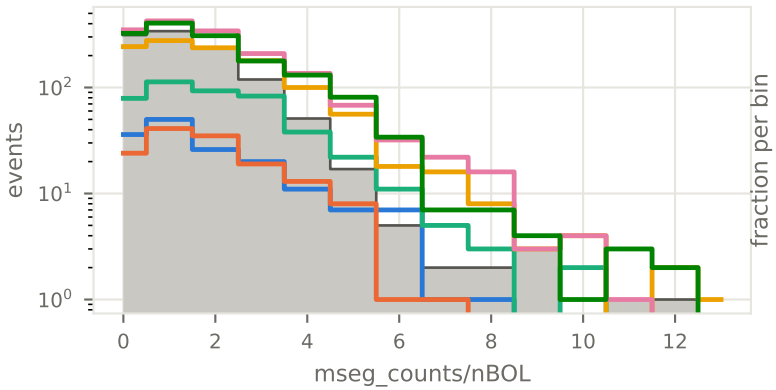
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

# mseg_counts/nBOL

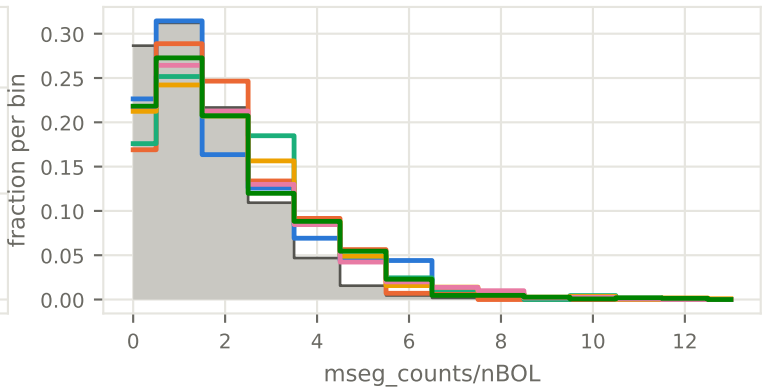
region: BARREL · rank 85 / 98

D_equal 0.132 (noise 0.051) D_pooled 0.132 min 0.111 (mH125_mS15) max 0.181 (mH200_mS50) bkg n = 1,089 sig n = 4,987

1a · each mass point vs background · raw

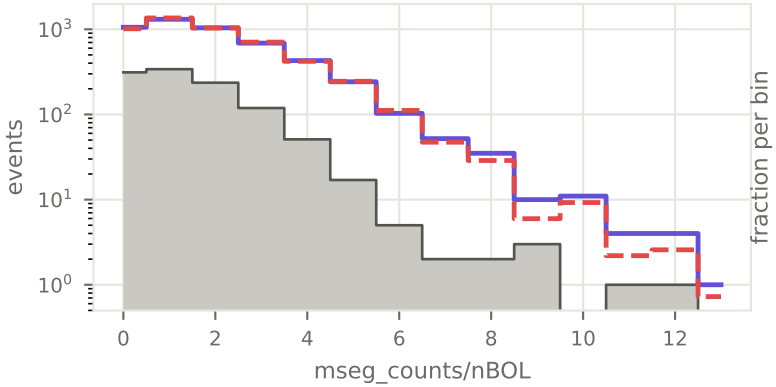



1b · each mass point vs background · normalized

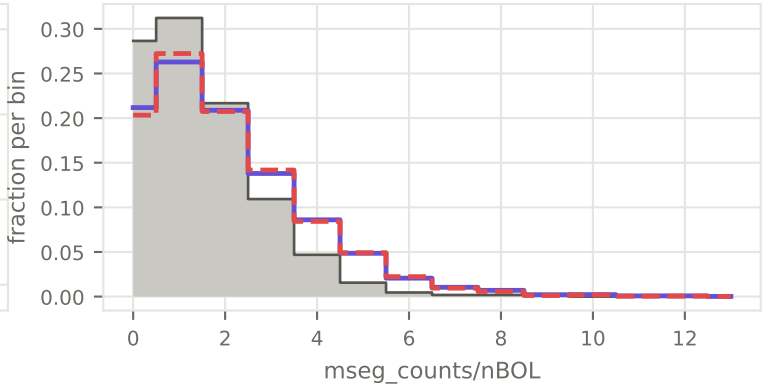



1 · each mass point
background n 1,089
mH125_mS15 D 0.111 n 159
mH125_mS55 D 0.141 n 142
mH200_mS50 D 0.181 n 449
mH400_mS100 D 0.154 n 1,144
mH600_mS150 D 0.121 n 1,608
mH1000_mS275 D 0.117 n 1,485

2a · combined signal vs background · raw

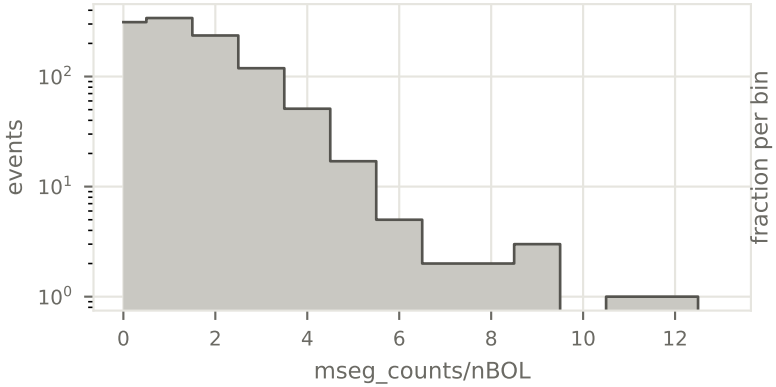



2b · combined signal vs background · normalized

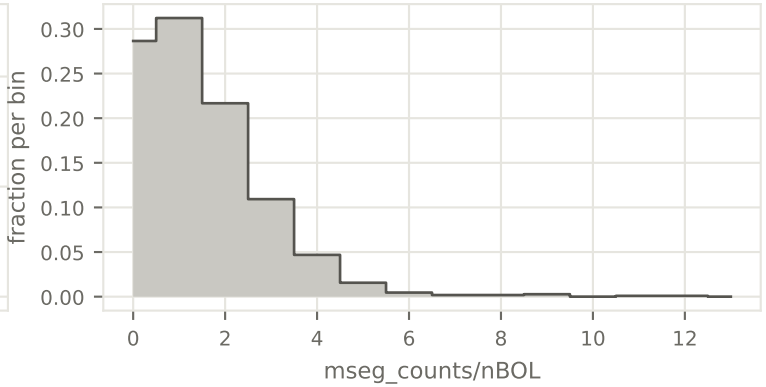



2 · combined signal
background n 1,089
combined, pooled D 0.132 n 4,987
equal weight per mass D 0.132

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0% −999 0.0%
background n 1,089

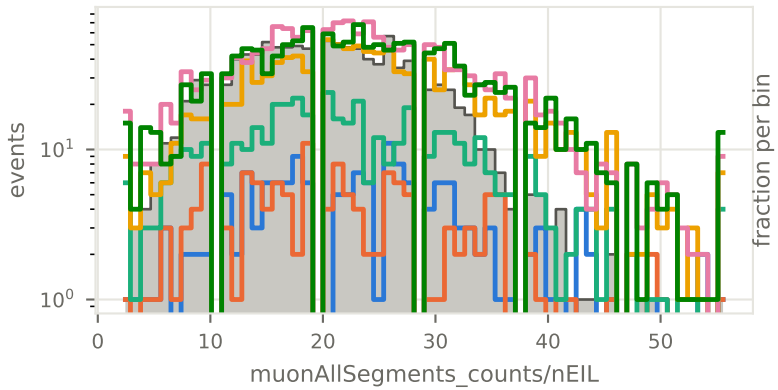
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

# muonAllSegments_counts/nEIL

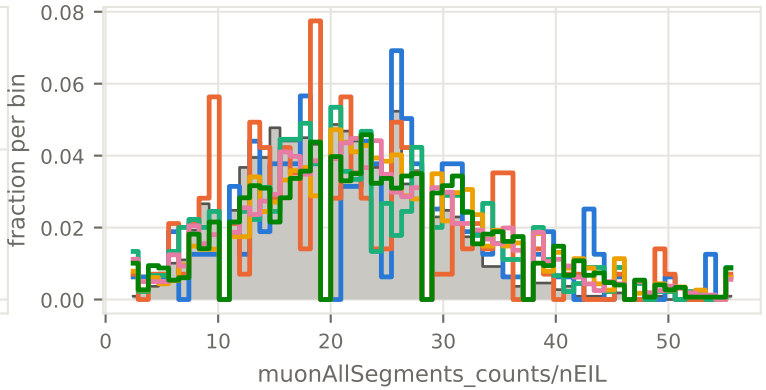
region: BARREL · rank 70 / 98

D_equal 0.132 (noise 0.051) D_pooled 0.143 min 0.109 (mH125_mS55) max 0.177 (mH125_mS15) bkg n = 1,089 sig n = 4,987

1a · each mass point vs background · raw

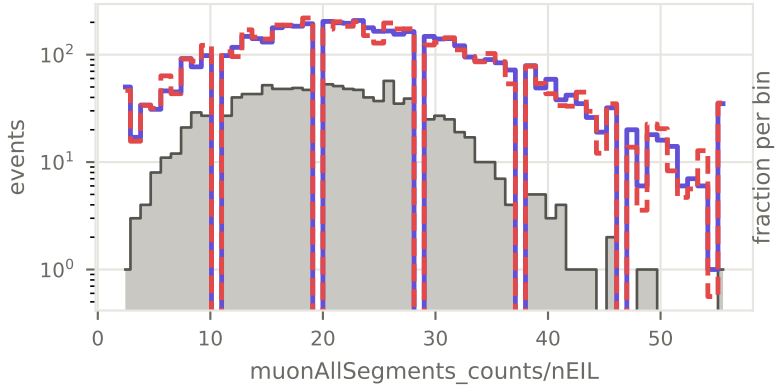



1b · each mass point vs background · normalized

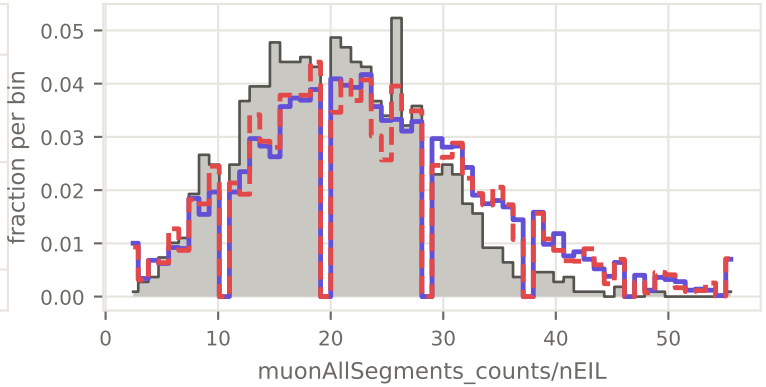



1 · each mass point
background n 1,089
mH125_mS15 D 0.177 n 159
mH125_mS55 D 0.109 n 142
mH200_mS50 D 0.132 n 449
mH400_mS100 D 0.161 n 1,144
mH600_mS150 D 0.132 n 1,608
mH1000_mS275 D 0.155 n 1,485

2a · combined signal vs background · raw

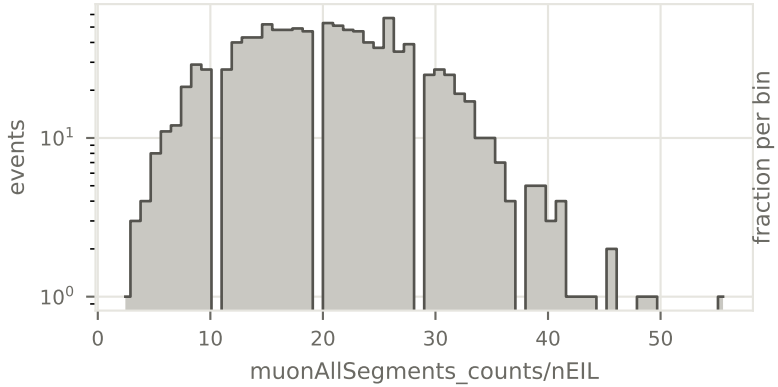



2b · combined signal vs background · normalized

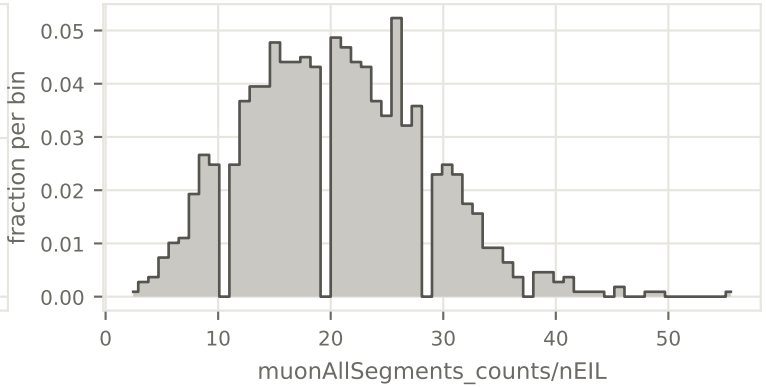



2 · combined signal
background n 1,089
combined, pooled D 0.143 n 4,987
equal weight per mass D 0.132

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0% −999 0.0%
background n 1,089

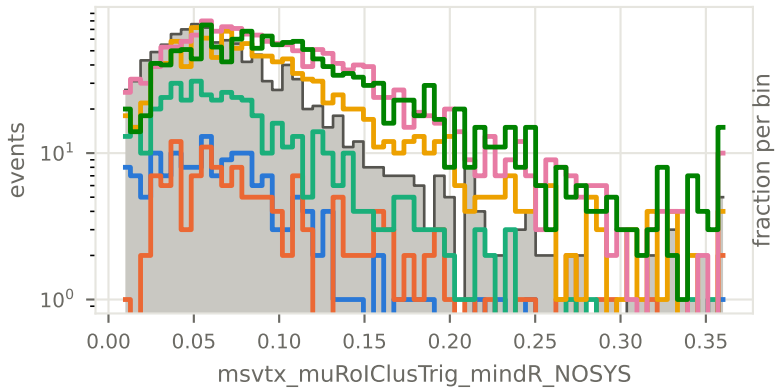
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

# msvtx_muRoIClusTrig_mindR_NOSYS

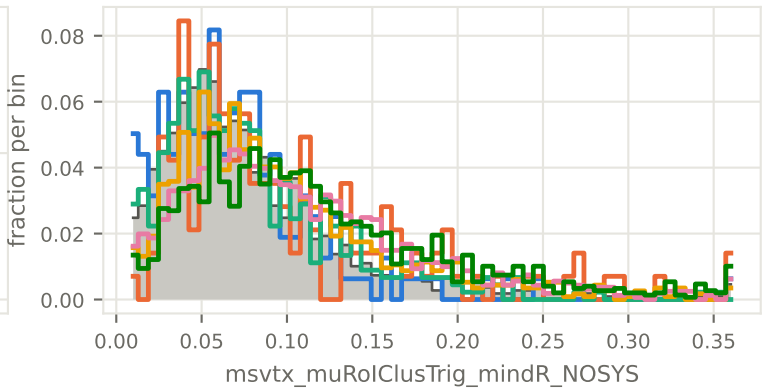
region: BARREL · rank 54 / 98

D_equal 0.119 (noise 0.051) D_pooled 0.182 min 0.035 (mH200_mS50) max 0.266 (mH1000_mS275) bkg n = 1,089 sig n = 4,987

1a · each mass point vs background · raw

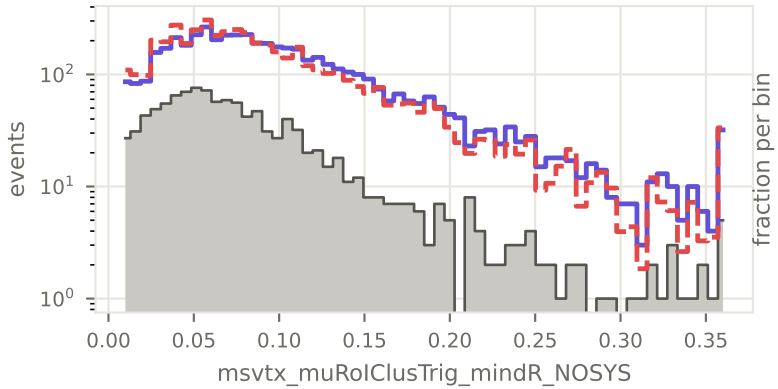



1b · each mass point vs background · normalized

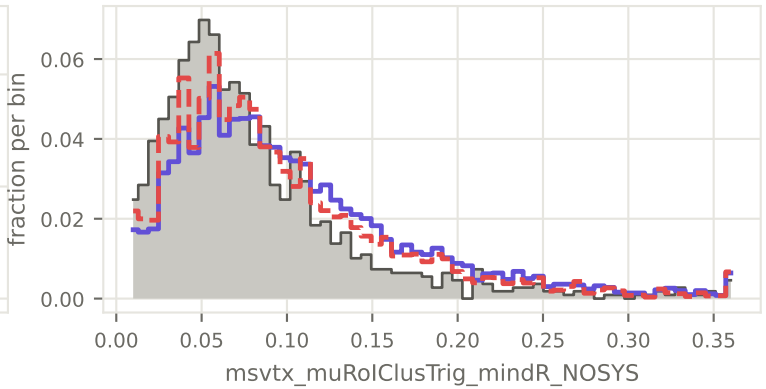



1 · each mass point
background n 1,089
mH125_mS15 D 0.078 n 159
mH125_mS55 D 0.140 n 142
mH200_mS50 D 0.035 n 449
mH400_mS100 D 0.142 n 1,144
mH600_mS150 D 0.211 n 1,608
mH1000_mS275 D 0.266 n 1,485

2a · combined signal vs background · raw

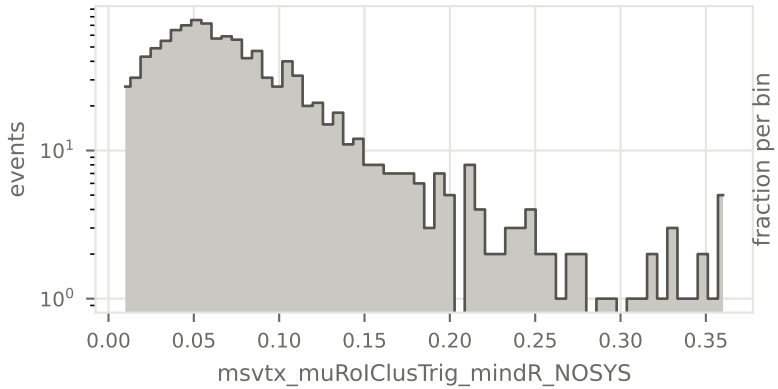



2b · combined signal vs background · normalized

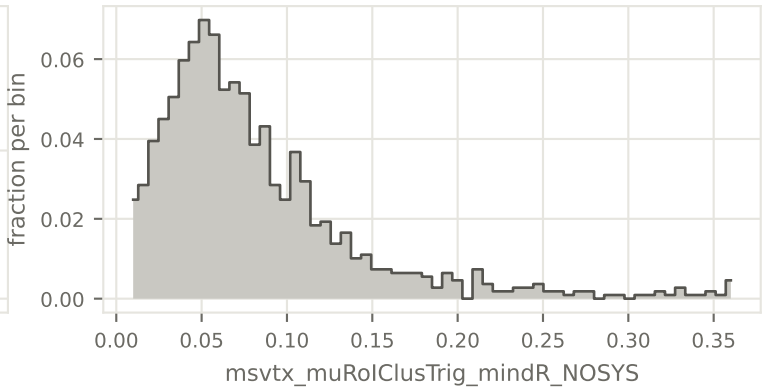



2 · combined signal
background n 1,089
combined, pooled D 0.182 n 4,987
equal weight per mass D 0.119

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0% −999 0.0%
background n 1,089

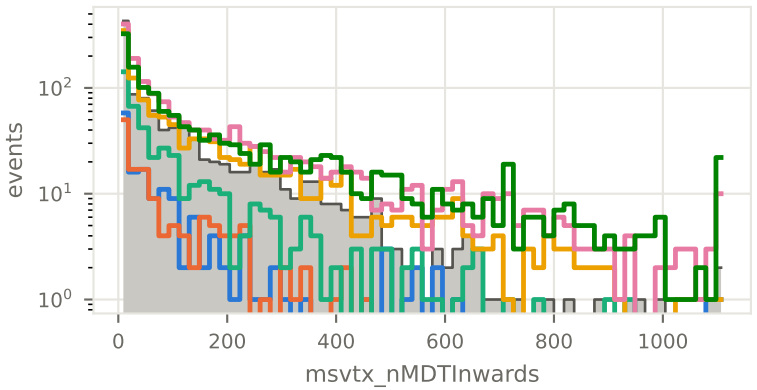
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

# msvtx_nMDTInwards

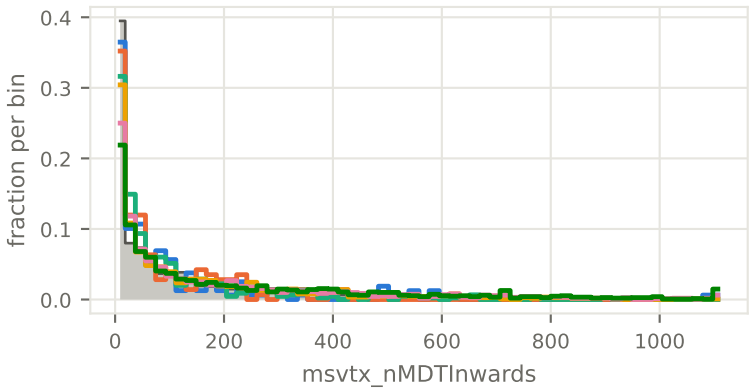
region: BARREL · rank 50 / 98

D_equal 0.114 (noise 0.051) D_pooled 0.137 min 0.092 (mH125_mS15) max 0.177 (mH1000_mS275) bkg n = 1,089 sig n = 4,987

1a · each mass point vs background · raw

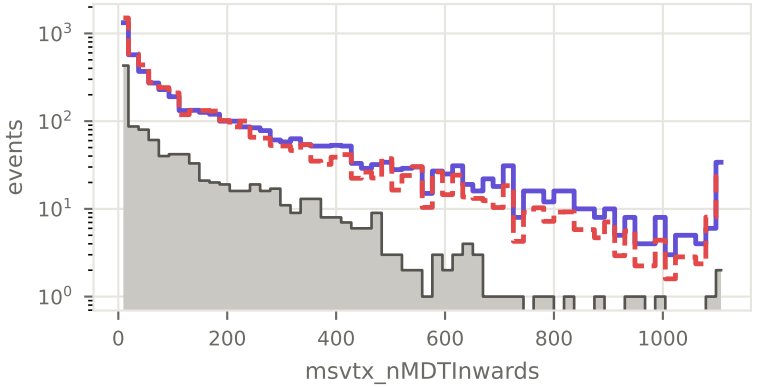



1b · each mass point vs background · normalized

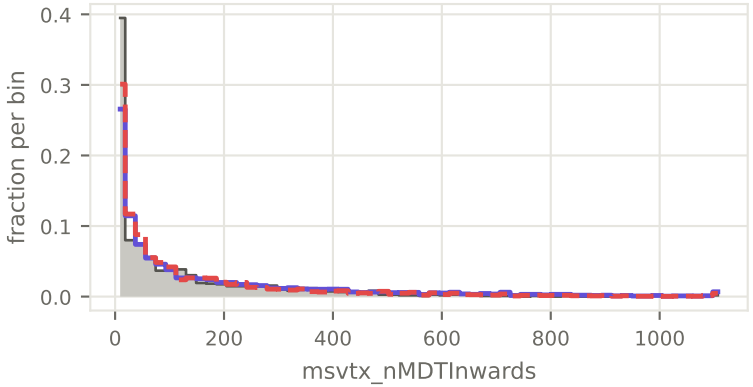



1 · each mass point
background n 1,089
mH125_mS15 D 0.092 n 159
mH125_mS55 D 0.097 n 142
mH200_mS50 D 0.106 n 449
mH400_mS100 D 0.095 n 1,144
mH600_mS150 D 0.159 n 1,608
mH1000_mS275 D 0.177 n 1,485

2a · combined signal vs background · raw

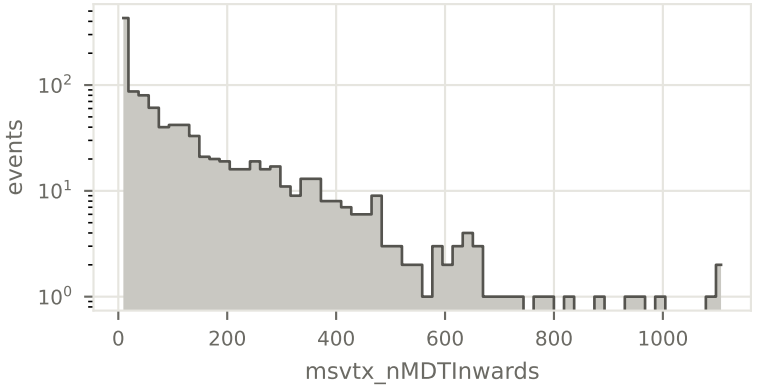



2b · combined signal vs background · normalized

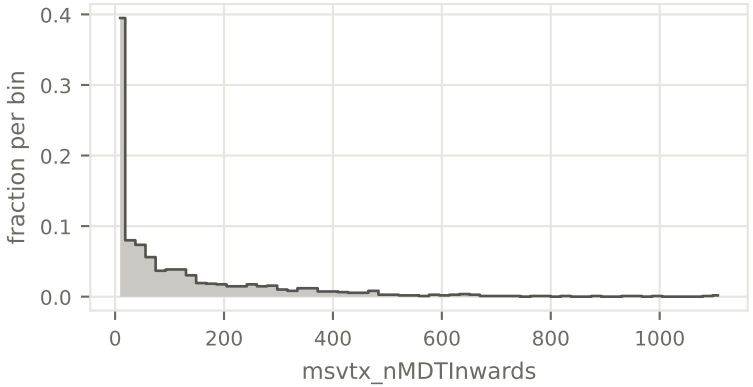



2 · combined signal
background n 1,089
combined, pooled D 0.137 n 4,987
equal weight per mass D 0.114

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0% −999 0.0%
background n 1,089

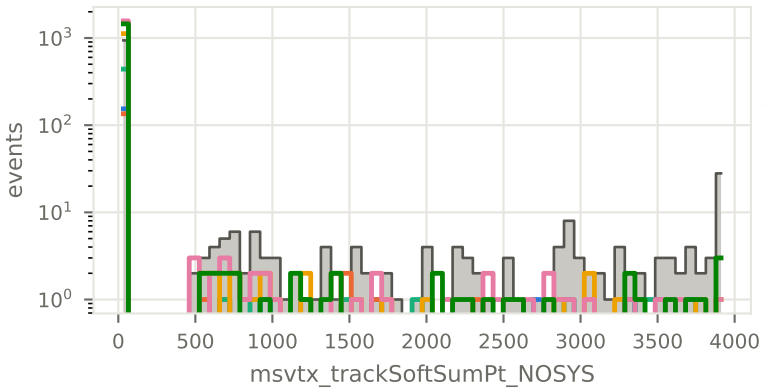
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

# msvtx_trackSoftSumPt_NOSYS

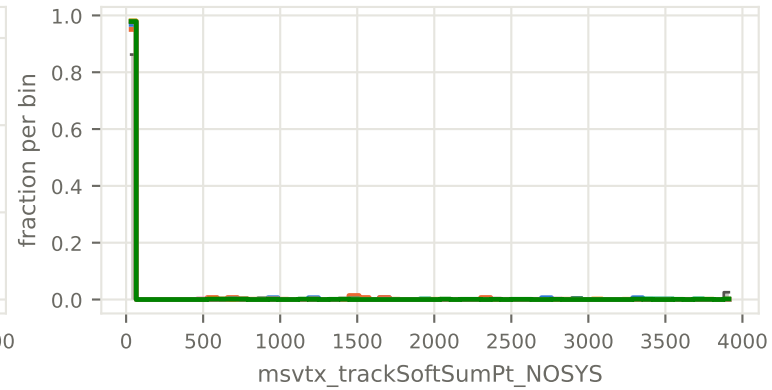
region: BARREL · rank 65 / 98 · in: NN1 barrel, NN1 endcap

D_equal 0.111 (noise 0.051)  D_pooled 0.116  min 0.092 (mH125_mS55)  max 0.119 (mH400_mS100)  bkg n = 1,089  sig n = 4,987

1a · each mass point vs background · raw

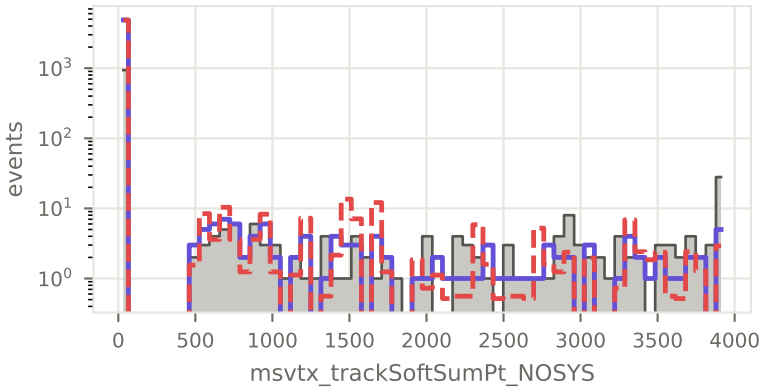



1b · each mass point vs background · normalized

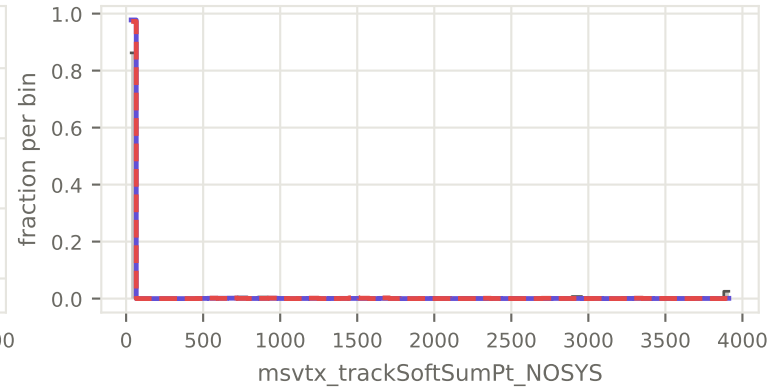



1 · each mass point
- background n 1,089
- mH125_mS15 D 0.106 n 159
- mH125_mS55 D 0.092 n 142
- mH200_mS50 D 0.118 n 449
- mH400_mS100 D 0.119 n 1,144
- mH600_mS150 D 0.116 n 1,608
- mH1000_mS275 D 0.116 n 1,485

2a · combined signal vs background · raw

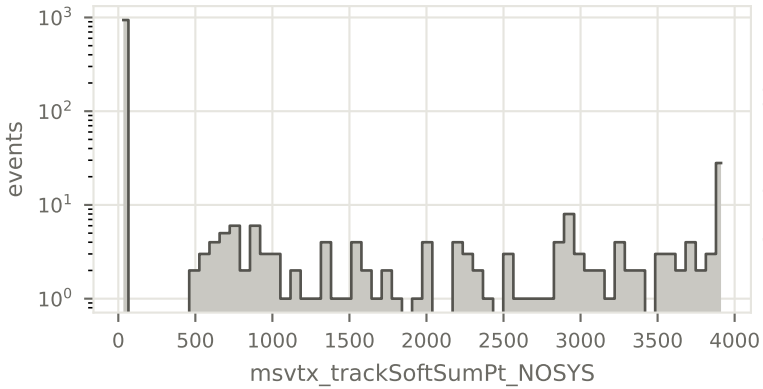



2b · combined signal vs background · normalized

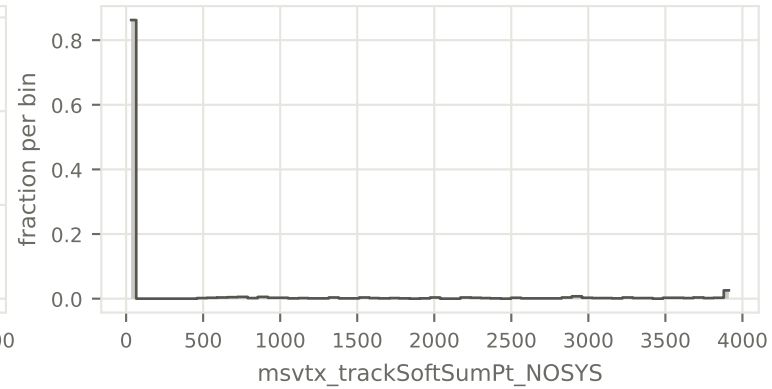



2 · combined signal
- background n 1,089
- combined, pooled D 0.116 n 4,987
- equal weight per mass D 0.111

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0%  −999 0.0%
- background n 1,089

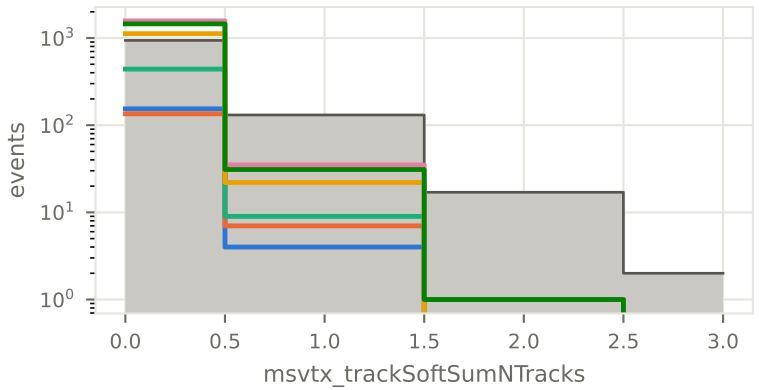
a = raw (log y)  b = normalized · edge bins include under/overflow · NaN and -999 removed

# msvtx_trackSoftSumNTracks

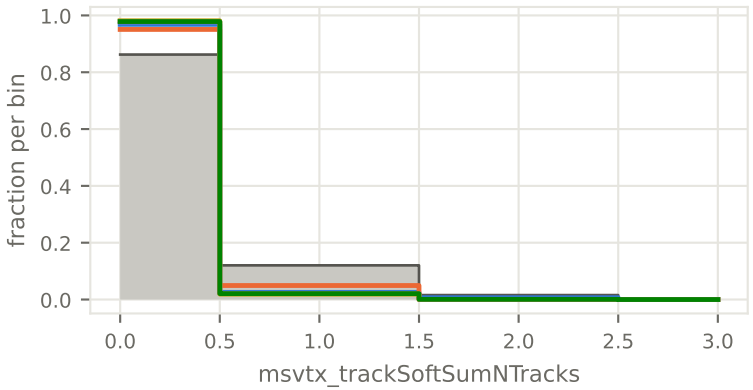
region: BARREL · rank 64 / 98

D_equal 0.110 (noise 0.051) D_pooled 0.115 min 0.088 (mH125_mS55) max 0.119 (mH400_mS100) bkg n = 1,089 sig n = 4,987

1a · each mass point vs background · raw

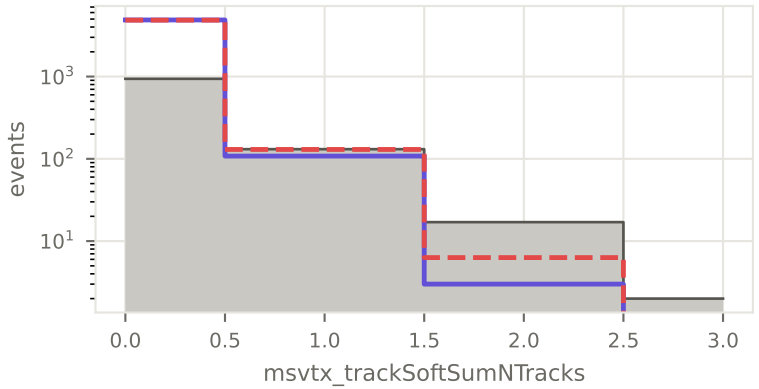



1b · each mass point vs background · normalized

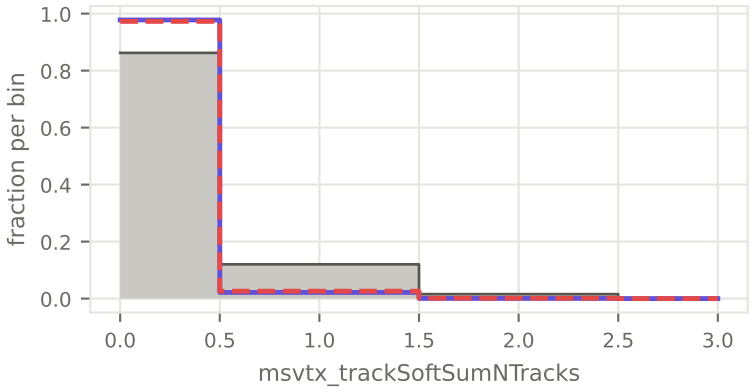



1 · each mass point
background n 1,089
mH125_mS15 D 0.106 n 159
mH125_mS55 D 0.088 n 142
mH200_mS50 D 0.118 n 449
mH400_mS100 D 0.119 n 1,144
mH600_mS150 D 0.115 n 1,608
mH1000_mS275 D 0.116 n 1,485

2a · combined signal vs background · raw

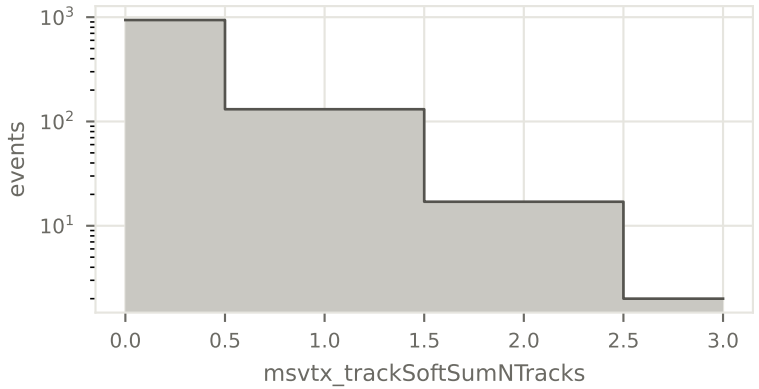



2b · combined signal vs background · normalized

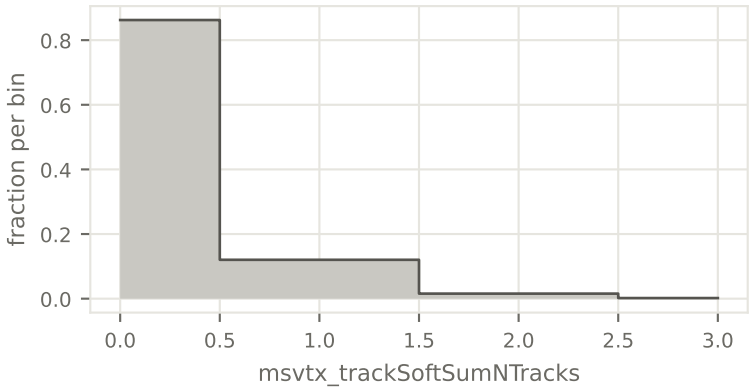



2 · combined signal
background n 1,089
combined, pooled D 0.115 n 4,987
equal weight per mass D 0.110

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0% −999 0.0%
background n 1,089

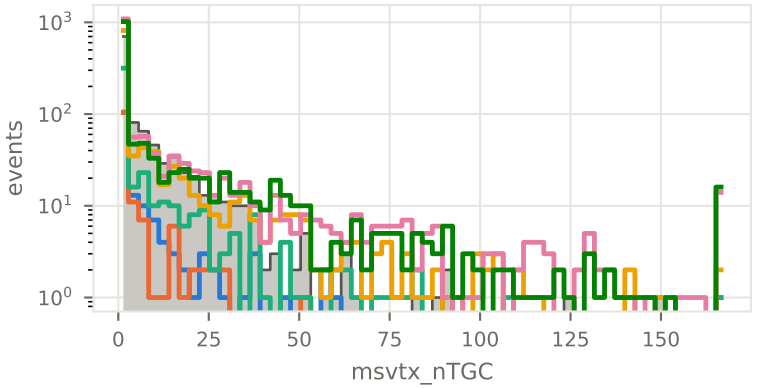
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

# msvtx_nTGC

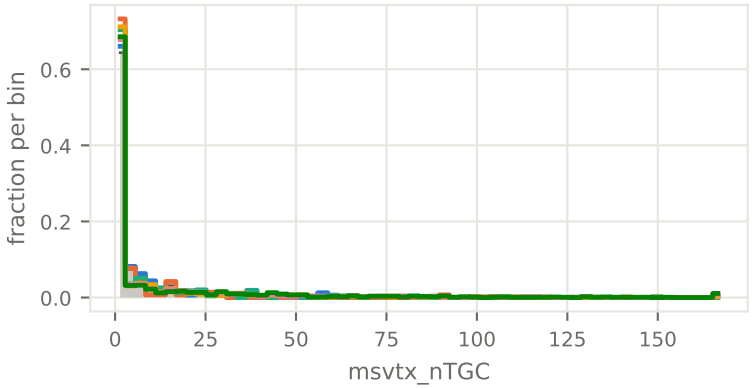
region: BARREL · rank 6 / 98

D_equal 0.108 (noise 0.051)  D_pooled 0.100  min 0.084 (mH600_mS150)  max 0.139 (mH125_mS55)  bkg n = 1,089  sig n = 4,987

1a · each mass point vs background · raw

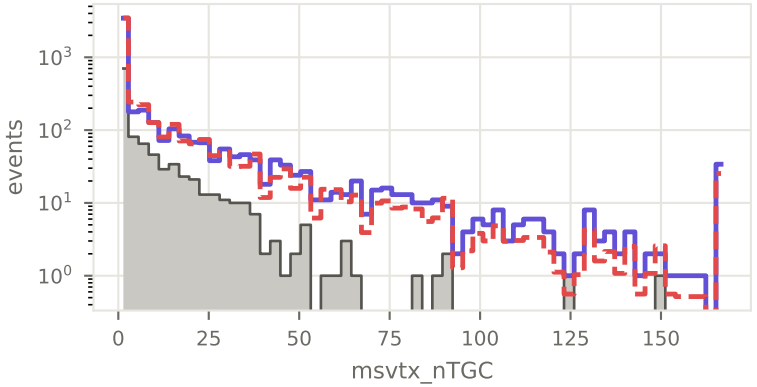



1b · each mass point vs background · normalized

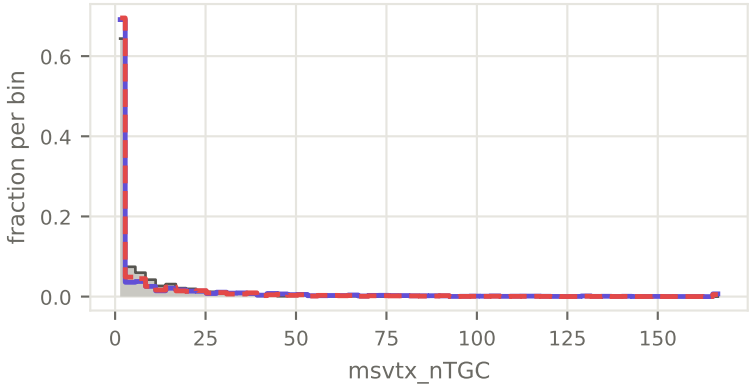



1 · each mass point
background n 1,089
mH125_mS15 D 0.089 n 159
mH125_mS55 D 0.139 n 142
mH200_mS50 D 0.121 n 449
mH400_mS100 D 0.118 n 1,144
mH600_mS150 D 0.084 n 1,608
mH1000_mS275 D 0.094 n 1,485

2a · combined signal vs background · raw

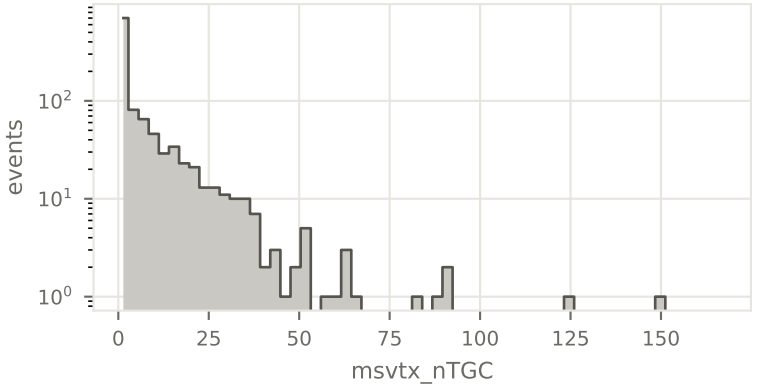



2b · combined signal vs background · normalized

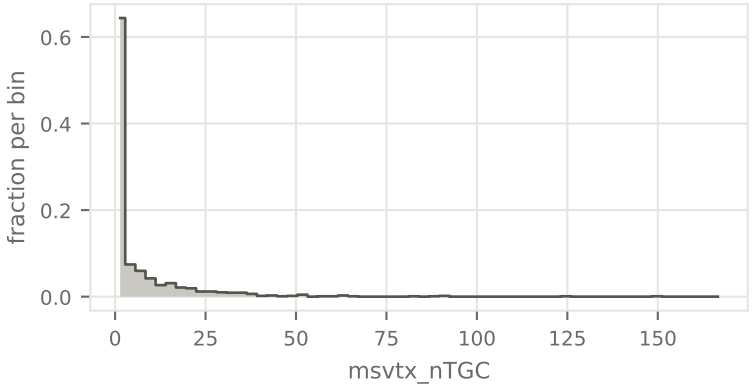



2 · combined signal
background n 1,089
combined, pooled D 0.100 n 4,987
equal weight per mass D 0.108

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0% −999 0.0%
background n 1,089

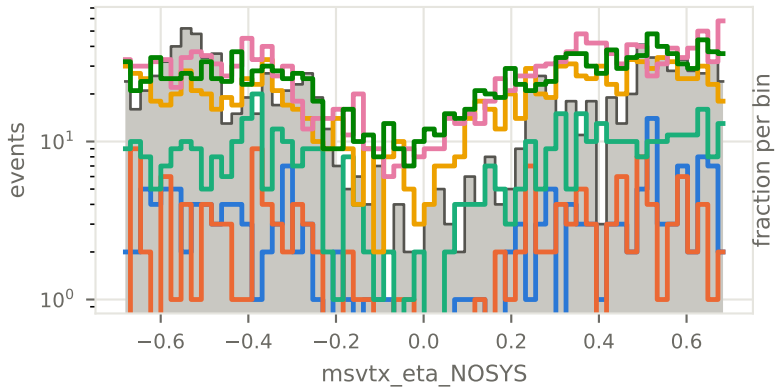
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

# msvtx_eta_NOSYS

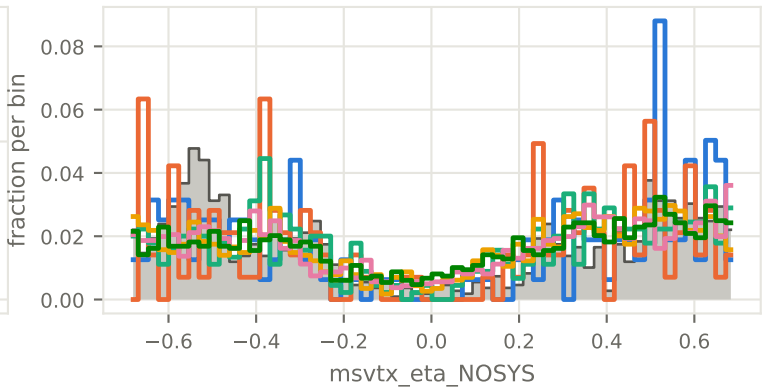
region: BARREL · rank 67 / 98

D_equal 0.104 (noise 0.051) D_pooled 0.116 min 0.095 (mH125_mS15) max 0.130 (mH1000_mS275) bkg n = 1,089 sig n = 4,987

1a · each mass point vs background · raw

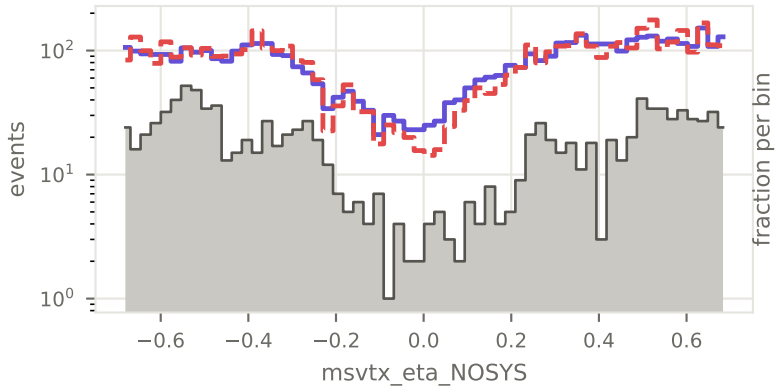



1b · each mass point vs background · normalized

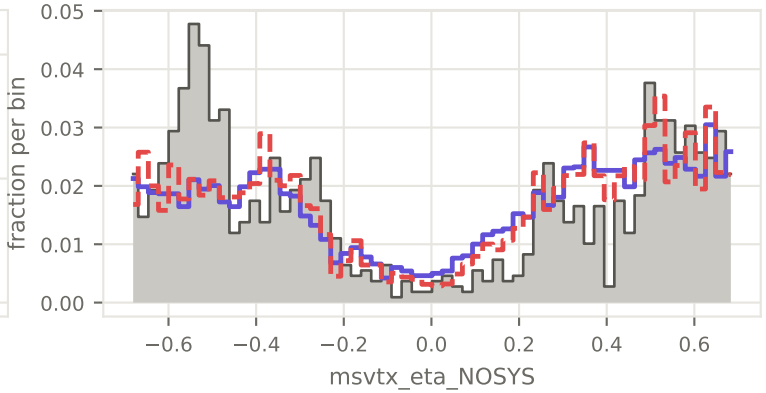



1 · each mass point
background n 1,089
mH125_mS15 D 0.095 n 159
mH125_mS55 D 0.105 n 142
mH200_mS50 D 0.128 n 449
mH400_mS100 D 0.125 n 1,144
mH600_mS150 D 0.118 n 1,608
mH1000_mS275 D 0.130 n 1,485

2a · combined signal vs background · raw

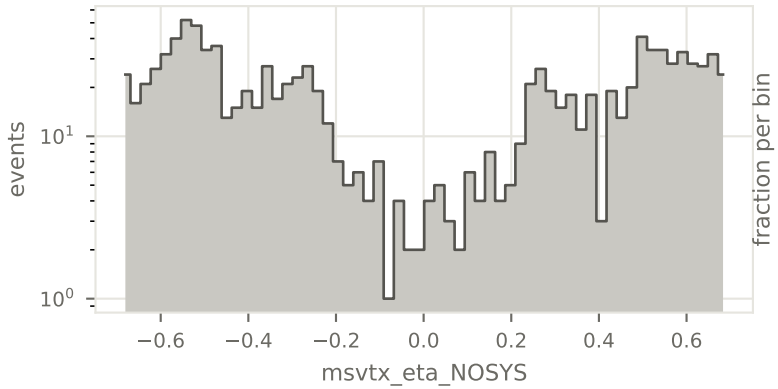



2b · combined signal vs background · normalized

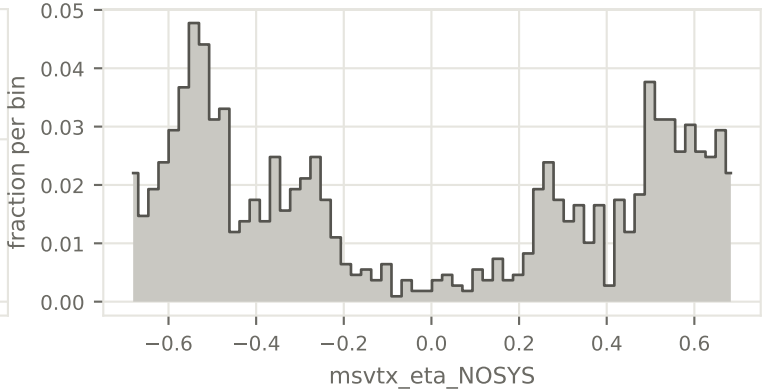



2 · combined signal
background n 1,089
combined, pooled D 0.116 n 4,987
equal weight per mass D 0.104

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0% −999 0.0%
background n 1,089

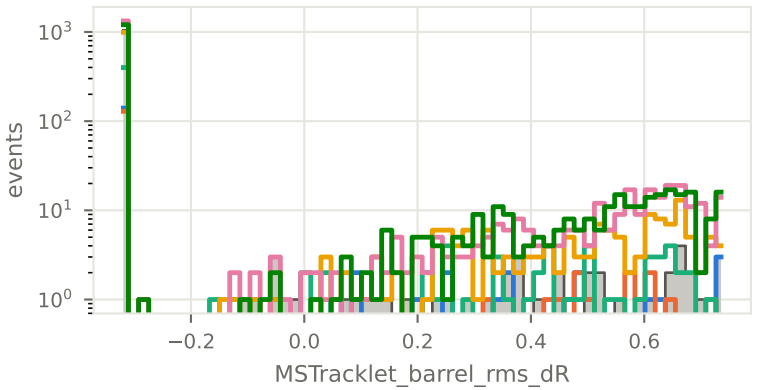
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

# MSTracklet_barrel_rms_dR

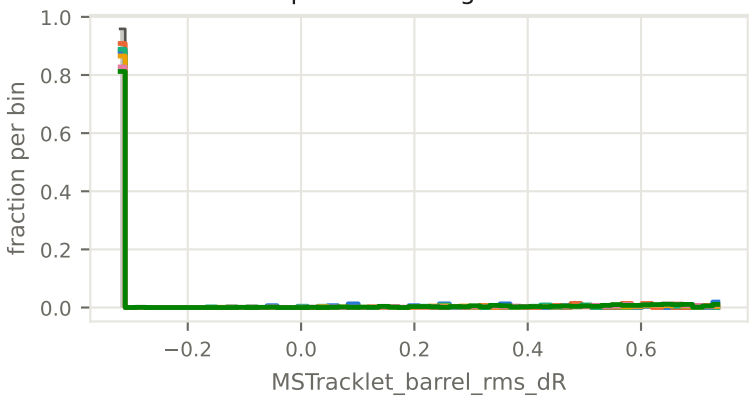
region: BARREL · rank 96 / 98 · in: NN2 barrel

D_equal 0.097 (noise 0.051) D_pooled 0.119 min 0.066 (mH125_mS55) max 0.150 (mH1000_mS275) bkg n = 1,089 sig n = 4,987

1a · each mass point vs background · raw

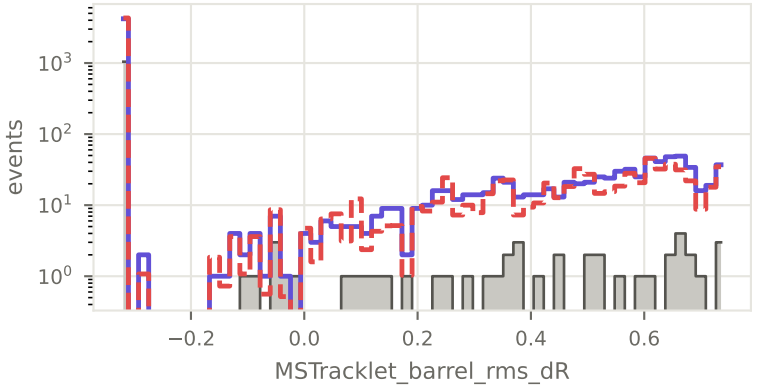



1b · each mass point vs background · normalized

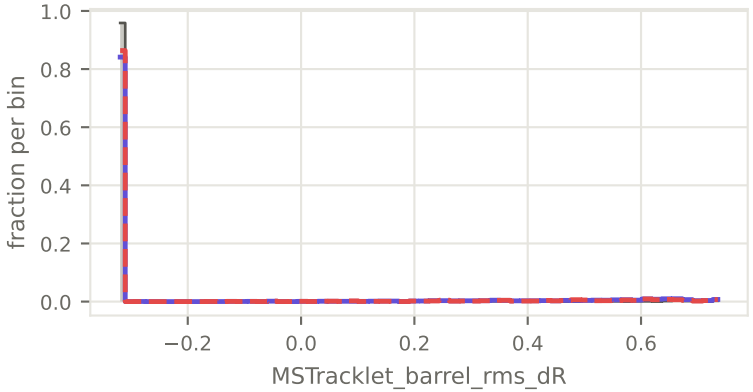



1 · each mass point
background n 1,089
mH125_mS15 D 0.083 n 159
mH125_mS55 D 0.066 n 142
mH200_mS50 D 0.070 n 449
mH400_mS100 D 0.096 n 1,144
mH600_mS150 D 0.132 n 1,608
mH1000_mS275 D 0.150 n 1,485

2a · combined signal vs background · raw

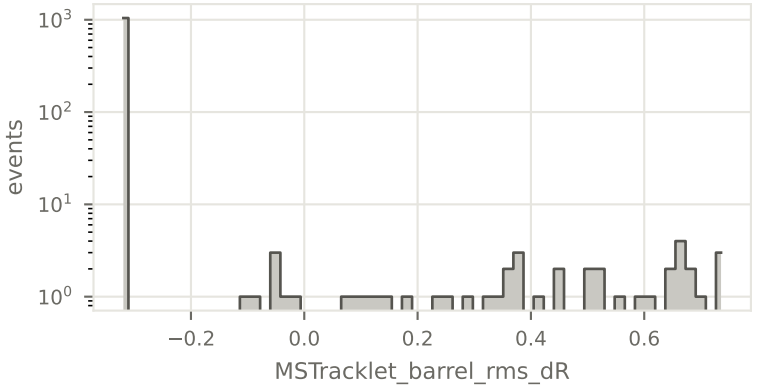



2b · combined signal vs background · normalized

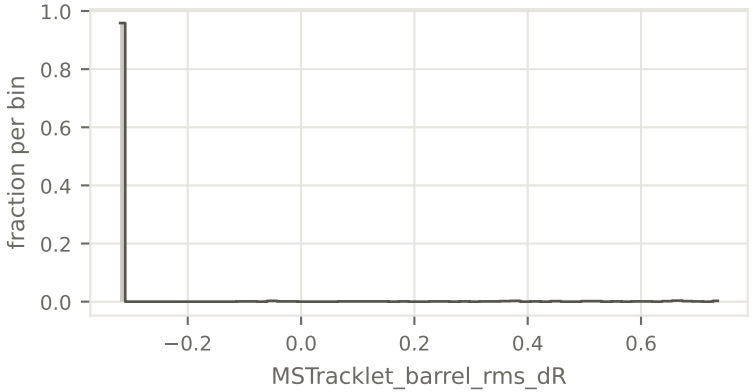



2 · combined signal
background n 1,089
combined, pooled D 0.119 n 4,987
equal weight per mass D 0.097

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0% −999 0.0%
background n 1,089

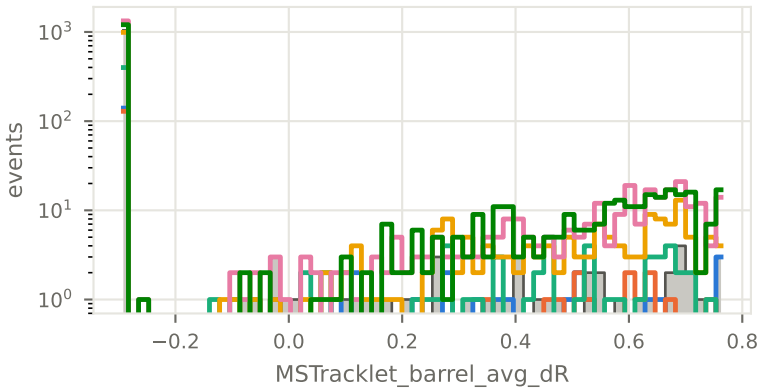
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

# MSTracklet_barrel_avg_dR

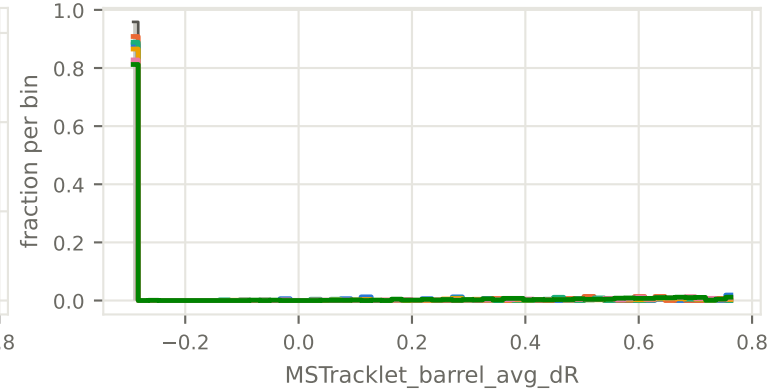
region: BARREL · rank 95 / 98 · in: NN2 barrel

D_equal 0.097 (noise 0.051) D_pooled 0.119 min 0.067 (mH125_mS55) max 0.150 (mH1000_mS275) bkg n = 1,089 sig n = 4,987

1a · each mass point vs background · raw

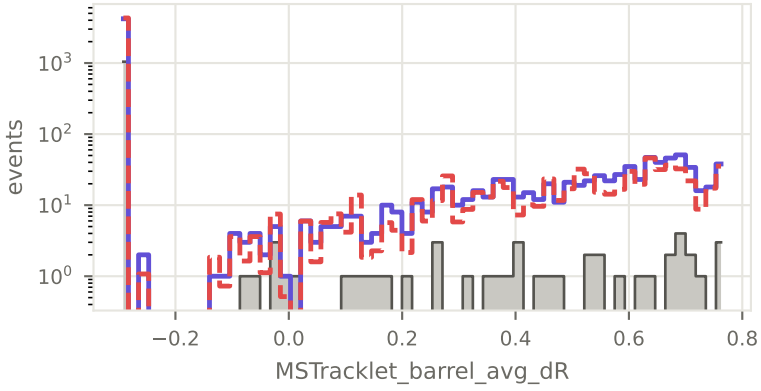



1b · each mass point vs background · normalized

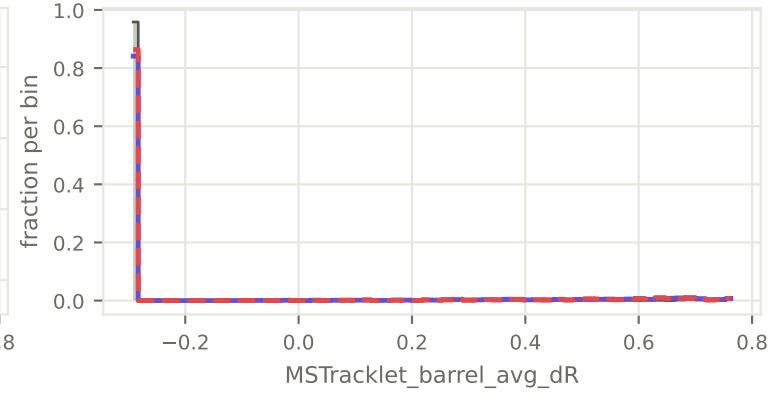



1 · each mass point
background n 1,089
mH125_mS15 D 0.083 n 159
mH125_mS55 D 0.067 n 142
mH200_mS50 D 0.070 n 449
mH400_mS100 D 0.096 n 1,144
mH600_mS150 D 0.132 n 1,608
mH1000_mS275 D 0.150 n 1,485

2a · combined signal vs background · raw

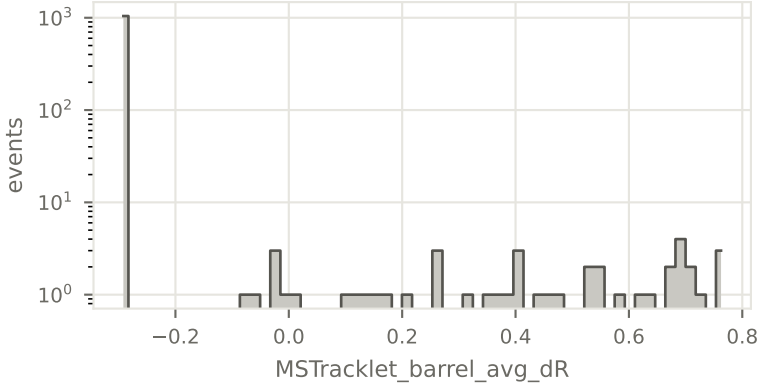



2b · combined signal vs background · normalized

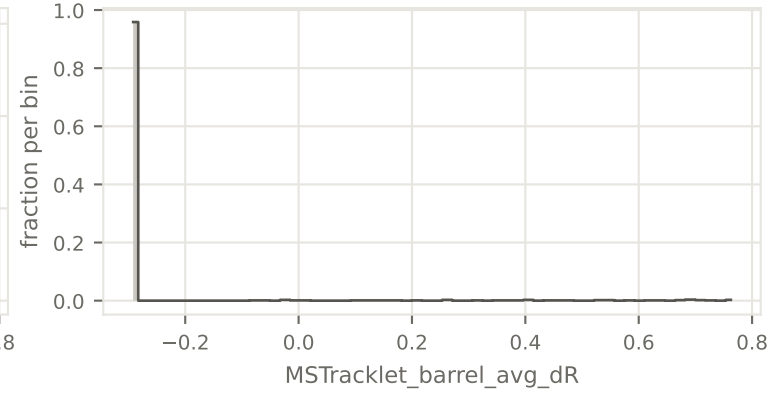



2 · combined signal
background n 1,089
combined, pooled D 0.119 n 4,987
equal weight per mass D 0.097

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0% −999 0.0%
background n 1,089

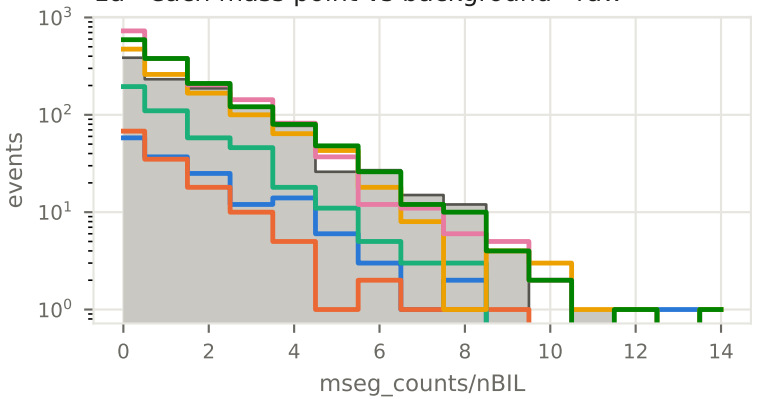
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

# mseg_counts/nBIL

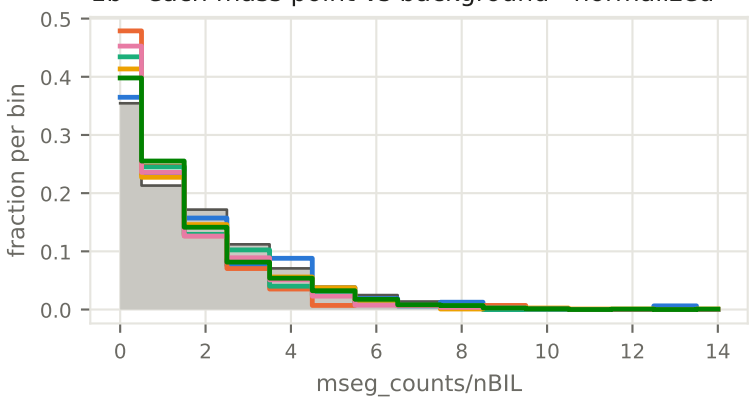
region: BARREL · rank 88 / 98

D_equal 0.097 (noise 0.051) D_pooled 0.097 min 0.030 (mH125_mS15) max 0.158 (mH125_mS55) bkg n = 1,089 sig n = 4,987

1a · each mass point vs background · raw

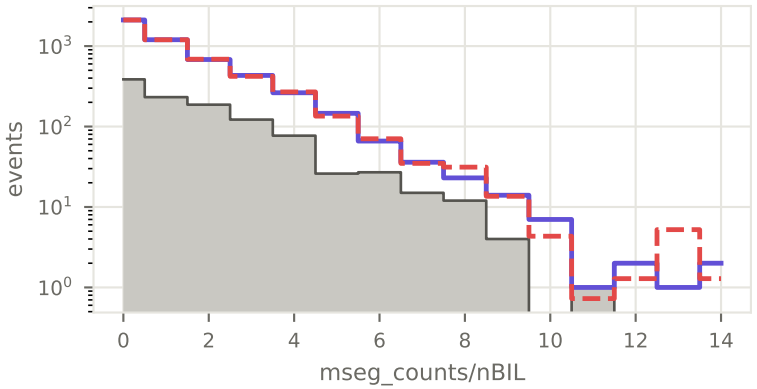



1b · each mass point vs background · normalized

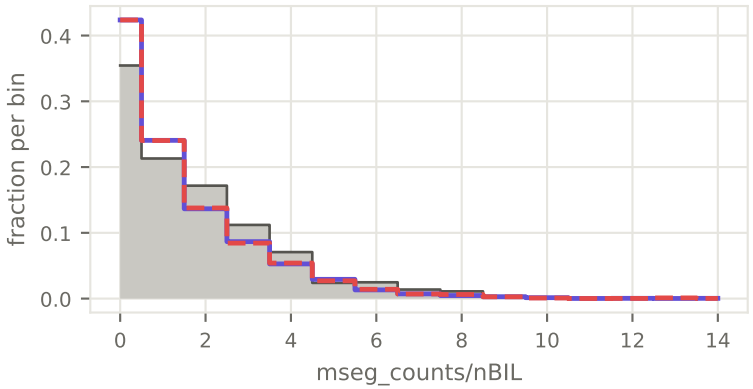



1 · each mass point
background n 1,089
mH125_mS15 D 0.030 n 159
mH125_mS55 D 0.158 n 142
mH200_mS50 D 0.112 n 449
mH400_mS100 D 0.073 n 1,144
mH600_mS150 D 0.121 n 1,608
mH1000_mS275 D 0.086 n 1,485

2a · combined signal vs background · raw

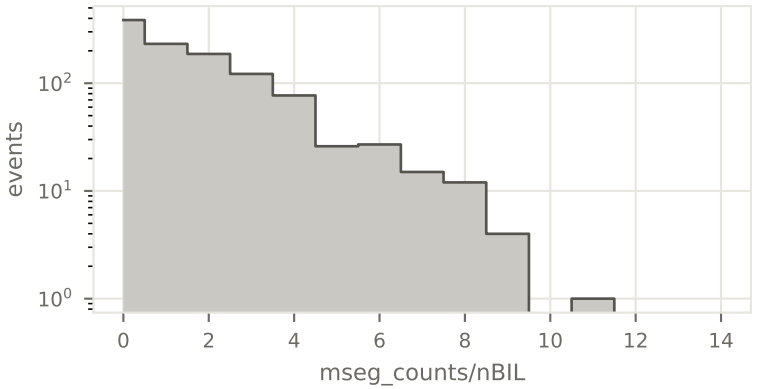



2b · combined signal vs background · normalized

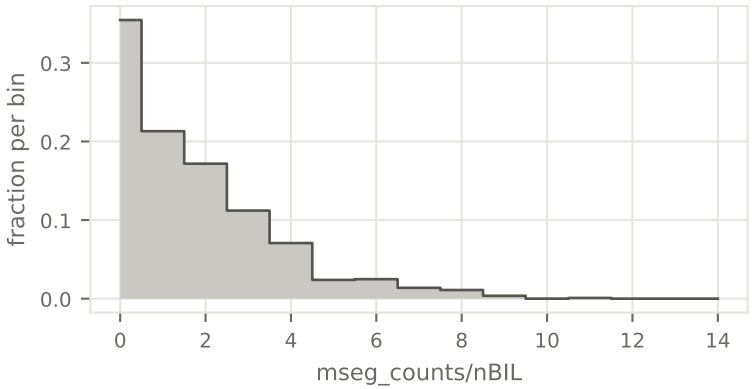



2 · combined signal
background n 1,089
combined, pooled D 0.097 n 4,987
equal weight per mass D 0.097

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0% −999 0.0%
background n 1,089

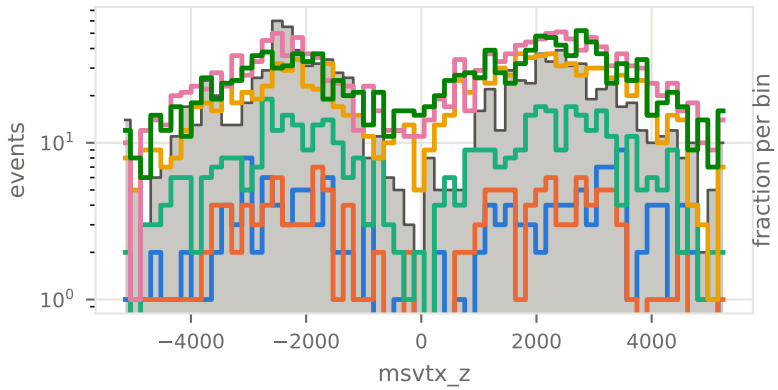
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

# msvtx_z

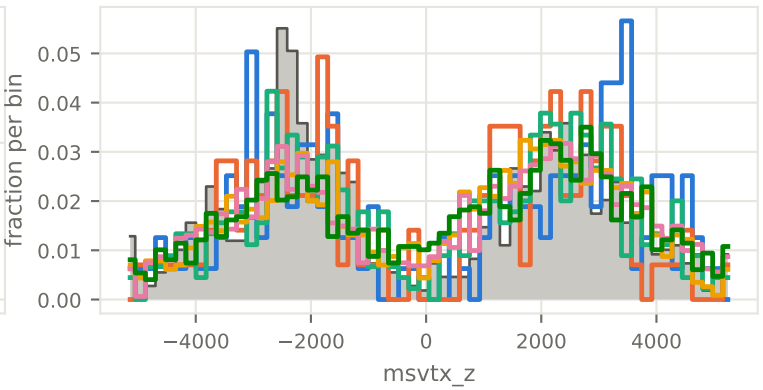
region: BARREL · rank 84 / 98

D_equal 0.096 (noise 0.051)  D_pooled 0.114  min 0.069 (mH125_mS55)  max 0.134 (mH125_mS15)  bkg n = 1,089  sig n = 4,987

1a · each mass point vs background · raw

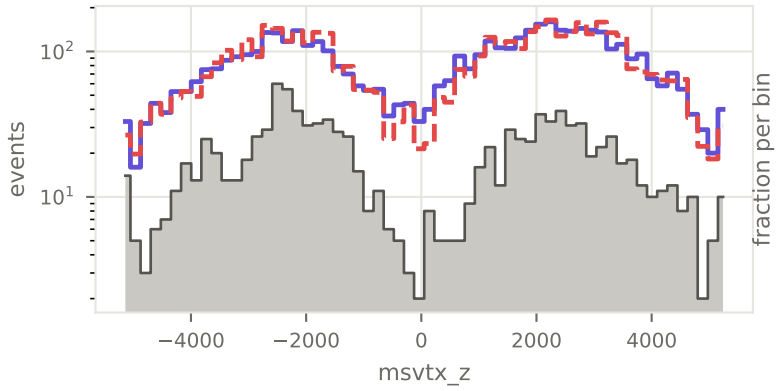



1b · each mass point vs background · normalized

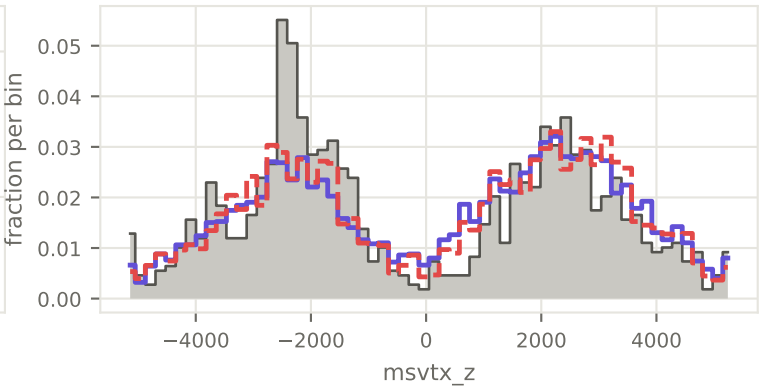



1 · each mass point
background n 1,089
mH125_mS15 D 0.134 n 159
mH125_mS55 D 0.069 n 142
mH200_mS50 D 0.088 n 449
mH400_mS100 D 0.121 n 1,144
mH600_mS150 D 0.115 n 1,608
mH1000_mS275 D 0.127 n 1,485

2a · combined signal vs background · raw

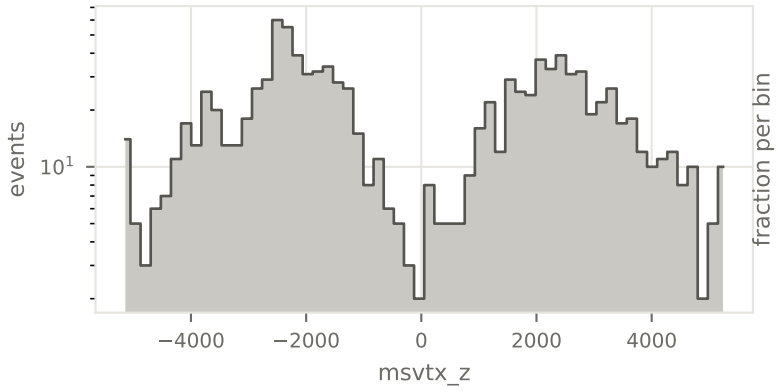



2b · combined signal vs background · normalized

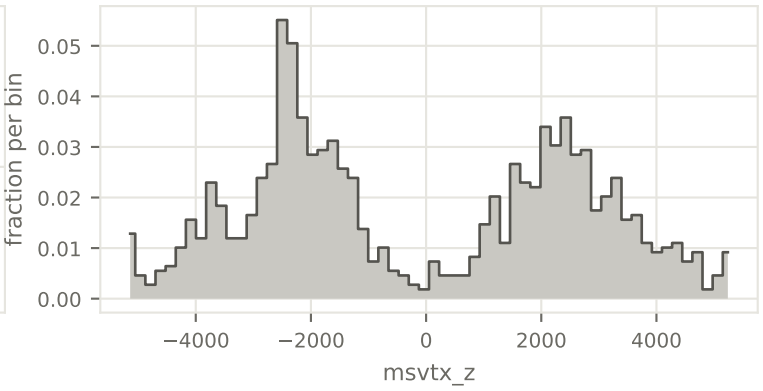



2 · combined signal
background n 1,089
combined, pooled D 0.114 n 4,987
equal weight per mass D 0.096

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0% −999 0.0%
background n 1,089

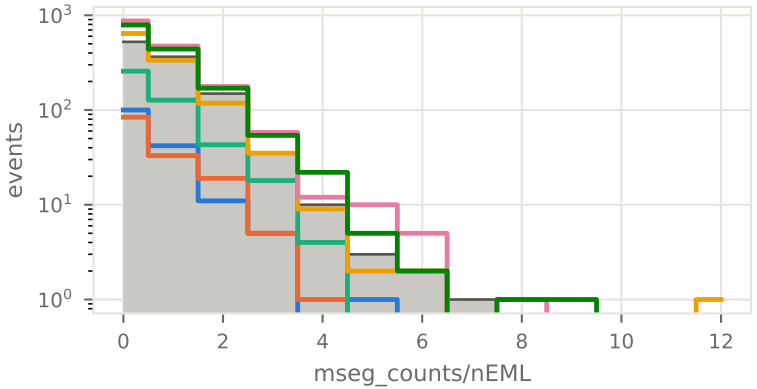
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

# mseg_counts/nEML

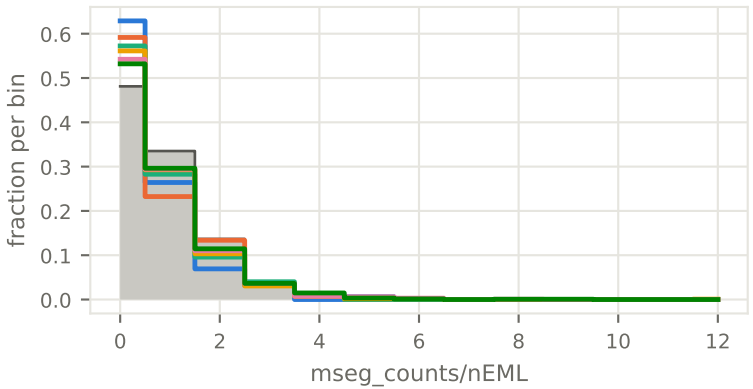
region: BARREL · rank 91 / 98

D_equal 0.090 (noise 0.051) D_pooled 0.069 min 0.051 (mH1000_mS275) max 0.148 (mH125_mS15) bkg n = 1,089 sig n = 4,987

1a · each mass point vs background · raw

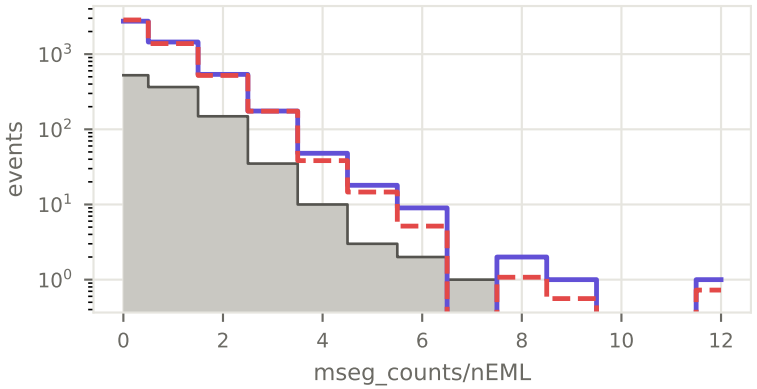



1b · each mass point vs background · normalized

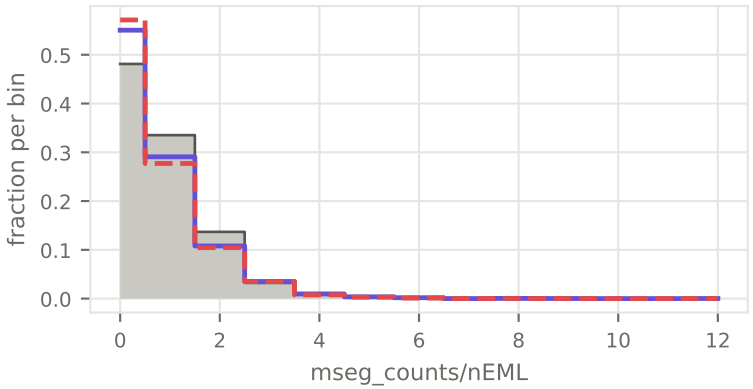



1 · each mass point
background n 1,089
mH125_mS15 D 0.148 n 159
mH125_mS55 D 0.110 n 142
mH200_mS50 D 0.091 n 449
mH400_mS100 D 0.080 n 1,144
mH600_mS150 D 0.061 n 1,608
mH1000_mS275 D 0.051 n 1,485

2a · combined signal vs background · raw

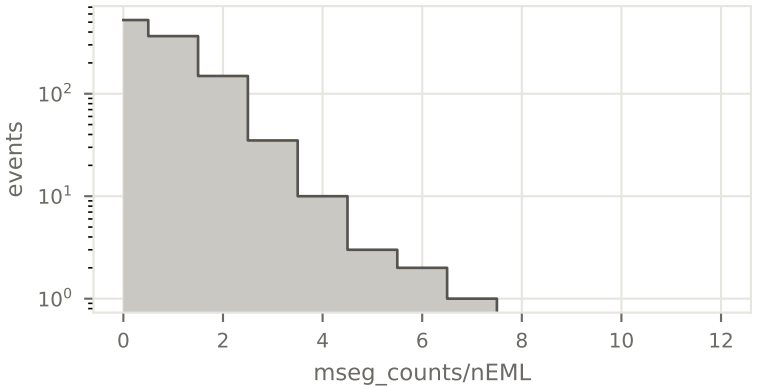



2b · combined signal vs background · normalized

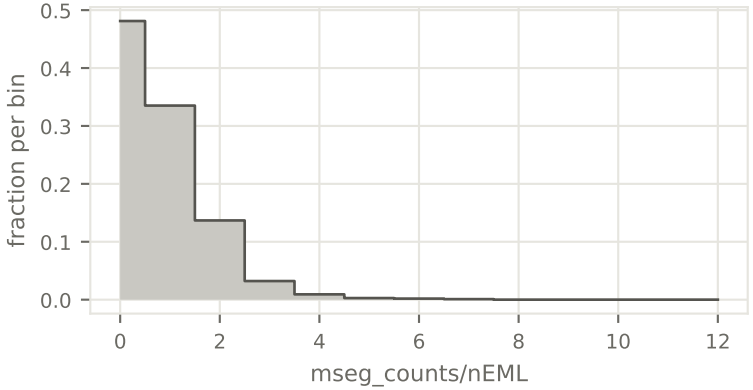



2 · combined signal
background n 1,089
combined, pooled D 0.069 n 4,987
equal weight per mass D 0.090

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0% −999 0.0%
background n 1,089

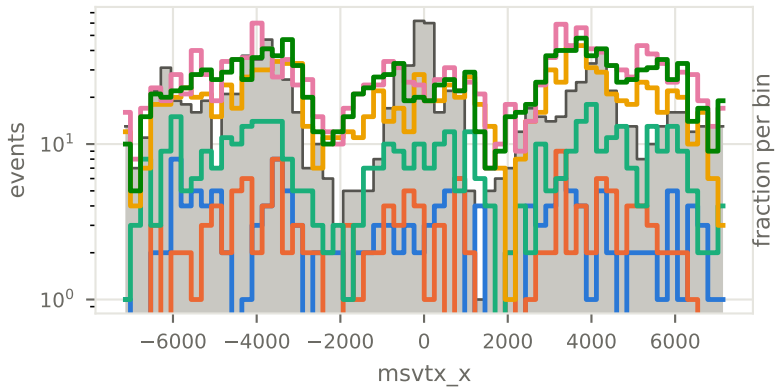
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

# msvtx_x

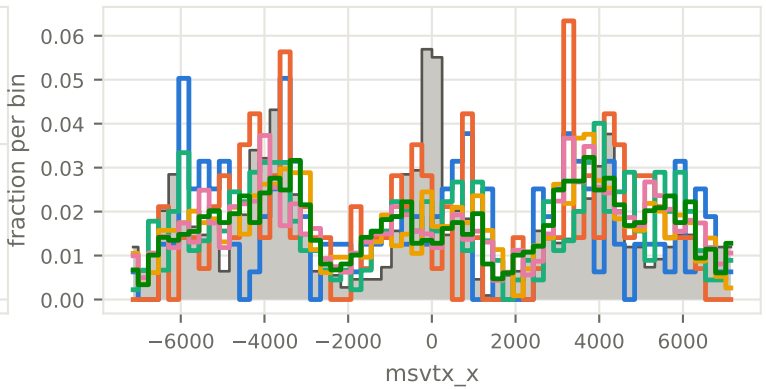
region: BARREL · rank 73 / 98

D_equal 0.087 (noise 0.051) D_pooled 0.098 min 0.073 (mH125_mS15) max 0.112 (mH600_mS150) bkg n = 1,089 sig n = 4,987

1a · each mass point vs background · raw

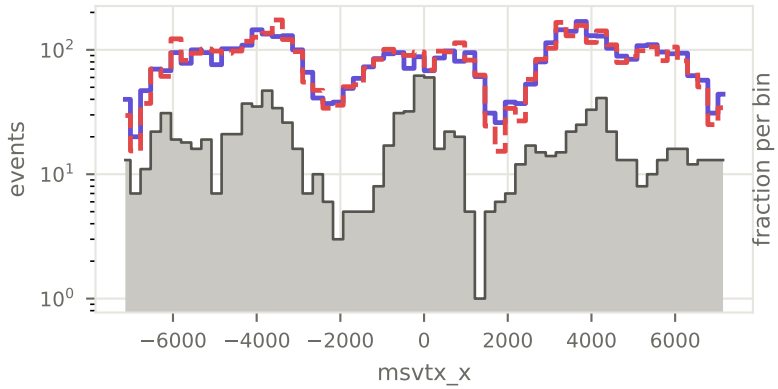



1b · each mass point vs background · normalized

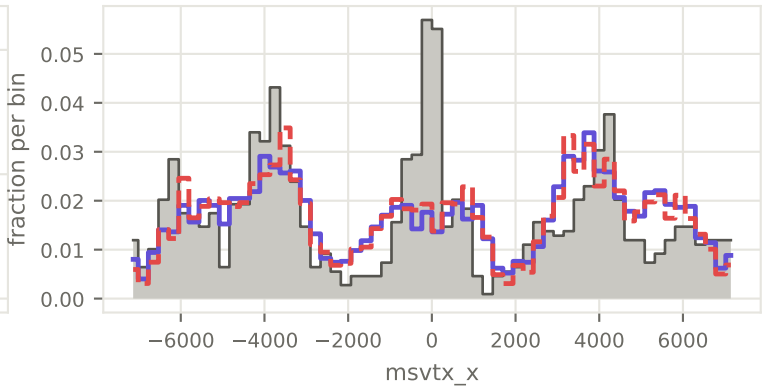



1 · each mass point
background n 1,089
mH125_mS15 D 0.073 n 159
mH125_mS55 D 0.077 n 142
mH200_mS50 D 0.095 n 449
mH400_mS100 D 0.096 n 1,144
mH600_mS150 D 0.112 n 1,608
mH1000_mS275 D 0.105 n 1,485

2a · combined signal vs background · raw

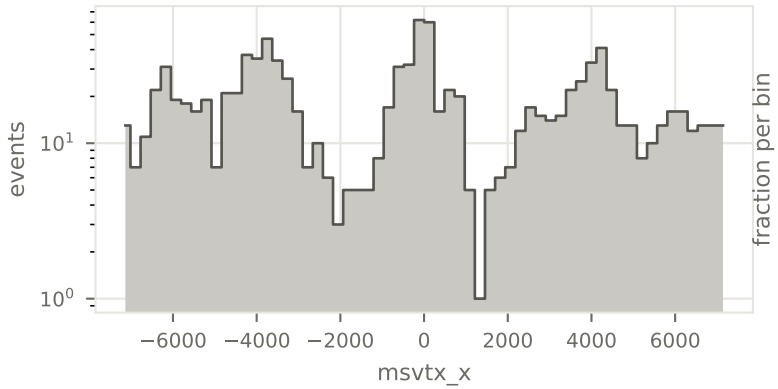



2b · combined signal vs background · normalized

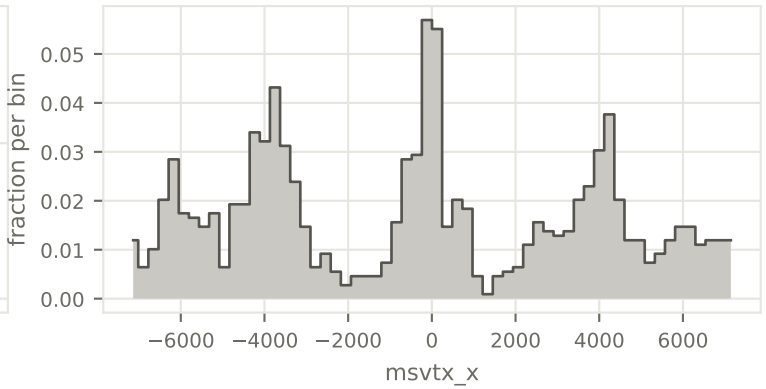



2 · combined signal
background n 1,089
combined, pooled D 0.098 n 4,987
equal weight per mass D 0.087

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0% −999 0.0%
background n 1,089

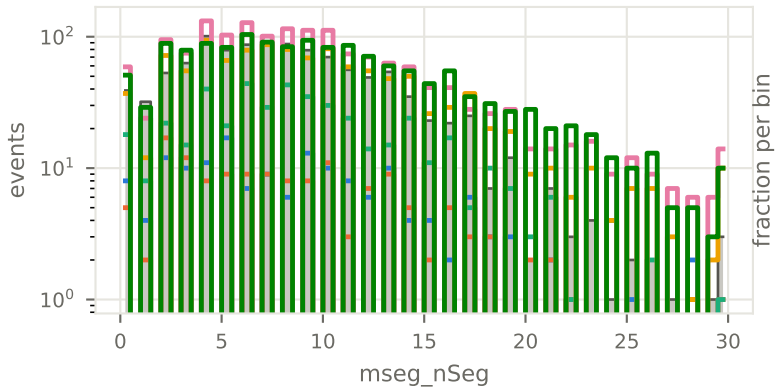
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

# mseg_nSeg

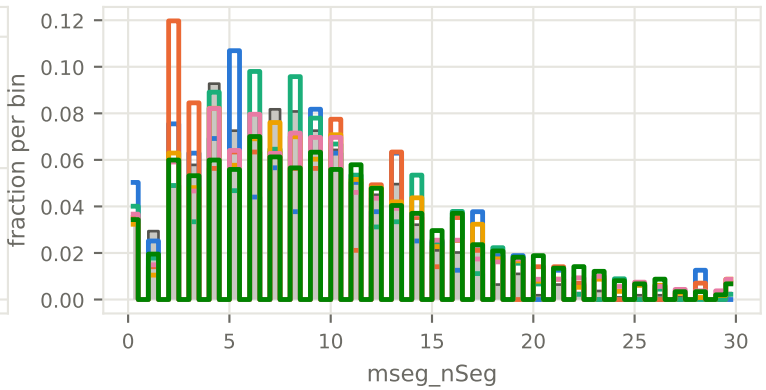
region: BARREL · rank 75 / 98

D_equal 0.071 (noise 0.051) D_pooled 0.089 min 0.053 (mH125_mS15) max 0.126 (mH1000_mS275) bkg n = 1,089 sig n = 4,987

1a · each mass point vs background · raw

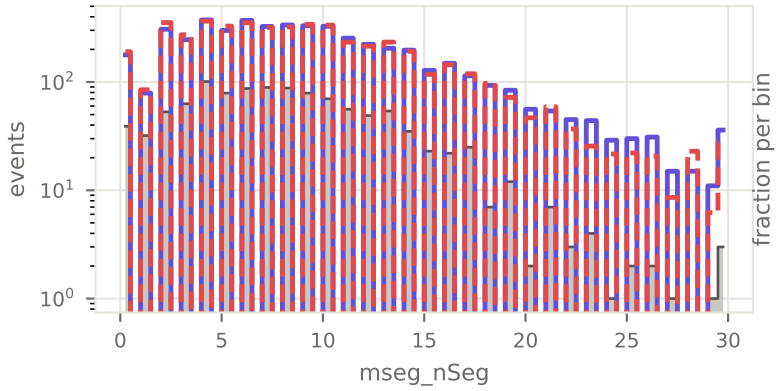



1b · each mass point vs background · normalized

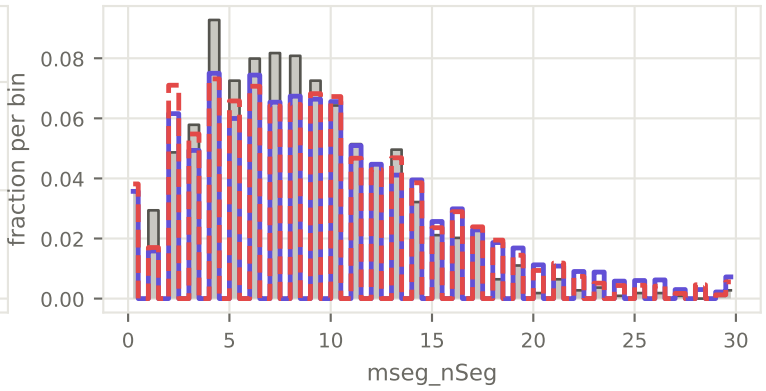



1 · each mass point
background n 1,089
mH125_mS15 D 0.053 n 159
mH125_mS55 D 0.082 n 142
mH200_mS50 D 0.065 n 449
mH400_mS100 D 0.083 n 1,144
mH600_mS150 D 0.077 n 1,608
mH1000_mS275 D 0.126 n 1,485

2a · combined signal vs background · raw

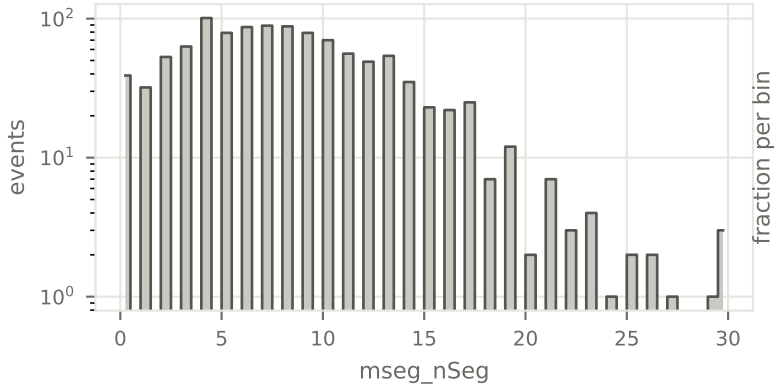



2b · combined signal vs background · normalized

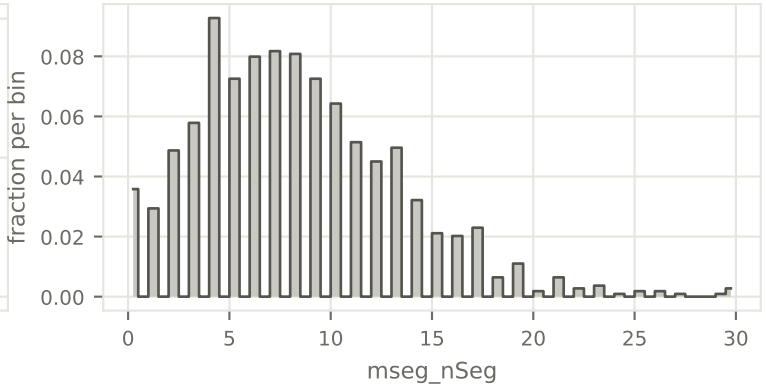



2 · combined signal
background n 1,089
combined, pooled D 0.089 n 4,987
equal weight per mass D 0.071

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0% −999 0.0%
background n 1,089

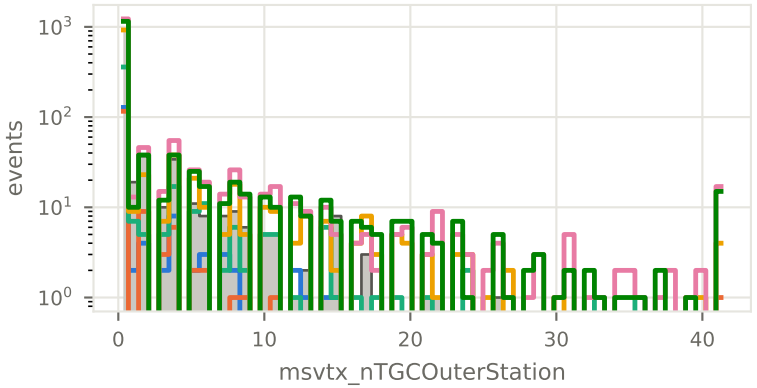
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

# msvtx_nTGCOuterStation

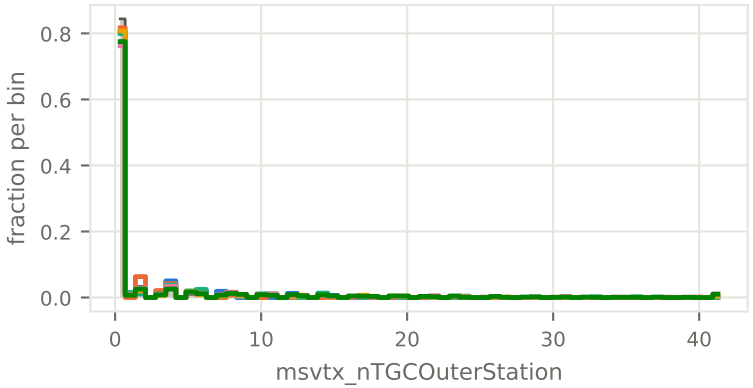
region: BARREL · rank 4 / 98

D_equal 0.063 (noise 0.051) D_pooled 0.080 min 0.044 (mH125_mS55) max 0.096 (mH600_mS150) bkg n = 1,089 sig n = 4,987

1a · each mass point vs background · raw

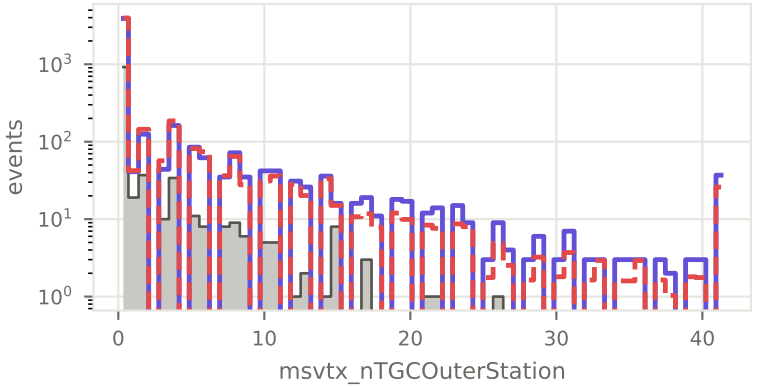



1b · each mass point vs background · normalized

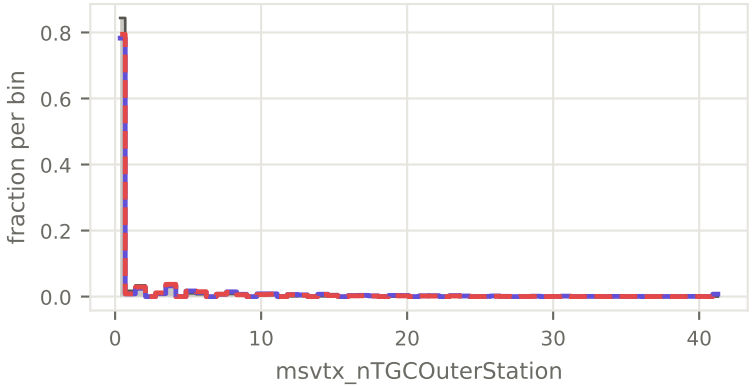



1 · each mass point
background n 1,089
mH125_mS15 D 0.053 n 159
mH125_mS55 D 0.044 n 142
mH200_mS50 D 0.069 n 449
mH400_mS100 D 0.063 n 1,144
mH600_mS150 D 0.096 n 1,608
mH1000_mS275 D 0.095 n 1,485

2a · combined signal vs background · raw

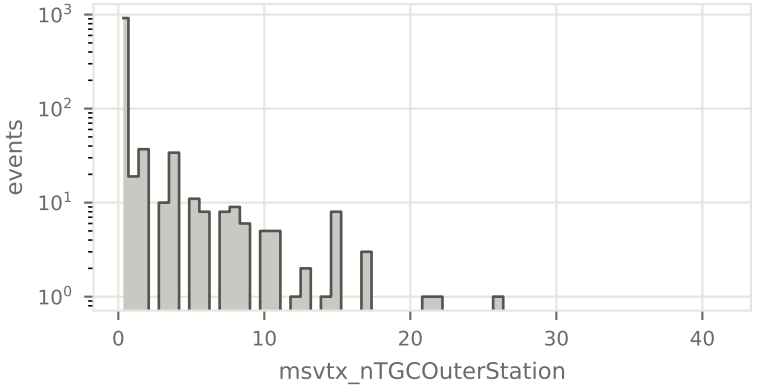



2b · combined signal vs background · normalized

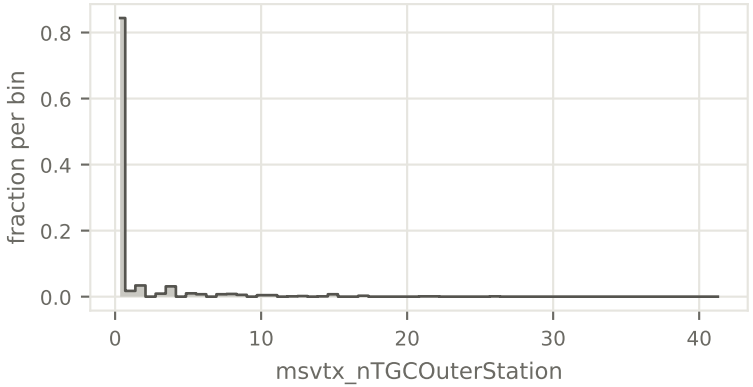



2 · combined signal
background n 1,089
combined, pooled D 0.080 n 4,987
equal weight per mass D 0.063

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0% −999 0.0%
background n 1,089

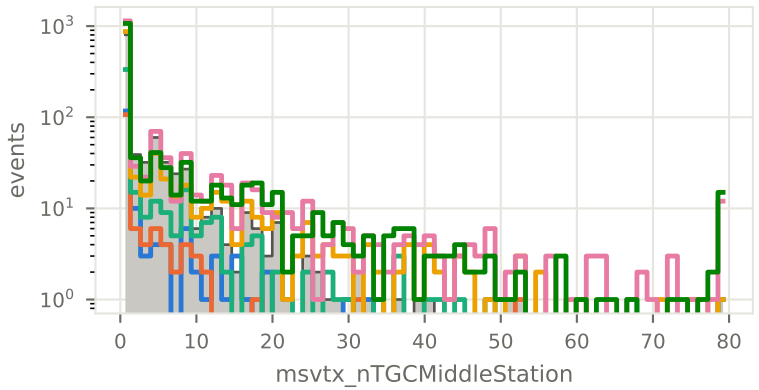
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

# msvtx_nTGCMiddleStation

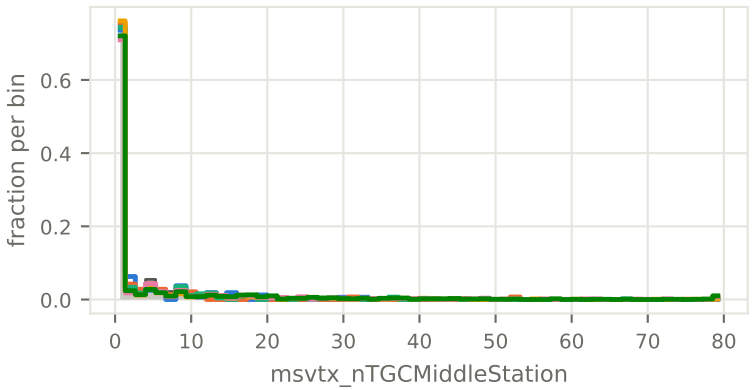
region: BARREL · rank 13 / 98

D_equal 0.059 (noise 0.051) D_pooled 0.081 min 0.045 (mH125_mS15) max 0.099 (mH1000_mS275) bkg n = 1,089 sig n = 4,987

1a · each mass point vs background · raw

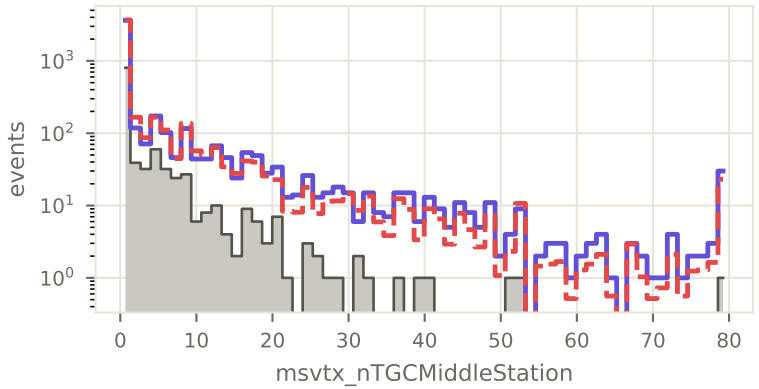



1b · each mass point vs background · normalized

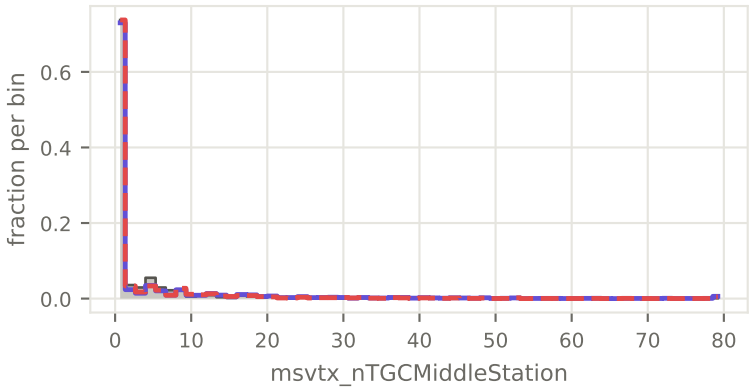



1 · each mass point
background n 1,089
mH125_mS15 D 0.045 n 159
mH125_mS55 D 0.061 n 142
mH200_mS50 D 0.056 n 449
mH400_mS100 D 0.061 n 1,144
mH600_mS150 D 0.096 n 1,608
mH1000_mS275 D 0.099 n 1,485

2a · combined signal vs background · raw

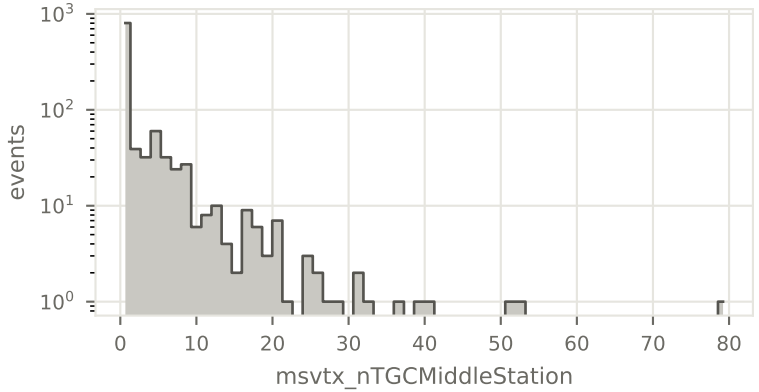



2b · combined signal vs background · normalized

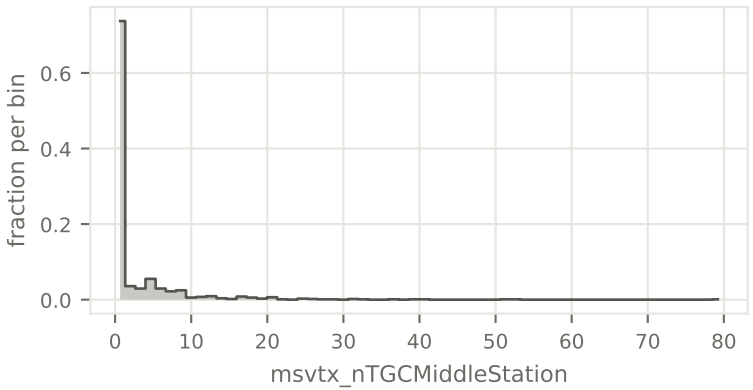



2 · combined signal
background n 1,089
combined, pooled D 0.081 n 4,987
equal weight per mass D 0.059

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0% −999 0.0%
background n 1,089

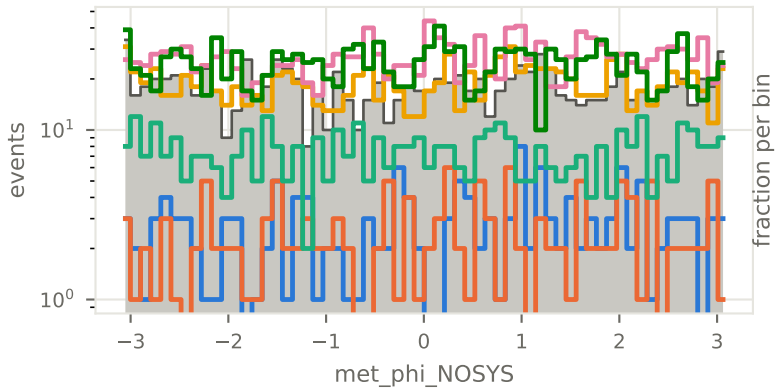
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

# met_phi_NOSYS

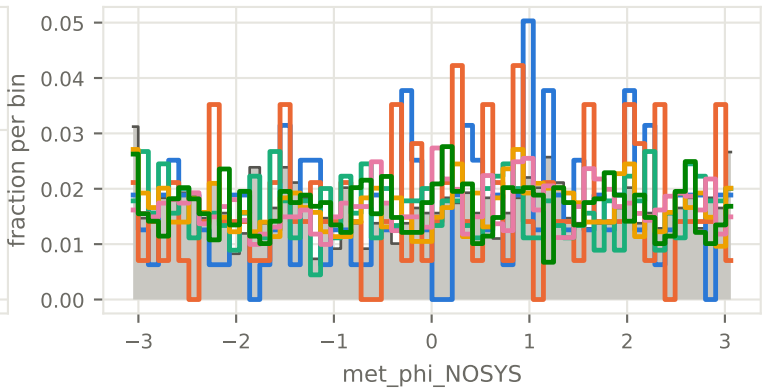
region: BARREL · rank 92 / 98

D_equal 0.041 (noise 0.051) D_pooled 0.036 min 0.028 (mH400_mS100) max 0.080 (mH125_mS15) bkg n = 1,089 sig n = 4,987

1a · each mass point vs background · raw

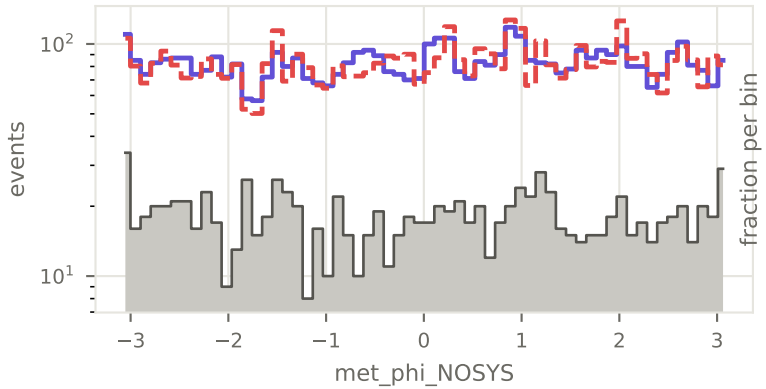



1b · each mass point vs background · normalized

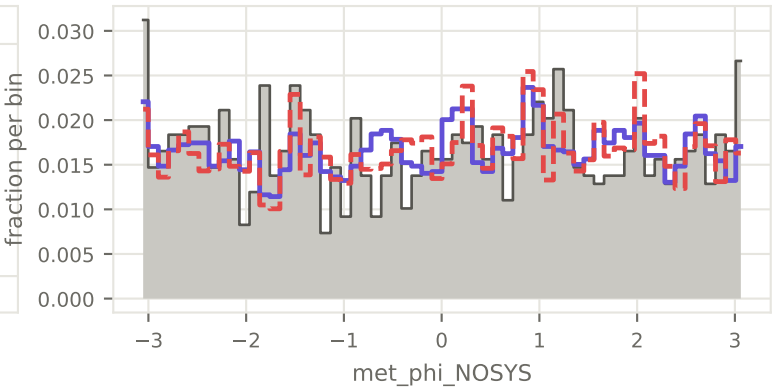



1 · each mass point
background n 1,089
mH125_mS15 D 0.080 n 159
mH125_mS55 D 0.070 n 142
mH200_mS50 D 0.030 n 449
mH400_mS100 D 0.028 n 1,144
mH600_mS150 D 0.054 n 1,608
mH1000_mS275 D 0.029 n 1,485

2a · combined signal vs background · raw

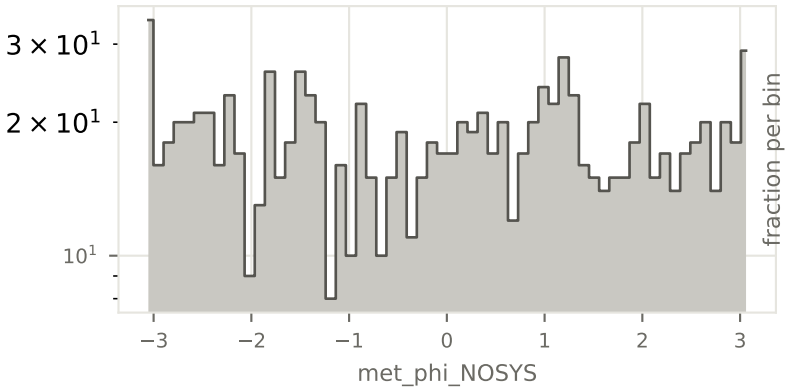



2b · combined signal vs background · normalized

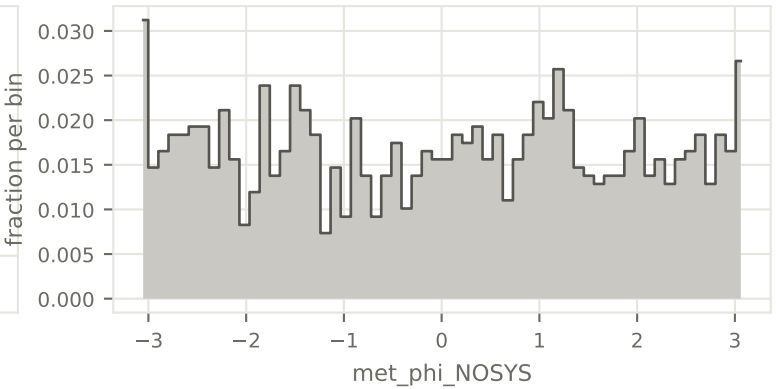



2 · combined signal
background n 1,089
combined, pooled D 0.036 n 4,987
equal weight per mass D 0.041

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0% −999 0.0%
background n 1,089

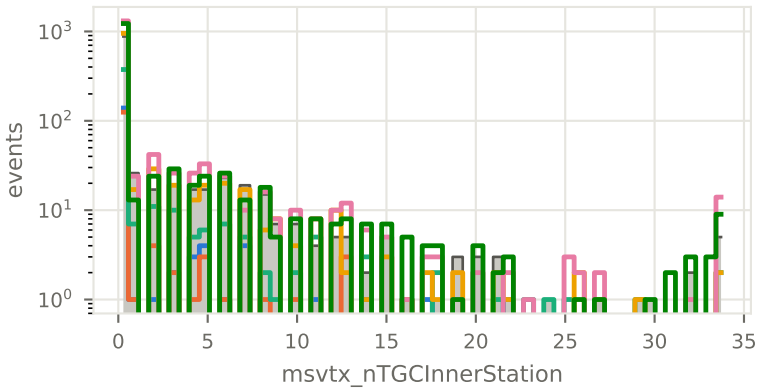
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

# msvtx_nTGCInnerStation

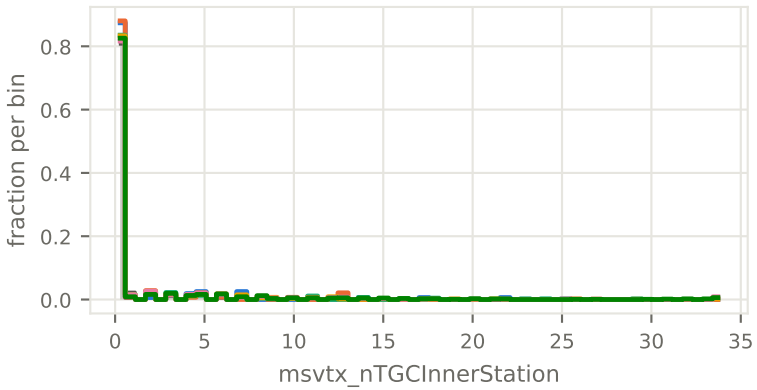
region: BARREL · rank 89 / 98

D_equal 0.038 (noise 0.051)  D_pooled 0.022  min 0.016 (mH600_mS150)  max 0.074 (mH125_mS55)  bkg n = 1,089  sig n = 4,987

1a · each mass point vs background · raw

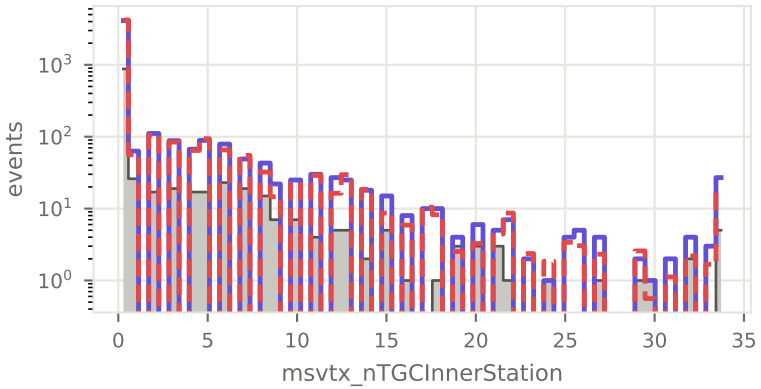



1b · each mass point vs background · normalized

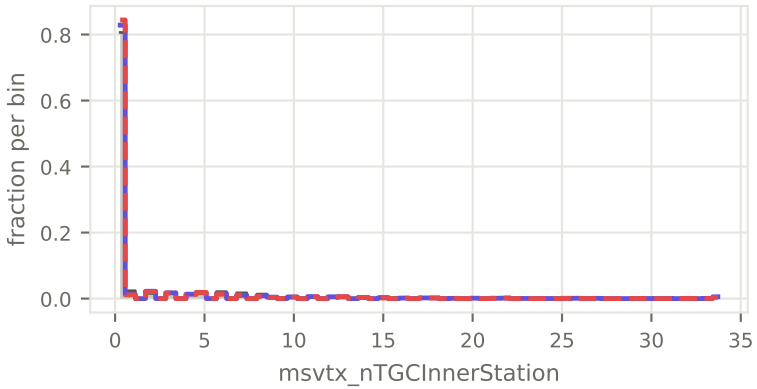



1 · each mass point
background n 1,089
mH125_mS15 D 0.068 n 159
mH125_mS55 D 0.074 n 142
mH200_mS50 D 0.034 n 449
mH400_mS100 D 0.028 n 1,144
mH600_mS150 D 0.016 n 1,608
mH1000_mS275 D 0.019 n 1,485

2a · combined signal vs background · raw

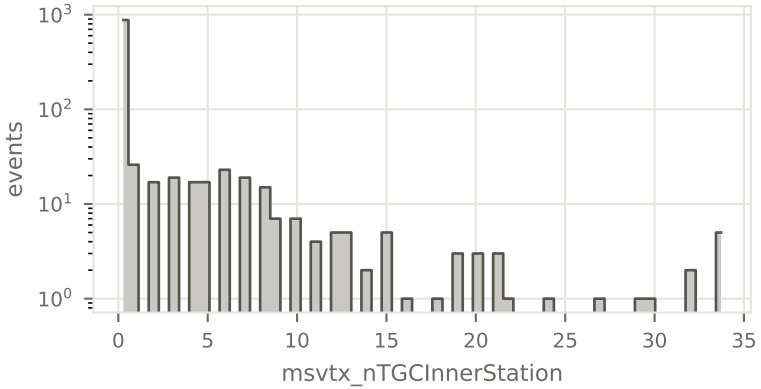



2b · combined signal vs background · normalized

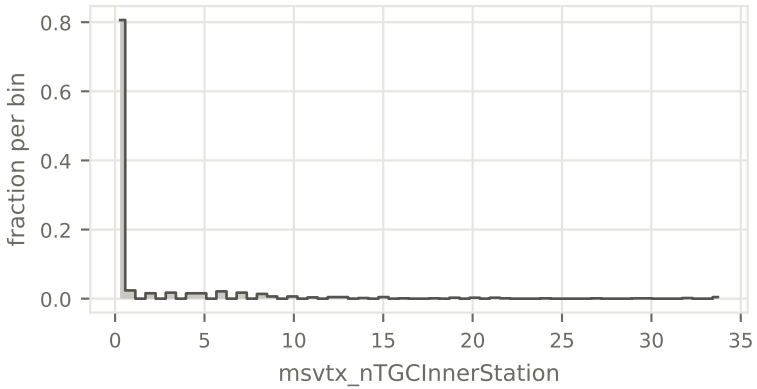



2 · combined signal
background n 1,089
combined, pooled D 0.022 n 4,987
equal weight per mass D 0.038

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0% −999 0.0%
background n 1,089

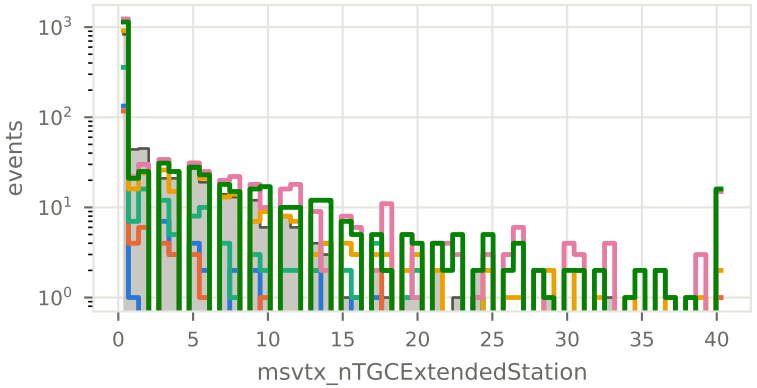
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

# msvtx_nTGCExtendedStation

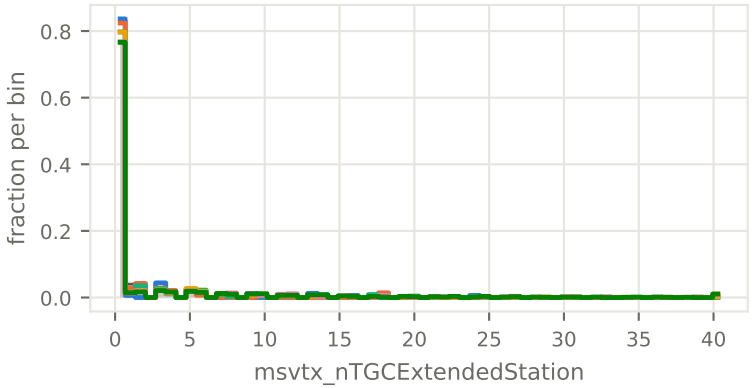
region: BARREL · rank 7 / 98

D_equal 0.034 (noise 0.051) D_pooled 0.041 min 0.033 (mH400_mS100) max 0.072 (mH125_mS15) bkg n = 1,089 sig n = 4,987

1a · each mass point vs background · raw

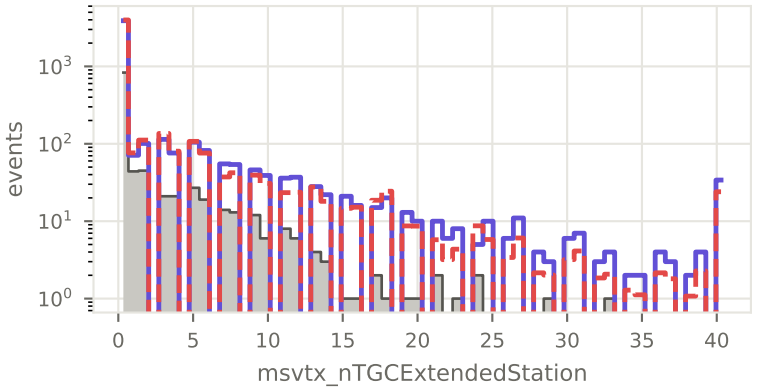



1b · each mass point vs background · normalized

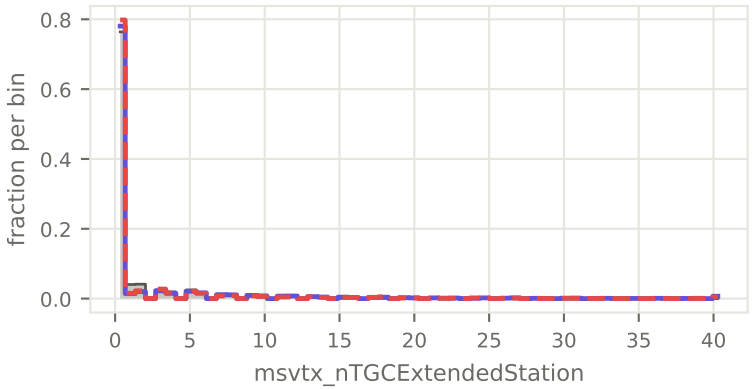



1 · each mass point
background n 1,089
mH125_mS15 D 0.072 n 159
mH125_mS55 D 0.060 n 142
mH200_mS50 D 0.033 n 449
mH400_mS100 D 0.033 n 1,144
mH600_mS150 D 0.056 n 1,608
mH1000_mS275 D 0.060 n 1,485

2a · combined signal vs background · raw

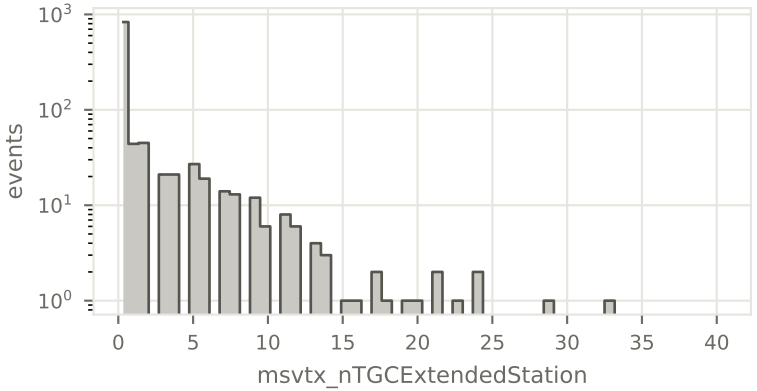



2b · combined signal vs background · normalized

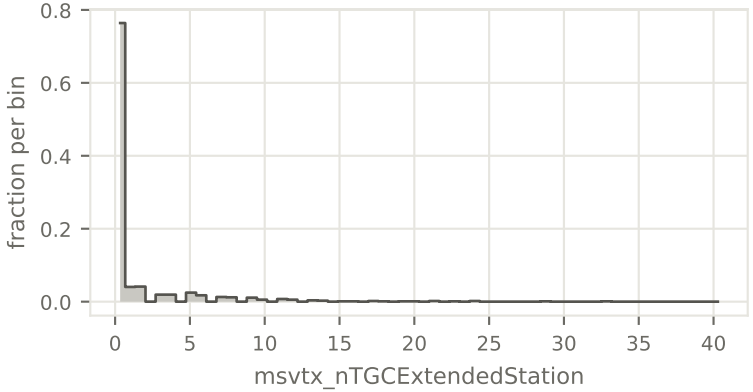



2 · combined signal
background n 1,089
combined, pooled D 0.041 n 4,987
equal weight per mass D 0.034

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0% −999 0.0%
background n 1,089

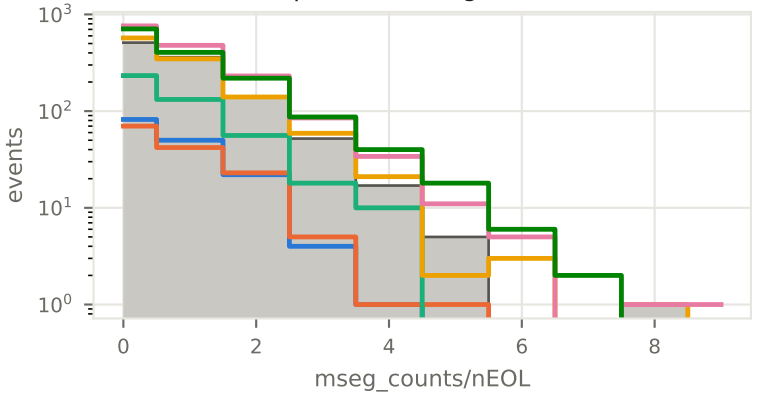
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

# mseg_counts/nEOL

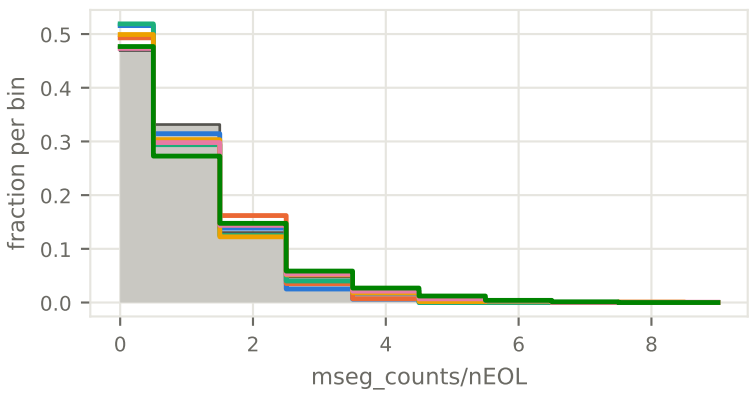
region: BARREL · rank 87 / 98

D_equal 0.027 (noise 0.051) D_pooled 0.023 min 0.024 (mH125_mS55) max 0.051 (mH1000_mS275) bkg n = 1,089 sig n = 4,987

1a · each mass point vs background · raw

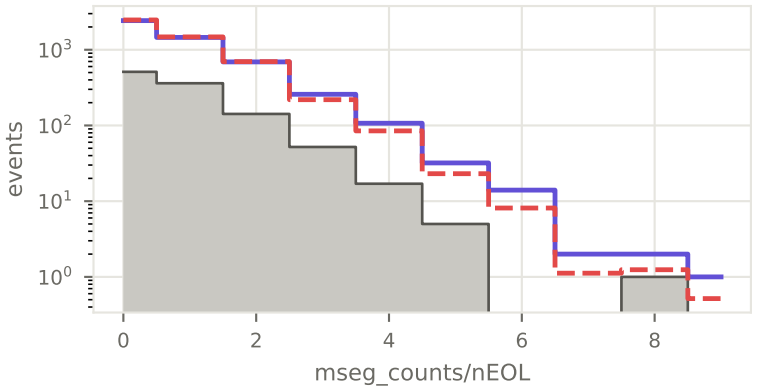



1b · each mass point vs background · normalized

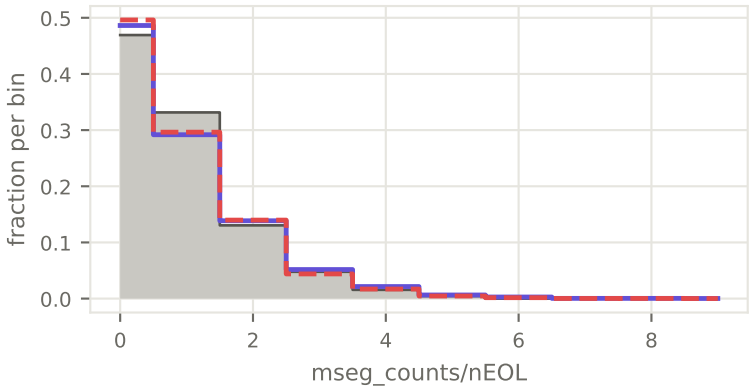



1 · each mass point
background n 1,089
mH125_mS15 D 0.046 n 159
mH125_mS55 D 0.024 n 142
mH200_mS50 D 0.050 n 449
mH400_mS100 D 0.030 n 1,144
mH600_mS150 D 0.030 n 1,608
mH1000_mS275 D 0.051 n 1,485

2a · combined signal vs background · raw

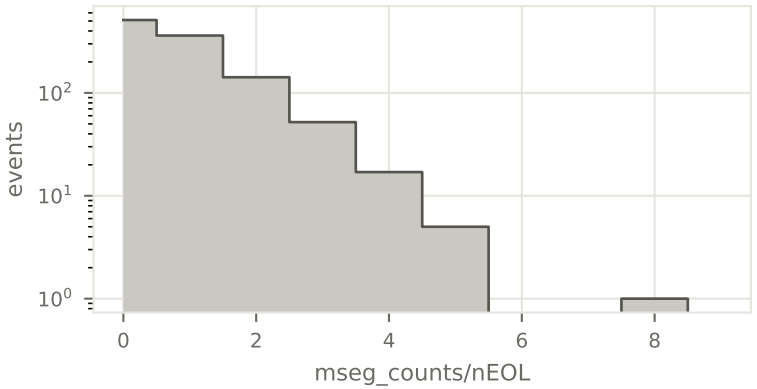



2b · combined signal vs background · normalized

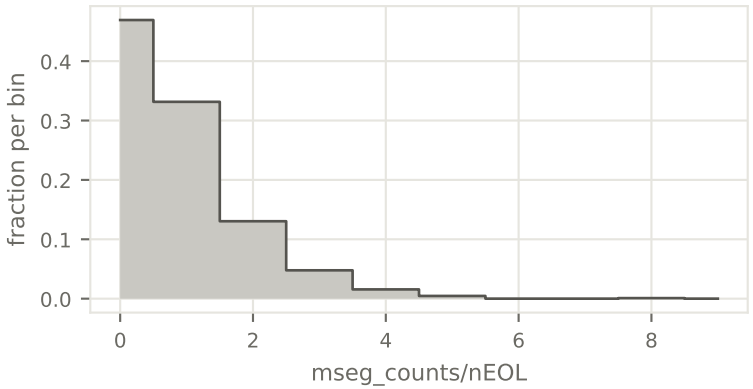



2 · combined signal
background n 1,089
combined, pooled D 0.023 n 4,987
equal weight per mass D 0.027

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0% −999 0.0%
background n 1,089

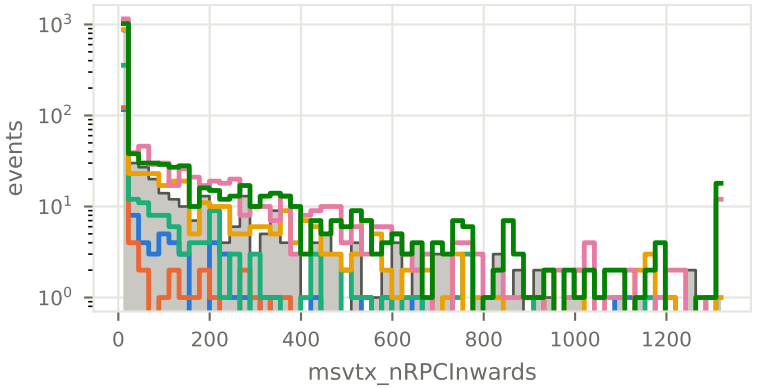
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

# msvtx_nRPCInwards

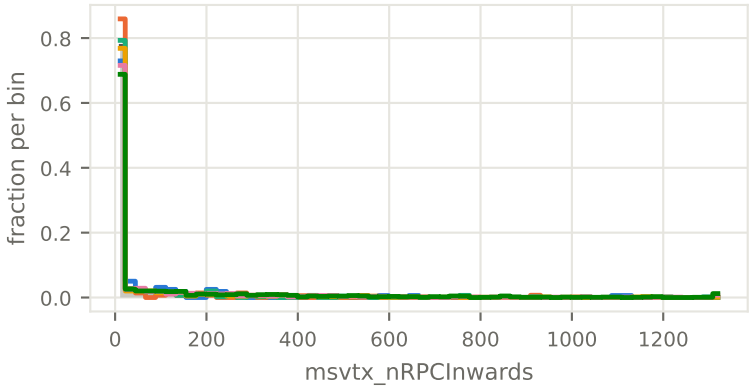
region: BARREL · rank 77 / 98

D_equal 0.024 (noise 0.051)  D_pooled 0.052  min 0.022 (mH400_mS100)  max 0.097 (mH1000_mS275)  bkg n = 1,089  sig n = 4,987

1a · each mass point vs background · raw

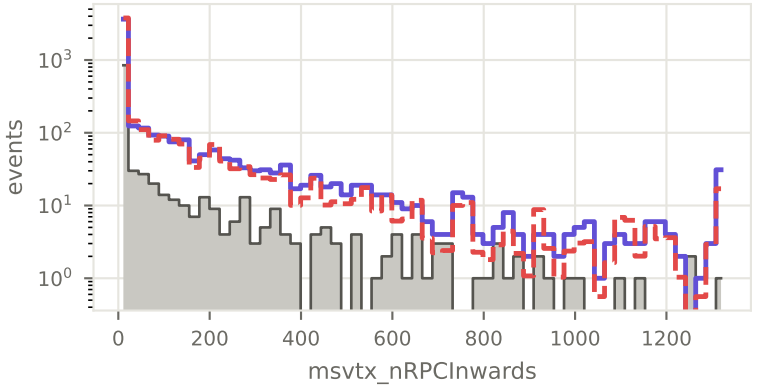



1b · each mass point vs background · normalized

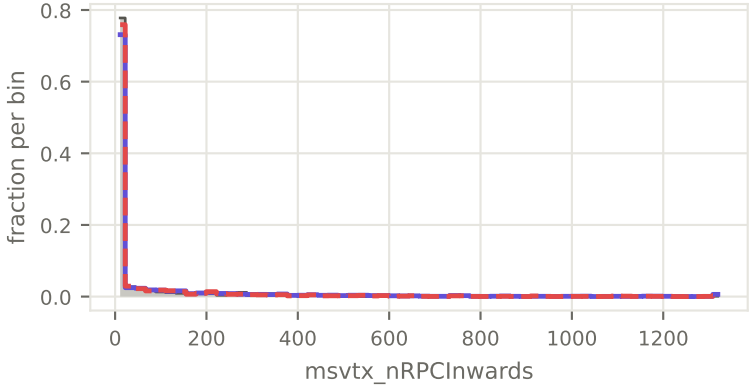



1 · each mass point
background n 1,089
mH125_mS15 D 0.060 n 159
mH125_mS55 D 0.097 n 142
mH200_mS50 D 0.030 n 449
mH400_mS100 D 0.022 n 1,144
mH600_mS150 D 0.069 n 1,608
mH1000_mS275 D 0.097 n 1,485

2a · combined signal vs background · raw

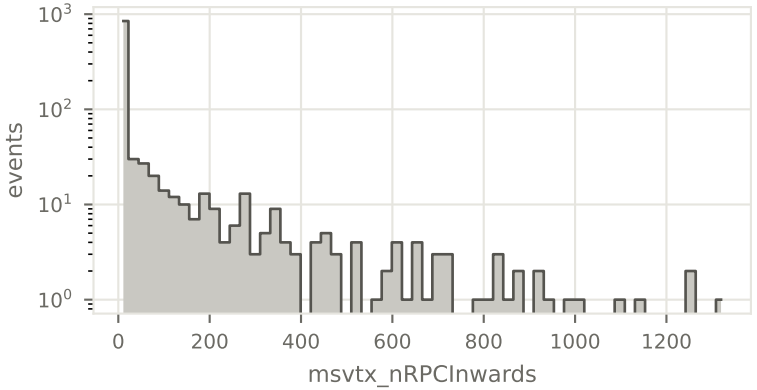



2b · combined signal vs background · normalized

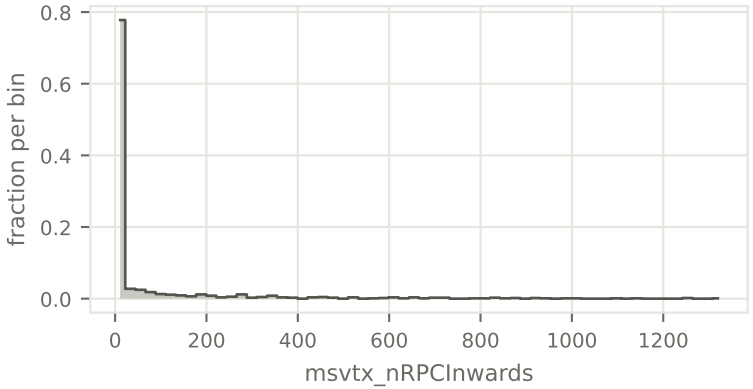



2 · combined signal
background n 1,089
combined, pooled D 0.052 n 4,987
equal weight per mass D 0.024

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0% −999 0.0%
background n 1,089

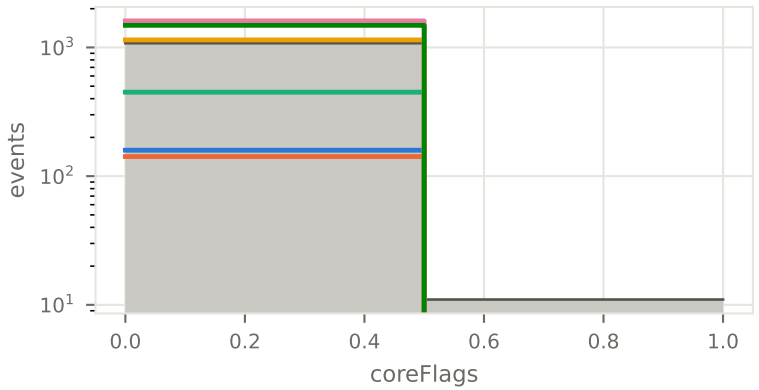
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

# coreFlags

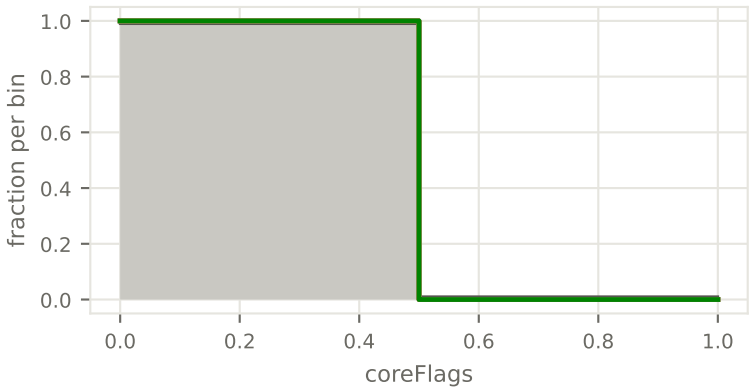
region: BARREL · rank 93 / 98

D_equal 0.010 (noise 0.051) D_pooled 0.010 min 0.010 (mH400_mS100) max 0.010 (mH1000_mS275) bkg n = 1,089 sig n = 4,987

1a · each mass point vs background · raw

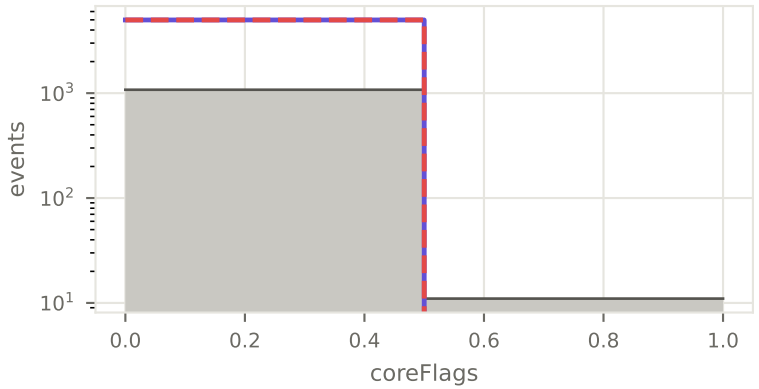



1b · each mass point vs background · normalized

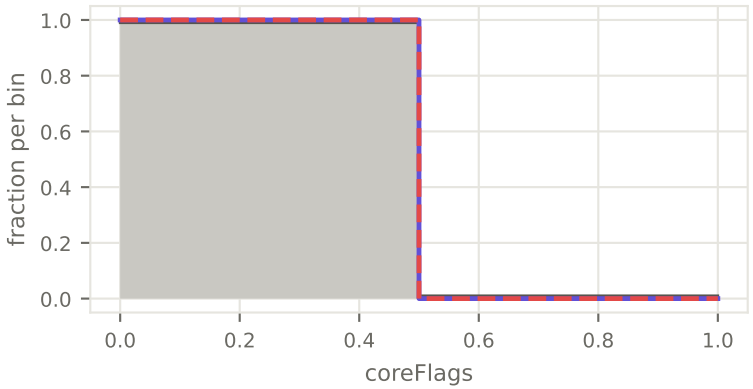



1 · each mass point
background n 1,089
mH125_mS15 D 0.010 n 159
mH125_mS55 D 0.010 n 142
mH200_mS50 D 0.010 n 449
mH400_mS100 D 0.010 n 1,144
mH600_mS150 D 0.010 n 1,608
mH1000_mS275 D 0.010 n 1,485

2a · combined signal vs background · raw

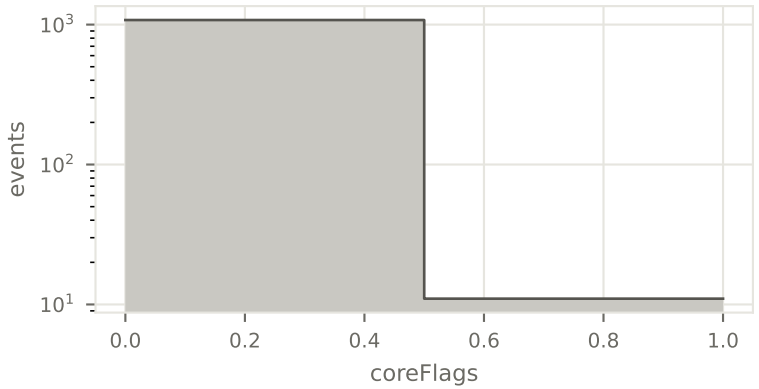



2b · combined signal vs background · normalized

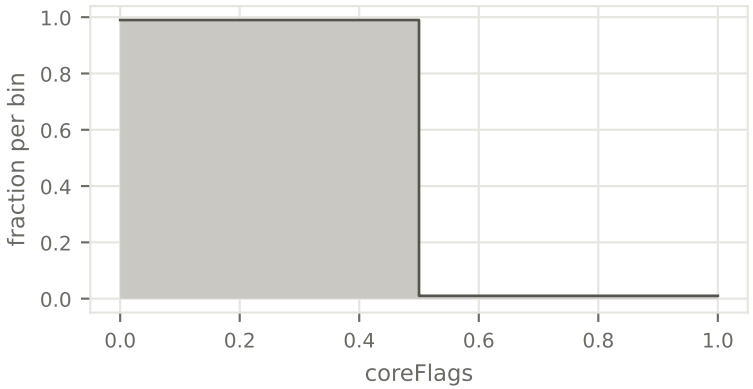



2 · combined signal
background n 1,089
combined, pooled D 0.010 n 4,987
equal weight per mass D 0.010

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0% −999 0.0%
background n 1,089

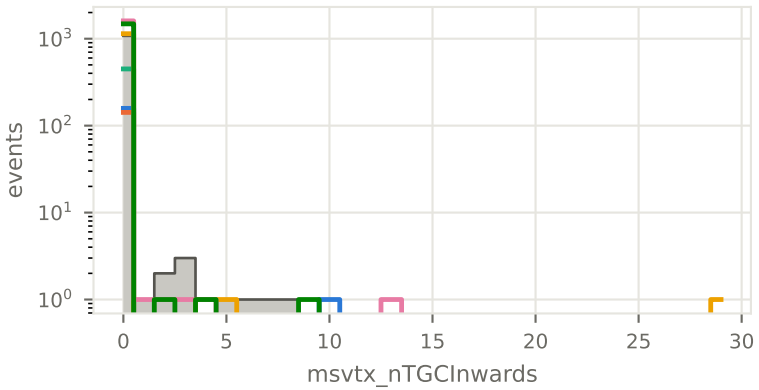
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

# msvtx_nTGCInwards

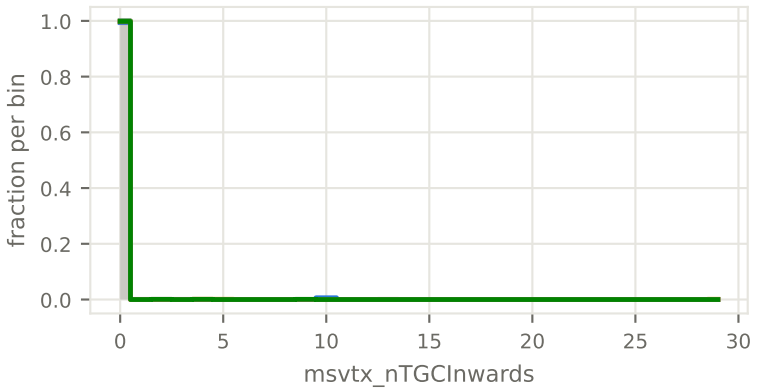
region: BARREL · rank 90 / 98

D_equal 0.007 (noise 0.051) D_pooled 0.007 min 0.006 (mH600_mS150) max 0.009 (mH125_mS55) bkg n = 1,089 sig n = 4,987

1a · each mass point vs background · raw

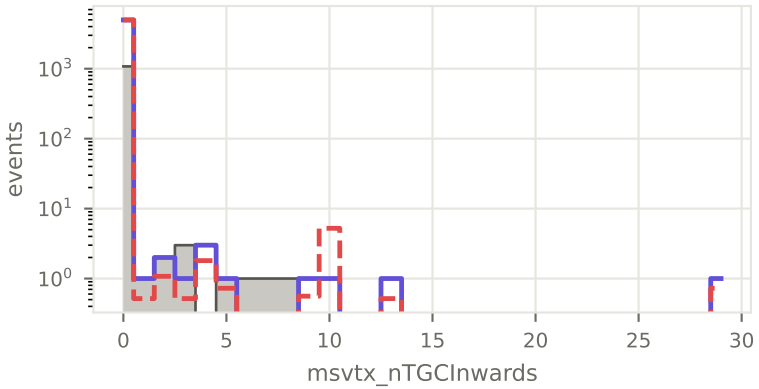



1b · each mass point vs background · normalized

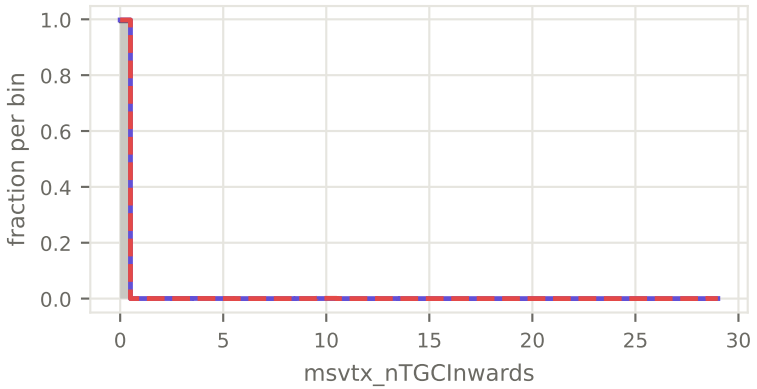



1 · each mass point
background n 1,089
mH125_mS15 D 0.006 n 159
mH125_mS55 D 0.009 n 142
mH200_mS50 D 0.009 n 449
mH400_mS100 D 0.007 n 1,144
mH600_mS150 D 0.006 n 1,608
mH1000_mS275 D 0.007 n 1,485

2a · combined signal vs background · raw

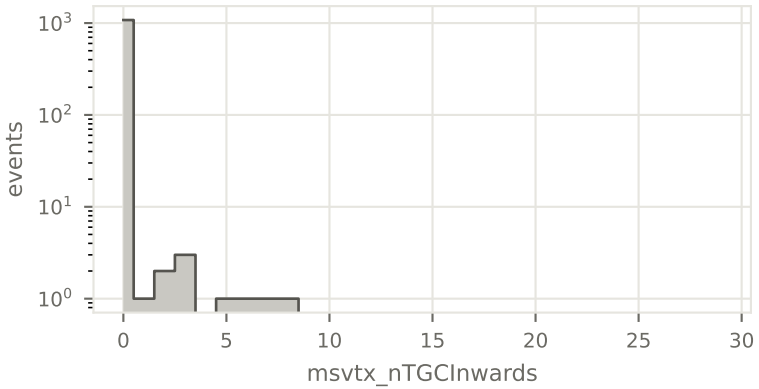



2b · combined signal vs background · normalized

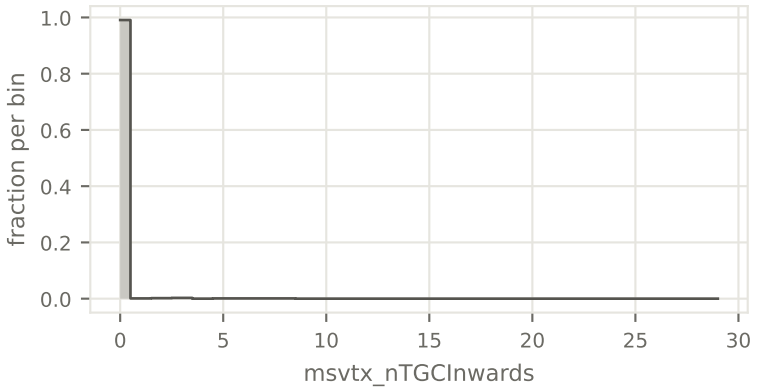



2 · combined signal
background n 1,089
combined, pooled D 0.007 n 4,987
equal weight per mass D 0.007

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0% −999 0.0%
background n 1,089

a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed